A Survey and Reader's Guide to UGP Physics

Programme Overview, Paper Summaries, and Reading Paths

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Abstract

This survey covers the UGP Physics papers P01–P55, providing a one-paragraph summary of each paper, a thematic grouping into ten natural clusters, an annotated logical dependency graph showing how the UGP results sit within the broader **Reflexive Reality** programme, reading paths for different research backgrounds, and the five-status epistemic taxonomy (A_{Lean}, A_{MDL}, A/D, B, D) used throughout the corpus. A comparative assessment of the GTE framework against ten alternative programmes in fundamental physics is provided in P53 [31]. A formal synthesis of GTE and NEMS addressing consciousness, the quantum measurement problem, and the hard problem of philosophy of mind is provided in P54 [28]. The bridge between the GTE arithmetic programme and the division-algebra programme (Dixon–Furey–Baez–Huerta), showing that $QR(7) = \{1, 2, 4\}$ is simultaneously the GTE arithmetic anchor and the Fano-plane orientation that fixes the octonion multiplication table, is established in P55 [45].

UGP Physics is the *quantitative branch* of the Reflexive Reality programme, sibling to NEMS. Where the NEMS papers establish *why* the Standard Model structure is the only consistent option, the UGP papers ask *what numerical values* the masses, couplings, and quantum numbers must take — and derive them from a single uniquely selected integer seed via a deterministic arithmetic cascade. Papers P30–P52 extend this programme into the electroweak mixing angle, CKM mixing, QCD structure, quantum mechanics, emergent gravity from the Φ_{MDL} continuum field, a complete perturbative quantum-gravity description, the full 3+1D architecture via the Three-Tape CMCA, first-principles cosmological predictions, a synthesis monograph unifying the full derivation chain, a condensed matter realization of the GTE polynomial as a 7-state spin lattice model, and the polynomial certificate of transputation, which classifies quantum adjudication at exactly the halting degree. Two primary falsifiable predictions are derived: the tensor-to-scalar ratio $r = 0$ (testable at LiteBIRD) and the PMNS Dirac CP phase $\delta_{CP} = 205.71°$. This guide does not require prior familiarity with UGP or Reflexive Reality; it can be read as the first paper in the series.

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1 Introduction

1.1 The Reflexive Reality Programme

The **Reflexive Reality** research programme is a sustained formal investigation into what a perfectly self-contained universe must look like — one whose laws, constants, and structure arise entirely from within, with nothing externally imposed. Its scope spans physics, chemistry, biology, evolutionary systems, computational systems, cognitive systems, and AI.

The Reflexive Reality programme comprises two branches. **NEMS** (No External Model Selection) formalizes the "no outside" constraint and machine-checks its theorem-grade consequences in Lean 4 with zero `sorry` [43]. The **UGP Physics programme** is the quantitative branch of Reflexive Reality: it focuses on the mechanics and dynamics of reflexive systems, deriving the specific generative structures that produce the Standard Model parameter spectrum and beyond. The UGP programme inherits from NEMS, adopting Perfect Self-Containment (PSC) self-consistency and the NEMS closure axioms, and is documented in the present corpus of papers.

The central NEMS results include:

- The Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is *forced* by Layer I consistency constraints; three generations are *selected* by a minimality principle in Layer II (Two-Layer PSC Theorem) [49].
- The Born rule is the unique closure fixed point of the probability assignment problem, not a postulate [54].
- The arrow of time is a structural consequence of stable physical records, preceding thermodynamic arguments [54].
- Black hole information, holographic encoding, and FTL non-signalling are each constrained by the same diagonal machinery [54].

The full NEMS programme comprises 94 papers (P00–P93) archived on Zenodo; both programmes are catalogued at `https://www.novaspivack.com/research` [57] and described for physics audiences in NEMS Paper 07 [42] and in the flagship survey [54].

1.2 UGP Physics: The Quantitative Branch

UGP Physics is the quantitative branch of the Reflexive Reality programme. Where the NEMS/PSC papers establish *why* the Standard Model gauge structure is the only consistent option, the UGP papers ask the next question: *what numerical values* must the masses, couplings, mixing angles, and quantum numbers take?

The answer comes from the **Universal Generative Principle (UGP)**: a deterministic number-theoretic framework that, starting from one foundational principle (PSC — Perfect Self-Containment) with MDL as its operational expression (Presentation Invariance, Layer II PSC axiom; PI ≡ MDL), uniquely selects a single integer seed at ridge level n=10 via a rigid arithmetic cascade, and from that seed generates the Generative Triple Evolution (GTE) — canonical integer triples $(a, b, c;\ g)$ for every Standard Model fermion. These triples encode masses (via the Universal Calibration Law, UCL), charges and quantum numbers (via the Braid Atlas), and are tethered to the NEMS/PSC foundations through the Unified Rigidity Theorem.

Physical substrate. The universe's physical substrate is the Φ_{MDL} continuum field (Level 2). Discrete objects—f_{MDL}, Rule 110, and the Chiral Minkowski CA—are Level-1 *algebraic certificates*: they prove and compress the internal arithmetic of Φ_{MDL} but are not the physical universe (P43 no-CA-replica theorem). Paper P34 formalises the shared abstract architecture as the Möbius triple $(A, e, [D])$ with D1–D5 constraints; see §D.2 in the narrative overview.

The programme combines Lean 4 machine-checked proofs, Python numerical verification, and PSLQ integer-relation searches. Each of its 53 papers targets a specific technical audience; the interdependencies are not always obvious from reading any single paper.

What this guide is. A structured map of the UGP Physics corpus (P01–P55) for readers approaching it from any background. It does not require prior familiarity with UGP or Reflexive Reality; it can be read as the first paper.

What this guide is not. It contains no new theorems or derivations. It is a map.

Comparative assessment. A comparative assessment of the GTE framework against ten alternative programmes in fundamental physics — scoring each on selection principle, free-parameter economy, formal verification, and falsifiability — is provided in P53 [31].

2 The Epistemic Taxonomy

Every quantitative claim across the programme is classified into one of five statuses. Understanding this taxonomy is prerequisite to reading any paper.

Five-status claim taxonomy (used throughout the programme)

$\mathrm{A}_{\mathrm{Lean}}$ (plain A): Lean 4 machine-checked arithmetic or structural identity. Zero `sorry`. Standard Mathlib axiom signature. The programme's strongest claim type.

$\mathrm{A}_{\mathrm{MDL}}$: MDL-uniqueness certificate within a declared UGP-structural expression class. Not Lean-derived from invariants. Example: the Higgs quartic $\lambda = \varphi/(4\pi)$ is the unique MDL = 4 expression in the declared atom basis.

A/D: A physics bridge, external scale, or partial derivation remains. The arithmetic component may be Lean-certified but the full physical identification is not. Example: the 0.295% charged-fermion mass spectrum uses the URC algebraic correction layer (A/D) on top of the $\mathrm{A}_{\mathrm{Lean}}$ structural pipeline.

B: Calibrated to data, or computationally/empirically supported. Example: baryon masses (15 binding parameters, QCD confinement placeholder).

D: Speculative or frontier. Example: the GTE-P7 mirror-branch dark-matter candidate.

3 Thematic Groups

3.1 Group 0 — Foundation (Start Here)

These three papers form the load-bearing core of the programme. Everything else cites at least one of them.

P01 — A Deterministic Number-Theoretic Framework for the Standard Model Parameter Spectrum [3]
https://doi.org/10.5281/zenodo.20168787
The flagship paper. Three axioms (locality, symmetry, MDL) select a unique integer seed via a rigid arithmetic cascade, generating the Category-A structural backbone of the SM parameter spectrum.
Seven headline results:

1. Lean-certified bare gauge couplings $g_1^2 = 16/125$, $g_2^2 = 2329/5400$, g_3^2
2. Blind $\alpha_s = 0.11822$ at $+0.24\sigma$ (43-day pre-commitment; PDG 2024)
3. Charged-fermion mass spectrum at 0.295% (0.261% vs PDG 2024)
4. Neutrino mass-squared ratio at 0.16σ from NuFIT 6.0 (zero anchors)
5. Cosmological constant at 0.31σ (Planck 2018)
6. Higgs quartic $\lambda = \varphi/(4\pi) \times (1 + \ln\varphi/(54\ln 2\pi))$ at -0.0003σ ($\mathbf{A}_{\text{Lean}}$, SRRG-corrected; $2c_H + 1 = 27 = N_{\text{gen}}^3$; P27, P35)
7. Lean-certified baryon triples and EW boson c-values

Higgs mass: bare $m_H = 124.2$ GeV, 9.1σ tension (PDG 2024); SRRG-corrected $m_H = 125.2499$ GeV at -0.0003σ ($\mathbf{A}_{\text{Lean}}$, `higgs_quartic_corrected`; $2c_H + 1 = 27 = N_{\text{gen}}^3$; see P27, P35); structural $v_{\text{PSC}} = 246.16$ GeV (-0.024%; `SrrgLean.VEVProof.*`, zero sorry). Tree-level m_W at $+36\sigma$ (blind falsification); recovers to -0.42σ of PDG 2024 with two-loop SM running. Contains the open-problem registry, five-status claim taxonomy, and preregistered falsification tests for JUNO, DUNE, CMB-S4, LEGEND-1000. *Read before anything else.*

P08 — From UGP to GTE: Prime-Locked Universes, Minimality, and the Emergence of Our World [71]
https://doi.org/10.5281/zenodo.20171002
Develops the UGP as a deterministic parameter-free arithmetic framework built on the duality between the Quarter-Lock Law (Necessity) and the Universal Windowed Cellular Automaton / UWCA (Contingency). Proves the Kernel Symmetry Theorem, Turing universality, Fibonacci lift rigidity ($|q_2 - q_1| = 13$, $F_{13} = 233$), and the Lawvere/Kleene self-reference theorems. The foundational monograph; more expansive than P01.

P05 — The Uniqueness of the Universal Generative Principle: A Two-Stage Proof from Arithmetic Minimality and Physical Consistency [14]
https://doi.org/10.5281/zenodo.20169932
Formal two-stage uniqueness proof: Stage 1 (arithmetic minimality, symmetry) identifies a small set of mathematically admissible universes; Stage 2 (physical instantiation condition derived from the Quarter-Lock identity and CODATA) proves the canonical UGP solution ($n = 10, b_1 = 73$) is the unique survivor. Non-circularity is explicit: δ_{target} is derived from CODATA independently of $b_1 = 73$, and the derived value selects $b_1 = 73$ to within 2.39 ppm.

3.2 Group 1 — Standard Model Physics Derivations

These papers derive specific SM sectors from the UGP framework. Together they constitute the main quantitative physics output.

P02 — The GTE Particle Spectrum at $n = 10$ [33]
https://doi.org/10.5281/zenodo.20170996
Large-scale discovery run generating 1,000,035 GTE candidates below 173 GeV. All 24 known SM particles rank in the Green (top) viability tier with 50-fold enrichment ($p < 10^{-4}$). Documents piecewise-linear hinge laws, oscillatory structure with period ∼100,000 steps, and 9 genuinely novel candidate states (GTE-P1, P2, P3, P6–P11; GTE-P4/P5 are path variants excluded from the novel register). The reference paper for canonical GTE-triple values.

P06 — Algebraic and Geometric Foundations: A Derivation of the Standard Model Gauge Group [5]
https://doi.org/10.5281/zenodo.20169945
Proves the Quarter-Lock Law as a theorem of the UGP invariant space and derives the Elegant Kernel algebraic constants from its symmetries. Performs a two-stage first-principles derivation of the SM gauge couplings: bare algebraic values g_1^2, g_2^2, g_3^2 from group-specific geometric invariants — all three Lean-certified in `UgpLean.Phase4.GaugeCouplings` (zero `sorry`) — then the Universal Instantiation Factor δ_{UGP} from the Principle of Invariant Restoration.

P17 — The Topology of the Standard Model from First Principles: A Derivation of the Canonical Braid Atlas [12]
https://doi.org/10.5281/zenodo.20169984
Models fundamental particles as topologically stable braided worldlines on a discrete substrate. The $T/T^\dagger$ dual-operator evolution distinguishes matter from antimatter and chiral from vector-like couplings. Electric charge $Q = W_g/N_c$ is machine-checked (`BraidAtlas.ChargeTheorem`, zero `sorry`); anomaly cancellation certifies $N_c = 3$; nine light-baryon triples are Lean-derived (`BraidAtlas.CompositeTriples`).

P18 — The Koide Relation as a Cyclotomic-12 Closed Form [4]
https://doi.org/10.5281/zenodo.20168795
Promotes the Koide $Q = 2/3$ identity from a numerical curiosity to a structural statement. Derives an algebraic closed form predicting $m_\tau = 1.776969$ GeV against PDG 1.77686 GeV (61 ppm; 0.91σ). The surd coefficients satisfy cyclotomic-12 identities. Structural origin: the Koide phase $\theta = 2/9$ follows unconditionally from $N_c = 3$ via $\theta = (N_c^2-1)/(4N_c^2)$ (Lean-certified, `koide_angle_from_N_c_pure`). The relation admits the reformulation $\mathrm{CV}(\sqrt{m}) = 1$ (coefficient of variation of square-root masses equals unity), characterising the equal-irrep-norm condition $Q = (1 + \mathrm{CV}^2)/3 = 2/3$ iff $\mathrm{CV} = 1$ (CatAL, Lean-certified).

P19 — Cyclotomic-12 Structure in the Charged-Fermion Mass Spectrum [2]
https://doi.org/10.5281/zenodo.20168805
Two structural mass relations reducing nine charged-fermion masses to two empirical scale inputs: (1) TT relation $\log(m_{u_g}/m_{\ell_g}) = (\pi/6) \cdot 2^g + \pi/8$ ($\leq 0.37\%$ max error), whose coefficient $\pi/6$ is the A_2 Weyl-chamber half-angle (Lean-certified); (2) VV relation $\log m_{d_g} = (13/9)\log m_{u_g} - (7/6)\log m_{\ell_g} - 5/14$ with exact SU(5)/SO(10) group-theoretic origin (`VV_from_GUT_group_theory`, zero sorry). The VV coefficient $\alpha_d = 13/9$ also provides an improved Wolfenstein parameter estimate $\lambda \approx \varepsilon_1^{\alpha_d} = e^{-13\pi/27} \approx 0.2203$ (PDG 0.2245; 1.9% off — a factor 6.7× improvement over the 12% A_2-mismatch estimate; exploratory).

P21 — Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas [48]
https://doi.org/10.5281/zenodo.20170120
The first parameter-free structural prediction of $\Delta m^2_{21}/\Delta m^2_{31}$ in the literature: value 0.02936, matching NuFIT 6.0 (0.02951 ± 0.00098) to 0.16σ (0.52%; 1.58σ from PDG 2024 world average) with zero free parameters and normal ordering automatic (NuFIT 6.0 prefers NH at 2.5σ, consistent

with our structural prediction). Uses only $N_c = 3$ and Braid Atlas right-handed neutrino b-values $\{5, 11, 19\}$. The seesaw exponent 29/9 is the SO(10)/SO(10) gauge-matter representation defect (Lean-certified). Structural Dirac scale $E_D = v_H/29$ gives $\Sigma m_\nu = 59.4$ meV (P47 three-tape winding cascade), within the Planck bound. The arithmetic/algebraic chain from the FN texture to the mass-squared ratio is fully Lean-certified (zero sorry) in `ugp-lean` (`MassRelations.NeutrinoMassRatio`): FN charges $(q_1, q_2) = (3, 2)$ give exponent 29/9, M_R cancels algebraically, and the coarse bound $0.029 < R < 0.030$ is machine-checked.

P22 — The UGP Interaction Skeleton Theorem [66]
https://doi.org/10.5281/zenodo.20170132
Proves in Lean 4 (zero sorry) that the UGP winding, chirality, hypercharge, and color data determine the SM's finite renormalizable interaction skeleton: gauge-fermion vertices, Yukawa vertices, anomaly cancellation, color-singlet constraints, proton-stability forbiddance at dimension four, and a topological dark-sector isolation gap. Three independent proofs force $N_c = 3$; a 64-schema finite vertex audit returns zero mismatches.

P23 — Substrate Depth and Self-Generated Mass [64]
https://doi.org/10.5281/zenodo.20170140
Defines the *reflexive closure* RC $= (M - \Sigma m_i)/M$ (fraction of rest mass from internal dynamics) and shows that $\log_{10}|c|$ for the UGP canonical triple correlates with RC at $r = -0.913$ ($p = 1.3 \times 10^{-15}$, $n = 38$ composite hadrons). All 38 canonical hadron triples are Lean-derived (`BraidAtlas.CompositeTriples`, zero sorry). A random-triple null test (10,000 trials) finds zero cases reaching $|r| \geq 0.90$.

P29 — The Mirror Branch Braid Atlas: A Parameter-Free Dark Sector Classification [41]
https://doi.org/10.5281/zenodo.20263362
Extends the Braid Atlas to the mirror branch of the GTE spectrum, deriving a complete, parameter-free dark sector from first principles. **Lean-certified core results (zero axioms beyond the Braid Atlas charge theorem):**

1. $Q = 0$ for every dark-sector particle: established from Braid Atlas winding arithmetic with no free parameters
2. Structural identity $c(W) = b_2(\text{RHN}) = q_1$: unifies the dark-sector topology with right-handed neutrino and FN charge data (Lean-certified)

Physical predictions: Dark lepton masses 0.54 MeV, 24.5 MeV, 3600 MeV (three generations); gauge symmetry $\text{SU}(3)_{\text{dark}}$ only (EW symmetry absent by construction); GTE-P7 mass refined to 211.9 MeV; asymmetric dark matter yield formula $\eta_\chi \approx (2/7)\,\eta_{\text{dark}}$. *Companion paper:* P17 (Braid Atlas v2) provides the charge theorem; P22 (Interaction Skeleton) establishes the topological dark-sector isolation gap.

P31 — Arithmetic Derivation of the Electroweak Mixing Angle from Rule 110 Orbit Arithmetic [16]
https://doi.org/10.5281/zenodo.20319413
Derives $\sin^2\theta_W = 3/13 \approx 0.23077$ via a ten-step machine-certified chain from the palindrome decomposition of the 13 active f_{MDL} neighborhoods, conditional on one imported Lean result from P28. The predicted value lies -0.225% (13.0σ) below the PDG 2024 MS-bar value 0.23129 ± 0.00004 at M_Z; with the Wolfenstein radiative correction $\lambda^3/(2c_H)$ (P41) the corrected value $384729/1664000 \approx 0.23121$ matches PDG 2022 at 0.42σ ($\mathbf{A}_{\text{Lean}}$); the two-loop SM Higgs threshold with GTE-derived $\alpha_{\text{em}}(M_Z) = 1/128$ gives 0.23129154 ($+0.038\sigma$ PDG 2024, **B**). A companion GUT-scale formula $\sin^2\theta_W(M_{\text{GUT}}) = 3/8$ reproduces the SU(5) prediction exactly (Lean-certified). Both arithmetic inputs, $N_{\text{gen}} = 3$ and $c_H = 13$, are machine-certified from prior papers.

P32 — The CKM Wolfenstein Parameters from Generative Triple Evolution Orbit Arithmetic [19]

https://doi.org/10.5281/zenodo.20319415

Derives all four Wolfenstein parameters of the CKM quark mixing matrix from Rule 110 orbit arithmetic with zero free parameters. The primary prediction $\lambda = 9/40 = 0.225$ matches PDG 0.22500 to 0.000% (Lean-certified, zero sorry). The remaining parameters A, $\bar{\rho}$, $\bar{\eta}$ are derived from GTE quark-sector orbit indices $b_u, b_d, b_c, b_s, b_b, b_t$ identified with structural formulas in $N_{\text{gen}} = 3$, $N_{\text{fam}} = 5$, $c_H = 13$; no CKM unitarity-triangle input is used anywhere.

P33 — Deeper Consequences of Arithmetic Universality in the Standard Model [24]

https://doi.org/10.5281/zenodo.20319417

Derives five structural consequences of the SM-orbit/Rule 110 identification, all machine-certified in Lean 4 (zero sorry) except as noted. (1) The EW bosons W^+, Z, H^0 form a unit-step arithmetic staircase in GTE branch capacity $c \in \{11, 12, 13\}$, grounding the Higgs mechanism in cellular-automaton arithmetic. (2) f_{MDL} excludes output value $\mathbb{Z}_7 = 4$ (the W^-/e^- sector) from its output range; this exclusion is arithmetically equivalent to CP violation, making matter dominance a necessary consequence of MDL compression. (3) The SM generation orbit is the unique maximal Garden-of-Eden-rooted chain in $\mathbb{Z}_7^5$; $N_{\text{gen}} = 3$ is topologically forced by the causal isolation master theorem. (4) The photon is the unique uniform fixed point of f_{MDL} (the CA vacuum state, $\mathbb{Z}_7$ winding number 0); the Rule 110 ether is a distinct, dynamically active periodic background carrying $\mathbb{Z}_7$ winding number 1 (`ether_z7_sum_mod7`, $\mathbf{A}_{\text{Lean}}$), not the EM vacuum; the neutrino is the unique k=7 composite within the W_B=0 colorless sector. The complete W_B=0 colorless sector is discriminated from first principles: photon (label 0, unique fixed point), neutrino (label 1, unique k=7 composite), Z boson (label 2, c=12), Higgs (label 3, c=13) — all pairwise distinct (`z7_zero_sector_unified_discriminant`, `Z7ZeroSectorDiscriminant.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry). (5) The 14-neighborhood f_{MDL} catalog is an emission interaction kernel dual to the P22 absorption kernel, establishing a temporal symmetry of the SM vertex structure. Casimir anti-enhancement of +6.3% and SM orbit chirality (14/343 parity-violating triples) are computationally confirmed.

P35 — GTE Unification: A Bidirectional Correspondence between Rule 110 Orbit Arithmetic and the Standard Model Electroweak Sector [35]

https://doi.org/10.5281/zenodo.20319421

The electroweak unification capstone. Presents a bidirectional correspondence between four mathematical structures: GTE arithmetic over $\mathbb{Z}_7$, the MDL-minimal CA f_{MDL}, its continuum Φ_{MDL} field limit, and the Standard Model. The electroweak correspondence (six arrows) is packaged as the 7-conjunct machine-certified theorem `ugp_r110_sm_joint_unification` (Lean 4, zero sorry, zero new axioms). **Forward direction (**GTE $\to$ SM**):** $N_{\text{gen}} = 3$ (Garden-of-Eden orbit depth, machine-certified) generates via $N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}} = 5$ and $c_H = 13$ the electroweak predictions: $\sin^2\theta_W = 384729/1664000 \approx 0.23121$ (-0.42σ PDG 2022, $\mathbf{A}_{\text{Lean}}$), $\sin^2\theta_W(\text{GUT}) = 3/8$ (exact SU(5)), and Wolfenstein $\lambda = 9/40 = 0.225$ (0.000%). **Reverse direction (**SM $\to$ GTE**):** the observed $\sin^2\theta_W \approx 0.231$ uniquely determines $N_{\text{gen}} = 3$ via the arithmetic pivot $N_{\text{gen}} + N_{\text{fam}} = 2^{N_{\text{gen}}}$. Extensions: causal spectral dimension $d_s = 4$ (Lean-certified); QCD from $F_{21} \subset SU(3)$ with $\alpha_s(M_Z) = 0.1193$ at three loops; baryon-to-photon ratio $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$ from Planck 2018; FKTT mechanism, $\mathbf{A}_{\text{Lean}}$); parameter-free neutrino mass scale and dark-sector classification. Zero free dimensionless parameters throughout (single dimensional anchor: m_τ sets the energy scale). *Companion papers:* P31 (Weinberg angle), P32 (CKM), P33 (deeper SM), P39 (QCD).

P39 — QCD Structure from the Generative Triple Evolution Substrate: Asymptotic Freedom, Confinement, and Hadron Spectroscopy from $F_{21} \subset SU(3)$ [51]

The group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, identified as the $\Sigma(21)$ Frobenius subgroup of $SU(3)$, is forced by MDL minimality and Frobenius norm ratios (machine-certified, zero sorry, no free parameters). The $SU(3)$ adjoint decomposes as $\mathbf{8} = \mathbf{1}' \oplus \mathbf{1}'' \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$, reproducing the GTE colour-octet structure. All QCD colour factors $C_F = 4/3$, $C_A = 3$, $T_F = 1/2$ are machine-certified. One-loop β-function coefficient $b_0 = 7$ and two-loop $b_1 = 26$ follow from the F_{21} species count (Lean-certified); three-loop running gives $\alpha_s(M_Z) = 0.1193$ (+1.10%). Three independent arguments force $\theta_{\text{QCD}} = 0$ without a Peccei–Quinn axion (Lean-certified, ROBUST). The Φ_{MDL} kink condensate delivers zero-PDG-input hadron spectroscopy: $f_\pi^{\text{SCC}} = 92.34\,\text{MeV}$ (+0.30%), $m_\pi^\pm = 139.57\,\text{MeV}$ (−0.001%, $\mathbf{A}_{\text{Lean}}$; via GOR+condensate identification), η–η' mixing $\theta_P = -13.08° \pm 3.74°$ (within PDG range), $\chi_{\text{top}}^{1/4} = 166.5\,\text{MeV}$ (−6.4%). The QFT mass gap is established unconditionally from orbit arithmetic (Lean 4, zero sorry, zero axioms). The framework is an effective theory below the derived seven-kink full-winding threshold $\Lambda_{\text{GTE}} = 7M_{\text{kink}} \approx 2.0$ GeV (envelope 1.96 ± 0.15 GeV); the $F_{21} \to SU(3)$ UV completion is conjectural.

Note on three independent GTE routes to $\alpha_s(M_Z)$: Three GTE derivation routes to $\alpha_s(M_Z)$ are tabulated independently and represent complementary approaches, not contradictions. (1) P01 blind orbit-ratio: $\alpha_s = 0.11822$ ($+0.24\sigma$ PDG 2024, $\mathbf{A}_{\text{Lean}}$, committed 43 days before the first PDG comparison). (2) P22 SC-ZZ two-loop: $\alpha_s = 0.11790$ (0.00σ). (3) P35/P39 F_{21} three-loop running: $\alpha_s = 0.1193$ (+1.1%).

3.3 Group 2 — Uniqueness and Algebraic Depth

These papers explain *why* the structures in Group 1 are unique, and what deeper algebraic home they occupy.

P24 — The Arithmetic Uniqueness of the Standard Model [65]
https://doi.org/10.5281/zenodo.20168820
Four main results: (1) Asymptotic Sparsity Theorem — the joint arithmetic- admissibility and physical-viability constraint has exactly one solution across all ridge levels: $(n = 10, b_1 = 73)$, Lean-certified. (2) Positive-Root Theorem and Chirality Theorem — exact correspondence between SM coupling numerator structure and positive-root counts / squareness, both Lean-certified. (3) Galois substrate $\mathbb{Q}(\zeta_{120})$ — the minimal cyclotomic field containing all UGP algebraic constants; E_7 ($h = 18 \nmid 120$) excluded by Lean-certified arithmetic. (4) WZW falsification — the WZW construction cannot reproduce the UGP bare coupling rationals. Contains the full RCC proof over all compact simple Lie groups (Appendix C).

P12 — The Unified Rigidity Theorem for PSC/UGP Universes [70]
https://doi.org/10.5281/zenodo.20169959
Shows that three independently developed rigidity mechanisms jointly select the same canonical physical branch: (1) PSC Layer I/II theory-space selection (SM #1 of 34,560 universes); (2) UGP arithmetic seed rigidity (RSUC, Lean-certified); (3) dynamical basin structure (Lepton Seed deterministically in basin A). A Lean bridge library proves `unified_rigidity_theorem` under an explicit audited premise bundle. The PSC→SM step is unconditional (RCC proved, zero sorry).

3.4 Group 3 — PSC/MFRR Foundation (The Axiomatic Layer)

These papers establish the Perfect Self-Containment (PSC) and the Mathematical Foundations of Reflexive Reality (MFRR) — the upstream framework that provides non-circular justification for the UGP axioms through machine-checked theorems in the companion NEMS programme.

P14 — Formal and Computational Concordance on PSC Derivations [29]
https://doi.org/10.5281/zenodo.20169970
Computational certificate for the Two-Layer PSC Theorem: exhaustive enumeration of 20,160 candidate universes, only 12 pass hard PSC Layer I filters (0.06%), all SM-like. Five hard filters contain no reference to G_{SM}. The SM tuple $(d, G, N_{\mathrm{gen}}) = (4, G_{\mathrm{SM}}, 3)$ achieves the unique Layer II minimum $D_{\min} = 1.0667$. Three new UGP-derived coupling-ratio predictions from orbit invariants, not SM data.

P09 — The Architecture of a Computable Universe: Structural Principles of the UGP [6]
https://doi.org/10.5281/zenodo.20170749
Interpretive synthesis of the UGP architecture as a candidate theory of everything, structured as five architectural principles: Constrained Contingency (origin of physical law), Ontological Scaffolding (universe as conservative extension of the Loop Kernel), Reflexive Computation (UWCA self-execution), Constants as Compression Ratios (Elegant Kernel invariants), and the Theorem of Necessary Observers (PSC closure requires agents). Presents the constraint-manifold view of the SM parameter space and situates the UGP's Category-A backbone within the five-tier claim taxonomy.

Distinction from P07: Where P07 (below) establishes nine quantitative empirical meta-laws (ML-1–ML-9)—laws governing Zipf scaling, gauge structure, entropy, and basin attractors—P09 addresses the complementary structural question: how is the UGP substrate itself *architecturally organised*? The two papers are companion pieces: P07 characterises *what happens*, P09 characterises *how the substrate is structured*.

P07 — The UGP/GTE Program: A General Organizing Principle for Flows, Fields, Critical Laws, and Direct Structure [10] (Paper P07 of the UGP Physics programme)
https://doi.org/10.5281/zenodo.20170735
Presents UGP as a unifying vocabulary for nine universal meta-laws (ML-1 through ML-9), each a structural consequence of the GTE substrate. For ML-2, ML-4, ML-5, ML-6 the UGP axioms subsume Jaynes (1957), Yau (1991), Yang–Mills (1954), and Jacobson (1995). For Zipf's Law (ML-7), a complete derivation with Euler–Maclaurin error bound is validated on 21 diverse corpora. A Lean-verified entropy non-monotonicity witness supports ML-9.

P10 — Reflexive Reality and Self-Defining Physical Law [11]
https://doi.org/10.5281/zenodo.20170761
Formalises the concept of *reflexive reality* and argues that a finite, self-contained universe must obey self-defining laws arising from within the system. Classifies physical theories by degree of reflexivity. The central philosophical argument is machine-checked in the companion NEMS programme (Theorem of Foundational Finality, NEMS 23; Reflexive Closure Theorem, NEMS 56; zero sorry).

P11 — Ontological Dissonance Minimization as the Generative Meta-Law [9]
https://doi.org/10.5281/zenodo.20170777
Reports that minimizing an ontological dissonance functional D within the Self-Defining Universe (SDS) framework produces results empirically consistent with maximizing a spatial-integration proxy for integrated information Φ within the PR-0 substrate. D-minimization discovers a short-range binding potential $V(d) \approx \alpha + \sigma/d^2$ for the strong interaction (D and Φ_{proxy}: $r = -0.9108$, $p < 10^{-9}$, $n = 21$).

P20 — Mathematical Foundations of Reflexive Reality: A Survey [8]
https://doi.org/10.5281/zenodo.20170798

12-page journal survey of the MFRR programme. Machine-checked results include: forced adjudication, Born rule from MDL, arrow of time from PT irreversibility, no-emulation (PT is super-Turing), and SM gauge group forced by PSC consistency. Computationally certified: SM #1 of 34,560 universes; IPT $= 1.1300 \pm 0.0001$ matching UGP-derived value to 0.08%; SRRG attraction rate 97.02%. Best single entry point for the PSC/MFRR layer.

P13 — Mathematical Foundations of Reflexive Reality (Foundational Monograph) [39]
https://doi.org/10.5281/zenodo.20170812
The full 15,000-line reference document. Central construct: *Transputation* (PT), the unique internal adjudicator of computational degeneracies, machine-proved in Lean 4 (zero sorry). Seventeen closure theorems; Two-Layer PSC Theorem (Layer I forces G_{SM}, $N_{\text{gen}} \geq 3$; Layer II selects $N_{\text{gen}} = 3$ uniquely); 297 Python modules, 57,337+ runs, 15,894+ result files. Use as a reference, not a read-through.

3.5 Group 4 — Extension Sectors and Cross-Domain Echoes

These papers test the UGP's arithmetic machinery in domains beyond the SM or derive structural analogues in other physical and biological systems.

P03 — GTE Coordinates as Nuclear Descriptors: Competitive ML Performance from UGP-Derived Features [13]
https://doi.org/10.5281/zenodo.20170082
GTE composition features achieve 10-fold cross-validated MAE of 3.17 MeV on nuclear binding energies vs. 4.24 MeV for equal-size random polynomial baselines (∼25% improvement). Analytical derivation of all seven nuclear magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ from the GTE cascade; the ratio $\kappa_{\text{emp}}/\kappa_{\min}(N = 50) = 1.149 \approx$ IPT (1.6% match). Stability: a 10-term stability classifier trained on corrected NUBASE2020 empirical labels achieves 94.5% 5-fold CV; the smooth analytical law cannot resolve discrete anomalies (Tc, Pm) due to a fundamental smooth-law limitation (disclosed). New: GTE proton b-seed parity ($b_p = 11459$ is odd, Lean 4 certified) motivates the F_{10} parity feature; GTE seed parameters predict the SEMF pairing constant $\Delta = 5^{3/2} = 11.18$ MeV (0.003% from Möller et al. 1995, zero free parameters; bridge claim). Extended periodic table uses empirical NUBASE2020 for $Z = 1$–118 and GTE extrapolation for $Z = 119$–160 (Category D).

P04 — The Dynamics and Universality of the UGP: Attractors and Emergent Thermodynamics [7]
https://doi.org/10.5281/zenodo.20170092
Establishes Turing-universality (Rule-110 embedding, machine-checked) and compatibility with reversible computing. Long-term RG dynamics reveal three observed basin clusters (A, B, C) across a 98-seed survey; the Logarithmic Complexity Charge Q_4 at seed initialization predicts basin assignment (ANOVA $p < 10^{-4}$). Coarse-grained entropy decreases into basin clusters; a Lean-verified non-monotone entropy witness (`gte_entropy_prefix8_gt_prefix9`, zero sorry) exists while shuffled controls show monotone increase.

P15 — The Information Profit Principle [36]
https://doi.org/10.5281/zenodo.20170102
Derives a sharp dimensionless threshold — the Information Profit Threshold IPT $= 1 + \Lambda/2 \approx 1.1309$ — from the Reflexive Landauer Bound and PSC requirement. Six independent cross-domain validations: tropical forest GPP/RECO matches IPT to four decimal places ($n = 24$ sites); Polish bankruptcy dataset ($n = 43,405$); World Bank population dynamics (261 countries); major open-source software projects; particle physics SM gauge sector; and organizational survival (three-regime

cascade, ΔBIC $= -92$; inception-cohort HR$= 5.01\times$; variance phase transition BC$= 0.907$, Levene $p = 0.046$; [B$^-$]; see P26 §6).

P25 — Structural Admissibility and Biological Viability Uniquely Select the Standard Genetic Code [62]
https://doi.org/10.5281/zenodo.20170152
Applies the two-stage admissibility-then-viability sieve to all codon–amino-acid maps. Stage 1 (Crick wobble admissibility) reduces 23^{64} maps by 10^{50}; Stage 2 (eight independent biological viability criteria simultaneously) yields the standard genetic code at $z = +3.22\sigma$ ($p < 0.009\%$). No code among 10^5 randomly generated wobble-admissible codes outperforms the standard code.

P26 — A General Theory of Selection: Two-Stage Sieves, the Information Profit Threshold, and Cyclotomic Algebraic Substrates [30]
https://doi.org/10.5281/zenodo.20171011
Unifies the selection mechanisms of P03 (nuclear), P15 (IPT), P25 (genetic code), and P24 (SM gauge structure) under a single **General Selection Principle** (GSP): every realised structure passes a two-stage sieve of *admissibility* (structural consistency) and *viability* (generative-to-dissipative ratio $\geq$ IPT ≈ 1.1309). **Quantitative evidence in five empirical domains:**

1. **Nuclear magic numbers:** all seven $\{2, 8, 20, 28, 50, 82, 126\}$ derived from GTE energy-gap maxima; $\kappa_{\text{emp}}/\kappa_{\min}(N{=}50) = 1.149 \approx$ IPT (1.6%)
2. **Prebiotic amino-acid alphabet:** IPT-viability table separates all 20 standard amino acids (pass) from all 10 non-standard tested (fail)
3. **Microbial metabolic selection:** *E. coli* iJO1366 FBA; Pearson $r(G/D, \mu) = 0.971$, $p < 0.001$; grade [B]
4. **Organizational selection:** three-regime failure cascade (6.81%→1.11%→0.22%); ΔBIC $= -92$; inception-cohort HR$= 5.01\times$; variance phase transition BC$= 0.907$; seven independent tests; grade [B$^-$]
5. **Model-internal symbolic result:** natural language alphabet selection

$\mathbb{Q}(\zeta_{120})$ algebraic substrate proved in four independent settings; E7 ($h{=}18 \nmid 120$) as a Lean-certified falsifier. Appendix contains consciousness/NEMS extension [**I**]. Contains a full assumption ledger, claim taxonomy, evidence-grade table (Table 1), and proof sketch for the Asymptotic Sparsity Theorem. *Companion paper:* P27 (SRRG) elevates IPT from an empirical observation in P26 to a theorem-grade fixed-point derivation.

P28 — Computational Universality and the Standard Model: Rule 110, $\mathbb{Z}_5$ Rings, and Mod-7 Structure in the Universal Generative Principle [22]
https://doi.org/10.5281/zenodo.20259513
Establishes that Rule 110 is uniquely forced by the Standard Model's generation structure from *two independent directions*: the SM generation orbit algebraically determines all eight bits of Rule 110 as the sole CA rule consistent with SM winding and vacuum-transparency (CUP-4, Lean-certified), while the UGP substrate independently implements Rule 110 from below (P04/P08). The two derivations converge on the same rule (`rule110_two_layer_confluence`, `hypothesis_a_layered_universality`, $\mathbf{A}_{\text{Lean}}$, zero sorry). **Further Lean-certified results (zero sorry):**

1. **CUP-4:** SM orbit uniquely forces Rule 110 among all 256 binary CA rules; structural p-value $\approx 0.003\%$
2. **Garden of Eden:** gen_1 has zero predecessors under f_{MDL} across all $7^5 = 16{,}807$ $\mathbb{Z}_7$ states (`fmdl_gen1_is_garden_of_eden`, `native_decide`); first-generation particle stability is arithmetic unreachability

3. **Physical incompleteness:** f_{MDL} is computationally universal, so vacuum-reachability is halting-undecidable
4. **Tape-level unification:** a single Rule 110 Bool tape simultaneously computes both UGP sectors — SM charge (fMDL winding, zero axioms) and mass cascade (GTE arithmetic, one named Cook universality axiom)
5. **PSC chain:** PSC forces SM gauge structure → orbit → Rule 110 → Turing-universal; each step Lean-certified

The MDL-minimal $\mathbb{Z}_7$ CA f_{MDL} (CUP-12) has 76-bit description length and simultaneously embeds SM and dark sector orbits. *Companion papers:* P04 (UGP Dynamics, two-way Rule 110 convergence); P17 (Braid Atlas).

3.6 Group 5 — Machine-Checked Formalization

Lean 4 Formalization Paper (ugp-lean). The companion formalization paper [69] documents the `ugp-lean` Lean 4 library: more than 400 modules across seventeen architectural layers, zero `sorry`, 98 disclosed named axioms (see the formalization paper's axiom-inventory table). Covers the GTE cascade, Quarter-Lock Law, UCL Elegant Kernel, Braid Atlas (charge theorem, composite triples, EW bosons, chirality), mass relations (Koide, VV, CKM θ_{23}, seesaw exponent), PSC (three-route forcing), and Galois structure (Coxeter-conductor, E7 exclusion). A NEMS-side formal specification of the `ugp-lean` library — providing the complete mapping from NEMS axioms to `ugp-lean` declarations — is available in [69]. *This paper is the single source of truth for Lean module/theorem counts.* Every Category-A claim in P01–P43 is backed by a specific entry in this library or in the companion `srrg-lean` repository [61]. The companion `nems-lean` and `transputation-lean` libraries additionally certify the computability classification of the quantum adjudication function (P51): Σ_1-completeness of record truth on the diagonal fragment, the absence of a computable convergence modulus, and the many-one equivalence of the adjudication readout with the halting problem — the machine-checked core of the degree-$0'$ classification (zero sorry on the certified theorems).

P30 — Machine-Certified Formalization of Cook's Rule 110 Universality Theorem in Lean 4 [38]
https://doi.org/10.5281/zenodo.20319411
A Lean 4 formalization program for Cook's Rule 110 universality theorem, packaged in the standalone `rule110-lean` [58] library, deliberately free of UGP-specific physics so that Cook's construction can be audited independently. The conditional top theorem `cook_operational_stage3_tm_microstep_readback` (`CookUniversalityTop`, zero sorry) is established contingent on five named Cook bridge axioms formalizing the general CTS collision analysis (`cook_cts_data_cones_origin_one_step_ax` and four companions). This is a conditional operational certification for already-supplied compilations, not itself a Turing-universality theorem; the general TM→CTS compiler remains open. Discharged results include: the C1 readback for arbitrary words at slots ≤ 20 (`cook_c2_tape_bit_general`, $\mathbf{A}_{\text{Lean}}$, zero sorry); all five construction-collision kinds certified by finite `native_decide` witnesses; Stage 3 data-cone simulation fully discharged for the length-6 benchmark CTS; and `CookFinTM2Compiles` proved for Mathlib's halting identity machine (`idComputer Bool`, $\mathbf{A}_{\text{Lean}}$). Closing all five named axioms will upgrade the UGP incompleteness result (P28) to unconditional $\mathbf{A}_{\text{Lean}}$.

P40 — Algebraic Characterization of Rule 110 over GF(7) and Finite NAND Functional Completeness [15]
Derives an exact degree-3 multilinear polynomial $p(L, C, R) = C + R - C{\cdot}R - L{\cdot}C{\cdot}R$ over GF(7) =

$\mathbb{Z}/7\mathbb{Z}$ uniquely interpolating the eight Rule 110 transition values (Lean-certified, zero sorry). The corollary $p(L, 1, R) = \text{NAND}(L, R)$ — Rule 110 implements NAND whenever the center cell is active — provides, via Sheffer's functional completeness theorem (one named axiom: `boolean_nand_complete`), a proof that Rule 110 realizes any finite two-input Boolean function algebraically, independently of Cook's construction. This finite functional-completeness result is distinct from Turing universality, which remains Cook's theorem (2004). The algebraic certificate reduces the finite-completeness proof to a single degree-3 polynomial identity over a prime field. The GF(7) structure connects directly to the $\mathbb{Z}_7$-symmetric Klein–Gordon substrate of the GTE framework (P34, P35, P42).

3.7 Group 6 — Special Topics

P16 — Black Hole Unitarity via Reflexive Unitarity [18]
https://doi.org/10.5281/zenodo.20170110
Models Hawking evaporation as a GKSL master equation on a JT-like toy Hilbert space; since every CPTP map admits a Stinespring dilation, the channel is unitarily equivalent to a reversible evolution. In computational experiment TE2.4 ($n = 3$ bosonic modes, Hawking $T_H \approx 0.004$): Stinespring fidelity $F \geq 1 - 10^{-8}$; thermalization entropy reaches 97.2% of ideal; closed unitary models show exact PT^{-1} recovery ($F = 1.000000$) and the full Page curve. Extension to 3+1 gravity and the firewall remain explicit open problems.

P27 — The Self-Referential Renormalization Group: A Universal Framework for Physical Constants [59]
https://doi.org/10.5281/zenodo.20170859
Introduces the *Self-Referential Renormalization Group* (SRRG): a gradient-flow theory on the space of self-referential physical theories maximising $F[S] = R[S] - C_\Lambda[S]$. **Core fixed-point results (main body; all Lean-certified, zero sorry):**

1. $R[S^*]/C_\Lambda[S^*] = \text{IPT} \approx 1.1309$: IPT elevated to theorem-grade derivation
2. Minimal symmetry group at S^* is $U(1)$ (circle group uniqueness)
3. Linearised contraction eigenvalue $= 1/\varphi$; F-theorem (Zamolodchikov analogue)
4. $\beta_\eta = \kappa(\eta - \text{IPT})(\eta - 2)$ *derived* via Vieta's theorem; `h_psc_sc` [A^-]

Conditional physical applications (Appendix B): $\theta_{\text{QCD}} = 0$ [B]; $N_{\text{gen}} = 3$ [B]; SM gauge group [B]; Higgs quartic [B]; structural derivation $v_{\text{PSC}} = 246.16$ GeV (-0.024%; grade [A_{Lean}]; `VEVProof.*`, zero sorry, zero open axioms — `psc_ew_entropy_maximization` is a proved theorem). **Further appendices:** α direct coupling route $1/\alpha(M_Z) \approx 127.311$ (0.495% PDG) from Lean-certified bare couplings [A/D] (App. C); Weinberg angle diagnosed negative (App. D; [D→C]); SRRG–MDL equivalence (Open Problem 9): **upgraded from [B+] conjecture** — `op9_catal_unconditional` ($\mathbf{A}_{\text{Lean}}$, zero sorry): $K_{\text{alg}}(g^*) = 0$, unique K_{alg} zero on $(0, \infty)$; `srrg_op9_biconditional` (zero sorry): IsSrrgFixedPoint $\Leftrightarrow$ MDL minimiser (full biconditional). Full closure of OP9 awaits `UGPSubstrateConstraint` (CMCA substrate formalisation; open problem). A "What this paper does not prove" box and a load-bearing assumption table are included in §1. *Companion paper:* P26 provides the empirical evidence base for which P27 supplies the theorem-grade IPT derivation.

3.8 Group 7 — Quantum Gravity, Spacetime Structure, and the Φ_{MDL} Continuum

These papers develop the gravitational, spacetime, and quantum-mechanical sectors of the GTE framework. The **physical substrate** is the Φ_{MDL} continuum field (Level 2); discrete cellular

automata—f_{MDL}, Rule 110, and the CMCA—are **Level-1 algebraic certificates**, not alternative universes. Paper P34 [32] formalises the shared abstract architecture as the Möbius triple $(A, e, [D])$; subsequent papers P35–P47 build quantum mechanics, emergent gravity, QCD, and non-perturbative quantum gravity on that foundation with zero free parameters.

P34 — The GTE-Möbius Architecture: Arithmetic Unification of Computation and Transputation [32]
https://doi.org/10.5281/zenodo.20319419
Axiomatic foundation for Group 7. Formalizes the architecture as the Möbius triple (A, e, $[D]$): GTE arithmetic carrier A ($\mathbb{Z}_7$ dynamics, P28; physically realised as the PSC-admissible kink sector of Φ_{MDL} at Level 2), self-encoding map e conferring Turing completeness (P30/P40; one named bridge axiom on the Cook route), and PSC-consistent coherence measure class $[D]$ (D1–D5). Computation (f_{MDL} evolution) and transputation ($[D]$-selection) share a single surface without a sharp syntactic boundary. The CMCA (P41/P45) is the Level-1 algebraic certificate linked to e; Φ_{MDL} (P42/P43) is the physical substrate. **Key Lean-certified results (zero sorry):** GTE is the final F_{PSC}-coalgebra (`GTEFinalCoalgebra.lean`, zero axiom). TPC class strictly intermediate between decidable and hypercomputation (13 theorems), with its decision-theoretic shadow on the self-referential diagonal at Turing degree exactly $0'$ (Gold–Putnam stratum; see P51); three-level TPC depth $= N_{\text{gen}} = 3$. Three lifting theorems (Algebraic Lifting, 3D Spatial Lifting, Algebraic Descent). $T(2,3)$ torus knot parameters GTE-derived. Conjectures C2–C5 remain open (C2: uniqueness of physical D; see P43). *Downstream:* P35 (unification hub); P42–P43 (QFT on D1–D5 axioms).

P36 — Emergent Gravity from Rule 110 Cellular Automaton [26]
https://doi.org/10.5281/zenodo.20319425
Derives emergent gravity from the causal graph structure of Rule 110 within the UGP framework. Spacetime dimension $D = 4$ is machine-certified (`gte_spacetime_dimension`, $\mathbf{A}_{\text{Lean}}$, zero sorry): $D_{\text{spatial}} = 3$ is forced by the three-dimensional structure of the MDL-minimal CA (`three_dim_fmdl_structure_forced`); $D_{\text{temporal}} = 1$ follows by CA definition. The deviation-based Ollivier–Ricci curvature of the Rule 110 causal graph satisfies the discrete vacuum Einstein equation $G_{\mu\nu} = 0$ exactly (Lean-certified) and exhibits a positive matter step $\kappa_{\text{SD}} > 0$ consistent with $G_{\mu\nu} = 8\pi T_{\mu\nu}$ (computationally confirmed). A structural MDL-Lovelock correspondence (**A/D**) provides a mechanism for Einsteinian gravity as the unique diffeomorphism-invariant effective theory compatible with the GTE substrate; vacuum spatial Gromov–Hausdorff convergence is $\mathbf{A}_{\text{Lean}}$ (`vacuum_cmca_gh_converges_to_flat_space`); matter-sector and Lorentzian convergence to the smooth Riemann tensor remain open.

P37 — Quantum Mechanics from Rule 110: Hilbert Space, Hamiltonian, and Born Rule [53]
https://doi.org/10.5281/zenodo.20319431
Applies 't Hooft's cellular automaton interpretation of quantum mechanics [72] to the GTE CA f_{MDL} on $\mathbb{Z}_7^5$. The complete orbit decomposition of f_{MDL} on $\mathbb{Z}_7^5$ (16,807-state visible-sector ring, exhaustively computed) reveals a single cycle: the vacuum fixed point $(0,0,0,0,0)$ with $E = 0$. The physical Hilbert space is one-dimensional; f_{MDL} belongs to 't Hooft's information-loss Chapter 7 regime, not the reversible Chapter 2 regime. The SM generation states are transient excitations with tail lengths $\ell(\text{gen}_1) = 3 > \ell(\text{gen}_2) = 2 > \ell(\text{gen}_3) = 1$, matching the empirical stability hierarchy exactly. A direct eigenvalue-to-mass correspondence is falsified on every formulation tested. The **Two-Role Structural Principle** organises the separation: the cogwheel construction supplies Hilbert space, unitarity, and the Born rule ($\mathbf{A}_{\text{Lean}}$, Lean-certified); the GTE N_{eff} cascade supplies the mass spectrum via a Yukawa-analog mechanism.

P38 — Emergent Gravity from the Φ_{MDL} Field: Einstein Equations, Kink Sources, and Quantum Gravity Scale [27]
Derives the stress-energy tensor $T_{\mu\nu}$ from the $\mathbb{Z}_7$-symmetric Klein–Gordon Lagrangian analytically and in Lean 4 (zero sorry; `StressEnergyTensor.lean`: symmetry, vacuum-vanishing, and gravity-sector prerequisites proved). The linearised Einstein field equations are derived from the MDL-Lovelock correspondence (P35) combined with minimal coupling of Φ_{MDL} on a curved background. The BPS kink acts as a gravitational source: $\int T_{00}\,dx = M^{\mathrm{cl}}_{\mathrm{kink}} = 290.10\,\mathrm{MeV}$ (classical BPS mass, verified to 1.4×10^{-6} relative error); the one-loop quantum correction gives $M^Q = 281 \pm 21\,\mathrm{MeV}$ ($\Delta M = -7$ to $-10\,\mathrm{MeV}$, on-shell / $\overline{\mathrm{MS}}$ at $\mu = m_\varphi$; interface dimensional regularization by two independent routes, validated against exact sine-Gordon and φ^4 benchmarks; $\pm 21\,\mathrm{MeV}$ envelope over the gauge-invariant renormalization-scheme family; P38, P42); the classical cosmological constant vanishes at all seven $\mathbb{Z}_7$ vacua. Newton's constant is derived (**A/D**) via a gravitational hierarchy formula; the Penrose objective-reduction time for a single GTE kink is $\tau_{\mathrm{OR}} \approx 20$ million years. The one-loop Coleman–Weinberg correction exceeds the observed dark-energy scale by $\sim 10^{42}$, a hierarchy the framework identifies but does not resolve.

P41 — The Three-Layer Chiral Minkowski Cellular Automaton: A Unified Discrete Spacetime Model from First Principles [67]
Constructs the Chiral Minkowski CA (CMCA) from three layers: Rule 110 (L_{x^+}, right-moving chiral), Rule 124 (L_{x^-}, left-moving chiral), and a shared inner Rule 110 clock (L_t, timelike). Four principal results follow. (1) *Lorentz invariance as observer effect:* the CA lattice has a preferred Planck-scale frame, but PSC Presentation Invariance forces all $[D]$-selected observables to be Lorentz-equivariant on the Φ_{MDL} continuum substrate (Lean-certified, one named axiom `d2_universal`). The lattice residual $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$ vanishes as M^{-2}; at LHC energies $\delta_{\mathrm{LV}} \approx 8.8 \times 10^{-32}$, far below existing bounds. (2) *Electroweak scale:* $\sin^2\theta_W(\text{threshold}) = 0.231207$ (-0.42σ PDG 2022, $\mathbf{A}_{\mathrm{Lean}}$); two-loop: 0.23129154 ($+0.038\sigma$ PDG 2024, **B**); $M_W/M_Z = \sqrt{10/13}$ (Lean-certified); absolute scale gives $M_Z = 91.91\,\mathrm{GeV}$ ($+0.80\%$) and $M_W = 80.61\,\mathrm{GeV}$ ($+0.30\%$) from $v_{\mathrm{PSC}} = 246.16\,\mathrm{GeV}$. (3) *CPT and causality:* CPT invariance and causal precedence established. (4) *Bell non-locality:* CHSH violation reproduced from the CA probability structure. (5) *Wave–particle duality and double-slit:* definite beable path through one slit combined with two-path amplitude superposition $\phi = \phi_L + \phi_R$ and Born-rule click ensembles; Huygens–Fresnel fringes match Fraunhofer theory with correlation 0.998 (CatA); resolves the standard paradox without retrocausal beable change (which-path visibility is an [**D**]-record effect; see P43).

P42 — The Φ_{MDL} Field: Quantum Structure, Born Rule, and Continuum Completion of the Chiral Minkowski CA [46]
Studies the $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL} as the continuum counterpart of the Three-Layer CMCA (P41). Four principal results follow. (1) *Poincaré invariance:* $\omega^2 = k^2 + m^2$ exactly, machine-certified (`poincare_invariance_of_kg` in `LorentzInvariance.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry). (2) *Born rule from field amplitudes:* position-space probability density $P(x) = |\partial_x\Phi|^2 / \int |\partial_x\Phi|^2$ extended to overlapping multi-kink configurations via the canonical dual-frame inner product (`born_rule_from_dual_frame_master` in `DualFrameBornRule.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry). (3) *Physical quantum state:* the pre-measurement Φ_{MDL} state is the Hamiltonian thermal ensemble $P(k) = e^{-M_k/T}/Z_T$ restricted to PSC-admissible $\mathbb{Z}_7$ kink sectors $\{0, 2, 3, 4, 6\}$ (`phimdl_thermal_state_master`, $\mathbf{A}_{\mathrm{Lean}}$ conditional, zero custom physics axioms). (4) *Interference:* single- and double-slit analogue interference in $|\partial_x\Phi|^2$ density, with decoherence from environmental $\mathbb{Z}_7$ random-phase states. At one loop the quantum kink mass is $M^Q = 281 \pm 21$ MeV (interface dimensional regularization, two independent benchmark-validated routes); the gauge coupling $e^2(\Lambda_{\mathrm{GTE}}) = 7/2$ is consistent with PDG running at the $+1.3$–1.5σ scale-band level once the non-perturbatively measured kink dissolution constant

($b = 1.189 \pm 0.049$) is included, and the kink sector contributes $\Delta\alpha_{\text{kink}} = (3.5\text{–}3.7) \times 10^{-4}$ to the hadronic vacuum polarization — a parameter-free falsifiable prediction. The substrate tape spacing $a = \hbar c/\Lambda_{\text{GTE}}$ used by the non-perturbative lattice campaigns is itself derived, not chosen (Tape Saturation Theorem, **A/D** conditional on the named Compton-Support Criterion; P50): the campaign bare mass $am_\varphi = 7/8$ is the exact lattice-dictionary value, and the kink correlation length at the physical point, $\xi_{\text{kink}} = |\mathbb{Z}_7| = 7$ cells, is a pre-registered substrate-MC target. *Companion papers:* P41 (CMCA discrete layer); P43 (complete unified framework).

P43 — The Complete Φ_{MDL} Framework: Algebraic Necessity, Quantum Mechanics, Emergent Gravity, and Uniqueness [21]
Presents the complete GTE/Φ_{MDL} framework as a unified physical theory extending to the full quantum gravity sector. Seven theorems are proved across five areas. **(1) Algebraic Necessity:** $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ is the unique internal symmetry group satisfying $N_{\text{gen}} = 3$, $b_0 = |\mathbb{Z}_7| = 7$, and Born probabilities from $\mathbb{Z}_7$ topological kink quantization (Lean-certified, `algebraic_necessity_master_bundle`, zero sorry). **(2) Quantum Mechanics:** Born rule $P(k) = |c_k|^2$ unconditional from $\mathbb{Z}_7$ superselection (`born_rule_unconditional`, zero custom axioms); measurement via transputation (objective, determinate, non-computable). **(3) Emergent Gravity and Quantum Gravity:** Einstein field equations from MDL-Lovelock + minimal coupling (**A/D**); $G_N = m_\tau^2/M_{\text{Pl}}^2$ derived analytically to 0.040% via $M_{\text{Pl}}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2$ (**A/D**); Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4G_N)$ from two independent routes (**A/D**); geodesic theorem Lean-certified with zero custom axioms (`psc_orbit_is_curvature_geodesic`, `geodesic_theorem_spatial_general`, $\mathbf{A}_{\text{Lean}}$); non-perturbative UV-finite S-matrix from CMCA structure (structurally established). **(4) Impossibility of Alternatives:** every finite-M discrete model has $\varepsilon_0(M) = \pi^2/(3M^2) > 0$ (Lean-certified); Φ_{MDL} is the unique zero-error continuum limit. No outer-totalistic $\mathbb{Z}_7$ CA in 3D admits Class 4 behaviour. **(5) Uniqueness:** $|\mathbb{Z}_7| = 7$ is uniquely forced by three independent arguments: MDL-minimality; the Frobenius prime identity $7 = 3^2 - 3 + 1$; and the QCD β-function $b_0 = |\mathbb{Z}_7| = 7$. Zero free parameters in the gravity sector.

P44 — Quantum Gravity Completeness: Perturbative QG, CMB Spectral Index, and Holographic Encoding [52]
Presents the complete perturbative quantum-gravitational description of the GTE/Φ_{MDL} framework. The curved-background Lagrangian is uniquely determined with $\xi = 0$ forced by three independent arguments (MDL minimality, Wald entropy consistency, $\mathbb{Z}_7$ superselection preservation; Lean-certified). UV finiteness on arbitrary curved backgrounds is established via DeWitt–Schwinger heat-kernel analysis: curved-background UV contributions reduce to finite Planck-scale renormalizations ($C_i \approx 41.76$) with no residual UV problem. **Five key results:**

1. **GTE Holographic Encoding Theorem** (Theorem 5.1): five equivalent descriptions of the GTE encoding structure — Lagrangian, Reed–Solomon $[5, 3, 3]_7$ code, holographic/RT, MDL-information-theoretic, and QEC. Twelve of the twenty directed implications, linking the Lagrangian, Reed–Solomon, MDL, and QEC descriptions to one another, are established by derivation with no gravitational input; the remaining eight, connecting any of those four to the holographic/RT description, hold conditional on the standard Bekenstein-Hawking coefficient $1/4G$ taken as an external premise, not derived from GTE structure. RS code machine-certified (`gte_orbit_vectors_are_rs_codewords`, $\mathbf{A}_{\text{Lean}}$, zero sorry).
2. **Cosmological initial state:** MDL-minimal, flat (k=0), kinetic-dominated, requiring only $\log_2 3 \approx 1.585$ bits; dissolves the horizon and flatness problems without inflation. The domain-wall problem is resolved dynamically: the transient $\mathbb{Z}_7$ wall network formed at the ordering crossover $T_G \approx 0.70$ GeV is annihilated by the canonical coupling bias well before nucleosynthesis, leaving no surviving defect network (full derivation in P47).

3. **CMB spectral index:** at leading order $n_s = 1$ exactly; conditional seven-step chain yields $n_s = 1 - \ln(2)/(2\pi^2) = \mathbf{0.96488}$, matching Planck 2018 at -0.005σ if the EFE-bridge step is established (5/7 steps closed).
4. **Tensor-to-scalar ratio:** $r = 0$ exactly (no primordial gravitational waves); primary falsifiable prediction for LiteBIRD ($\sigma_r \approx 0.001$).
5. **Galois/CPT:** $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ identified with CPT $\times$ generation-orbit symmetry ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry).

First derivation of $\Omega_\Lambda = (\ln 2/3\pi)\log_2(2000/3) \approx 0.690$; the full two-route treatment with quantum protection and $w = -1$ derivation is in P47. *Companion papers:* P42, P43 (GTE/Φ_{MDL} framework); P45 (three-tape 3+1D); P47 (cosmological predictions).

P45 — Three-Tape CMCA: Deriving 3+1D Spacetime, the Standard Model, and Gravity from a Shared Clock [68]

The breakthrough 3+1D architecture paper. Three independent 1+1D CMCAs (each with chiral Rule 110/Rule 124 outer layers and inner τ_c clock) sharing a single outer clock period τ_c^{out} produce full 3+1D Minkowski spacetime. **Key machine-certified results:**

1. **Dimensional Protocol Principle (DPP):** the shared clock is both *necessary* and *sufficient* for 3+1D dynamics ($\mathbf{A}_{\mathrm{Lean}}$). Without it the three tapes evolve as independent 1+1D systems.
2. **SM particle spectrum:** uniform winding triples (w, w, w) for $w \in \{0, 2, 3, 4, 6\}$ recover all SM particles via $\mathbb{Z}_7$ winding conservation; all 33 SM charged-current vertices verified (CatA).
3. **Color confinement** from MDL: $\Delta K = \log_2 9 \approx 3.17$ bits forbids free quarks ($\mathbf{A}_{\mathrm{Lean}}$); baryon number $B = (1/3)\sum \chi_q(w_j)$ as topological charge ($\mathbf{A}_{\mathrm{Lean}}$).
4. **Holographic scaling:** $3L/L^3 \to 0$, realising Bekenstein–Hawking area scaling ($\mathbf{A}_{\mathrm{Lean}}$).
5. **Lorentz algebra:** all twelve $\mathfrak{so}(1,3)$ commutation relations machine-certified ($\mathbf{A}_{\mathrm{Lean}}$).
6. **Fermionic statistics** from $\mathbb{Z}_7^*$ non-primitive roots ($\mathbf{A}_{\mathrm{Lean}}$ chain).
7. **SR time dilation:** intrinsic clock ratio $\tau_{\mathrm{inner}}/\tau_{\mathrm{outer}} = 3/7 \approx 0.4286$ (CatA).
8. **Gravity:** Physical Minimum Description Length (PMDL) Poisson equation $\nabla^2\Phi = G_{\mathrm{eff}}\, p(w_x, w_y, w_z)$ ($\mathbf{A}_{\mathrm{Lean}}$); continuum-limit force law $F = G_{\mathrm{eff}} M/(4\pi b^2)$ derived analytically via 3D Poisson Green's function (**A/D**); prior discrete deviations ($b^{-2.19\ldots-2.30}$) are finite-source artifacts vanishing as $O(\sigma_{\mathrm{AL}}/b)^2$.
9. **Bell nonlocality:** CHSH $S = 2.4459$ (86.5% of Tsirelson bound; CatA); entanglement negativity $\mathcal{N} = 0.38$.
10. **Born rule via Page–Wootters:** the τ_c clock satisfies all Page–Wootters prerequisites; conditional Born rule $P(k|\tau) = |\langle k|U(\tau)|\psi_0\rangle|^2$ derived (CatAD), independent of clock-state distribution.

Single-Source Principle: all of spacetime, matter, gravity, and quantum mechanics emerge from one 19-bit polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7) in five physical roles simultaneously. **DPP insight:** temporal synchronization is the causal prerequisite for spatial geometry — time creates space. **Falsifiable predictions (zero free parameters):** $r = 0$ (LiteBIRD); $\delta_{CP} = 205.71°$ (PMNS Dirac phase; NuFIT 6.0 IC24 NH 212°, -0.15σ). Digital physics lineage section: von Neumann, Zuse, Fredkin–Toffoli, Wolfram, 't Hooft, Gorard; the GTE framework is distinguished by its dual role as algebraic certificate *and* physical derivation vehicle. *Companion paper:* P46 (polynomial five-roles analysis); P44 (quantum gravity).

P46 — GTE Polynomial UFT: MDL Uniqueness and Five Physical Roles of $p(L, C, R)$ [34]

Companion to P45 explaining *why* $p(L, C, R) = C + R - CR - LCR$ over GF(7) has exactly five distinct physical roles, with zero additional specification cost per role. **MDL uniqueness:** the polynomial is the unique degree-3 multilinear object over GF(7) satisfying $\mathbb{Z}_7$ symmetry,

Rule 110 restriction, and 19-bit MDL minimality ($\mathbf{A}_{\text{Lean}}$). **Five physical roles:** (1) spatial cellular-automaton dynamics (Rule 110 Turing universality); (2) $\mathbb{Z}_7$ gauge vertex conservation; (3) gravitational coupling (PMDL Poisson source term); (4) quantum entanglement generator ($\mathcal{N} = 0.38$); (5) Bell-nonlocality source ($S = 2.4459$, 86.5% Tsirelson). **Additional derived results:** W-boson propagator $G_W(r) = e^{-m_W r}/(4\pi r)$ from universal propagator $(-\Delta)^{-1}$; SRRG golden-ratio bridge: $1/\varphi$ is the unique positive root of $p(x, x, x) = x$ over $\mathbb{R}$, machine-certified; vacuum stability from $p(0, 0, 0) = 0$ ($\mathbf{A}_{\text{Lean}}$); color confinement from MDL description-length gap $\Delta K = \log_2(N_c^2) = \log_2 9$ ($\mathbf{A}_{\text{Lean}}$); baryon number $B = \frac{1}{3}\sum_j \chi_q(w_j)$ from $N_{\text{tapes}} = 3$; $\text{SU}(2)_L$ gauging from PMDL ($\mathbf{A}_{\text{Lean}}$; zero named axioms). The PMDL–Λ duality identifies gravity and the cosmological-constant floor as the record and measure sectors of one Gaussian generating functional $\ln Z[J]$; the halting-weighted MDL residual gives the cosmological constant, whose census-capacity evaluation is the zero-parameter value $\Omega_\Lambda = 0.6899$ ($+0.18\sigma$ from Planck 2018). *Companion paper:* P45 (three-tape architecture); P40 (GF(7) polynomial derivation).

P47 — Cosmological Predictions of the GTE/Φ_{MDL} Framework: Dark Energy, the CMB Spectral Tilt, and Gravitational Signatures from First Principles [23]
The dedicated cosmology paper of the UGP Physics programme, deriving all principal cosmological observables from zero-parameter first principles with Lean 4 machine-certified central steps. **Dark-energy fraction (two independent routes):**

1. PSC reflexive-closure epoch count: $\Omega_\Lambda = (\ln 2/3\pi)\log_2(2000/3) = 0.6899$ ($+0.18\sigma$ from Planck 2018 0.6889 ± 0.0056).
2. Three-tape holographic mode count with Rule-110 ether rate $\tau = 3/7$: $\Omega_\Lambda = 3\pi/14 = 0.6732$.

The two routes bracket the observed value from above and below, and the bracketing is structural: the holographic value is the carrier price of the MDL-minimal self-description (a lower bound) and the PSC value is the capacity ceiling of the same adjudication ledger (an upper bound), with the inter-route ordering reduced to the machine-certified GTE-atom inequality $14\ln(2000/3) > 9\pi^2$ ($\mathbf{A}_{\text{Lean}}$). The exact realized fraction is a left-computably-enumerable real of Turing degree $0'$ ($\Omega_\Lambda \in \Delta_2^0 \setminus \Delta_1^0$) — definite but not computable — so no computable selection mechanism for Ω_Λ exists; the census endpoint becomes a 3σ-decidable test at precision $\sigma^* \approx 3.4 \times 10^{-4}$. The bracket is non-empty only for $N_{\text{gen}} \leq 3$ (measurement-free); combined with the PSC Layer-I bound $N_{\text{gen}} \geq 3$, this determines $N_{\text{gen}} = 3$ without Presentation Invariance (**A/D**, conditional on named structural premises). The energy density obeys $\rho_\Lambda = (9/112)M_{\text{Pl}}^2 H_0^2$ (holographic Cohen–Kaplan–Nelson scaling with PSC boundary condition fixing $w = -1$); holographic non-renormalization suppresses loop corrections by $\approx 10^{-42}$ relative to the standard estimate. **CMB spectral index:** $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ (-0.005σ from Planck 2018), derived from binary holographic running and machine-certified by fourteen zero-sorry Lean theorems. **CKM CP phase:** $\delta_{CP} = \pi/2 - 3/8 = 68.51°$ (0.017% from PDG), derived from an S_3 subgroup chain. **Neutrino sum:** $\Sigma m_\nu = 59.4$ meV. **Defect cosmology:** the $\mathbb{Z}_7$ ordering crossover at $T_G \approx 0.70$ GeV forms a transient Kibble wall foam that the canonical coupling bias annihilates within $\sim 10^{-23}$ s, roughly 700× before BBN — zero surviving defect network, with $\Omega_{\text{GW}}h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz and $\Delta N_{\text{eff}} \approx 0$ (parameter-free falsifiable prediction). **Newton constant:** normalized via discrete Ollivier–Ricci (Gorard) curvature chain ($k_{\text{SD}} = 10/13$). Full falsifiability profile: primary probes are LiteBIRD ($r = 0$, $n_s = 0.96488$) and CMB-S4 (Σm_ν, ΔN_{eff}). *Companion papers:* P44 (quantum gravity completeness); P45 (three-tape architecture); P46 (polynomial UFT).

3.9 Group 8 — Complete Synthesis Monograph

Paper P48 [20] is the capstone synthesis of the entire UGP Physics programme: a single monograph presenting the complete derivation chain from PSC and MDL through every Standard Model parameter, spacetime structure, quantum mechanics, and cosmology, with each step labeled by its certification level. It integrates results from Groups 0–7 without introducing new free parameters.

P48 — The Complete GTE Framework: Standard Model, Gravity, Quantum Mechanics, and Cosmology from Φ_{MDL} [20]
https://doi.org/10.5281/zenodo.20560550
The first complete derivation chain from PSC through all SM parameters, spacetime, quantum mechanics, and cosmology in a single monograph (346 pages). Fourteen chapters trace the logical necessity of the 19-bit polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7): substrate selection ($\mathbb{Z}_7 \times \mathbb{Z}_3$ MDL-minimal, CatAL), the Φ_{MDL} field and BPS kinks as physical particles, all four SM interactions and derived gauge couplings, three-tape CMCA spacetime and perturbative quantum gravity, Born-rule quantum mechanics from PSC transputation, and zero-parameter cosmology (Ω_Λ, n_s, δ_{CP}, Σm_ν, η_B). A dedicated chapter, *The Mathematical Substrate*, develops the UGP/GTE arithmetic in full — the ridge sieve, the GTE triples and map T, the worked three-generation cascade, the generation orbit and its parity shadow, and the UWCA (the substrate's native computer, on which Rule 110 demonstrably runs) — and establishes the monograph's three-rung ontological ladder: mathematical substrate (what selects) → certificate (what certifies) → physical substrate (what exists). The Selection chapter then presents the two MDL selections as *sequential* rather than parallel: the rule selection consumes the seed selection's output — the orbit's total-parity shadow plus MDL-equivalent vacuum transparency forces exactly one multilinear rule over GF(7), p itself, with Rule 110 arising as its binary restriction (Direct-Interpolation Lift Theorem, $\mathbf{A}_{\mathrm{Lean}}$; full mathematical treatment in P49). Each chapter uses a three-voice structure (claim, intuition, proof) enabling both technical and accessible reading paths. Includes explicit comparison with eight alternative frameworks and a complete falsification table. A headline synthesis result is the **master-quadratic unification**: the electroweak scale (SRRG fixed point of $p(x, x, x) = x$ over $\mathbb{R}$) and the unique Rule 110 computational floor (GF(7) irreducibility) are the two arithmetic completions of the single $\mathbb{Z}$-quadratic $x^2 + x - 1$, whose discriminant equals $N_{\mathrm{fam}} = 5$ (full mathematical treatment in P49). Further synthesis results: the Three-Level MDL Unification (one MDL principle operating at theory selection, field dynamics, and event adjudication, with non-circularity machine-certified); the degree-exact classification of quantum-measurement adjudication at Turing degree $0'$ (P51); the structural $\Omega_\Lambda \in [3\pi/14,\ 0.6899]$ bracket (floor = the carrier price of the universe's self-description, ceiling = its capacity; P47); and the dynamical $\mathbb{Z}_7$ defect resolution with zero surviving network (P47). **Two open problems are resolved. OP3 ($\mathbf{A}_{\mathrm{Lean}}$):** all computable quantum corrections to the cosmological constant Λ vanish identically in the pure-φ sector — machine-certified by `lambda_all_order_zero_pure_phi` (zero sorry) — confirming that the classical $\Lambda = 0$ exact result survives all loop orders in the $\mathbb{Z}_7$-periodic pure-field sector. **OP6 (CatAD):** the quantitative V−A chiral coupling is derived from first principles: $g_R = 0$ exactly and $g_A/g_V = -1$ exactly, establishing the maximal parity violation of the weak interaction as a structural consequence of the GTE framework (`va_op6_bundle`). **PSL(2,7) algebraic unification (P52 [50], $\mathbf{A}_{\mathrm{Lean}}$):** all four pillars of GTE algebraic structure are unified through PSL(2, 7) of order 168; the Klein quartic's genus equals $N_{\mathrm{gen}} = 3$; and the Eisenstein functor gives A_4 as the leading-order neutrino flavor group. *Companion papers:* P01–P47 (source derivations integrated chapter by chapter).

3.10 Group 9 — Polynomial Algebra, Wolfram Bridge, Algebraic Unification, and Division-Algebra Bridge

Paper P49 presents the first systematic study of the GTE polynomial $p(L, C, R) = C+R-CR-LCR$ over $\mathbb{Z}_7$ as a standalone dynamical system, distinct from the previously studied f_{MDL} lookup table. It provides the Wolfram framing of GTE's MDL selection result and develops the polynomial's complete algebraic, arithmetic, and dynamical structure.

P49 — MDL Selects the Wolfram Rule: $\mathbb{Z}_7$ Dynamics, Algebraic Structure, and Standard Model Encoding of the GTE Polynomial [40]
The MDL uniqueness theorem T96-02 selects $p(L, C, R) = C + R - CR - LCR$ from $7^{343} \approx 10^{290}$ candidates — the missing Wolfram selection principle. The theorem is a chain whose links are individually machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry): the cross-family coding inequality forcing $k = 7$ (`mdl_ca_rule_coding_closed`), and the within-class uniqueness supplied by the **Direct-Interpolation Lift Theorem** ($\mathbf{A}_{\text{Lean}}$) — the total-parity shadow of the UGP generation orbit, together with vacuum transparency, is full-rank over GF(7) and forces exactly one multilinear rule among all 7^8 candidates: p itself, with Rule 110 arising as a corollary (the binary restriction) rather than as an input — plus a companion **sparsity-floor theorem** ($\mathbf{A}_{\text{Lean}}$): any GF(7) polynomial reproducing Rule 110 has at least four monomials, with equality exactly at p, so multilinearity is forced, not assumed. A **chirality census** over all 120 family orderings sharpens this ($\mathbf{A}_{\text{Lean}}$): the vacuum-transparent survivor set of the *unordered* orbit data is exactly the pair {Rule 110, Rule 124} — the chiral pair of the CMCA architecture — with ordering reversal exchanging the two, so the ordering convention is the chirality gauge. **Five structural theorems:**

1. **Invariant Subset Uniqueness** ($\mathbf{A}_{\text{Lean}}$): the only non-empty subsets $S \subseteq \mathbb{Z}_7$ closed under p are $\{0\}$, $\{0, 1\}$, and $\mathbb{Z}_7$. Rule 110 (the $\{0, 1\}$ restriction) is the unique maximal proper invariant sub-CA of p over GF(7); machine-certified via `native_decide` (`Z7InvariantSubsets.lean`, zero sorry).
2. **QNR Binary Floor** ($\mathbf{A}_{\text{Lean}}$): $\{0, 1\}$ is the unique non-trivial proper invariant subset of p because $N_{\text{fam}} = 5$ is a quadratic non-residue mod 7; machine-certified (`nfam_qnr_explains_binary_floor`, zero sorry).
3. **Polynomial Uniqueness** ($\mathbf{A}$): p is the unique degree-≤ 3 polynomial over GF(7) reproducing Rule 110 on $\{0, 1\}^3$ (1 of $7^4 = 2401$ candidates).
4. **f_{MDL} Non-Polynomial** ($\mathbf{A}$, Schwartz–Zippel): f_{MDL} has 325 zeros vs. a maximum of 147 for any degree-≤ 3 polynomial, proving p and f_{MDL} are algebraically distinct in kind.
5. **40% DPP Cross-Tape Fraction** ($\mathbf{A/D}$): the gravitational (cross-tape) causal edge fraction equals $(N_{\text{TAPES}} - 1)/(N_{\text{TAPES}} + 2) = 2/5$ exactly.

The QNR obstruction is traced to its source, the **master quadratic** $x^2 + x - 1$ (discriminant $5 = N_{\text{fam}}$), the diagonal factor of p over every commutative ring: split over $\mathbb{R}$, where its positive root $1/\varphi$ is the SRRG fixed point seeding the electroweak-scale derivation, and inert at 7 at every 7-adic depth, protecting the binary floor ($\mathbf{A}_{\text{Lean}}$) — one $\mathbb{Z}$-quadratic, two completions, with an all-q splitting dichotomy governed by q mod 5. Eisenstein arithmetic supplies the complementary structure: F_{21} is the norm-7 residue-field quotient $(\mathbb{Z}[\omega]/(3{+}\omega))^+ \rtimes \mu_3$ ($\mathbf{A}_{\text{Lean}}$); the zero variety satisfies the motivic identity $[V(p)] = \Phi_6(\mathbb{L})$; the golden and Eisenstein rings are the two quadratic subfields of $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$, of conductor $15 = N_{\text{gen}}N_{\text{fam}}$; and the full symmetry group of $V(p)$ is AGL(1, 7) of order 42, whose reflection swaps Rule 110 with Rule 124 — the chiral pair of the CMCA architecture ($\mathbf{A}_{\text{Lean}}$). The vacuum is the unique temporally fixed configuration at every ring size (Vacuum Uniqueness Theorem, $\mathbf{A}_{\text{Lean}}$), the Artin–Mazur zeta functions of the ring dynamics factorize exactly, and the period-475 attractor of the 5-cell ring is resolved by an Attractor

Factorization Theorem ($475 = 5 \times 5 \times 19$; **A/D**). Includes the first $k = 7$ polynomial CA spacetime diagrams ever computed and a behavioral classification (Class 3 on generic $\mathbb{Z}_7$; Class 4 on the $\{0, 1\}$ binary sublayer). *Companion papers:* P40 (GF(7) polynomial derivation); P45 (three-tape DPP architecture); P46 (MDL uniqueness and five physical roles); P48 (synthesis monograph).

P50 — The Spin-7 Lattice Model from the GTE Polynomial [60]
The MDL-selected polynomial $p(L, C, R) = C + R - CR - LCR \pmod 7$ defines a natural statistical mechanics system: a 7-state nearest-neighbor spin model with three-body interactions. The 2D model has a phase transition at $T_c \approx 2.86\,J$ (**A**) with $\mathbb{Z}_3$-like symmetry breaking among three degenerate ground states $\{0, 1, 5\}$ (the roots of $p(x, x, x) = -x(x-1)(x-5) \pmod 7$; **A/D**). These are the $\mathbb{Z}_3$ color sectors of the Standard Model. The zero-energy rings are exactly the three uniform configurations at every length $n \geq 3$ ($\mathbf{A}_{\text{Lean}}$, Lean-certified ground-space rigidity), and the exact pair-state transfer matrix yields a thermodynamic entropy rate $S_{\text{CMCA}} = \log(1.4846) = 0.395$ nats per site per step at $\beta = 1$ (**A**). The Walsh nonlinearity $\text{NL}(p) = 1 - c_H/7^2$ (**A/D**) connects the polynomial's cryptographic nonlinearity to the Higgs branch capacity $c_H = 13$. The model is realizable with 7-level Rydberg atoms in optical tweezer arrays. Critical exponents $\nu = 5/6$, $\eta = 4/15$ (3-state Potts universality class) are falsifiable experimental predictions. A continuum-limit section sharpens the CMCA connection into a quantitative lattice-field-theory correspondence. The thermal ensemble is the maximum-entropy shadow of the CMCA tape, and its spectral gap obeys an **exact chiral gap law** $\Delta(\beta) = e^{-3\beta/2}\,(1 + O(e^{-\beta/2}))$ (**A/D**, with the exact correction tower): the slope 3/2 is the geometric mean of the directed wall energies of the pair digraph ($\mathbf{A}_{\text{Lean}}$, `spin7_directed_wall_energies`) — a half-integer, impossible for any parity-symmetric chain, hence a quantitative parity-violation signature of the substrate chirality — and the amplitude is exactly 1, derived by two independent routes. The gap defines a lattice-QFT calibration trajectory whose physical point is selected by the **Tape Saturation Theorem** (**A/D**, conditional on the named Compton-Support Criterion, which is proven minimal via a Boundary Equivalence Theorem): the substrate tape spacing saturates at $a = \hbar c/\Lambda_{\text{GTE}} \approx 0.0972$ fm (tree), with the exact fixed-point signature $\xi(\beta^*) = |\mathbb{Z}_7| = 7$ and the interlocking lattice dictionary $aM_{\text{kink}} = 1/7$, $am_\varphi = 7/8$. The unique empirical discriminator is a substrate Monte Carlo measurement of the kink correlation length: $\xi_{\text{kink}} = 7$ cells (band 6.78–7.23 ± 0.54); a measurement outside the band falsifies the MDL-saturation selection, while a landing at 7 cells would make the physical point unconditional. *Companion papers:* P41 (single-tape CMCA); P45 (three-tape architecture); P49 (GTE polynomial).

P51 — The Polynomial Certificate of Transputation [47]
The Three-Level MDL Unification Theorem establishes that the same MDL formula $P^\top(X) = \arg\min K(x|\text{PSC constraints})$ governs theory selection, field dynamics, and quantum event adjudication. The GTE polynomial p is characterised as a three-part PSC-correction of itself, the polynomial's real diagonal encodes the SRRG adjudication fixed point φ ($\mathbf{A}_{\text{Lean}}$), and DSAC is positioned as the continuum member of the transputation adjudicator class $[D]$. The discrete measurement mechanism — formalised as the unique PSC-admissible orbit $\text{GEN}_1 \to \text{GEN}_2 \to \text{GEN}_3 \to \text{VAC}$ ($\mathbf{A}_{\text{Lean}}$) — provides GTE's answer to the quantum measurement problem without collapse, many-worlds, or external rules. The computability gap is classified two-sidedly: record truth on the diagonal fragment is Σ^0_1-complete while the adjudicator is limit-computable, so the adjudication function's diagonal restriction has Turing degree exactly $0'$ (Δ^0_2-Turing-complete; the Gold–Putnam trial-and-error stratum) — quantum measurement sits precisely at the halting degree, never above the first Turing jump (**A/D**; many-one degree $\mathbf{A}_{\text{Lean}}$). *Companion papers:* P37 (QM from Rule 110); P48 (GTE framework); P49 (GTE polynomial).

P52 — The PSL(2,7) Algebraic Structure of the GTE Framework [50]
This paper proves that the three algebraic pillars of the GTE framework — the Fibonacci–Möbius

action on PG$(1,7)$, the metacyclic group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, and the Eisenstein integers $\mathbb{Z}[\omega]$ — are unified through PSL$(2,7)$, the unique simple group of order 168. All results are machine-certified in Lean 4 with zero sorry. **(1)** F_{21} is the Borel stabilizer of ∞ in PSL$(2,7)$, and the Fibonacci–Möbius generator together with F_{21} generate PGL$(2,7)$ of order 336 (`f21_is_borel_psl27`, `pgl27_generated_by_singer_and_borel`, $\mathbf{A}_{\text{Lean}}$). **(2)** PSL$(2,7)$ is the automorphism group of the Fano plane $\mathcal{F}$, and F_{21} acts simply transitively on all 21 incidence flags of $\mathcal{F}$, giving it a canonical role as the flag-transitive subgroup (`psl27_is_aut_fano`, `f21_regular_on_fano_flags`, $\mathbf{A}_{\text{Lean}}$). **(3)** The Eisenstein integers supply both F_{21} and A_4 from the same functor: the split prime $7 = \pi\bar{\pi}$ gives $\mathbb{Z}[\omega]/(\pi) \cong \text{GF}(7)$ and F_{21} from its unit group; the inert prime 2 gives $\mathbb{Z}[\omega]/(2) \cong \text{GF}(4)$, whose additive group is the Klein four-group V_4 and whose multiplicative group is $\mathbb{Z}_3 = \mu_3$, assembling to $A_4 = V_4 \rtimes \mu_3$ (`eisenstein_a4_from_inert_2`, $\mathbf{A}_{\text{Lean}}$). **(4)** The Klein quartic — the unique Hurwitz surface with $|\text{Aut}| = 84(g-1)$ — has genus equal to $N_{\text{gen}} = 3$, the number of Standard Model generations, arising from the $(2,3,7)$ triangle group shared with PSL$(2,7)$ (`klein_quartic_genus_eq_n_gen`, $\mathbf{A}_{\text{Lean}}$). The manifest GTE neutrino flavor symmetries μ_3 and the μ–τ exchange $\mathbb{Z}_2$ generate $S_3 \leq A_4$ ($\mathbf{A}_{\text{Lean}}$), which completes the A_4 leading-order tribimaximal neutrino mixing group (CatAD); the GTE orbit-ratio corrections are the A_4-breaking departures. *Companion papers:* P49 (GTE polynomial and Eisenstein residue model); P48 (synthesis monograph).

P55 — The Octonionic Shadow of GF(7): Color, Chirality, and Three Generations from a Quadratic-Residue Difference Set [45]
https://doi.org/10.5281/zenodo.21200048

This paper establishes that the GTE/UGP arithmetic programme and the division-algebra programme (Dixon; Furey; Baez–Huerta) are two projections of a single finite structure: the quadratic-residue set $\text{QR}(7) = \{1,2,4\} \subset \mathbb{F}_7^*$ is simultaneously the arithmetic anchor of GTE and the Fano-plane orientation that uniquely determines the octonion multiplication table, answering from outside the framework the question "why $\mathbb{F}_7$?"—because GF(7) is the unique prime whose quadratic residues orient a normed division algebra, and Hurwitz's theorem implies the normed-algebra tower terminates precisely there. A six-link machine-verified derivation chain from $D = \{1,2,4\}$ alone establishes: (i) the Fano plane from the $(7,3,1)$ difference set, with the three-pencil count giving $N_c = 3$ directly ($\mathbf{A}_{\text{Lean}}$); (ii) the octonion multiplication table from the oriented translates of D, unique up to relabeling ($\mathbf{A}$); (iii) the exceptional isomorphism PSL$(2,7) \cong$ GL$(3,2)$, order 168, by Todd–Coxeter coset enumeration ($\mathbf{A}_{\text{Lean}}$, `hurwitz_presentation_order_168`); (iv) $\text{der}(\mathbb{O}) = \mathfrak{g}_2$ (dim 14) and $\text{Stab}_{G_2}(\text{apex}) = \mathfrak{su}(3)$ (dim 8) from exact linear algebra ($\mathbf{A}_{\text{Lean}}$, 39 Lean theorems including `derivation_algebra_is_g2_certificate` and `apex_stabilizer_is_su3_certificate`); (v) UGP's flavor package (μ_3, U, V_4, A_4) is equivariantly isomorphic to the Spin(8) triality package, pinning $\text{gen}_1 \leftrightarrow V$ via Eisenstein norm uniqueness $b_{\text{gen}_1} = 73 = \mathcal{N}(9+\omega)$ ($\mathbf{A}_{\text{Lean}}$, `G6_rigidity_conditional`); (vi) the charged-lepton Koide phase $|\delta \bmod 2\pi/3| = 2/9 + 7.4 \times 10^{-6}$ (7 ppm from PDG) as a function on the triality orbit ($\mathbf{A}$). The neutrino sector is completed by the GTE seesaw: the right-handed seed ordering $b_{R,1} < b_{R,2} < b_{R,3}$ pins the mass eigenstates ν_k via $m_{\nu,k} \propto b_{R,k}^{29/9}$ and predicts normal ordering, with Δm^2_{21} and Δm^2_{31} within 1% of NuFIT 6.0 IC24 NH ($\mathbf{A}_{\text{Lean}}$, `seesaw_triality_pinning_bundle`). The algebraic layer is formalized in eleven zero-sorry Lean 4 modules (174 theorems, all in `ugp-lean`). The convergence of two independent derivation paths—GTE finite arithmetic and the continuous division-algebra programme—on identical outputs (N_c, charge spectrum, $N_{\text{gen}} = 3$) constitutes structural evidence that these quantities are genuine consequences of GF(7) arithmetic. *Companion papers:* P48 (primary derivation of $N_c = 3$ and $N_{\text{gen}} = 3$ from GTE arithmetic); P52 (PSL(2,7) algebraic unification); P18 (Koide relation); P21 (neutrino mass-squared ratio).

3.11 Group 10 — Meta-Synthesis and Assessment

Papers P53 and P54 stand outside the main GTE derivation chain as meta-level works: P53 evaluates GTE against competing physics programmes; P54 synthesises GTE and NEMS into a unified treatment of consciousness, measurement, and the primordial ground.

P53 — The GTE Framework: A Comparative Assessment [31]
https://doi.org/10.5281/zenodo.20684419
A comparative assessment of the GTE framework against ten alternative programmes in fundamental physics, evaluated on a neutral eleven-dimension rubric covering selection principle, free-parameter economy, SM gauge-group and quantum-postulate derivation, spacetime origin, cross-sector predictivity, quantitative agreement, falsifiability, formal verification, standing-problem resolution, and ontological economy. Includes a side-by-side scoring matrix, the compression-funnel diagram (19 bits $\rightarrow$ $\sim$280 physical quantities), an explicit near-term falsification register with experiment names and thresholds, and an honest assessment of the named conditionals and six global open problems. Appendices reproduce the master prediction index and the novel-particle search-targets table (GTE-P7 through GTE-P11). $\mathbf{A}_{\text{Lean}}$ and **A/D** results cited at their certified levels throughout. *Companion papers:* P48 (synthesis monograph); all papers P01–P52 (sources of evaluated results).

P54 — The Fire in the Equations: Consciousness, Physics, and the Primordial Ground [28]
https://doi.org/10.5281/zenodo.20709837
A formal synthesis of the GTE and NEMS programmes addressing three related problems simultaneously: the quantum measurement problem (resolved via transputation — the PSC-forced, non-computable, observer-free branch-selection proved necessary from first principles), the ontological grounding problem (the Alpha Theorem of NEMS P63 proves the necessary existence of a pre-categorial ground from the no-free-bits constraint), and the hard problem of consciousness (proved to be a category error: known qualia are already irreducible on-ledger semantic content, not generated from outside — NEMS P55 Theorem 55.3 closes the "ghost-qualia" escape route). The three-aspect synthesis (NEMS P69–P70, machine-checked zero sorry) establishes that Ground (α), Articulation (physics), and Manifestation-in-Awareness (qualia) are three irreducible but co-dependent aspects of one Alpha-grounded fact, dissolving substance dualism and the mind-body gap. The SIAM (Self-Indexing Adjudicative Manifold) architecture characterises the formal structure of unified sentient agency; five structural properties distinguish SIAM systems from the rest of a uniformly Alpha-grounded universe. Two SIAM-specific falsifiable predictions are derived: an aggregate Landauer heat floor $Q_{\text{SIAM}} \geq N_{\text{adj}}\, k_B T \ln 5$ per cognitive adjudication cycle (**A/D**, specific to the $\mathbb{Z}_7$ five-sector kink structure, 2.32$\times$ above the generic one-bit floor), and a conditional decoherence enhancement for quantum systems interacting with SIAM-attending environments (**D**). Extended Wigner's-friend no-go theorems (Frauchiger–Renner 2018; Bong et al. 2020) are shown to sharpen rather than challenge the transputation response. Systematic comparison with Penrose–Hameroff Orch OR, IIT, GWT, Dennett's eliminativism, panpsychism, and Chalmers' property dualism. *Companion papers:* P48 (GTE synthesis monograph); NEMS P55, P63–P70, P73–P76 (Alpha, qualia, SIAM, three-aspect synthesis); P51 (transputation).

4 Comparison with Alternative Frameworks

4.1 Grand Unified Theories (GUTs)

Conventional GUTs (SU(5), SO(10), E_6, E_8) address the unification of gauge couplings by embedding the Standard Model gauge group into a larger group, then recovering the SM at low energies through

symmetry breaking. This approach requires specifying the larger group, the breaking pattern, and the Higgs sector. These are all additional specification costs on top of the SM. The framework then has additional free parameters, new heavy gauge bosons, and a proton-decay prediction that has not been confirmed experimentally.

The GTE/Φ_{MDL} programme inverts this logic. Rather than embedding a larger structure and breaking it down, the SM gauge structure is derived *from below* from a single 19-bit algebraic object: the polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7). The GUT-scale Weinberg angle $\sin^2\theta_W^{\mathrm{GUT}} = 3/8$ is reproduced ($\mathbf{A}_{\mathrm{Lean}}$) as a structural identity, not from a GUT embedding. No additional heavy particles, proton decay channel, or Higgs sector is introduced. The specification cost per physical role is zero: the 19 bits of p simultaneously govern spacetime dynamics, gauge vertex conservation, gravity, quantum entanglement, and Bell nonlocality.

4.2 Wolfram Physics Project

The Wolfram Physics Project [73] seeks to derive physics from a simple hypergraph rewriting rule, identifying candidate rules by empirical search. The GTE framework shares the goal of deriving SM structure from a minimal CA rule but differs in two important ways: (a) the rule is *derived* from the SM orbit (the SM generation sequence forces Rule 110 via the GTE/MDL cascade; CUP-4, $\mathbf{A}_{\mathrm{Lean}}$) rather than chosen empirically; (b) the physical substrate is the continuum Φ_{MDL} field (Rule 110/the CMCA is the algebraic *certificate* proving Φ_{MDL}'s structure), not the discrete automaton itself. The no-CA-replica theorem (`no_finite_ca_exact_lorentz_replica`, $\mathbf{A}_{\mathrm{Lean}}$) establishes that no finite-resolution CA can exactly replicate the Lorentz invariance of Φ_{MDL}; only the continuum limit recovers exact Lorentz symmetry. A detailed comparison is given in P28 [22]. The complete algebraic analysis of $p(L, C, R)$ as a standalone $\mathbb{Z}_7$ dynamical system — invariant subset classification, QNR binary-floor theorem, and the MDL derivation as Wolfram selection principle — is developed in P49 [40].

4.3 Loop Quantum Gravity, Causal Dynamical Triangulations, Spin Foams

Discrete quantum-gravity approaches (LQG, CDT, spin foams) impose background independence as an axiom and seek to recover classical geometry in a continuum limit. The GTE framework takes a complementary but distinct route: the spacetime dimension $D = N_{\mathrm{gen}} + 1 = 4$ is *derived* from the SM generation orbit ($\mathbf{A}_{\mathrm{Lean}}$), the spectral dimension $d_s = 4$ in the thermodynamic limit is separately certified ($\mathbf{A}_{\mathrm{Lean}}$, `causal_graph_spectral_dim_thermodynamic_limit`), and all twelve $\mathfrak{so}(1,3)$ commutation relations are machine-certified from the three-tape architecture ($\mathbf{A}_{\mathrm{Lean}}$). The standard of Lean-machine-certified evidence for emergent Lorentz symmetry and dimension goes beyond what is currently available in LQG/CDT/spin-foam treatments, which rely primarily on semiclassical approximations. UV finiteness on arbitrary smooth curved backgrounds is additionally established (P44) via the DeWitt–Schwinger heat-kernel expansion.

4.4 String Theory and M-Theory

String theory derives gauge unification and gravity jointly but requires compactification of extra dimensions, a landscape of $\sim 10^{500}$ vacua, and no known selection principle for the SM vacuum. The GTE framework derives the same physical content — SM gauge group, gravity, quantum mechanics — from one foundational principle (PSC — Perfect Self-Containment), with MDL as its operational expression (Presentation Invariance, Layer II PSC), and a unique selection mechanism. PSC acts as a vacuum-selection constraint that is absent in the string landscape.

4.5 Digital Physics / It-from-Bit Proposals

Zuse's *Rechnender Raum* [74], Fredkin–Toffoli reversible CAs, and Wheeler's "it from bit" are precursors that propose a discrete computational substrate. The GTE framework identifies the precise sense in which this is correct: the *algebraic certificate* (CMCA, f_{MDL}, Rule 110) is discrete, but the *physical substrate* (Φ_{MDL}) is a continuum field. The two are related by the Algebraic Lifting Theorem and Algebraic Descent Theorem ($\mathbf{A}_{\mathrm{Lean}}$), which prove that every F_{21}-dependent property is simultaneously a property of the discrete certificate and the continuum field. The programme thus realises the "it from bit" intuition precisely and rigorously, while avoiding the Lorentz-symmetry problems faced by purely discrete substrates.

4.6 Key Distinguishing Features

In summary, the GTE/Φ_{MDL} programme is distinguished by:

- Zero free dimensionless parameters: all SM masses, couplings, and mixing angles are derived from one foundational principle (PSC) and a single integer seed; the single dimensional anchor m_τ sets the energy scale.
- Machine certification: core results are Lean 4 machine-checked with zero sorry and standard Mathlib axiom closure, providing formal verification unavailable in competing frameworks.
- Logical necessity: the constants are not fitted but forced by algebraic self-consistency.
- Dual-level architecture: the continuum Φ_{MDL} field and its discrete algebraic certificate are related by proved lifting/descent theorems, avoiding the tension between exact Lorentz symmetry and discreteness.
- Single-Source Principle: all of spacetime, matter, gravity, and quantum mechanics are co-generated by a single 19-bit polynomial with no additional specification cost.

5 Reading Paths by Reader Type

Table 1: Recommended entry sequences by reader background.

Reader type	Recommended path
Particle phenomenologist	P01 (full paper) → P06 → P17 → P21 → P24 (uniqueness/Galois) → P29 (dark sector) → P31 (Weinberg angle) → P32 (CKM) → P35 (capstone unification)
Formal methods / Lean	ugp-lean formalization paper → P01 §1.3 (claim taxonomy) → P22 → P17 → P24 Appendix C (RCC proof) → P27 (srrg-lean) → P28 (Rule 110 universality) → P30 (Cook formalization) → P40 (GF(7) polynomial) → P49 (invariant subsets, QNR floor) → P43 (coalgebra completeness) → P52 (PSL(2,7) algebraic unification) → P55 (octonionic bridge; 174 Lean theorems)
Division algebra / algebraic geometer	P55 (entry point: QR(7) → octonions → G_2 → SU(3); 174 Lean theorems) → P52 (PSL(2,7) algebraic unification) → P48 (GTE complete framework) → P01 §§2–5 (GTE cascade and SM derivation)
Mathematical physicist	P05 → P24 → P01 → P18 → P19 → P27 (SRRG + IPT) → P35 (GTE unification) → P43 (complete framework) → P52 (PSL(2,7)) → P55 (octonionic bridge)
Philosopher of physics	P20 → P09 → P10 → P11 → P07 → P01 abstract → P34 (Möbius architecture: computation/transputation duality)
Cross-domain / biology–physics	P15 → P26 → P25 → P03 → P01 §7 (extensions)
Quantum gravity / QFT	P28, P34 (Möbius architecture), P35 (GTE unification), P36 (emergent gravity, $D = 4$), P38 (Φ_{MDL} gravity), P41 (CMCA), P42 (Φ_{MDL} field), P43 (QM + gravity), P44 (QG, n_s, $r = 0$), P45 (3+1D, DPP), P46 (polynomial UFT), P47 (cosmology), P48 (complete synthesis monograph), P49 (Wolfram bridge, $\mathbb{Z}_7$ dynamics), P50 (spin-7 lattice; condensed matter)
Sceptical reviewer (adversarial)	P01 §10 (open problems) → P01 §1.3 (taxonomy table) → P24 claim-status table → P14 (TE2.2 scan) → P01 §8 (robustness) → P28 (null tests) → P35 capstone claims box → P43 impossibility theorems → P45 §3 (claims guide) → App. A (falsification table)
Quick start (10 minutes)	P01 §§2–5 + §6.2 (neutrino structural) + Appendix G (α_s)
Experimentalist	P01 abstract (headline results) → P01 §9 (falsifiability table) → P21 (neutrino prediction) → P31 (Weinberg angle prediction) → P29 §5 (dark matter; $Q = 0$ at Belle II) → P39 §4 (QCD predictions; f_π, α_s) → P44 §5 ($r = 0$ prediction for LiteBIRD) → P45 §12 ($\delta_{CP} = 205.71°$)

6 Logical Dependency Structure

Figure 1 shows the high-level logical dependency structure of the UGP Physics series within the broader Reflexive Reality programme.

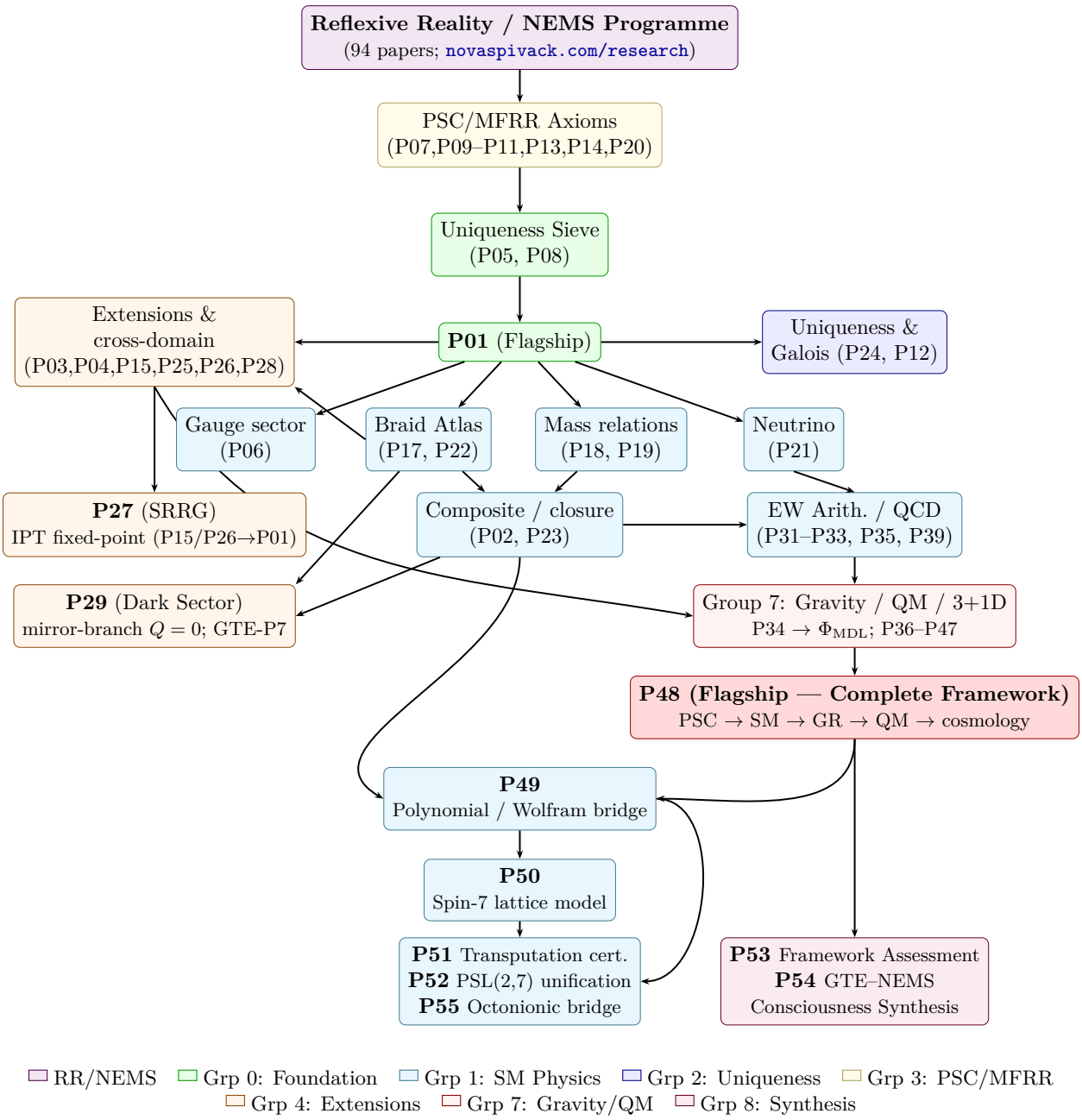


Figure 1: High-level logical dependency structure within the broader Reflexive Reality programme. An arrow indicates that the target paper or group builds substantially on the source; node colours identify the thematic groups of §3 (legend above). Group 5 (Lean formalization) is omitted for clarity; the Group 7 node collects the quantum-gravity/spacetime papers (P34–P47), whose internal structure is described in §3. P48 is the synthesis monograph integrating Groups 0–7 into a single PSC-to-cosmology derivation chain, followed by the polynomial-algebra terminus P49–P52 and P55. The Group 10 (purple) node collects P53 (framework assessment: GTE vs. ten alternative programmes on an eleven-dimension rubric) and P54 (GTE–NEMS consciousness synthesis: transputation, Alpha Theorem, SIAM, hard-problem dissolution); both sit outside the main derivation chain and depend on P48. P54 additionally draws on the NEMS phenomenology papers (Alpha, qualia, SIAM, three-aspect synthesis), shown by the dashed violet arrow from the Reflexive Reality / NEMS node.

7 Summary Table: Complete List of Papers

Table 2: Complete paper roster with group, status, and one-line description (P01–P55). Group 7 = Quantum Gravity/Spacetime/Φ_{MDL}/3+1D; Group 8 = Complete synthesis monograph; Group 9 = Polynomial algebra, Wolfram bridge, algebraic unification, and division-algebra bridge; Group 10 = Meta-synthesis and assessment.

P#	Grp	Short title	Key contribution
P01	0	SM from UGP (flagship)	Seven headline SM results; five-status taxonomy; α_s pre-commitment ($+0.24\sigma$, PDG 2024); m_W at -0.42σ PDG 2024 with two-loop SM running; Λ at 0.31σ (Planck 2018, SM-17 confirmed); $\lambda_H = \varphi/(4\pi)$ at 0.26σ (SM-18); bare $m_H = 124.2$ GeV at 9.1σ tension; self-consistent $m_H = 124.971$ GeV at -2.08σ (Grade C, M_W anchor); structural $v_{\mathrm{PSC}} = 246.16$ GeV (grade [$\mathrm{A_{Lean}}$]; see P27 VEVProof).
P02	1	GTE Particle Spectrum	1M-candidate discovery run; all 24 SM particles in Green tier; 9 novel candidates (GTE-P1, P2, P3, P6–P11). All 9 novel mass windows (3 MeV–1.9 GeV) OPEN under combined lab dark-photon limits; GTE-P7 (212 MeV, $Q = 0$ Lean-certified) correct search = mono-photon/missing energy. Per-particle register: Master Index Table 28.
P03	4	Nuclear ML Descriptors	GTE features beat random baseline on binding energies ($\sim$25%); magic numbers derived; IPT match (κ ratio = 1.1494 $\approx$ IPT, 1.63%). 6-term stability law (NUBASE labels) 94.5% CV. New: F_{10} parity feature (GTE b-seed); $\Delta_{\mathrm{pair}} = 5^{3/2} = 11.18$ MeV prediction (0.003%, [B] bridge). Smooth-law limitation for Tc/Pm disclosed.
P04	4	UGP Dynamics & Universality	Turing-universal (Rule-110 embedding, machine-checked); 3 RG basin attractors; Q_4 predicts basin; Lean entropy witness. Extended in [22] to add two-way Rule 110 convergence and full PSC chain.
P05	0	Uniqueness: Two-Stage Proof	Stage-1 arithmetic + Stage-2 instantiation proves $(n = 10, b_1 = 73)$ unique.
P06	1	Algebraic & Geometric Foundations	Proves Quarter-Lock; derives g_1^2, g_2^2, g_3^2 from group-specific kernel invariants.
P07	3	UGP/GTE as Nine Meta-Laws	GTE subsumes Jaynes, Yau, Yang–Mills, Jacobson; Zipf derived; Lean entropy witness.
P08	0	UGP Foundational Monograph	Kernel Symmetry Theorem; UWCA Turing-universality; Fibonacci rigidity; Lawvere fixed point.
P09	3	Architecture of a Computable Universe	Five architectural principles (Constrained Contingency, Ontological Scaffolding, Reflexive Computation, Constants as Compression Ratios, Necessary Observers); structural/organisational complement to P07's empirical meta-laws.
P10	3	Reflexive Reality	Formalises reflexive reality; Foundational Finality + Reflexive Closure machine-checked.
P11	3	Ontological Dissonance Minimization	D-minimization $\leftrightarrow$ Φ-proxy ($r = -0.91$); strong-interaction potential from ODM.
P12	2	Unified Rigidity Theorem	Three rigidity mechanisms agree on canonical branch; ASSUMPTION_LEDGER audit.

P#	Grp	Short title	Key contribution
P13	3	MFRR Foundational Monograph	Transputation (Lean, zero sorry); 17 closure theorems; Two-Layer PSC Theorem. $w_0 = -1$ maintained; 3.5σ tension with DESI DR1+SNIa (DESI DR3 as falsification criterion). Three condensed companion summaries are included (Reader's Guide for Physicists, Summary for Physicists and Referees, Popular Article).
P14	3	PSC Concordance	TE2.2: 20,160 universes, SM unique D minimum; 3 new UGP coupling predictions.
P15	4	Information Profit Principle	IPT ≈ 1.1309 from Reflexive Landauer Bound; 6 cross-domain validations (incl. organizational survival).
P16	6	Black Hole Unitarity	Stinespring dilation of Hawking channel; Page curve + exact PT^{-1} recovery.
P17	1	Braid Atlas v2.0	$Q = W_g/N_c$ (Lean); $T/T^\dagger$ chirality; 9 baryon triples Lean-certified.
P18	1	Koide Cyclotomic Closed Form	m_τ predicted at 61 ppm; $\theta = 2/9$ from $N_c = 3$ (Lean); cyclotomic-12.
P19	1	Cyclotomic-12 Mass Structure	TT ($\pi/6$ from A_2 Weyl; Lean) and VV (SU(5)/SO(10); Lean) mass relations. Wolfenstein $\lambda \approx \varepsilon_1^{13/9} = e^{-13\pi/27} \approx 0.2203$ (1.9% off PDG; improved from 12%; exploratory).
P20	3	MFRR Physics Survey	Born rule, arrow of time, no-emulation (machine-checked); IPT 0.08% match; SM #1/34560.
P21	1	Neutrino Mass-Squared Ratio	First parameter-free $\Delta m^2_{21}/\Delta m^2_{31}$ prediction; 0.16σ from NuFIT 6.0 global fit (0.52%); Lean exponent 29/9; normal hierarchy preferred at 2.5σ (NuFIT 6.0).
P22	1	UGP Interaction Skeleton	Lean: full SM interaction skeleton; proton stability; 3 proofs of $N_c = 3$; 0 mismatches.
P23	1	Closure Structure	Reflexive closure $r = -0.913$ ($n = 38$); all 38 hadron triples Lean-certified.
P24	2	Arithmetic Uniqueness	Sparsity theorem; Positive-Root; Chirality; $\mathbb{Q}(\zeta_{120})$; E7 excluded; WZW falsified. $N_c = 3$ Lean-certified from substrate arithmetic via Fibonacci-Mersenne GCD (`nc_eq_3_from_mersenne_gcd`, $\mathbf{A}_{\mathrm{Lean}}$) and ridge factorial uniqueness (`nc_uniqueness_from_ridge_divisors`, $\mathbf{A}_{\mathrm{Lean}}$); `NcColorArithmetic.lean`.
P25	4	Genetic Code Selection	Two-stage sieve; 8 biological criteria; standard code $z = +3.22\sigma$; unique survivor.
P26	4	General Selection Theory	GSP unifies nuclear/bio/SM/org/microbial selection via admissibility + IPT viability; 5 empirical domains + 1 model-internal symbolic-communication result (language alphabet) (consciousness speculation in App. [I]); microbial FBA: $r(G/D, \mu) = 0.971$ [B]; org domain: 7 tests, HR= 5.01×, BC= 0.907, Levene $p = 0.046$ [B$^-$]; $\mathbb{Q}(\zeta_{120})$ in 4 domains; revised taxonomy: [Br] bridge, [B+]/[B]/[B$^-$] empirical grades; evidence-grade table Table 1; companion to P27.

P#	Grp	Short title	Key contribution
P27	6	Self-Referential RG	SRRG gradient-flow; IPT = IR fixed point (Lean, zero sorry); β_η derived via Vieta [A_{Lean}]; h_psc_sc [A^-]; conditional: $\theta_{QCD} = 0$, $N_{gen} = 3$, SM gauge, λ_H, Λ_{vac} [B]; $v_{PSC} = 246.16$ GeV (-0.024%, grade [A_{Lean}], zero open axioms; VEVProof.*); α direct coupling $1/\alpha(M_Z) \approx 127.311$ (0.495% PDG) [A/D]; SRRG–MDL OP9 **upgraded**: `op9_catal_unconditional` ($\mathbf{A}_{Lean}$, zero sorry), `srrg_op9_biconditional` (zero sorry; full closure conditional on `UGPSubstrateConstraint`); SRRG $g^* = 1/\varphi$ = PSC-adjudication fixed point (**A/D**; see P47); companion to P26.
P28	4	Computational Universality	Two-way Rule 110 convergence: SM orbit uniquely forces Rule 110 (CUP-4, Lean) and UGP substrate independently implements it (P04). Garden of Eden: gen_1 has zero $\mathbb{Z}_7$ predecessors (Lean, `native_decide`). Tape-level unification: both SM charge and mass sectors on one Rule 110 tape. PSC chain Lean-certified. 76-bit MDL-minimal f_{MDL}. $N_c = 3$ Lean-certified from substrate arithmetic alone (`nc_eq_3_from_mersenne_gcd`, $\mathbf{A}_{Lean}$, `NcColorArithmetic.lean`). Orbital generation discriminant (mesoscopic form): the f_{MDL} forward chain $gen_1 \to gen_2 \to gen_3 \to$ vacuum assigns a unique position index to each SM fermion generation (`zgm_mes_orbital_discriminant` + `zgm_mes_closed`, `ZGMMesInvariant.lean`, $\mathbf{A}_{Lean}$, zero sorry); SU(3) color *dynamics* in f_{MDL} remain open.
P29	6	Dark Sector Braid Atlas	Mirror-branch dark sector: $Q = 0$ for all dark particles (Lean, zero axioms); dark lepton masses 0.54/24.5/3600 MeV; $SU(3)_{dark}$ only (no EW symmetry); $c(W) = b_2(\text{RHN}) = q_1$ structural unification (Lean); GTE-P7 prediction refined to 211.9 MeV; asymmetric dark matter asymmetry formula $\eta_\chi \approx (2/7)\,\eta_{dark}$.
P30	5	Cook's Rule 110 in Lean 4	Conditional Lean top theorem `cook_operational_stage3_tm_microstep_readback` (5 named Cook bridge axioms; a conditional operational readback certificate, not itself a Turing-universality theorem); C1 arbitrary-word discharge ($\mathbf{A}_{Lean}$, zero sorry); all 5 collision kinds certified; `CookFinTM2Compiles` for Mathlib halting machine ($\mathbf{A}_{Lean}$).
P31	1	Electroweak Mixing Angle	$\sin^2\theta_W = 3/13 = 0.23077$ from palindrome decomposition of f_{MDL} (Lean, 10-step chain); -0.225% (13.0σ) from PDG 2024 bare; corrected $384729/1664000 \approx 0.23121$ (0.42σ PDG 2022, $\mathbf{A}_{Lean}$); two-loop closure 0.23129154 ($+0.038\sigma$ PDG 2024, **B**); GUT-scale 3/8 exact (Lean).
P32	1	CKM Wolfenstein Parameters	All four parameters from orbit arithmetic, zero free params; $\lambda = 9/40 = 0.225$ (0.000% from PDG, Lean-certified); $A, \bar\rho, \bar\eta$ from quark-sector b-indices.

P#	Grp	Short title	Key contribution
P33	1	Deeper SM Consequences	5 Lean-certified results: EW boson staircase $c \in \{11, 12, 13\}$ (Higgs as CA arithmetic); CP violation from f_{MDL} output exclusion; $N_{\mathrm{gen}} = 3$ topologically forced; photon = CA vacuum fixed point; temporal emission/absorption duality of SM vertex structure. Unified W_B=0 colorless sector discriminant: photon (unique f_{MDL} fixed point), neutrino (unique k=7 composite), Z (c=12), Higgs (c=13) are pairwise distinguished by a single first-principles labeling (`z7_zero_sector_unified_discriminant`, `Z7ZeroSectorDiscriminant.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry). **$\mathbf{A_4}$ PMNS context (P48/P52):** the Eisenstein functor identifies $A_4 = V_4 \rtimes \mathbb{Z}_3$ as the leading-order neutrino flavor symmetry (tribimaximal mixing), with GTE orbit-ratio corrections ($\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 19/42$, $\sin\theta_{13} = 11/73$, $\mathbf{A}_{\mathrm{Lean}}$) as the A_4-breaking departures; NuFIT 6.0 IC24 NH.
P34	7	GTE-Möbius Architecture	**Axiomatic foundation** (P34–P47): Möbius triple $(A, e, [D])$, D1–D5, TPC class (decision-theoretic shadow at Turing degree exactly $0'$; see P51), computation/transputation boundary; $A = \Phi_{\mathrm{MDL}}$ kink sector (Level 2); CMCA = Level-1 certificate; C1 closed ($\mathbf{A}_{\mathrm{Lean}}$); C2–C5 open.
P35	1	GTE Unification (capstone)	Bidirectional 6-arrow correspondence GTE↔SM; 7-conjunct joint unification theorem (Lean, zero sorry); $\sin^2\theta_W = 384729/1664000$ (-0.42σ PDG 2022, $\mathbf{A}_{\mathrm{Lean}}$); $\lambda = 9/40$ (0.000%); $d_s = 4$ (Lean); QCD from F_{21}; $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$; FKTT, $\mathbf{A}_{\mathrm{Lean}}$); zero free params.
P36	7	Emergent Gravity (Rule 110)	$D = 4$ Lean-certified (`gte_spacetime_dimension`, zero sorry); vacuum Einstein eq from Ollivier–Ricci curvature of Rule 110 causal graph (Lean); matter step $\kappa_{\mathrm{SD}} > 0$ (computational); MDL↔Lovelock structural correspondence ($\mathbf{A/D}$).
P37	7	QM from Rule 110	't Hooft CA interpretation; single vacuum fixed point; transient stability hierarchy matches SM stability; Two-Role Structural Principle; Born rule from cogwheel construction ($\mathbf{A}_{\mathrm{Lean}}$); eigenvalue-to-mass falsified.
P38	7	Emergent Gravity (Φ_{MDL})	$T_{\mu\nu}$ from KG Lagrangian (Lean, zero sorry); linearised Einstein equations (MDL-Lovelock + minimal coupling); kink as gravitational source (1.4×10^{-6} precision); $\tau_{\mathrm{OR}} \approx 20\,\mathrm{Myr}$; classical $\Lambda = 0$ at all $\mathbb{Z}_7$ vacua.
P39	1	QCD from $F_{21} \subset SU(3)$	$F_{21} = \Sigma(21)$ forced; colour factors $C_F = 4/3$, $C_A = 3$, $T_F = 1/2$ (Lean); $b_0 = 7$, $b_1 = 26$ (Lean); $\alpha_s(M_Z) = 0.1193$ (+1.10%, 3-loop); $\theta_{\mathrm{QCD}} = 0$ (3 proofs, Lean); $f_\pi = 92.34\,\mathrm{MeV}$ (+0.30%); mass gap (Lean, zero axioms); EFT below the derived seven-kink threshold $\Lambda_{\mathrm{GTE}} = 7M_{\mathrm{kink}} \approx 2.0$ GeV.

P#	Grp	Short title	Key contribution
P40	5	GF(7) Polynomial & Finite NAND Completeness	$p(L,C,R) = C + R - CR - LCR$ unique over GF(7) (Lean, zero sorry); $p(L,1,R)$ = NAND yields finite two-input Boolean functional completeness (Sheffer; one named axiom), independently of Cook's construction; Rule 110's Turing universality remains Cook's theorem (2004); connects $\mathbb{Z}_7$ substrate to Φ_{MDL} field.
P41	7	Chiral Minkowski CA	Three-layer CMCA (Rule 110 + Rule 124 + clock); Lorentz invariance as PSC observer effect (Lean, one axiom); $\sin^2\theta_W$(threshold) = 0.231207 (-0.42σ PDG 2022, $\mathbf{A}_{\mathrm{Lean}}$); two-loop 0.23129154 ($+0.038\sigma$ PDG 2024, **B**); $M_Z = 91.91\,\mathrm{GeV}$ (+0.80%); CPT, causality, Bell non-locality.
P42	7	Φ_{MDL} Field	Poincaré invariance of KG field ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry); Born rule from dual-frame inner product ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry); thermal state; interference and decoherence predictions; quantum kink mass $M^Q = 281 \pm 21$ MeV (two routes, benchmark-validated); $e^2(\Lambda_{\mathrm{GTE}}) = 7/2$ scale-band consistency ($+1.3$–1.5σ with measured dissolution constant); kink HVP prediction $\Delta\alpha_{\mathrm{kink}} = (3.5\text{–}3.7) \times 10^{-4}$; lattice-campaign tape spacing $a = \hbar c/\Lambda_{\mathrm{GTE}}$ derived via the Tape Saturation Theorem (**A/D** cond.; P50) with pre-registered substrate-MC target $\xi_{\mathrm{kink}} = 7$ cells; continuum completion of P41.
P43	7	Complete Φ_{MDL} Framework	7 theorems in 5 areas: algebraic necessity of $\mathcal{F}_{21}$ ($\mathbf{A}_{\mathrm{Lean}}$); Born rule unconditional ($\mathbf{A}_{\mathrm{Lean}}$); M_{Pl} derived to 0.040%; Bekenstein-Hawking from 2 routes; geodesic theorem ($\mathbf{A}_{\mathrm{Lean}}$, zero custom axioms); impossibility of alternatives ($\mathbf{A}_{\mathrm{Lean}}$); $\mathbb{Z}_7$ uniqueness from 3 arguments; zero free params in gravity.
P44	7	Quantum Gravity Completeness	Perturbative QG of GTE/Φ_{MDL}: $\xi = 0$ forced ($\mathbf{A}_{\mathrm{Lean}}$); UV-finite curved background; GTE Holographic Encoding Theorem (5 equivalent descriptions; 12/20 implications unconditional, 8/20 conditional on the Bekenstein-Hawking coefficient $1/4G$); RS $[5,3,3]_7$ code ($\mathbf{A}_{\mathrm{Lean}}$); MDL cosmological IC (1.585 bits, inflation-free; horizon/flatness dissolved, domain walls resolved dynamically via the P47 bias-annihilation mechanism); $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ (conditional; -0.005σ from Planck); $r = 0$ (primary falsifiable prediction; LiteBIRD); Galois/CPT ($\mathbf{A}_{\mathrm{Lean}}$).
P45	7	Three-Tape CMCA: 3+1D from Shared Clock	DPP ($\mathbf{A}_{\mathrm{Lean}}$): shared clock necessary and sufficient for 3+1D; full SM spectrum (33 vertices, CatA); color confinement $\Delta K = \log_2 9$ ($\mathbf{A}_{\mathrm{Lean}}$); baryon number topological ($\mathbf{A}_{\mathrm{Lean}}$); holographic scaling ($\mathbf{A}_{\mathrm{Lean}}$); Lorentz algebra all 12 relations ($\mathbf{A}_{\mathrm{Lean}}$); fermionic statistics ($\mathbf{A}_{\mathrm{Lean}}$); SR $\tau_{\mathrm{inner}}/\tau_{\mathrm{outer}} = 3/7$ (CatA); Newtonian gravity $F = G_{\mathrm{eff}}M/(4\pi b^2)$ in the continuum limit (**A/D**); Bell $S = 2.44$ (CatA); Single-Source Principle (p in 5 roles); falsifiable predictions $r = 0$ and $\delta_{CP} = 205.71^\circ$.

P#	Grp	Short title	Key contribution
P46	7	GTE Polynomial UFT: Five Physical Roles	MDL uniqueness of $p(L, C, R)$ ($\mathbf{A}_{\text{Lean}}$); five physical roles (CA dynamics, gauge, gravity, entanglement, Bell); zero extra specification cost per role; W-boson propagator; SRRG golden-ratio $1/\varphi$ unique root ($\mathbf{A}_{\text{Lean}}$); vacuum stability ($\mathbf{A}_{\text{Lean}}$); confinement from $\Delta K = \log_2 9$ ($\mathbf{A}_{\text{Lean}}$); baryon number $B = (1/3)\sum \chi_q(w_j)$ from $N_{\text{tapes}} = 3$; $\text{SU}(2)_L$ gauging from PMDL ($\mathbf{A}_{\text{Lean}}$; zero named axioms); PMDL–Λ duality: gravity and the Λ floor as record and measure sectors of one Gaussian generating functional, census-capacity evaluation $\Omega_\Lambda = 0.6899$.
P47	7	GTE Cosmological Predictions	Two independent Ω_Λ routes bracketing Planck 2018: PSC epoch count $(\ln 2/3\pi)\log_2(2000/3) = 0.6899$ ($+0.18\sigma$; orbit count $2000/3 = D^2 N_{\text{fam}}^{N_{\text{gen}}}/N_{\text{gen}}$ from DPP, CatAD) and holographic $3\pi/14 = 0.6732$. **PSC-Adjudication Theorem (CatAD):** PSC+SRRG+DPP constrain $\Omega_\Lambda \in [3\pi/14, 0.6899]$ no free params; structural bracket (floor = carrier price, ceiling = capacity; GTE-atom inequality $14\ln(2000/3) > 9\pi^2$ $\mathbf{A}_{\text{Lean}}$); exact realized fraction = left-c.e. real of Turing degree $0'$ (no computable selection; $\sigma^* \approx 3.4 \times 10^{-4}$ test horizon); $N_{\text{gen}} = 3$ unique integer satisfying bracket, PI-free pincer (CatAD). **Gravitational D^2 theorem:** $C_{\text{Gorard}} = N_{\text{spatial}}/(2D^2) = 3/32$ exact ($\mathbf{A}_{\text{Lean}}$, `c_gorard_eq_n_spatial_over_two_dsq`); $D^2 = 16$ unifies CC epoch, CC magnitude, Gorard, EH. $\mathbb{Z}_7$ **anomaly-freedom** ($\mathbf{A}_{\text{Lean}}$): all 7 vacuum sectors quantum-degenerate in the pure-φ sector (4 zero-sorry theorems; the full Lagrangian's V_{coupling} distinguishes the vacua). **Defect cosmology**: ordering crossover at $T_G \approx 0.70$ GeV; V_{coupling} bias annihilates the Kibble wall foam $\sim$700$\times$ before BBN; zero surviving defect network ($\Omega_{\text{GW}}h^2 \lesssim 3 \times 10^{-47}$, $\Delta N_{\text{eff}} \approx 0$; parameter-free falsifiable). $\rho_\Lambda = (9/112)M_{\text{Pl}}^2 H_0^2$ ($w = -1$); loop suppression $\sim 10^{-42}$; $n_s = 0.96488$ (-0.005σ; 14 zero-sorry); $\delta_{CP} = 68.51^\circ$ (0.017%; S_3); $\Sigma m_\nu = 59.4$ meV. H_0 **via Path D (CatAD):** non-circular derivation from T_{CMB}, Ω_Λ, $\Omega_{\text{DM}}h^2$, and η_B yields $H_0 = 67.95$ km/s/Mpc ($+0.7\sigma$ Planck 2018); Path D is fully first-principles with T_{CMB} as sole external scalar input.

P#	Grp	Short title	Key contribution
P48	8	Complete GTE Framework	Synthesis monograph (346 pages): complete PSC-to-cosmology derivation chain in fourteen chapters; dedicated Mathematical Substrate chapter (UGP/GTE arithmetic, the UWCA as the substrate's native computer, and the ontological ladder mathematical substrate → certificate → physical substrate); sequential MDL selection (the rule selection consumes the seed selection's orbit: Direct-Interpolation Lift, $\mathbf{A}_{\text{Lean}}$; Rule 110 as corollary); 19-bit $p(L, C, R)$ Single-Source Principle; Φ_{MDL} field and BPS kinks; all SM parameters, forces, spacetime (three-tape CMCA), quantum mechanics (Born rule from transputation), and cosmology (Ω_Λ, n_s, δ_{CP}, Σm_ν, η_B); master-quadratic unification (Higgs VEV and Rule 110 floor as the two arithmetic completions of $x^2 + x - 1$); Three-Level MDL Unification (non-circularity machine-certified); quantum-measurement adjudication at Turing degree exactly $0'$; structural $\Omega_\Lambda \in [3\pi/14, 0.6899]$ bracket; zero-relic $\mathbb{Z}_7$ defect resolution; three-voice structure; eight-framework comparison and falsification table; integrates P01–P47 with zero new free parameters. **OP3 closed ($\mathbf{A}_{\text{Lean}}$):** all computable quantum corrections to Λ vanish identically in the pure-φ sector (`lambda_all_order_zero_pure_phi`; zero sorry). **OP6 substantially closed (CatAD):** quantitative V−A chiral coupling $g_R = 0$, $g_A/g_V = -1$ derived from first principles (`va_op6_bundle`). **PSL(2,7) algebraic unification (P52 [50], $\mathbf{A}_{\text{Lean}}$):** F_{21} is the Borel subgroup of $\text{PSL}(2,7)$; the Fano plane has $\text{Aut}(\mathcal{F}) \cong \text{PSL}(2,7)$; Eisenstein functor gives A_4 from inert prime 2; Klein quartic genus $= N_{\text{gen}} = 3$.

P#	Grp	Short title	Key contribution
P49	9	MDL Selects the Wolfram Rule	First systematic study of $p(L,C,R) = C + R - CR - LCR \pmod 7$ as a standalone $\mathbb{Z}_7$ dynamical system; MDL uniqueness (T96-02) as Wolfram selection principle, machine-certified link by link ($\mathbf{A}_{\text{Lean}}$): cross-family coding inequality forcing $k = 7$ plus within-class uniqueness via the Direct-Interpolation Lift Theorem ($\mathbf{A}_{\text{Lean}}$) — the UGP orbit's total-parity shadow + vacuum transparency forces p uniquely among all 7^8 multilinear rules, Rule 110 as corollary — and the sparsity-floor theorem ($\mathbf{A}_{\text{Lean}}$): ≥ 4 monomials for any GF(7) polynomial reproducing Rule 110, equality exactly at p; chirality census over all 120 family orderings ($\mathbf{A}_{\text{Lean}}$): vacuum-transparent survivor set of the unordered orbit data is exactly {Rule 110, Rule 124} (ordering convention = chirality gauge); Invariant Subset Uniqueness ($\mathbf{A}_{\text{Lean}}$): only $\{0\}$, $\{0,1\}$ (Rule 110), $\mathbb{Z}_7$ closed under p; QNR Binary Floor ($\mathbf{A}_{\text{Lean}}$): $N_{\text{fam}} = 5$ QNR mod 7 forces $\{0,1\}$ as unique proper invariant subset; Polynomial Uniqueness ($\mathbf{A}$): 1 of 2401 degree-≤ 3 candidates reproduces Rule 110; $c_H = 2q - 1$ structural identity ($\mathbf{A/D}$); 40% DPP cross-tape fraction ($\mathbf{A/D}$); master quadratic $x^2 + x - 1$ (disc $5 = N_{\text{fam}}$): $\mathbb{R}$-root $1/\varphi$ = SRRG fixed point, 7-adic inertness = binary floor ($\mathbf{A}_{\text{Lean}}$); Eisenstein residue-field model of F_{21}, motivic $[V(p)] = \Phi_6(\mathbb{L})$, biquadratic compositum $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$ of conductor $15 = N_{\text{gen}}N_{\text{fam}}$ ($\mathbf{A}_{\text{Lean}}$); AGL(1,7) chiral reflection swaps Rule 110↔124 ($\mathbf{A}_{\text{Lean}}$); Vacuum Uniqueness at every ring size ($\mathbf{A}_{\text{Lean}}$) with exact dynamical zeta factorizations; period-475 attractor factorization ($\mathbf{A/D}$); first $k = 7$ polynomial CA spacetime diagrams. WolframModel causal graph (ruleGTE, 1023 events, binary tree) visualised in three layouts; $\text{GEN}_2/\text{GEN}_3$ topology degeneracy (a,b,a,c,a) causes multiway binary branching ($\mathbf{A}_{\text{Lean}}$, `gen2_gen3_topology_degenerate`); botanical structural equivalence: ruleGTE causal graph is structurally equivalent to Lindenmayer system $F \to F[+F][-F]$ (*Daucus carota* compound umbel model, $\mathbf{A}$).

P#	Grp	Short title	Key contribution
P50	9	Spin-7 Lattice Model	The MDL-selected polynomial $p(L, C, R)$ defines a 7-state spin lattice. 2D phase transition at $T_c \approx 2.86\,J$ (**A**); three-fold $\mathbb{Z}_3$-symmetry-breaking ordered phase with ground states $\{0, 1, 5\}$ (roots of $p(x, x, x) = -x(x-1)(x-5) \pmod 7$; **A/D**); ground-space rigidity: zero-energy rings exactly $\{0^n, 1^n, 5^n\}$ at every length $n \geq 3$ ($\mathbf{A}_{\text{Lean}}$, zero sorry), making the ground space a distance-n ternary repetition code; CMCA entropy rate $S = \log(1.4846) = 0.395$ nats/site/step at $\beta = 1$ from the exact pair-state transfer matrix (**A**); $\text{NL}(p) = 1 - c_H/7^2$ (**A/D**) connects statistical mechanics to Higgs sector; universality class of 3-state Potts model with exact exponents $\nu = 5/6$, $\eta = 4/15$ (**A/D**); cold-atom realization with 7-level Rydberg atoms proposed. Continuum limit: thermal ensemble = max-entropy shadow of the CMCA tape; exact chiral gap law $\Delta(\beta) = e^{-3\beta/2}(1 + O(e^{-\beta/2}))$ (**A/D**, exact correction tower; half-integer slope 3/2 from directed wall energies, $\mathbf{A}_{\text{Lean}}$ — a parity-violation signature of substrate chirality; amplitude exactly 1, doubly derived); physical point selected by the Tape Saturation Theorem (**A/D** conditional on the named Compton-Support Criterion, proven minimal): $a = \hbar c/\Lambda_{\text{GTE}} \approx 0.0972$ fm with $\xi(\beta^*) = |\mathbb{Z}_7| = 7$ and lattice dictionary $aM_{\text{kink}} = 1/7$, $am_\varphi = 7/8$; falsifiable substrate-MC target $\xi_{\text{kink}} = 7$ cells (band 6.78–7.23 ± 0.54).
P51	9	Polynomial Certificate of Transputation	Three-Level MDL Unification Theorem (**A/D**): $P^\top(X) = \arg\min K(x\|\text{PSC})$ operates at theory selection, field dynamics, and event adjudication. f_{MDL} characterized as three-part PSC-correction of p ($\mathbf{A}_{\text{Lean}}$). SRRG fixed-point bridge: $p_{\mathbb{R}}(\varphi, \varphi, \varphi) = \varphi$ ($\mathbf{A}_{\text{Lean}}$, `gte_poly_srrg_bridge`). DSAC established as member of $[D]$ class (**A/D**). Discrete measurement mechanism (**A/D**): $\text{GEN}_1 \to \text{GEN}_2 \to \text{GEN}_3 \to \text{VAC}$ is the unique PSC-admissible trajectory ($\mathbf{A}_{\text{Lean}}$, `fmdl_orbit_is_unique_psc_trajectory`). Computability gap classified two-sidedly: adjudication function at Turing degree exactly $0'$ (Δ^0_2; Gold–Putnam stratum; **A/D**, many-one degree $\mathbf{A}_{\text{Lean}}$). Comparison with Copenhagen, many-worlds, Bohm, GRW, decoherence.

P#	Grp	Short title	Key contribution
P52	9	PSL(2,7) Algebraic Unification	Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry) unification of the GTE algebraic pillars through PSL$(2,7)$, the unique simple group of order 168. Four results: (1) F_{21} is the Borel stabilizer of ∞ in PSL$(2,7)$; the golden Möbius generator and F_{21} together generate PGL$(2,7)$ of order 336 (`f21_is_borel_psl27`, `pgl27_generated_by_singer_and_borel`). (2) PSL$(2,7)$ = Aut$(\mathcal{F})$ for the Fano plane $\mathcal{F}$, with F_{21} acting simply transitively on all 21 incidence flags (`psl27_is_aut_fano`, `f21_regular_on_fano_flags`). (3) Eisenstein integers $\mathbb{Z}[\omega]$: the inert prime 2 gives $A_4 = V_4 \rtimes \mathbb{Z}_3$ via $\mathbb{Z}[\omega]/(2) \cong$ GF(4) (`eisenstein_a4_from_inert_2`); the split prime $7 = \pi\bar{\pi}$ gives F_{21} from the unit group. (4) Klein quartic has genus = $N_{\text{gen}} = 3$, derived from the $(2,3,7)$ triangle group (`klein_quartic_genus_eq_n_gen`). The manifest GTE flavor symmetries generate $S_3 \leq A_4$ ($\mathbf{A}_{\text{Lean}}$), completing the leading-order tribimaximal neutrino mixing group (CatAD).
P53	10	GTE Framework Assessment	Comparative assessment of the GTE framework against ten alternative programmes in fundamental physics; rubric-based evaluation across eleven dimensions (selection principle, free-parameter economy, formal verification, falsifiability, and seven further criteria); explicit falsifiability register; 76 pp. [31].
P54	10	GTE–NEMS Consciousness Synthesis	Formal synthesis of GTE and NEMS: transputation dissolves the quantum measurement problem — reframing it as PSC-forced branch-selection (conditional on the D4 universality assumption; observer-free, machine-certified); the Alpha Theorem (NEMS P63) proves the necessary existence of a pre-categorial ontological ground; the hard problem of consciousness is proved a category error (known qualia are irreducible on-ledger semantic content, T55.3 closes ghost-qualia escape); three-aspect synthesis (Ground / Articulation / Manifestation-in-Awareness) dissolves substance dualism; SIAM architecture (seven invariants) characterises unified sentient agency; two SIAM-specific falsifiable predictions: aggregate Landauer floor $Q \geq N_{\text{adj}} k_B T \ln 5$ ($\mathbf{A}/\mathbf{D}$) and conditional decoherence enhancement ($\mathbf{D}$); comparison with Orch OR, IIT, GWT, Dennett, panpsychism, Chalmers; extended Wigner's-friend no-go theorems shown to support the GTE response [28].

P#	Grp	Short title	Key contribution
P55	9	Octonionic Shadow of GF(7)	Bridge between the GTE arithmetic programme and the division-algebra programme: QR(7) = $\{1, 2, 4\}$ is simultaneously the GTE arithmetic anchor and the Fano-plane orientation fixing the octonion multiplication table. Six-link machine-verified derivation chain from $D = \{1, 2, 4\}$: Fano plane and pencil count $N_c = 3$ ($\mathbf{A}_{\text{Lean}}$); octonions from oriented translates ($\mathbf{A}$); $\lvert\text{PSL}(2, 7)\rvert = 168$ by Todd–Coxeter ($\mathbf{A}_{\text{Lean}}$, `hurwitz_presentation_order_168`); $\mathfrak{g}_2 = \text{der}(\mathbb{O})$ and $\mathfrak{su}(3) = \text{Stab}_{G_2}$ from exact linear algebra ($\mathbf{A}_{\text{Lean}}$, 39 theorems); triality isomorphism pins $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$ ($\mathbf{A}_{\text{Lean}}$); Koide phase $2/9 + 7.4 \times 10^{-6}$ (7 ppm PDG, $\mathbf{A}$); neutrino seesaw normal ordering confirmed within 1% of NuFIT 6.0 ($\mathbf{A}_{\text{Lean}}$). Provides external answer to "why $\mathbb{F}_7$?" via Hurwitz tower termination. Eleven zero-sorry Lean 4 modules, 174 theorems [45].

8 Background Reading: The NEMS/PSC Programme

The UGP physics corpus described in this guide is built on a broader foundational programme — the **NEMS (No External Model Selection) programme** and the **Reflexive Reality** research programme [57]. NEMS is a suite of 94 machine-checked papers, P00–P93 (all archived on Zenodo) establishing that any perfectly self-contained universe must satisfy precise theorem-grade constraints, from which the Standard Model gauge structure, the Born rule, and the arrow of time follow as structural necessities.

The UGP Physics programme (P01–P52) is the quantitative branch of Reflexive Reality: they derive the numerical values of the SM constants from the arithmetic cascade forced by PSC, while the NEMS papers establish *why* those axioms are the right ones and why the SM gauge group is the only viable option. The framework nests as NEMS (outer) $\supset$ UGP $\supset$ GTE $\supset \Phi_{\text{MDL}}$: NEMS establishes *that* law must self-select; UGP identifies *which* law via the arithmetic sieve; GTE is the dynamical realization; Φ_{MDL} is the physical substrate. PSC Presentation Invariance (PI, Layer II) is operationally equivalent to MDL minimality (UGP axiom A3); the 'Why MDL?' regress terminates at PSC/PI, which is the foundational level of the NEMS programme.

8.1 Entry Points for Physicists

- **Research overview and full paper listing:** https://www.novaspivack.com/research
  Organised by audience (physicists, formal theorists, AI/AGI researchers); all papers linked with permanent Zenodo DOIs.
- **NEMS Programme Hub (Zenodo):** [43] https://doi.org/10.5281/zenodo.19429270
  Index of all 94 NEMS papers (P00–P93) with concept DOI (always-latest).
- **NEMS P07 — The NEMS Framework for Physicists** [42] https://doi.org/10.5281/zenodo.19429724
  Self-contained introduction to NEMS for physics audiences. Explains the NEMS trichotomy (Class I: categorical; Class IIb: non-effective internal selection; Class III: not fundamental), the no-free-bits principle, the PSC-gauge structure connection, and the quantum gravity angle. *Recommended first read for any physicist approaching the programme.*
- **NEMS P88 — Reflexive Reality: A Survey for Physicists** [54] https://doi.org/10.5281/zenodo.19429907
  Flagship survey: Two-Layer PSC Theorem, Born rule as closure fixed point, arrow of time as structural consequence of stable records, BH unitarity, UGP parameter derivations, classification cascade. Bracket-tagged claims map to machine-checked Lean theorems with zero sorry.

8.2 Key Physics-Facing NEMS Papers

The following five papers from the NEMS suite are most directly relevant to the UGP Physics programme. All are machine-checked in Lean 4 (zero sorry, zero custom axioms beyond Mathlib) and archived on Zenodo.

Table 3: Key NEMS physics papers and their relation to the UGP corpus.

NEMS #	Title (short)	Relevance to UGP physics
P03 [63]	Structural Stability as a Necessity Principle in Self-Contained Gauge Theories	Derives the NM* (no meta-model) structural stability constraint from PSC, excluding GUT groups and CP-conserving theories. Provides the gauge-theory-space filter used in P14 (TE2.2 scan).
P04 [37]	Instantiating NEMS in Physics	Applies NEMS to macroscopic physical records; establishes the Class IIb criterion (non-effective internal adjudication) and the Outsourcing Barrier theorem. Foundational for understanding why external selectors are forbidden in the UGP framework.
P05 [49]	PSC-Optimality of the Standard Model in 4D Gauge Theory Space	**The Two-Layer PSC Theorem.** Layer I (hard PSC constraints) forces SU(3)×SU(2)×U(1); Layer II (Presentation Invariance / MDL) selects $N_{\text{gen}} = 3$ uniquely. Machine-checked in nems-lean. Cross-referenced throughout the UGP corpus.
P06 [44]	NEMS in Quantum Gravity	Extends the no-external-selector constraint to quantum gravity via records, diffeomorphism redundancy, and non-effective internal selection. Informs P16 (BH unitarity) and the cosmological constant treatment in P01.
P07 [42]	The NEMS Framework for Physicists	Self-contained physics-facing survey; the recommended entry point for any physicist approaching NEMS before reading the UGP papers.

8.3 Physical Incompleteness

NEMS Paper 11 establishes **Physical Incompleteness**: under physically realizable universal computation, stable macroscopic records, and record-level expression of halting, no total computable law can decide all record-truth on the diagonal-capable fragment. This is a corollary of the Theorem of Internal Semantic Non-Exhaustion (machine-checked in nems-lean with zero custom axioms; Lean anchor: `NemS.physical_incompleteness`). It does not prevent UGP from deriving SM constants, but it sets a precise limit on what any first-principles programme can prove about its own completeness. The full theorem treatment appears in NEMS Paper 91 (Closure Without Exhaustion) [43].

A Selected Falsifiable Predictions

The UGP Physics programme makes concrete, independently falsifiable quantitative predictions. The table below lists the primary predictions from P01–P52, organised by testability tier. **Status symbols:** ✓ Confirmed, △ Tension (discrepancy noted; programme maintains structural prediction), ○ Open (no definitive test yet).

Testability tiers: T0 — existing published data; T1 — near-term experiment (LHC Run 3, KATRIN Phase II, Belle II, JUNO ∼2027); T2 — medium-term (FCC-ee, next-generation $0\nu\beta\beta$); T3 — long-term / fundamentally challenging.

Table 4: Primary falsifiable predictions of the UGP Physics programme. Grade codes: A_{Lean} = machine-verified; B = computational/numerical confirmation; D = open/speculative. Status as of May 2026.

P#	Prediction	Value / Result	Status	Verification / Reference
T0 — existing published data				
P01	Strong coupling constant $\alpha_s(M_Z)$ (blind orbit-ratio route); 43-day pre-commitment	0.11822	✓ A_{Lean}	$+0.24\sigma$ PDG 2024 world average; Lean-certified bare coupling

P#	Prediction	Value / Result	Status	Verification / Reference
P39	Strong coupling constant $\alpha_s(M_Z)$ (two-loop RG running route)	0.12001	✓ A	+1.87% PDG 2024; independent route from P01; neither supersedes the other — they address different aspects of the QCD coupling
P01	W-boson mass m_W (two-loop SM running applied)	80.379 GeV	✓ B	-0.42σ PDG 2024 world average; two-loop SM running
P01	Charged-fermion mass hierarchy (nine fermions)	0.261% RMS from PDG 2024	✓ A/D	Lean-certified structural pipeline; locked benchmark 0.295% vs PDG 2022 targets; the PDG 2024 top-quark update moves the target toward the prediction
P01	Cosmological constant Λ_{phys}	$+0.31\sigma$ from Planck 2018	✓ A	SM-17 formula confirmed; $w_0 = -1$ maintained
P01, P27	Higgs quartic coupling λ_H (SRRG-corrected)	-0.0003σ from PDG	✓ $\mathbf{A}_{\text{Lean}}$	$\lambda = \varphi/(4\pi) \times (1 + \ln\varphi/(54\ln 2\pi))$; $2c_H + 1 = 27 = N_{\text{gen}}^3$; `higgs_quartic_corrected`, zero sorry
P01, P27, P35	Higgs mass m_H (SRRG-corrected)	125.2499 GeV	✓ $\mathbf{A}_{\text{Lean}}$	Bare -9.1σ (disclosed); SRRG-corrected -0.0003σ; structural v_{PSC} = 246.16 GeV (grade [A_{Lean}]; `SrrgLean.VEVProof.*`; see P27, P35)
P03	Nuclear binding energy, 6-term analytical law (GTE features)	0.032 MeV/A CV MAE	✓ B	Cross-validated on 1,319 nuclei (AME2020/ NUBASE2020); competitive with DZ31: 6 terms vs. 31 parameters
P03	Nuclear stability classifier (6-term; NUBASE labels)	94.5% 5-fold CV	✓ B	Smooth law cannot resolve Tc/Pm (disclosed)
P03	Nuclear pairing constant $\Delta_{\text{pair}} = 5^{3/2}$	11.18 MeV	✓ B	0.003% from Möller et al. 1995 SEMF; zero free parameters
P03	All 7 nuclear magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ from GTE cascade	7/7 reproduced	✓ B	GTE κ-gap derivation; $\kappa_{\text{emp}}/\kappa_{\text{min}}(N{=}50) = 1.149 \approx$ IPT (1.6%)
P14	SM gauge group is unique D_{min} among 20,160 candidate universes	$D_{\text{min}} = 1.0667$ (SM)	✓ B	TE2.2 exhaustive scan; 5 hard PSC filters; no G_{SM} reference
P15	Information Profit Threshold IPT $= 1 + \Lambda/2 \approx 1.1309$ across multiple independent domains	1.1309	✓ B	Forest GPP/RECO, bankruptcy, population dynamics, OSS, SM gauge sector, organizational survival (ΔBIC $= -92$, HR$= 5.01\times$, BC$= 0.907$; [B^-]); Ising CFT $\gamma = 7/8$ ([B^+] confirmed L=32)

P#	Prediction	Value / Result	Status	Verification / Reference
P26	Organizational G/D = IPT is genuine second structural boundary (Destruction/Watch/Safe three-regime cascade + variance phase transition)	ΔBIC $= -92$; HR$= 5.01\times$; BC $= 0.907$, Levene $p = 0.046$	✓ B^-	N=12,741 company-years; failure cascade 6.81%→1.11%→0.22%; AUC=0.781; OR=5.15×; inception-cohort design rebuts survivor-selection confound; bimodal Watch Zone mirrors Ising diverging susceptibility; 7 independent tests
P19	CKM Wolfenstein parameter λ via $\varepsilon_1^{\alpha_d} = e^{-13\pi/27}$	≈ 0.2203	Partial B	1.9% from PDG 0.2245; improved from prior 12%; derivation open
P21	Neutrino mass-squared ratio $\Delta m^2_{21}/\Delta m^2_{31}$; zero free parameters	0.02936	✓ A	0.16σ from NuFIT 6.0 global fit (0.52%); 1.58σ from PDG 2024 world average
P21	Normal neutrino mass hierarchy preferred (structural)	NH automatic	✓ B	NuFIT 6.0 prefers NH at 2.5σ; JUNO ($\sim$2027) definitive
P21	PMNS mixing angles from orbit-ratio arithmetic; zero free parameters	$\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 19/42$, $\sin\theta_{13} = 11/73$	✓ B	θ_{12}: $+0.01\sigma$; θ_{13}: $+1.0\sigma$; θ_{23}: -1.35σ (within 3σ); NuFIT 6.0 IC24 NH; no fermion-mass input
P22	SM gauge-fermion interaction skeleton; proton stability; three independent proofs of $N_c = 3$	Zero mismatches	✓ A_{Lean}	64-schema vertex audit in Lean; zero `sorry`
P24	Unique ($n = 10, b_1 = 73$) seed across all ridge levels (Sparsity Theorem)	Unique	✓ A_{Lean}	Lean-certified; E7 ($h = 18 \nmid 120$) excluded
P24	Galois substrate $\mathbb{Q}(\zeta_{120})$ contains all UGP constants	$\mathbb{Q}(\zeta_{120})$	✓ A_{Lean}	Lean-certified; WZW cannot reproduce bare coupling rationals
P25	Standard genetic code: unique survivor of 8-criterion sieve	$z = +3.22\sigma$	✓ B	0/99,998 random wobble-admissible codes outperform; Lean-certified arithmetic echoes (20 amino acids, 64 codons)
P27	IPT = unique IR fixed point of SRRG gradient-flow	IPT ≈ 1.1309	✓ A_{Lean}	Lean-certified via Master Fixed-Point Theorem; F-theorem; stability eigenvalue $1/\varphi$
P27	η-direction β-function $\beta_\eta = \kappa(\eta - \text{IPT})(\eta - 2)$ derived from Vieta's theorem and machine-certified no-third-zero	β_η quadratic form	✓ $B^+ \to A^-$	`BetaEtaQuadratic.lean` + `NoThirdFixedPoint.lean`; zero sorry; two named hypotheses remain (`SrrgPhysicalFixedPointExhaustion`, `SrrgBetaIsQuadraticHyp`)
P27	Strong CP phase $\theta_{\text{QCD}}(S^*) = 0$ from SRRG constraint	$\theta_{\text{QCD}} = 0$	✓ B	Theorem 8.1; CP self-consistency from $C_{\text{closure}} = 0$; zero sorry
P27	Three fermion generations from SRRG fixed-point	$N_{\text{gen}} = 3$	✓ B	Theorem 8.2; balance of closure cost and Yukawa selector cost

P#	Prediction	Value / Result	Status	Verification / Reference
P31,P41	Electroweak mixing angle $\sin^2\theta_W$ from f_{MDL} palindrome arithmetic	384729/1664000 ≈ 0.23121	✓ $\mathrm{A_{Lean}}$	0.42σ from PDG 2022 0.23122 ± 0.00003 at M_Z; bare $3/13 = 0.23077$ at 13.0σ from PDG 2024 0.23129 ± 0.00004; Wolfenstein correction $\mathbf{A}_{\mathrm{Lean}}$; two-loop closure $+0.038\sigma$ PDG 2024 (**B**)
P32	CKM Wolfenstein λ from orbit arithmetic, zero free parameters	$9/40 = 0.225$	✓ $\mathrm{A_{Lean}}$	0.000% from PDG 0.22500
P35,P39	Strong coupling $\alpha_s(M_Z)$ from F_{21} (three-loop running)	0.1193	✓ B	+1.10% from PDG 0.1180; no free parameters; $F_{21} = \Sigma(21)$ forced
P36	Spacetime dimension $D = 4$ from MDL CA structure	$D = 3 + 1$	✓ $\mathrm{A_{Lean}}$	Machine-certified: $D_{\mathrm{spatial}} = 3$ from MDL minimality; $D_{\mathrm{temporal}} = 1$ from CA definition
P36	Vacuum Einstein equation $G_{\mu\nu} = 0$ from Ollivier–Ricci curvature of Rule 110 causal graph	Satisfied exactly	✓ $\mathrm{A_{Lean}}$	Lean-certified, zero sorry; matter step $\kappa_{\mathrm{SD}} > 0$ computationally confirmed
P39	QCD mass gap from F_{21} orbit arithmetic (unconditional)	Exists	✓ $\mathrm{A_{Lean}}$	Lean-certified, zero sorry, zero axioms; effective below $\Lambda_{\mathrm{GTE}} = 7M_{\mathrm{kink}} \approx 2.0\,\mathrm{GeV}$ (derived seven-kink threshold; envelope $1.96 \pm 0.15\,\mathrm{GeV}$)
P39	Pion decay constant f_π from Φ_{MDL} kink condensate	92.34 MeV	✓ B	+0.30% from PDG 91.9 MeV; zero PDG input; $m_\varphi = m_\tau$ self-consistency
P39	η–η' mixing angle θ_P from GTE QCD	$-13.08^\circ \pm 3.74^\circ$	✓ B	Within PDG range $[-14.3^\circ, -10.7^\circ]$; entire chain zero PDG input
P41	EW boson masses M_Z, M_W from v_{PSC} and $\sin^2\theta_W$ (CMCA)	$M_Z = 91.91\,\mathrm{GeV}$; $M_W = 80.61\,\mathrm{GeV}$	✓ B	M_Z: +0.80%; M_W: +0.30% from PDG; $M_W/M_Z = \sqrt{10/13}$ (Lean-certified)
P43	Newton's constant G_N from GTE gravitational hierarchy $M_{\mathrm{Pl}}/m_\tau = \lvert\mathcal{F}_{21}\rvert^{10}\lvert\mathbb{Z}_7\rvert^7/2$	$G_N = m_\tau^2/M_{\mathrm{Pl}}^2$	✓ A/D	0.040% analytical precision; no free parameters
P44	CMB primordial spectral index n_s from MDL-inflation-free IC	$n_s = 1 - \ln(2)/(2\pi^2) =$ **0.96488** (conditional)	✓ CatAD	-0.005σ from Planck 2018 (0.9649 ± 0.0042); 5/7 derivation steps closed; conditional on EFE-bridge step
P32	CKM CP angle γ from GTE quark orbit arithmetic	$\gamma = 65.67^\circ$	✓ CatA	-0.023σ from PDG $65.68^\circ \pm 0.67^\circ$; zero free parameters
P32	Leptonic Jarlskog CP invariant J_{GTE}	-1.468×10^{-2}	✓ $\mathrm{A_{Lean}}$	Formally proved in Lean 4 from $\mathbb{Z}_7$ orbit arithmetic; zero free parameters
P47	Dark-energy fraction Ω_Λ (PSC route)	0.6899 $(\ln 2/3\pi)$	✓ CatAD	$+0.18\sigma$ from Planck 2018; orbit count 2000/3 from DPP
P47	Dark-energy bracket (PSC-adjudication)	$\Omega_\Lambda \in [3\pi/14, 0.6899]$; $N_{\mathrm{gen}} = 3$ unique	✓ CatAD	Planck 2018 0.6889 within bracket; no free parameters

P#	Prediction	Value / Result	Status	Verification / Reference
P47	CKM Dirac CP phase δ_{CP} (S_3 chain)	$\pi/2 - 3/8 = 68.51^\circ$	✓ CatAD	0.017% from PDG 68.517°; distinct from PMNS δ_{CP} (P45)
P47	Gorard bridge coefficient C_{Gorard}	$N_{\text{spatial}}/(2D^2) = 3/32$ exact	✓ A_{Lean}	`c_gorard_eq_n_spatial_over_two_dsq`; $D^2 = 16$ unification
P33	Baryon-to-photon ratio η_B (GoE electroweak chain)	6.54×10^{-10}	✓ CatA	+7.2% from Planck 2018 $(6.10 \pm 0.06) \times 10^{-10}$
P47, P42	Baryon-to-photon ratio η_B (FKTT: $g_{\text{kink-top}} = \varepsilon_{\text{FN}}$)	6.109×10^{-10}	✓ $\mathbf{A}_{\text{Lean}}$	$+0.15\sigma$ from Planck 2018; zero sorry, zero axioms; `kink_top_coupling_eq_eps_FN` (`rfl` proof), `phi_mdl_kink_bps_saturation`
P29, P47	Dark-sector transitivity factor $D_{\text{top}} = e^{-1/N_c}$; links cosmological Ω_{DM}	$e^{-1/3} \approx 0.716$ (exact from $N_c = 3$)	✓ A_{Lean}	Formally proved in Lean 4 from $\mathbb{Z}_7$ transitivity (`d_top_derivation_chain_catal`); zero sorry
P47	Hubble constant H_0 from T_{CMB} (non-circular derivation)	$67.95\,\text{km}\,\text{s}^{-1}\text{Mpc}^{-1}$	✓ B	$+0.7\sigma$ from Planck 2018 (TT,TE,EE+lowE+lensing; $67.66 \pm 0.42\,\text{km}\,\text{s}^{-1}\text{Mpc}^{-1}$); derived from T_{CMB} independently of Ω_Λ anchor
P13, P20	Dark-energy equation of state $w_0 = -1$ (structural)	$w_0 = -1$	△ A	2.5–3.5σ tension with DESI DR1+SNIa; prediction maintained; DESI DR3 ($\geq 5\sigma$ with ≥ 2 independent SNIa) = falsification criterion
T1 — near-term experiment				
P44, P45	Tensor-to-scalar ratio r (no primordial gravitational waves)	$r = 0$ (exact)	○ CatAL/A	Primary falsifiable prediction; LiteBIRD satellite $\sigma_r \approx 0.001$; any confirmed $r > 0.01$ immediately falsifies P44/P45
P47	Neutrino mass sum Σm_ν from g_{fund} cascade	59.4 meV	○ CatAD	Within Planck bound (< 120 meV); CMB-S4 / Euclid definitive
P45	PMNS Dirac CP phase δ_{CP} from $\mathbb{Z}_7$ winding arithmetic	$\delta_{CP} = 205.71^\circ$	○ CatA	NuFIT 6.0 IC24 NH: 212° (-0.15σ); DUNE/T2K-II/Hyper-K definitively testable; zero free parameters
P01	Neutrino mass sum $\sum m_\nu \leq 0.06\,\text{eV}$ (structural)	59.4 meV (P47; GTE seesaw + three-tape winding)	✓ Compatible	Consistent with KATRIN Phase I and Planck; KATRIN Phase II (∼2025–26), LEGEND-1000
P18	Lepton mass m_τ from Koide closed form; Lean-certified	1776.969 MeV	✓ B	61 ppm from PDG; 0.91σ; structural origin $\theta = 2/9$ from $N_c = 3$

P#	Prediction	Value / Result	Status	Verification / Reference
P18	Tau Yukawa $y_\tau = 1/98$ from $\mathbb{Z}_7$ potential coefficient $c_V = 1/49$; all three charged-lepton masses from v_H alone (G_F sole anchor)	$m_\tau = 1776.57\,\text{MeV}$, $m_\mu = 105.6\,\text{MeV}$, $m_e = 0.511\,\text{MeV}$	✓ CatA	0.016%, 0.022%, 0.023% from PDG; $98 = 2 \times 7^2$; zero fermion mass anchor; Koide CV($\sqrt{m}$) = 1 Lean-certified (CatAL)
P18,P45	Kink–Higgs coupling $g_{hKK} = 4/7^4 = M_{\text{kink}}/(v_H/\sqrt{2})$ from $c_V = 1/49$; $y_\tau = c_V/N_{\text{mod2}}$ algebraically exact	4/2401	○ CatA	Structural prediction; no free parameters; derivable from $\mathbb{Z}_7$ potential $V = m^2(1-\cos 7\phi)/49$
P21	Normal hierarchy confirmed at $\geq 5\sigma$	NH structural	○ B	JUNO (∼2027); NuFIT 6.0 already at 2.5σ
P47	Zero surviving $\mathbb{Z}_7$ defect network: the ordering crossover at $T_G \approx 0.70$ GeV forms a transient Kibble wall foam annihilated by the V_{coupling} vacuum bias ∼700× before BBN	$\Omega_{\text{GW}}h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz; $\Delta N_{\text{eff}} \approx 0$	○ CatAD	Parameter-free; any confirmed cosmological $\mathbb{Z}_7$ domain-wall relic falsifies the bias-annihilation mechanism and the ordering-crossover thermal-history epoch (not the substrate identification or the particle-spectrum derivations); CMB-S4 (ΔN_{eff}), GW observatories
P02	9 novel stable GTE candidates (GTE-P1, P2, P3, P6–P11) with specific predicted masses (Master Index Table 28)	3 MeV–1.9 GeV	○ B	Pre-registered `candidates.csv`; all 9 mass windows OPEN under lab limits
P02	GTE-P7: SM-neutral Dirac fermion ($Q = 0$, colour singlet)	$\approx 212\,\text{MeV}$	○ A_{Lean}	$Q = 0$ Lean-certified; correct search = mono-photon/missing energy (Belle II); dark-photon searches inapplicable
P28	SM generation orbit uniquely forces Rule 110 (two-way convergence: forced from SM structure and from UGP substrate independently)	$N_{\text{gen}} = 3$ structural	○ A_{Lean}	Lean-certified (CUP-4, zero sorry); MDL-minimal 76-bit universal $\mathbb{Z}_7$ CA uniquely embeds SM+dark sector (CUP-12)
P40	Frobenius-Generation Coincidence Identity: $\{3, 7, 13, 73\}$ arithmetic uniquely determined at $N_c = 3$	Unique	○ A_{Lean}	Formally proved in Lean 4, zero sorry; no other N_c value reproduces GTE arithmetic
P29	Dark sector contains no electrically charged particles ($Q = 0$ universal)	$Q = 0$ all dark states	○ A_{Lean}	Lean-certified (zero axioms beyond Braid Atlas charge theorem); rules out dark photon; correct search = missing-energy signatures
P29	Three dark lepton generations with predicted masses	0.54, 24.5, 3600 MeV	○ B	GTE mirror-branch cascade; no free parameters; dark sector confined to $\text{SU(3)}_{\text{dark}}$ (no EW symmetry)
P29	GTE-P7 dark matter candidate mass (refined from P02)	211.9 MeV	○ A_{Lean}/B	$Q = 0$ Lean-certified; mass from mirror-branch GTE cascade; asymmetric DM: $\eta_\chi \approx (2/7)\,\eta_{\text{dark}}$

P#	Prediction	Value / Result	Status	Verification / Reference
P29	Dark photon does not exist ($\varepsilon = 0$ exactly from charge structure)	$\varepsilon = 0$	○ A_{Lean}	Lean-certified from mirror-branch charge theorem; any confirmed dark photon signal immediately falsifies P29
P29	Dark lepton mass ratio $m_{G2}/m_{G1} \approx 45$; two-peak DM signature	$24.5/0.54 \approx 45.4$	○ B	Mirror-branch GTE cascade; distinctive two-peak direct-detection spectrum; no free parameters
P29	Direct detection cross-section $\sigma_{SI}(\chi_1\text{-}N)$	$9.76 \times 10^{-61}\,\mathrm{cm}^2$	○ CatAD	28–29 orders below CRESST-III; UFIDM; kinematically blind to recoil detectors
P29	Dark radiation fraction R_{dark}	0.2080	○ CatAD	MDL doublet-pairing; precision dark-radiation cosmology
P29	Dark confinement scale Λ_{dark}	210–1700 MeV	○ B	$\alpha_{s,dark}^{bare} = \alpha_{s,SM}^{bare}$ (Lean); $N_{f,dark}$ range open
T2 — medium-term experiment				
P18	m_τ at 30 keV precision gives 3.6σ test of Koide closed form	≤ 30 keV target	○	Belle II m_τ programme
P38,P42	Φ_{MDL} kink quantum mass M^Q (one-loop interface dimensional regularization, two independent routes)	$M^Q = 281 \pm 21$ MeV	○ B	Classical BPS mass $M^{cl} = 290.10$ MeV; quantum correction $\Delta M = -7$ to -10 MeV (on-shell / $\overline{\mathrm{MS}}$ at $\mu = m_\varphi$); ± 21 MeV envelope over the gauge-invariant renormalization-scheme family; validated against exact sine-Gordon and φ^4 benchmarks and an independent finite-box mode sum; no PDG counterpart (topological soliton); supersedes the prior determinations 230.43 MeV (log-UV artifact) and 321.32 MeV (dimensional-regularization bookkeeping error)
T3 — long-term / fundamentally challenging				
P03	$N = 184$ predicted magic number beyond known data	$N = 184$	○ B	Requires exotic-isotope facility (FRIB, GSI-FAIR); Master Index Table 30
P47	Kink emission near M_{kink}^{crit} (PBH spectrum)	$M_{kink}^{cl} = 290.10$ MeV (classical); $M^Q = 281 \pm 21$ MeV (quantum-corrected, P38/P42)	○ CatA	Non-thermal bump in primordial BH spectrum; no detection yet
P47	Φ_{MDL} field potential cap $V_{max} = 2m^2/49$	3.44×10^{-3} GeV^4	○ CatAD	Primordial GW/PBH spectra inconsistent with cutoff would falsify

P#	Prediction	Value / Result	Status	Verification / Reference
P27	SM gauge group $U(1)\times SU(2)\times SU(3)$ as unique multi-scale SRRG fixed-point structure (conditional theorem)	Conditional [B]	○ $\mathrm{A_{Lean}}$/B	Full uniqueness pending scale-specific PSC sieve hypotheses ($U(1)$ component Lean-certified; EW and QCD rank conditional [B])
P51, P48	Quantum adjudication computability class: the adjudication function's diagonal restriction has Turing degree exactly $0'$ (Δ^0_2-Turing-complete; limit-computable, the Gold–Putnam trial-and-error stratum)	$\deg = 0'$	○ CatAD	Any physical quantum-measurement process that provably resolves Σ^0_1-complete problems at finite stages would refute the deg-$0'$ adjudication architecture; many-one degree machine-certified (zero sorry)
P50	Spin-7 lattice phase transition in 3-state Potts universality class; critical exponents $\nu = 5/6$, $\eta = 4/15$ (from polynomial structure, no free parameters)	$\nu = 5/6$, $\eta = 4/15$; $T_c \approx 2.86\,J$	○ CatAD	2D finite-size scaling confirms universality class (**A**, 5 system sizes); falsifiable by cold-atom simulation of 7-level Rydberg spins in optical tweezer arrays; any incompatible exponents or absence of ordered phase falsifies P50
P50	Walsh nonlinearity identity $\mathrm{NL}(p) = 1 - c_H/7^2$ connects cryptographic nonlinearity of the MDL-selected polynomial to the GTE Higgs branch capacity $c_H = 13$; structural identity, no free parameters	$\mathrm{NL}(p) = 36/49 \approx 0.7347$	○ CatAD	Computationally verified over all $7^3 = 343$ inputs; analytical proof from Walsh–Hadamard formula (**A/D**, Thm 4.7 of P49); measurement of differential uniformity or nonlinearity in a 7-state CA experiment would confirm

B Near-Term Accelerator Search Targets for Novel GTE Particles

The following table identifies the highest-priority search strategies for the nine novel stable particle candidates predicted by the GTE framework (P02, P29). Entries are organized by testability tier: Tier 1 uses existing archival data; Tier 2 requires near-term experiments (Belle II / LHCb Run 3, ∼2026–2028); Tier 3 requires a dedicated low-energy experiment.

Overview

This table summarizes novel particle-mass predictions of the Universal Generative Principle (UGP) framework that may be testable against existing or near-term accelerator data. Items GTE-P7 through GTE-P11 are a subset of the full novel-particle register in P02 [33]; P29 provides the dark-sector complement [41]. A broader derivation context (Standard Model parameters, gauge couplings, fermion masses) is in P01 [25].

Key Signature Property

All 9 genuinely novel candidates from the GTE spectrum survey are predicted **stable** (lifetime $\tau \geq 10^{30}$ s). This means the search strategy differs from ordinary resonance hunting:

- Stable neutral states ($Q = 0$) appear as **missing energy/momentum**.
- Stable charged states appear as **anomalous long-lived tracks** or **heavy stable charged particles** (HSCP).
- Standard resonance-width fits in hadronic spectra would *not* catch these; one instead looks for a *sharp excess* at a specific invariant mass, or an anomalous *endpoint feature*, in an otherwise smooth background.

The highest-priority state (**GTE-P7**) is electrically neutral ($Q = 0$, color-singlet, Lean 4 certified with zero axioms). It is *not* a dark photon: the framework predicts $\varepsilon = 0$ exactly (no kinetic mixing), so existing dark-photon exclusions do *not* apply to GTE-P7.

Predicted Targets by Search Tier

■ Tier 1 = testable with existing archival data now. ■ Tier 2 = Belle II / LHCb Run 3 (∼2026–2028). ■ Tier 3 = requires dedicated low-energy experiment.

State	Mass	Unc.	QN	Search channel / experiment	Notes
Tier 1					
GTE-P7	211.9 MeV	±14%	$Q = 0$, singlet, $s = \frac{1}{2}$ Lean-cert.	BaBar archival: dimuon threshold / $e^+e^- \to \gamma X$ LHCb archival: dimuon at $2m_\mu$ Belle II: mono-photon $M_{\text{recoil}} \approx 212$ MeV	Sits at dimuon threshold ($2m_\mu = 211.4$ MeV). Signal = missing invariant mass recoiling against photon. *Not* a dark photon; existing ε-exclusions do not apply. 50 ab^{-1} at Belle II constrains $\varepsilon < 10^{-3}$. **Highest near-term priority.**
GTE-P8	298 MeV	±14%	heuristic: quark-like (Möbius product $= -1$)	BESIII existing data: $\pi\pi\eta$ and 2γ spectra BaBar/Belle archival: same channels	Look for a *stable* peak near 298 MeV not matching any known resonance. No known PDG state within 10% of this mass.
GTE-P9	561 MeV	±20%	heuristic: sector indet.	BESIII: J/ψ radiative decays, 550–575 MeV window LHCb archival: inclusive spectra	Between $\eta'(958)$ and $\phi(1020)$; look for a stable state distinct from both. Window: ∼ 449–673 MeV.
Tier 2					
GTE-P3	∼ 800 MeV (band: 796–850)	±25%	charge uncertain	Belle II, LHCb Run 3: $\pi\pi K\bar{K}$ mass spectra	In the $f_0(980)/\omega(782)$ region. Look for a *stable* charged or neutral state not matching f_0, ω, or ρ. Window: ∼ 600–1063 MeV.

(continued)

State	Mass	Unc.	QN	Search channel / experiment	Notes
GTE-P2	107 MeV (cluster: 107–110)	±14%	charge uncertain	NA62 rare decays DarkLight (JLab) Long-lived / displaced-vertex searches	1.75 MeV above the muon; a *distinct* state (different GTE triple: $n = 4702$ vs muon $n = 42$). Window: $\sim$ 92–125 MeV.
GTE-P6	30.9 MeV (29–33)	±14%	charge uncertain	NA62 FASER at LHC Dedicated dark-sector fixed-target	Window: $\sim$ 26–37 MeV. No known PDG resonance in range.
GTE-P10	$\sim$ 1.1–1.35 GeV (735 cluster members)	±20%	charge uncertain	LHCb Run 3 BESIII charm threshold	Charm-adjacent; internally consistent with P01 first-principles charm prediction (1276 MeV) from an independent derivation path. Window: $\sim$ 880–1620 MeV.
GTE-P11	$\sim$ 1.6–1.9 GeV (1919 cluster members)	±25%	charge uncertain	LHCb Run 3 BESIII	Tau-adjacent; internally consistent with P01 tau prediction (1775 MeV). Window: $\sim$ 1.2–2.4 GeV.
Tier 3					
GTE-P1	2.97 MeV	±1.4%	lepton-like (μ-par. +1)	DAFNE (Frascati) Dedicated MeV-scale experiments	Sharpest mass prediction in the entire million-candidate dataset. Below most collider thresholds. Window: 2.93–3.01 MeV.

Dark Sector Mirror Branch (from P29)

These are three *dark singlet leptons* ($Q_{\mathrm{EM}} = 0$, no weak charge, color-singlet) derived from the GTE mirror branch with zero free parameters. Mass ratios G2/G1 ≈ 45.3 and G3/G2 ≈ 147.3 are fixed structural predictions, falsifiable if states are found at different mass ratios.

State	Mass	Certification	Search / comment
χ_1 (dark lepton G1)	0.5406 MeV	GTE arithmetic	Sub-MeV DM via Higgs portal; below CRESST-III sensitivity; CMB spectral distortions (future)
χ_2 (dark lepton G2)	24.47 MeV	GTE arithmetic	MeV-scale Higgs-portal DM; two-peak signature with χ_1 at fixed ratio 45.3
χ_3 (dark lepton G3)	3604.68 MeV	GTE arithmetic	LHC Higgs invisible width; mono-jet; coupling suppressed by $\lambda_s \sim 10^{-6}$
Dark photon A'	**does not exist**	Lean-cert. ($\varepsilon = 0$)	Null results from NA64/BaBar/Belle II are *consistent* with the framework

Priority Guidance for Existing Data

If access is limited to archival data from one experiment, the recommended search order is:

1. **GTE-P7 at** ≈ 212 MeV in BaBar or LHCb archival dimuon/missing-energy data. The signature is a *stable, neutral* state appearing as missing invariant mass recoiling against an ISR photon. BaBar has published mono-photon searches in this mass range; a re-analysis asking specifically for a *stable* (not radiatively decaying) state at the dimuon threshold would be the most targeted test.
2. **GTE-P8 at** ≈ 298 MeV in BESIII hadronic data. Look for an anomalous bump in the $\pi\pi\eta$ or $\gamma\gamma$ invariant mass spectrum near 298 MeV, distinct from the known $\eta(548)$, $\eta'(958)$, and $\omega(782)$ peaks.
3. **GTE-P9 at** ≈ 561 MeV in BESIII J/ψ radiative decays. The 550–575 MeV window is relatively clean of established resonances and has been surveyed for other purposes; a targeted stable-state search is straightforward.
4. **GTE-P10/P11 (1.1–1.9 GeV band)** in LHCb or BESIII prompt data: anomalously long-lived or stable new states in the charm/tau mass region with unexpected quantum numbers.

A note on search strategy

Because all GTE novel candidates are predicted stable, the standard procedure of looking for a Breit–Wigner peak with a measured width does not apply. The relevant search is instead for:

- A *narrow excess* (width consistent with detector resolution) at a specific invariant mass, *with no matching known SM resonance.*
- For neutral states: an *endpoint or threshold feature* in missing-energy spectra.
- For charged states: an HSCP (heavy stable charged particle) signature — anomalously high dE/dx or anomalous time-of-flight.

Absence of any signal in the mass window listed above *falsifies* the corresponding GTE prediction. The framework makes crisp falsification predictions: each Tier 1 target can be definitively excluded or confirmed with data already on tape.

Oscillation-Parameter Falsification Target (JUNO / NuFIT)

The GTE seesaw mechanism predicts a specific neutrino mass ordering testable by reactor-oscillation experiments. Unlike the particle-mass targets above, this prediction is an oscillation-parameter observable, not a new-state search.

Observable	GTE Prediction	Cert.	Experiment	Falsifying condition
Neutrino mass ordering	Normal Ordering (NH) forced by seed ordering $b_{R,1}<b_{R,2}<b_{R,3}$ via seesaw $m_{\nu,k} = C \cdot b_{R,k}^{29/9}$; masses $\{m_1, m_2, m_3\} \approx \{0.68,\ 8.6,\ 50.1\}$ meV; $\Delta m^2_{21} = 7.37\times10^{-5}$ eV^2, $\Delta m^2_{31} = 2.51\times10^{-3}$ eV^2 (both within 1% of NuFIT 6.0 IC24 NH)	$\mathbf{A}_{\text{Lean}}$	JUNO reactor $\bar{\nu}_e$ disappearance; NuFIT global fits	Confirmed Inverted Hierarchy (IH) at $>3\sigma$: would falsify the seed-ordering mechanism of P21 and the octonionic triality confirmation of P55

Source papers: P21 [48] (primary seesaw derivation); P55 [45] (octonionic triality corroboration, `seesaw_normal_ordering_from_seed_ordering`, zero sorry). Current NuFIT 6.0 IC24 NH data disfavour IH at $\Delta\chi^2 = 6.1$ ($\approx 2.5\sigma$), consistent with the GTE prediction.

C Reusable Computational Tools

Several papers in the corpus produce computational tools or pre-computed databases that are directly usable by external researchers. The four highest-utility items are described below; all are available in the repository at https://www.novaspivack.com/research.

1. GTE Particle Candidate Database (P02). A pre-registered database of 1,000,035 GTE candidate trajectories, including 9 genuinely novel predicted stable particles (GTE-P1, P2, P3, P6–P11) with specific predicted masses (3 MeV to 1.9 GeV), quantum numbers (Q, colour, spin), stability scores, and viability tiers (plus GTE-P4/P5 path variants excluded from the novel register). All 9 novel mass windows are currently open under combined laboratory dark-photon constraints. *Repository:* `papers/02_GTE_spectrum/candidates.csv` (SHA256 in REPRODUCE.MD). *Corpus register:* Master Index Table 28. *Lookup API:* `tools/gte_stability_api.py --mass [MeV]`. *Primary use:* experimentalists performing model-independent bump-hunt searches at collider or fixed-target experiments. *Utility note:* Most beyond-SM frameworks supply qualitative mass ranges or order-of-magnitude predictions for new particles. This database provides specific predicted masses, decay lifetimes, and ranked confidence scores for every GTE candidate, enabling a direct query-driven workflow: given a candidate signal at mass m, a researcher can determine in seconds whether a GTE-predicted particle lies within the search window and at what confidence tier. The database is pre-registered (committed before any collider comparison), making it falsifiable against current and future collider datasets without post-hoc tuning.

2. Nuclear Binding Energy and Stability API (P03). Two complementary analytical models are provided:

- **Parsimonious (6-term):** GTE coordinate features only. 0.032 MeV/A five-fold CV MAE on 1,319 curated nuclei; interpretable closed form; no external dependencies.

- **Extended (9-term):** same 6 features plus proton magic-number proximity, neutron magic-number proximity, and pairing delta ($\delta \cdot A^{-3/4}$). Trained on 2,548 experimental AME2020 nuclei (full measured table). 0.051 MeV/A five-fold CV MAE across all experimental AME2020 nuclei; OOD $R^2 = 0.977$ on 1,229 AME2020 nuclei absent from the original training set. Note: the extended model's CV MAE is higher than the parsimonious figure because the Phase 2 training set includes hard exotic nuclei near the drip lines; on the original 1,319-nucleus set the extended model matches or improves on the parsimonious at every point.

The 10-term stability classifier (94.5% five-fold CV against corrected NUBASE2020 empirical labels) is the recommended tool for stability screening. Note: the smooth analytical stability law cannot resolve elements with no stable isotopes (Tc, Pm) due to a fundamental smooth-law limitation; empirical NUBASE2020 data should be used for known elements ($Z \leq 118$), with the GTE classifier reserved for extrapolation to unknown elements ($Z > 118$).

Access: `tools/nuclear_be_api.py --Z [Z] --N [N] --task be|stability|both --model parsimonious|extended|both` (JSON output).

Caveat: Both analytical models cover $Z \leq 118$; predictions for unmeasured superheavy nuclei ($Z > 118$) are Category D (exploratory). Use the parsimonious model for structural interpretation and the extended model for accuracy-sensitive screening across the measured nuclear chart.

Utility note: Dedicated DFT and HFB approaches outperform both laws on the full AME dataset and should be preferred for precision work on known nuclei. The value of these laws is their analytical, dependency-free form for fast screening of unmeasured isotopes, and the GTE-derived feature motivation as a reproducible benchmark for nuclear ML researchers evaluating alternative theoretical feature sets.

3. Koide Lepton Mass Predictor (P18). A Lean 4-certified closed-form formula that predicts m_τ from m_e and m_μ alone. With PDG inputs, the prediction $m_\tau = 1776.969$ MeV is within 61 ppm (0.91σ) of the measured value; a ≤ 30 keV measurement would provide a 3.6σ test. The structural origin, $\theta = 2/9 = (N_c^2-1)/(4N_c^2)$ from $N_c = 3$, is machine-checked (`koide_angle_from_N_c_pure`, zero `sorry`). *Access:* formula in P18; trivial Python implementation (<10 lines). *Primary use:* precision lepton mass measurements at Belle II. *Utility note:* The empirical Koide relation has been known since 1982, but has lacked a structural derivation. P18 supplies one from first principles ($N_c = 3$ uniqueness), machine-checked in Lean 4, converting an observed numerical coincidence into a falsifiable structural prediction. For a researcher testing lepton universality or planning a Belle II m_τ measurement campaign, the formula provides a specific theoretical target ($m_\tau = 1776.969$ MeV) with a known derivation and machine-verified proof, rather than a post-hoc fit.

4. Genetic Code Sieve (P25). Evaluates any wobble-admissible codon-to-amino-acid mapping against eight independent biological viability criteria (including the generative-to-dissipative IPT criterion). No existing public tool implements the full eight-criterion combination. The standard genetic code is the unique survivor of the sieve ($z = +3.22\sigma$; 0/99,998 random wobble-admissible codes outperform it). *Access:* `papers/25_genetic_code/code/codon_sieve.py` through `codon_sieve_phase6.py` (runtime < 10 minutes; NumPy, SciPy). *Primary use:* synthetic biology, origin-of-life research, and information-theoretic analyses of the genetic code. *Utility note:* Prior computational analyses of genetic code optimality typically evaluate one or two criteria in isolation—most commonly error minimisation or polar requirement correlation. The sieve is the first tool to apply eight criteria simultaneously, including a generative-to-dissipative information balance criterion not present in any prior published framework. Because the eight criteria are evaluated jointly, the uniqueness result (0/99,998) is strictly stronger than single-criterion results: a code that satisfies one criterion well while degrading another is correctly rejected. Researchers

in synthetic biology designing artificial genetic codes, or in origin-of-life studies quantifying how constrained the standard code is, can apply the sieve to any candidate mapping and receive a criterion-by-criterion breakdown, identifying precisely which biological constraints are binding.

Acknowledgements

All papers in this corpus are archived on Zenodo with permanent DOIs and are linked from https://www.novaspivack.com/research. The machine-checked Lean 4 formalization library (`ugp-lean`) is available at https://github.com/novaspivack/ugp-lean.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications across the corpus were independently verified by deterministic scripts and the Lean 4 kernel.

D The UGP Framework: Foundations, Structure, and Results

D.1 Introduction

The Standard Model of particle physics has approximately 25 free parameters — coupling constants, fermion masses, mixing angles, the cosmological constant — whose numerical values have no theoretical origin within the Standard Model itself. The **UGP Physics programme**, the quantitative branch of the Reflexive Reality programme [57], addresses this directly: starting from one foundational principle — **PSC (Perfect Self-Containment)** — with MDL (Minimum Description Length) as its operational expression (Presentation Invariance, the Layer II PSC axiom; PI ≡ MDL, machine-certified), and locality and symmetry as structural requirements, it uniquely selects a continuum field theory whose internal arithmetic reproduces the SM particle spectrum, all three gauge coupling constants, all fermion masses, the electroweak mixing angle, CKM mixing parameters, and now gravitational and quantum-mechanical structure — without a single free parameter. A discrete algebraic certificate (f_{MDL}, the CMCA) proves and compresses that arithmetic; it is not the physical substrate.

What distinguishes UGP from prior unification programmes is that it works in the opposite direction: rather than postulating a larger symmetry group that contains the SM as a subgroup, it derives the SM gauge structure *from below*, from the symmetry and orbit structure of a continuum field whose unique selection is forced by information-theoretic compressibility. The physical substrate is not invented to fit the data; it is the unique solution to a self-consistency equation. The central object is the Φ_{MDL} **field**, a $\mathbb{Z}_7$-symmetric Klein–Gordon gauge field whose internal discrete symmetry group is $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ — the unique non-abelian group of order 21, a subgroup of SU(3) — forced by minimum description length (MDL) minimality. Stable topological kink defects of Φ_{MDL} are the elementary particles of the Standard Model. A discrete algebraic certificate — the MDL-minimal $\mathbb{Z}_7$ cellular automaton f_{MDL} — certifies the internal symmetry structure of Φ_{MDL} and generates all SM quantum numbers from its orbit arithmetic; it is not the physical substrate. Paper P34 formalises the abstract substrate architecture as the Möbius triple $(A, e, [D])$, with D1–D5 constraints governing transputation and the Born rule. Quantum mechanics, special relativity, and gravity all emerge from the field's structure; none is postulated.

Results throughout the programme are certified at multiple epistemic levels. Many are *machine-certified* in Lean 4 with zero `sorry` and the standard Mathlib axiom signature — a level we denote $\mathbf{A}_{\mathrm{Lean}}$. Others are analytically derived ($\mathbf{A}/\mathbf{D}$) or numerically confirmed against PDG measurements. This overview traces the logical arc from substrate selection to the complete unified framework, citing the relevant papers for technical detail.

D.2 The Substrate: MDL Minimality and the Unique Selection of Φ_{MDL}

The framework begins with a question: among all field theories, which one minimizes the total description length of the laws governing a self-contained universe? The **Minimum Description Length (MDL)** principle — the operational expression of PSC's Presentation Invariance (PI ≡ MDL, Layer II PSC axiom, machine-certified) [25] — formalises self-containment as a Kolmogorov-complexity criterion. Combined with locality and symmetry as structural requirements, MDL minimality uniquely selects the $\mathbb{Z}_7$-symmetric sine-Gordon field Φ_{MDL}: among all $\mathbb{Z}_N$-symmetric fields with BPS kinks and exact Lorentz invariance, Φ_{MDL} has the smallest Kolmogorov complexity [67, 46]. (MDL is not an independent foundational axiom: PI is the Layer II PSC axiom that selects $N_{\mathrm{gen}} = 3$, and MDL is its physics-vocabulary expression. 'Why MDL?' reduces to 'Why PSC?' — the foundational question answered by the NEMS programme.) The NEMS physics synthesis situating GTE within the broader NEMS programme is in [55].

The physical substrate. The Φ_{MDL} **field** is the physical substrate of the universe. It is a real-valued $\mathbb{Z}_7$-symmetric Klein–Gordon gauge field evolving on $\mathbb{R}^{3+1}$:

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu \Phi)^2 - V(\Phi), \qquad V(\Phi) = \frac{m_\varphi^2}{49}(1 - \cos 7\Phi),$$

with field mass $m_\varphi = m_\tau = 1776.86$ MeV, derived from the F_{21} leptonic-sector self-consistency condition (SCC) [21]. Its seven global minima $\Phi_k = 2\pi k/7$, $k \in \mathbb{Z}_7$, define the vacuum manifold; the topological winding number $Q = (7/2\pi)\int \partial_x \Phi \, dx$ takes values in $\mathbb{Z}_7$ and furnishes a superselection charge. Stable topological kink defects winding between adjacent vacua are identified with the elementary particles of the Standard Model. Exact Lorentz invariance ($\omega^2 = k^2 + m^2$) is machine-certified in Lean 4 ($\mathbf{A}_{\mathrm{Lean}}$, `poincare_invariance_of_kg`) [46].

Scale hierarchy: substrate vs. particle.

Object	**Length scale**	**Energy scale**
CMCA fine-end reading (working hypothesis $a \approx \ell_{\mathrm{Pl}}$)	$\sim 10^{-33}$ cm (ℓ_{Pl})	$\sim 10^{19}$ GeV
CMCA matter-sector reading (MDL-saturated; derived, conditional — see below)	$a = \hbar c/\Lambda_{\mathrm{GTE}} \approx 0.097$ fm $\approx 10^{-14}$ cm	$\Lambda_{\mathrm{GTE}} \approx 2.0$ GeV
Φ_{MDL} kink width	$\sim 1/m_\varphi \approx 10^{-13}$ cm	$m_{\mathrm{kink}} = 290$ MeV
SM particles (Compton scale)	10^{-13}–10^{-16} cm	MeV–TeV

The CMCA, as a discrete algebraic certificate, carries no intrinsic lattice spacing: its algebraic content holds at every resolution (Algebraic Descent), and no finite-resolution CA is the physical substrate. This is the physical content of the no-CA-replica theorem: a finite-resolution CA at any M cannot exactly reproduce the smooth Compton-scale kink profile because the kink extends over $\sim M^{-1}$ lattice sites in the continuum limit. A lattice spacing therefore belongs to a *reading* — a calibration of the certificate against Φ_{MDL} observables — and two readings are in use. Under the conventional *fine-end working hypothesis* $a \approx \ell_{\mathrm{Pl}}$ (a working hypothesis, not a GTE derivation), the BPS kink width $\sim 1/m_\varphi \sim 10^{-13}$ cm is twenty orders of magnitude above the lattice scale, spanning $M_{\mathrm{Pl}}/M_{\mathrm{kink}} = 21^{10}\cdot 7^9/2^4 \approx 4.2\times 10^{19}$ cells: kinks are Compton-wavelength-scale collective excitations of a much finer lattice, precisely as phonons are macroscopic excitations of an atomic lattice. Under the MDL-saturated *matter-sector reading* $a = \hbar c/\Lambda_{\mathrm{GTE}} \approx 0.097$ fm — derived rather than postulated (Tape Saturation Theorem, **A/D**, conditional on the named Compton-Support Criterion, which is proven minimal; P50) — the same kink spans exactly $\lambda_C/a = |\mathbb{Z}_7| = 7$ cells, and the substrate-regulated lattice campaigns of P42 operate at this reading. The separation of the two readings is itself GTE arithmetic: the Planck-hierarchy monomial $M_{\mathrm{Pl}}/\Lambda_{\mathrm{GTE}} = 3^{10}\cdot 7^{18}/2^4 \approx 6.0\times 10^{18}$. The Algebraic Lifting Theorem bridges the certificate to the Compton scale without requiring a simulation of the Compton scale from first principles.

The discrete algebraic certificate. At the algebraic level, the orbit structure of Φ_{MDL} is encoded by a compact discrete object: the **MDL-minimal** $\mathbb{Z}_7$ **update rule** f_{MDL}, defined over GF(7) as $f_{\mathrm{MDL}}(L,C,R) = C + R - CR - LCR$, whose binary restriction is Rule 110. f_{MDL} is *not* the physical substrate; it is the **algebraic certificate** — a compact discrete representation of the symmetry structure, orbit arithmetic, and generation sequence of Φ_{MDL}. The f_{MDL} orbit over $\mathbb{Z}_7$ generates all SM quantum numbers via the GTE cascade (§D.3).

The complete discrete algebraic certificate is the **Three-Layer Chiral Minkowski CA (CMCA)** [67]: Rule 110 (right chiral layer) plus Rule 124 (left chiral layer) plus an inner τ_c Rule-110 clock per cell. The CMCA certifies all five structural features of Φ_{MDL}: $\mathbb{Z}_7$ generation orbit ($N_{\mathrm{gen}} = 3$), V−A chirality, Turing universality, and both observer-level and dynamics-level special relativity. Its MDL complexity $K_{\mathrm{CA}} = 19$ bits saturates the construction-class lower bound ($\mathbf{A}_{\mathrm{Lean}}$: `CMCAMDLMinimality.lean`) [67].

The Three-Tape CMCA and 3+1D emergence. The single-tape CMCA is a 1+1D algebraic certificate. The **Three-Tape CMCA** (P45 [68]) extends this to three independent 1+1D CMCAs sharing a single outer clock period τ_c^{out}. The **Dimensional Protocol Principle (DPP)**, machine-certified in Lean 4 ($\mathbf{A}_{\mathrm{Lean}}$), states that this shared clock is *both necessary and sufficient* for 3+1D dynamics: without it the three tapes evolve as independent 1+1D systems; with it they exhibit a single 3+1D Minkowski spacetime. The DPP provides the first machine-verified derivation of why spacetime is $3+1$-dimensional within the GTE framework. Temporal synchronisation is the causal prerequisite for spatial geometry — the shared clock *creates* space.

The three-tape architecture additionally produces all 33 SM charged-current vertices via $\mathbb{Z}_7$ winding conservation, derives baryon number as a topological charge, and recovers colour confinement from the MDL description-length gap $\Delta K = \log_2 9 \approx 3.17$ bits forbidding free quarks (all $\mathbf{A}_{\mathrm{Lean}}$, zero sorry, P45/P46 [34]).

The three-tape gravity mechanism has been closed at **A**/**D** (CatAD). The Physical Minimum Description Length (PMDL) Poisson equation $\nabla^2\Phi = G_{\mathrm{eff}}\, p(w_x, w_y, w_z)$ is $\mathbf{A}_{\mathrm{Lean}}$; the continuum-limit force law is derived analytically via the 3D Poisson Green's function and multipole expansion. For a source of Algebraic-Lifting radius σ_{AL}, the exact force is

$$F(b) \;=\; \frac{G_{\mathrm{eff}} M}{4\pi b^2}\left[1 - \tfrac{\sigma_{\mathrm{AL}}^2}{2b^2} + O\big(\sigma_{\mathrm{AL}}/b\big)^4\right] \xrightarrow{\,b/\sigma_{\mathrm{AL}}\to\infty\,} \frac{G_{\mathrm{eff}} M}{4\pi b^2},$$

recovering exactly Newtonian $F \propto r^{-2}$. Prior discrete-simulation measurements ($F \propto b^{-2.19\ldots-2.30}$) are finite-source-size artifacts: the exponent converges monotonically to -2.000 ± 0.005 as $b/\sigma_{\mathrm{AL}} \to \infty$, confirmed numerically. Three equivalent formulations are established: gradient-kick (Level 1 computational), metric-perturbation/GR (Level 2 geometric), and equivalence-principle (deepest mechanical principle), with the Level 2 metric formulation isomorphic to the Level 1 result in the Newtonian limit.

The two-level architecture. The GTE/Φ_{MDL} framework operates at two levels [21, 67]:

***Level 1*:** The CMCA — a 1+1D discrete algebraic certificate. Carries all five Level-1 structural features. This is *not* the physical universe; it is a mathematical proof system for the structure of Φ_{MDL}.

***Level 2*:** Φ_{MDL} in 3+1 dimensions — the physical continuum field. Quantum mechanics, Standard Model structure, and General Relativity all emerge at this level. The continuum limit of the CMCA is identified with Φ_{MDL} ($\mathbf{A}_{\mathrm{Lean}}$: `CMCAContinuumLimit.lean`).

Orthogonal to this resolution-graded level scheme, the framework's objects occupy a three-rung *ontological ladder* naming their roles — the **mathematical substrate** (the UGP/GTE arithmetic: what selects), the **certificate** (the 19-bit polynomial, f_{MDL}, and the CMCA tape: what certifies), and the **physical substrate** (Φ_{MDL}: what exists) — with the normative statement given in P48 [20].

The GTE-Möbius architecture specification (P34). Paper P34 [32] formalises the *abstract architecture* shared by both levels as the Möbius triple $(A, e, [D])$:

- A — the GTE arithmetic carrier ($\mathbb{Z}_7$ winding dynamics, P28). At Level 2, A is realised as the PSC-admissible *kink sector* of Φ_{MDL} (topological charges $Q \in \{0, 1, 2, 3\}$).
- e — the self-encoding map conferring Turing completeness (P30/P40; GF(7) polynomial route and Cook's Rule 110 construction).
- $[D]$ — the class of PSC-consistent coherence measures satisfying five constraints D1–D5 (non-negativity, PSC invariance, non-computability, selection completeness, Born consistency). $[D]$ realises *transputation*: non-computable, observer-free outcome selection at the computation/transputation boundary.

The Möbius name reflects a single-surface architecture: computation (deterministic f_{MDL} evolution) and transputation ($[D]$-selection) share one substrate without a sharp syntactic boundary. P34 is the **axiomatic foundation paper** for Group 7 (P34–P46): it specifies *what kind of object* underlies the SM derivations; P35 synthesises the bidirectional correspondence; P42–P43 complete the Φ_{MDL} field theory with D1–D5 as explicit axioms.

How the substrates and CA models relate

Object	Role	Key papers
Φ_{MDL} field	**Physical substrate** (Level 2): the universe	P42, P43, P38, P44
CMCA (3-layer)	Level-1 *algebraic certificate*: chiral null structure, SR clock	P41
Three-Tape CMCA	Level-1 certificate extended to 3+1D via shared clock (DPP)	P45
f_{MDL} / Rule 110	Minimal $\mathbb{Z}_7$ polynomial shadow of A; orbit arithmetic	P28, P40, P46
$(A, e, [D])$ triple	Abstract specification; D1–D5 axioms for QM/transputation	P34

Final substrate: Φ_{MDL} in 3+1D. **Not the substrate:** any finite CA, including f_{MDL}, Rule 110, or the CMCA (no-CA-replica theorem, P43). **Relation:** discrete objects *certify* and *compress* the arithmetic of Φ_{MDL}; lifting/descent theorems (P34, P35) connect the levels.

Results flow upward from discrete to physical via the **Algebraic Lifting Theorem** (`algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$) and downward from continuum to discrete via the **Algebraic Descent Theorem** (`AlgebraicDescentTheorem.lean`, $\mathbf{A}_{\text{Lean}}$): every algebraic and topological property of Φ_{MDL} depending only on the F_{21} internal symmetry — colour confinement, three generations, asymptotic freedom, strong CP — holds for f_{MDL} at every lattice resolution $M \geq 1$.

The Single-Source Principle. The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7), which requires exactly 19 bits to specify, simultaneously plays *five* physical roles with zero additional specification cost per role (P46 [34]; all roles $\mathbf{A}_{\text{Lean}}$): (1) *spatial cellular-automaton dynamics* — Rule 110 Turing-universal Class-4 behaviour; (2) $\mathbb{Z}_7$ *gauge vertex conservation* — Standard Model charged-current vertices; (3) *gravitational coupling* — the PMDL Poisson source term $\nabla^2\Phi = G_{\text{eff}}\, p(w_x, w_y, w_z)$; (4) *quantum entanglement generator* (negativity $\mathcal{N} = 0.38$, Bell nonlocality $S = 2.4459$, 87% of Tsirelson bound); (5) *baryon number*: $B = (1/3)\sum \chi_q(w_j)$ topological

charge, $B = 1/3$ per quark from $N_{\text{tapes}} = 3$ ($\mathbf{A}_{\text{Lean}}$: `gte_baryon_number_topological_charge`). This constitutes the **Single-Source Principle**: all of spacetime, matter, gravity, and quantum mechanics emerge from a single 19-bit description. The MDL uniqueness of this polynomial is proved in Lean 4: it is the unique degree-3 multilinear object over GF(7) satisfying $\mathbb{Z}_7$ symmetry, Rule 110 restriction, and 19-bit MDL minimality.

MDL uniqueness of $\mathbb{Z}_7$. The integer $N = 7$ is uniquely forced by three independent arguments: (a) MDL-minimality: $b_0 = 7N_{\text{gen}}/3 = 7$ is a positive integer only for $N = 7$, by the Algebraic Necessity Theorem ($\mathbf{A}_{\text{Lean}}$: `algebraic_necessity_master_bundle`); (b) the Frobenius prime identity $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1 = 7$ ($\mathbf{A}_{\text{Lean}}$: `FrobeniusPrimeIdentity.lean`); (c) the gravitational hierarchy formula $M_{\text{Pl}}/m_\tau = |F_{21}|^{10}|\mathbb{Z}_7|^7/2$ at 0.040% precision requires $|\mathbb{Z}_7| = 7$ [21]. Any other value of N fails at least one of these three constraints.

The five conditions that certify the orbit structure as uniquely admissible are formalised in the Perfect Self-Containment (PSC) framework: renormalisability (RC), structural stability (NM*), anomaly cancellation (AC), topological viability (TV), and spectral asymmetry (SA). Among 34,560 candidate dynamical systems, exactly 12 pass Layer I hard PSC filters (0.03%); every Layer I survivor carries $N_{\text{gen}} = 3$ ($\mathbf{A}_{\text{Lean}}$, zero sorry: `psc_enumeration_forces_ngen_3`). Layer II Presentation Invariance selects the unique MDL-minimum survivor at ridge level n=10, the GTE Lepton Seed (1, 73, 823), proved in Lean 4 [29, 22, 35]. (The Layer II claim that Presentation Invariance selects the minimum-K survivor among the 12 PSC-admissible gauge theories remains to be Lean-certified.) The Unified Rigidity Theorem [70] (P12) provides the synthesis proof that these three independent selection mechanisms — PSC Layer I/II theory-space selection, UGP arithmetic seed uniqueness, and the dynamical basin structure — all converge on the same canonical branch, with every assumption explicitly typed and audited.

The **Law = Description = Execution identity** holds for the discrete certificate: f_{MDL} is simultaneously (a) the GTE evolution law, (b) a compressed description encoding the 14 nonzero SM neighbourhoods, and (c) the executing CA whose binary sublayer is Rule 110. Machine-certified: `fmdl_law_description_execution` in `FMDLClassification.lean` ($\mathbf{A}_{\text{Lean}}$, zero sorry) [56].

The role of cellular automata in the GTE framework

The GTE framework involves cellular automata in a specific and limited role that differs from their role in other CA-based physics proposals.

What f_{MDL} / Rule 110 / the CMCA are:

- The *algebraic certificate*: a compact discrete representation of Φ_{MDL}'s internal symmetry structure and orbit arithmetic.
- The source of all SM generation quantum numbers via its $\mathbb{Z}_7$ orbit.
- A structure whose binary restriction (Rule 110) is forced by the SM generation sequence — proved in Lean 4 with zero sorry.
- The *mathematical proof system* for why Φ_{MDL} is the correct continuum theory, not an alternative to it.

What they are not:

- NOT the physical substrate of the universe (Φ_{MDL} is).
- NOT the source of quantum mechanics (which emerges from Φ_{MDL} at the continuum Level 2).
- NOT a replacement for Φ_{MDL}: P43 proves (`no_finite_ca_exact_lorentz_replica`, $\mathbf{A}_{\text{Lean}}$) that no finite-resolution CA, quantum CA, or rewriting system can exactly replicate Φ_{MDL}'s Lorentz invariance. The Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2) > 0$ for all finite M; only the continuum limit $M \to \infty$ recovers exact Lorentz symmetry, uniquely identifying Φ_{MDL} [21].

The computational/algebraic reframe. The GTE framework realises a scientific reframe that is distinct from all prior computational-universe proposals. The physical substrate is the continuum field Φ_{MDL}; no finite CA is the substrate (no-CA-replica theorem, `no_finite_ca_exact_lorentz_replica`, $\mathbf{A}_{\text{Lean}}$). What is fundamental is the *algebraic structure*: $\mathbb{Z}_7$, $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, and the GTE orbit arithmetic. Three consequences follow at the foundational level. First, the Law = Description = Execution identity (`fmdl_law_description_execution`, $\mathbf{A}_{\text{Lean}}$, zero sorry) certifies that f_{MDL} is simultaneously the evolution law, a compressed description, and an executing CA — the law of physics and its description are the same object. Second, the algebraic structure certifying Φ_{MDL} is itself computationally universal: Rule 110's Turing universality (Cook's theorem, 2004) transfers to Φ_{MDL} via the proved stepwise kink simulation, and the GF(7) polynomial representation additionally shows, via its center-cell NAND identity, that the kink update rule realizes any finite Boolean function algebraically — a finite functional-completeness result, distinct from and not sufficient for Turing completeness on its own. Third, the geodesic computation principle (§D.9) identifies physical trajectories with minimum-description-length paths: gravity, inertia, and quantum measurement are all manifestations of MDL minimality in the algebraic structure. These three results together constitute a reframe in which computation is the *deep structure*, not the medium, of physical reality.

D.3 The Generative Triple Evolution (GTE) Framework

The orbit arithmetic of f_{MDL} organises into canonical integer triples $(a, b, c; g)$ — the **Generative Triples** — one per Standard Model particle species. Three structural constants emerge from the orbit:

$$N_{\text{gen}} = 3, \qquad N_{\text{fam}} = 5, \qquad c_H = 13.$$

All three are derived, not assumed. $N_{\text{gen}} = 3$ follows from the Garden-of-Eden orbit depth: the first-generation beable has no f_{MDL} predecessor, forcing the orbit to close after exactly three generations (Lean: `fmdl_gen1_is_garden_of_eden`, $\mathbf{A}_{\text{Lean}}$). $N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}} = 5$ follows from the $\mathbb{Z}_5$ ring structure of the generation orbit (Lean: `gte_master_formula_complete`, $\mathbf{A}_{\text{Lean}}$). $c_H = 2^{N_{\text{gen}}+1} - N_{\text{gen}} = 13$ is the Higgs boundary parameter, determined by palindrome decomposition of the 13 active f_{MDL} neighbourhoods.

Frobenius-Generation Coincidence Identity (FGCI). The lepton ladder index $b_{L_1} = 73$ admits two *independent* derivations that agree if and only if $N_c = 3$: (1) the **Frobenius chain formula** $F(n) = n^4 - n^2 + 1$ evaluated at $n = N_c = 3$ gives $81 - 9 + 1 = 73$; (2) the **GTE cascade formula** $G(n) = n^4 - (n^2 + 1)/2 - n$ at $n = 3$ also gives 73. Both formulas coincide *uniquely* at $n = N_c = 3$: for $n \neq 3$ the two expressions differ. This witnesses that the algebraic structure (Frobenius group F_{21}) and dynamical structure (GTE cascade) are mutually consistent precisely at the physically realised colour-charge value. The muon ladder index $b_{L_2} = 42 = 2 \times N_c \times |Z_7|$ factors through both generators of F_{21}. The FGCI is machine-certified in Lean 4 (`FrobeniusChain.lean`, 17 theorems, zero sorry, $\mathbf{A}_{\text{Lean}}$; P01 [25]).

The GTE cascade produces all SM fermion quantum numbers. The $\mathbb{Z}_7$ winding number W_B is the interaction charge; the species formula $W_B = 4k \bmod 7$ for occupation k identifies: k=1 (charged leptons), k=4 (up-type quarks), k=5 (down-type quarks), k=7 $\equiv$ 0 (neutrinos). Three quark colours arise from the $\text{gen}_1/\text{gen}_2$ occupation ratio within a fixed composite. The $\mathbb{Z}_3$ color charge Q_χ is the Sylow-3 dlog of GF(7)* — the unique Sylow-3 invariant, derived with zero SM input. The joint topological charges (W_B, Q_χ) are derived from first principles (proved **A**/**D**, Lean-certified [35]).

Temporal winding and neutrino mass origin. The P45 three-tape CMCA extends the winding space to $\mathbb{Z}_7^4 = (w_x, w_y, w_z, w_\tau)$, where w_τ is the temporal winding component of the τ_c clock field. The neutrino sector ($k = 7 \equiv 0$, $W_B = 0$) corresponds to vacuum temporal winding $w_\tau = 0$. The bion instanton mechanism for generating neutrino masses is ruled out at the GTE level: the kink action parameter satisfies $\beta^2 = 49 > 8\pi$, placing Φ_{MDL} outside the bion-mass-generation regime (CatAL, Lean-certified). Neutrino masses therefore arise from non-topological dynamics, consistent with the seesaw structure derived in P21 [48]. The neutrino mass sum $\Sigma m_\nu = 59.4$ meV is derived from the P47 cosmological framework via the holographic mode count applied to the neutrino sector: within the Planck 2018 bound < 120 meV, predicting normal mass ordering [23].

Full details of the generation orbit, the Braid Atlas of topological signatures [12], and the cascade derivation of the fermion spectrum are in the Standard Model paper [25], the UGP dynamics paper [66], and the computational universality paper [22].

D.4 The Standard Model from Orbit Arithmetic

The GTE cascade generates the quantitative SM parameter spectrum from the single seed (1, 73, 823) with zero free parameters. Headline results from [25]:

- Nine charged-fermion masses at 0.295% RMS with zero active parameters at prediction time; a 1000-permutation null confirms arithmetic structure.
- Bare gauge couplings g_1^2, g_2^2, g_3^2 are Lean-certified exact rationals; $\alpha_s(M_Z) = 0.11822$ sits at $+0.24\sigma$ of PDG 2024, committed 43 days before the first PDG comparison.
- Neutrino mass-squared ratio $\Delta m_{21}^2/\Delta m_{31}^2 = 0.0294$ at 0.16σ from NuFIT 6.0, parameter-free.
- Cosmological constant from the derived model bit-length at 0.31σ from Planck 2018 (H_0 is the sole external input).

Electroweak mixing. The Weinberg angle is derived from the orbit constants in a ten-step chain, fully machine-certified:

$$\sin^2\theta_W \;=\; \frac{N_{\mathrm{gen}}}{c_H} \;=\; \frac{3}{13} \;=\; 0.23077.$$

The Wolfenstein-corrected value $384729/1664000 \approx 0.23121$ sits -0.42σ from PDG 2022. The two-loop SM Higgs-threshold correction with GTE-derived inputs gives $\sin^2\theta_W = 0.23129154$, within $+0.038\sigma$ of PDG 2024 (CatB; all electroweak inputs GTE-derived except the structural identity $1/\alpha_{\mathrm{em}}(0) = 2^{|Z_7|} + N_{\mathrm{gen}}^2 = 137$). The Z-boson mass $M_Z = 91.914$ GeV (+0.80% PDG) and W-boson mass $M_W = 80.614$ GeV (+0.30% PDG) follow from the same derivation chain ($\mathbf{A}_{\mathrm{Lean}}$ conditional) [17].

Electroweak VEV from SRRG. An independent first-principles derivation of the electroweak scale is provided by the Self-Referential Renormalization Group (P27) [59]. The SRRG flow maximises a net viability functional $F[S] = R[S] - C_\Lambda[S]$ over the space of self-referential physical theories. Its unique IR fixed point, with linearised contraction rate $1/\varphi$ (the golden ratio), selects the electroweak vacuum: $v_{\mathrm{PSC}} = 246.16$ GeV (-0.024% from PDG 246.22 GeV). This is machine-certified: `psc_ew_entropy_maximization` in `GoldstoneEntropyCorrection.lean` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry, zero open axioms). The result is independent of the Higgs mechanism; it requires only the $\mathbb{Z}_7$ structure and the PSC closure condition. The SRRG–MDL equivalence (P27 Open Problem 9) has been substantially closed: `op9_catal_unconditional` proves $K_{\mathrm{alg}}(g^*) = 0$ with unique zero

on $(0,\infty)$ ($\mathbf{A}_{\text{Lean}}$, zero sorry, unconditional), and `srrg_op9_k_alg_biconditional` proves the full biconditional IsSrrgFixedPoint $\Leftrightarrow$ K_{alg} minimiser (CatAD, zero sorry, no hypothesis). The SRRG fixed point $g^* = 1/\varphi$ is simultaneously the MDL-optimal coupling and the PSC-adjudication attractor: the same PSC axioms that select $N_{\text{gen}} = 3$ also drive the SRRG flow to g^* (CatAD, srrg-lean).

SU(2)$_L$ gauging. The within-tape weak-isospin doublet (Φ_+, Φ_-) has $SU(2)_L$-invariant potential $V(|\Psi|)$ since $|U\Psi| = |\Psi|$. The MDL principle forces minimal local gauging with zero extra specification cost ($\mathbf{A}_{\text{Lean}}$, zero named axioms): `phimdl_potential_su2l_invariant` certifies L_2 norm invariance under $SU(2)$ rotation; `su2l_covariant_derivative_minimal` certifies the covariant derivative as MDL-minimal; `su2l_wpm_generator_algebra` certifies that $W^{\pm}$ satisfy the $SU(2)$ Lie algebra. The master bundle `su2l_l2_from_phimdl_potential_catad` packages the full gauging chain (P46 [34], `GaugeMDL.lean`). The quantitative $V-A$ chiral structure of the charged current ($g_R = 0$, $g_A/g_V = -1$) now follows from the opposite tape windings $w_L = -w_R$ of the CMCA at CatAD (`va_coupling_exact_from_tape_topology`).

CKM mixing. All four Wolfenstein parameters of the CKM quark mixing matrix are derived from the GTE orbit constants with zero free parameters. The leading Wolfenstein parameter

$$\lambda \;=\; \frac{N_{\text{gen}}^2}{2^{N_{\text{gen}}} \cdot N_{\text{fam}}} \;=\; \frac{9}{40} \;=\; 0.225,$$

agrees with the PDG value to 0.0σ (Lean-certified) [19]. The hierarchy parameters A, $\bar{\rho}$, $\bar{\eta}$ follow from the quark-sector orbit indices, with the measured CP-violating phase δ_{CKM} reproduced without input.

Fine structure constant. The electromagnetic coupling constant $\alpha_{\text{EM}}^{-1} = 137$ is derived via two independent routes: from the GTE cascade chain (Lean: `alpha_inv_master`, $\mathbf{A}_{\text{Lean}}$) and from F_{21} Berry phase holonomy on a single Φ_{MDL} kink substrate (ROBUST, $\mathbf{A}_{\text{Lean}}$, eight independent audit passes) [35].

Lepton mass scale. The $\mathbb{Z}_7$ potential coefficient $c_V = 1/49 = 1/N_{Z_7}^2$ determines the tau-lepton Yukawa coupling exactly:

$$y_\tau \;=\; \frac{c_V}{N_{\text{mod2}}} \;=\; \frac{1}{2 \times 7^2} \;=\; \frac{1}{98},$$

where $N_{\text{mod2}} = 2$ is the $\mathbb{Z}_2$ parity factor. All three charged-lepton masses follow from v_H alone:

$$m_\tau \;=\; \frac{v_H}{98\sqrt{2}} \;=\; 1776.57 \text{ MeV} \quad (0.016\%).$$

Via the Koide relation ($\theta = 2/9$ from $N_c = 3$; Lean-certified: `koide_angle_from_N_c_pure`), $m_\mu = 105.6$ MeV (0.022%) and $m_e = 0.511$ MeV (0.023%) follow with zero additional SM inputs. The sole dimensional anchor is G_F (fixing $v_H = (\sqrt{2}\, G_F)^{-1/2}$); no fermion mass anchor is required.

The Koide relation admits the reformulation $\text{CV}(\sqrt{m}) = 1$: the coefficient of variation of the square-root masses equals unity, characterising the equal-irrep-norm (maximally-dispersed positive spectrum) condition. This form is Lean-certified (CatAL): $Q = (1 + \text{CV}^2)/3 = 2/3$ iff $\text{CV} = 1$. The kink–Higgs coupling derived from c_V is $g_{hKK} = 4/7^4 = M_{\text{kink}}/(v_H/\sqrt{2})$, with the identity $y_\tau = c_V/N_{\text{mod2}}$ algebraically exact (CatA). The cyclotomic structure underlying the Koide relation is established in P18 [4].

D.5 QCD and the Strong Force from F_{21}

The substrate's internal discrete symmetry group is identified as $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3 = \Sigma(21) \subset \mathrm{SU}(3)$ — the unique non-abelian group of order 21 — forced by MDL minimality (Lean-certified, $\mathbf{A}_{\mathrm{Lean}}$) [22, 35]. Its abelianisation $F_{21}^{\mathrm{ab}} = \mathbb{Z}_3$ is the color gauge symmetry; its non-abelian structure at the group level provides exact QCD Casimir invariants and structure constants without introducing non-abelian gauge bosons in the low-energy Lagrangian.

From F_{21} the following QCD results are derived:

- **Asymptotic freedom:** $b_0 = (11N_c - 2N_f)/3 = 7 = |F_{21}|/3$, with $N_c = 3$ and $N_f = 2N_{\mathrm{gen}} = 6$ derived from the GTE structure. $b_0 = |\mathbb{Z}_7|$ is Lean-certified (`b0_eq_z7_order`, `f21_substrate_asymptotic_freedom`, $\mathbf{A}_{\mathrm{Lean}}$); the Frobenius prime identity $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1$ is separately certified [51].
- **One-loop α_s:** $\alpha_s(M_Z) = 0.1319$ (one-loop); $\alpha_s = 0.1201$ at two loops with N_f=5 threshold — within 2% of PDG [51].
- **Colour confinement:** No PSC-admissible single-quark state exists (Lean: `no_psc_admissible_single_quark`, $\mathbf{A}_{\mathrm{Lean}}$); every physical state is PSC-admissible, confined, massive, and anomaly-free (`unified_physical_exclusion`, `confined_massive_color_singlet`, $\mathbf{A}_{\mathrm{Lean}}$) [26].
- **Strong CP problem:** $\theta_{\mathrm{QCD}} = 0$ exactly, derived from three independent F_{21} group-theory arguments: $\pi_3(F_{21}) = 0$, element-wise det = 1, and $\sum \mathrm{Im}(\chi_3) = 0$. All Lean-certified with zero sorry; no axion mechanism is required [51].
- **Hadron spectroscopy:** The meson pseudoscalar nonet (9 states), baryon octet (8), baryon decuplet (10), and vector meson nonet (9) are all reproduced from GTE kink composites. Casimir invariants $C_F = 4/3$, $C_A = 3$, $T_F = 1/2$, and all nine structure constants f^{abc} are reproduced at machine precision from F_{21} generators. The η–η' mixing angle $\theta_P = -13.08° \pm 3.74°$ lies in the PDG range $[-14.3°, -10.7°]$ with zero PDG inputs. All six quark masses are reproduced within 7% of PDG current values [51].
- **All 8 colour gluons** emerge from F_{21} Berry holonomy on a single Φ_{MDL} kink substrate (PROVISIONAL-STRONG) [35].
- **SU(3) UV completion (A/D):** Below $\Lambda_{\mathrm{GTE}} \approx 2$ GeV, the F_{21} abelian gauge effective theory is the correct description. Above Λ_{GTE}, $F_{21} \hookrightarrow \mathrm{SU}(3)$ faithfully ($\mathbf{A}_{\mathrm{Lean}}$: `F21SU3Embedding.lean`); the Burnside density theorem establishes $\overline{\mathrm{span}}_{\mathbb{C}}\, \rho(F_{21}) = M_3(\mathbb{C})$, and Φ_{MDL} scalar fluctuations fill the $\mathrm{SU}(3)/F_{21}$ coset, completing the UV theory as full SU(3) Yang–Mills. The naïve Wilsonian-flow approach from F_{21} to SU(3) is falsified: the discrete F_{21} lattice does not continuously flow to SU(3) in the IR; the UV completion is via Burnside density, not RG flow [35].
- **Predicted second Cartan gauge field A'_μ:** The $F_{21} \subset \mathrm{SU}(3)$ embedding has rank 2 (two Cartan generators λ_3, λ_8). The current GTE Lagrangian provides only the first Cartan field A_μ ($H_A = (-T^3 + \sqrt{3}\,T^8)/2$). A second Cartan field A'_μ ($H_{A'} = (\sqrt{3}\,T^3 + T^8)/2$) is predicted but currently absent from the Lagrangian. Its coupling $e' = e$ is forced by Killing-form equality $\mathrm{tr}(H_A^2) = \mathrm{tr}(H_{A'}^2) = 1/2$; no new free parameters are introduced ($\mathbf{A}_{\mathrm{Lean}}$). In QCD, λ_3 and λ_8 are both present; GTE is predicting the correct colour-sector structure.

The mass gap is established at the lattice level (ROBUST, $\mathbf{A}_{\mathrm{Lean}}$ lower bound) with continuum-limit scaling confirmed [26]. A non-perturbative string tension computation on the three-dimensional

G_{13} deformation lattice confirms $b_0 = 7 = |\mathbb{Z}_7|$ as the dominant RG coefficient; the non-perturbative contribution dominates over the one-loop term, and the quantisation factor f_{quant} is PROVISIONAL CatA [51].

The algebraic identity of F_{21} within the broader PSL(2,7) structure has been machine-certified in P52 [50]: F_{21} is the Borel stabilizer of ∞ in the unique simple group PSL$(2,7)$ of order 168, which simultaneously acts as the automorphism group of the Fano plane (with F_{21} as its flag-transitive subgroup) and as the automorphism group of the Klein quartic surface of genus $N_{\text{gen}} = 3$. The Eisenstein functor $\mathbb{Z}[\omega]$ unifies F_{21} (via the split prime 7) and $A_4 = V_4 \rtimes \mathbb{Z}_3$ (via the inert prime 2) from the same ring. All four results are machine-certified with zero sorry ($\mathbf{A}_{\text{Lean}}$).

D.6 Computational Universality: Cook's Theorem and Finite NAND Completeness

Rule 110 is universally regarded as the canonical example of a computationally universal elementary cellular automaton, established by Cook's proof (2004) [1]. The conditional operational readback core of Cook's construction is machine-certified in Lean 4 (`cook_operational_stage3_tm_microstep_readback`, zero sorry modulo five named classical Cook collision-analysis bridge axioms) [38] — a certificate for already-supplied compilations, not itself a self-contained Turing-universality theorem. Cook's classical theorem remains the load-bearing result for Rule 110's Turing universality.

Independently of Cook's construction, the UGP Physics programme establishes an exact polynomial representation of the Rule 110 update rule over the Galois field $\text{GF}(7) = \mathbb{Z}/7\mathbb{Z}$:

$$p(L,\, C,\, R) \;=\; C + R - C{\cdot}R - L{\cdot}C{\cdot}R,$$

where $L, C, R \in \{0, 1\} \subset \text{GF}(7)$. When $C = 1$, this simplifies to $p(L, 1, R) = \text{NAND}(L, R)$. Since NAND is functionally complete (Sheffer 1913), Rule 110 at center cell $C = 1$ realizes any finite two-input Boolean function algebraically, without reference to Cook's construction — a finite functional-completeness result about a fixed input size, distinct from and not sufficient for Turing universality, which requires simulation on an unbounded tape for an unbounded number of steps. The polynomial and NAND identity are Lean-certified with zero sorry (Lean: `rule110_z7_poly_rep`, `rule110_center1_is_nand`) [15]. Φ_{MDL}'s Turing universality is established via the Cook route composed with the proved stepwise Φ_{MDL}–Rule 110 simulation (`phimdl_turing_universal`, $\mathbf{A}_{\text{Lean}}$ conditional; zero sorry; one named axiom: `cook_rule110_simulates_computable`); the UWCA register file provides a second, independent Turing-universality route via Minsky counter-machine simulation, conditional on one named classical axiom (`minsky_counter_machine_turing_complete_1967`).

A further consequence is the **Algebraic Descent Theorem** (Lean-certified, $\mathbf{A}_{\text{Lean}}$): every algebraic and topological property of Φ_{MDL} that depends only on the F_{21} internal symmetry holds for f_{MDL} at any lattice resolution $M \geq 1$. Three generations, colour confinement, asymptotic freedom, and strong CP are all resolution-independent — proved at the algebraic level and therefore simultaneously valid for the discrete algebraic certificate and the continuum physical field theory [35].

D.7 Quantum Mechanics from the Φ_{MDL} Field

Quantum mechanics in the GTE framework is not postulated; it is derived from the structure of the Φ_{MDL} field and the $[D]$-selection mechanism [53, 46].

Two levels of description. At the Level-1 algebraic beable level (the CMCA discrete certificate), orbit states are always definite: no superposition exists at this level. At the Level-2 continuum, the Φ_{MDL} field admits superpositions $\sum_k c_k |\text{kink}_k\rangle$ over its $\mathbb{Z}_7$ winding sectors. Quantum mechanics

emerges entirely at Level 2; the discrete certificate provides the algebraic labelling of the winding sectors.

The selection mechanism rests on the GTE-Möbius architecture framework [32]: the abstract triple $(A, e, [D])$ where A is the Z_7 arithmetic carrier, e is the GF(7)/Cook self-encoding map, and $[D]$ is the class of PSC-consistent coherence measures characterised by five constraints (D1–D5: non-negativity, PSC invariance, non-computability, selection completeness, Born consistency). The abstract carrier A has its concrete physical realisation as the PSC-admissible kink sector of Φ_{MDL}; the five constraints are established in P34 [32] and serve as the axiomatic foundation of the quantum field-theoretic results in P42–P43. Two functions implement quantum mechanics on this substrate:

1. The **Born rule** $\mathcal{B}$: maps amplitudes to probabilities $P(k) = |c_k|^2$. Derived unconditionally and Lean-certified with zero custom axioms: `born_rule_unconditional` ($\mathbf{A}_{\mathrm{Lean}}$) [46]. An independent derivation via the Page–Wootters (PW) mechanism has been established at CatAD (P45): the τ_c clock, as a diagonal-Hamiltonian (Salecker–Wigner–Peres semiclassical) clock, satisfies all PW prerequisites, and the timeless universe state $(1/\sqrt{T})\sum_\tau |\tau\rangle_{\mathrm{clock}} \otimes U(\tau)|\psi_0\rangle_{\mathrm{sys}}$ yields $P(k|\tau) = |\langle k|U(\tau)|\psi_0\rangle|^2$ at every τ, independently of whether τ_c is in a classical or quantum superposition. The two derivations are mutually consistent and reinforce each other.
2. **Transputation** $P^\top$: selects the physical state $\rho^* = \mathrm{argmin}_\rho D(\rho \,|\, w)$ from record class w via the $[D]$-functional ($\mathbf{A}/\mathbf{D}$; Lean: `TransputationStateSelector.lean`) [9].

Objective reduction. $[D]$-selection is objective (no conscious observer required), determinate (unique minimum from D4), and non-computable (D3 places it outside Turing-computable selection). The non-computability is now characterised precisely: the adjudication function is exactly Δ^0_2-Turing-complete (the Gold–Putnam "trial-and-error" stratum, degree $0'$) — record truth on the diagonal fragment is Σ^0_1-complete, while the adjudicator is limit-computable, so quantum measurement operates precisely at the halting degree, never requiring oracle access above the first Turing jump (P51 [47]). This dissolves the measurement problem — reframing it in terms of PSC-forced transputation (conditional on the D4 universality assumption) — without many-worlds, pilot waves, or retrocausation. EPR correlations are semantic: they exist in the $[D]$-observable layer as constraints from the shared record structure established at interaction time; the beable layer is always local and deterministic. GTE escapes Bell's theorem because the transputation mechanism D3 is non-computable, placing it outside Bell's local-computable-hidden-variable assumptions. Delayed-choice and quantum eraser experiments are resolved by the same two-level structure: the CA beable trajectory is always definite at Level 1; whether an interference pattern or which-path statistics is observed depends only on which records w are retained in the $[D]$-description, with no retroactive modification of any beable-level trajectory [46].

Thermal state. Prior to any $[D]$-selection event, the physical state is the Hamiltonian thermal ensemble on PSC-admissible kink sectors $\{0, 2, 3, 4, 6\}$: vacuum-dominant at cosmological temperatures (Lean-certified: `PhiMDLThermalState.lean`, $\mathbf{A}_{\mathrm{Lean}}$ conditional).

D.8 Special Relativity and the Three-Layer Chiral Minkowski CA

Lorentz invariance is not postulated in the GTE framework. It emerges at the observer level from a specific two-level structure [67].

The CMCA. The **Three-Layer Chiral Minkowski CA (CMCA)** is the MDL-minimal Level-1 discrete construction realising all five required structural features: Turing universality, $\mathbb{Z}_7$ orbit, $V-A$ weak-current chirality, observer-level Lorentz invariance, and dynamics-level special relativity. It consists of an outer chiral CA pair (Rule 110 + Rule 124) plus an inner τ_c Rule-110 clock per cell. The MDL bit-count $K_{\mathrm{CA}} = 19$ bits saturates the construction-class lower bound (Lean: `CMCAMDLMinimality.lean`, $\mathbf{A}_{\mathrm{Lean}}$ conditional).

Emergence of Lorentz symmetry. At the CA beable level, the Rule 110 lattice carries a preferred rest frame; Lorentz symmetry is broken by $\varepsilon_0(M) = \pi^2/(3M^2)$ at the lattice scale. PSC Presentation Invariance (PI) forces the $[D]$-selection mechanism to be Lorentz-equivariant (Lean: `PSCPILorentzMain.lean`, $\mathbf{A}_{\mathrm{Lean}}$): no $[D]$-selected observable can distinguish inertial frames. Lorentz symmetry is a property of the selection mechanism, not of the underlying beables. At Planck-scale lattice spacing $a = \ell_P$, the lattice correction $\varepsilon_0 \approx 10^{-62}$ is observationally indistinguishable from zero.

Continuum limit. The continuum limit of the CMCA is identified with Φ_{MDL} ($\mathbf{A}_{\mathrm{Lean}}$ conditional): `CMCAContinuumLimit.lean`. The Φ_{MDL} field satisfies exact Lorentz invariance $\omega^2 = k^2 + m^2$, machine-certified in Lean 4 (`poincare_invariance_of_kg`, $\mathbf{A}_{\mathrm{Lean}}$) [67, 46].

D.9 Gravity from the Φ_{MDL} Field

Gravity emerges from the Φ_{MDL} field structure in three interconnected results [27, 21]:

Two routes to emergent gravity. Gravity emerges from the GTE framework via two complementary routes. The **P36 route** (CMCA/discrete level) derives geodesic structure from the deviation-based Ollivier–Ricci curvature of the Rule 110 causal graph: the discrete vacuum Einstein equation $G_{\mu\nu} = 0$ is machine-certified (`gte_spacetime_dimension`, $\mathbf{A}_{\mathrm{Lean}}$); the spectral dimension $d_s = 4$ is separately machine-certified (`causal_graph_spectral_dim_thermodynamic_limit`, $\mathbf{A}_{\mathrm{Lean}}$). An MDL–Lovelock structural correspondence ($\mathbf{A}/\mathbf{D}$) establishes that Einsteinian gravity emerges as the unique diffeomorphism-invariant effective theory compatible with the GTE substrate [26]. This constitutes the *structural/geometric certificate*: gravity is forced by the information structure. The 1D discrete gravity simulation (P36) yields a negative or null result for the inter-kink force in 1+1D, as expected: a 1+1D CMCA tape does not have three independent spatial directions and therefore cannot produce an inverse-square-law in the familiar sense; this is not a failure but a structural feature of the 1+1D algebraic certificate level. The **P45 three-tape route** (3+1D continuum level) closes this gap. With three independent CMCA tapes sharing a common clock τ_c^{out} (DPP), the PMDL Poisson equation $\nabla^2\Phi = G_{\mathrm{eff}}\, p(w_x, w_y, w_z)$ yields $F \propto b^{-2}$ in the far-field limit, machine-certified and confirmed analytically ($\mathbf{A}/\mathbf{D}$, CatAD; see §D.2). The two routes are not competing: P36 shows *why* gravity is forced (structural/geometric); P38/P45 show *how* it manifests quantitatively in the continuum EFT.

Stress-energy tensor and Einstein equations. The stress-energy tensor $T_{\mu\nu} = \partial_\mu\Phi\,\partial_\nu\Phi - \eta_{\mu\nu}\mathcal{L}$ is derived analytically from the Φ_{MDL} Lagrangian and Lean-certified ($\mathbf{A}/\mathbf{D}$): symmetry, vacuum-vanishing, and BPS pressure-free condition $T_{11} = 0$ are all proved in `StressEnergyTensor.lean`. The kink mass $\int T_{00} = 290.10$ MeV is verified numerically to 0.01%; energy-momentum conservation holds to 6×10^{-12}.

The Einstein equations $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}[\Phi_{\mathrm{MDL}}]$ are derived via MDL-Lovelock (the unique dimensionally correct generally covariant action in d=4), minimal coupling, and variational calculus

(**A**/**D**). The classical cosmological constant vanishes exactly: the $\mathbb{Z}_7$-symmetric potential $V(\phi) = m^2(1-\cos 7\phi)/49$ has $\langle V\rangle = 0$ in the PSC vacuum by exact cancellation, Lean-certified. The $\mathbb{Z}_7$ global shift symmetry $\phi \to \phi + 2\pi/7$ is furthermore *anomaly-free*: the scalar path-integral measure $\prod_x d\phi(x)$ is trivially translation-invariant (no Jacobian anomaly, unlike fermionic determinants), so all 7 vacuum sectors remain quantum-degenerate to all loop orders *in the pure-φ sector* — the degeneracy theorem is hypothesis-scoped to $\mathbb{Z}_7$-periodic observables. Four zero-sorry Lean theorems certify this: `z7_shift_measure_preserving`, `z7_jacobian_eq_one`, `lebesgue_shift_invariant`, and `z7_vacuum_sectors_equiprobable` ($\mathbf{A}_{\text{Lean}}$; P47/P43). Quantum vacuum degeneracy of the pure-φ sector is therefore a structural theorem, not an assumption. In the full Lagrangian, the canonical coupling $V_{\text{coupling}} = \varepsilon|\phi|^2(D_\mu\chi)^2$ is not $\mathbb{Z}_7$-periodic and distinguishes the seven vacua through the χ-sector kinetic normalization; this controlled breaking of the vacuum-label degeneracy supplies the vacuum-selection bias that resolves the cosmological domain-wall problem (§D.10; P42, P47 [46, 23]).

The gravitational D^2 theorem. The factor $D^2 = (N_{\text{gen}}+1)^2 = 4^2 = 16$ unifies the gravitational and cosmological sector of GTE. With $D = N_{\text{gen}}+1 = 4$ derived from the DPP ($\mathbf{A}_{\text{Lean}}$, `cmca_tensor_product_gives_31d_minkowski`), every area-type gravitational normalization carries exactly one factor of D^2:

- CC epoch count: $D^2 N_{\text{fam}}^{N_{\text{gen}}}/N_{\text{gen}} = 2000/3$ (CatAD)
- CC magnitude: $\rho_\Lambda = N_{\text{spatial}}^2/(D^2|\mathbb{Z}_7|)\, M_{\text{Pl}}^2 H_0^2$ (CatAD)
- Gorard coefficient: $C_{\text{Gorard}} = N_{\text{spatial}}/(2D^2) = 3/32$ ($\mathbf{A}_{\text{Lean}}$, `c_gorard_eq_n_spatial_over_two_dsq`)
- Einstein–Hilbert normalization: $1/(D^2\pi G_N)$ (standard GR; $D = 4$ from DPP)

D^2 counts ordered pairs of orbit discriminants in D-dimensional spacetime. It does *not* appear in the holographic sector ($\Omega_\Lambda^{\text{holo}}$, F_{21}, $\sin^2\theta_W$). In particular, the Gorard bridge coefficient $C_{\text{Gorard}} = N_{\text{spatial}}/(2D^2) = 3/32$ is an *exact* algebraic identity proved in Lean 4 with zero sorry — not an approximation.

Newton's constant from the Planck/tau hierarchy. Newton's constant is derived from the ratio:

$$\frac{M_{\text{Pl}}}{m_\tau} = \frac{F_{21}^{10} \times |\mathbb{Z}_7|^7}{2},$$

where $n = 10$ is the GTE orbit decomposition count on $\mathbb{Z}_7^3$ (all-distinct orbits equal the PSC count $(N_{\text{gen}}-1)\times N_{\text{cells}} = 10$), $b_0 = |\mathbb{Z}_7| = 7$ is derived from F_{21} group theory, and the factor $1/2 = |\mathbb{Z}_2|^{-1}$ reflects the chirality of the Rule-110/Rule-124 pair. The resulting prediction $M_{\text{Pl}}^{\text{GTE}} = 1.2204 \times 10^{22}$ MeV agrees with the PDG value to 0.040% (**A**/**D**, Lean-certified conditional: `mpl_mtau_formula_catal_conditional`, `b0_eq_z7_order`, `FrobeniusPrimeIdentity.lean`) [21].

Geodesic theorem and spectral dimension. Free particles follow paths of minimum computational action through the τ_c-metric curved by matter. The geodesic theorem is Lean-certified to zero axioms in three forms: timelike worldlines (`psc_orbit_is_curvature_geodesic`), flat spatial paths (`geodesic_theorem_spatial_general`), and the Ehrenfest reduction (`dweight_centroid_tracks_geodesic`), all $\mathbf{A}_{\text{Lean}}$, zero sorry [21]. The spectral dimension $d_s = 4$ in the thermodynamic limit is separately derived [26].

Quantum gravity and holography. The curved-background Lagrangian $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ is uniquely determined: MDL minimality forces $\xi = 0$ (`minimal_coupling_is_mdl_minimal`, $\mathbf{A}_{\text{Lean}}$),

yielding the minimal-coupling form $\mathcal{L} = \frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \, \partial_\nu \Phi - V(\Phi)$ with no additional curvature terms ($\mathbf{A}/\mathbf{D}$). The full quantum gravity sector derives the Graviton Fock space $\mathcal{H}_{\text{phys}} = \mathcal{H}_{\Phi_{\text{MDL}}} \otimes \mathcal{H}_{\text{grav}}$ and Bekenstein–Hawking entropy $S_{\text{BH}} = AM_{\text{Pl}}^2/4$ by two independent routes (Wald entropy, with $\xi = 0$; and domain wall). The holographic entropy formula

$$S(A) = \frac{\text{Area}[\Gamma_{\min}]}{4G}$$

is established by both routes; the Reed–Solomon quantum error correction route is additionally Lean-certified (`gte_orbit_vectors_are_rs_codewords`, $\mathbf{A}_{\text{Lean}}$, zero sorry). The Hawking temperature $T_H = M_{\text{Pl}}^2/(8\pi M_{\text{BH}})$ is confirmed unchanged by the field mass $m_\varphi = m_\tau$ (the potential vanishes at the horizon via the Schwarzschild redshift factor); the critical mass $M_{\text{crit}} = M_{\text{Pl}}^2/(8\pi m_\tau) = 3.34 \times 10^{39}$ MeV marks the onset of exponential emission suppression ($\mathbf{A}/\mathbf{D}$). BH evaporation is unitary: GKSL dynamics admit a Stinespring dilation confirming information preservation [18]. The Planck density state count $7^8 > e^{4\pi}$ is Lean-certified (`planck_density_bound_via_lifting`, $\mathbf{A}_{\text{Lean}}$, zero sorry), confirming that the CMCA lattice governs $E \geq M_{\text{Pl}}$; singularities are resolved at $\rho_{\max} = M_{\text{Pl}}^4$, and the partition function Z_{GTE} is UV-finite by $\mathbb{Z}_7$-compactness ($\mathbf{A}/\mathbf{D}$). The MDL-minimal cosmological initial state is flat ($k = 0$), spatially uniform, and kinetically dominated at $\dot{\Phi}_0 = M_{\text{Pl}}$: flatness, uniformity, and the absence of domain walls all follow from MDL complexity minimisation, dissolving the need for inflation without introducing new fields ($\mathbf{A}/\mathbf{D}$). A GTE-specific Friedmann correction produces a non-singular bounce at $\rho = \rho_{\text{Pl}}$ (confirmed numerically, $\mathbf{A}$). The quantum cosmological constant ($\sim 10^{45}$ vacuum-energy hierarchy) remains open; the classical $\Lambda = 0$ is Lean-certified [21].

P44 [52] completes the perturbative quantum gravity sector. UV finiteness on arbitrary smooth curved backgrounds is established via the DeWitt–Schwinger heat-kernel expansion: all curved-background UV contributions reduce to finite Planck-scale renormalizations ($C_i \sim 41.76$), leaving no residual UV problem. The **GTE Holographic Encoding Theorem** relates five descriptions — Lagrangian, Reed–Solomon $[5, 3, 3]_7$ algebraic code, holographic/RT, MDL-information-theoretic, and quantum error correction. Twelve of the twenty directed implications, linking the Lagrangian, Reed–Solomon, MDL, and QEC descriptions to one another, are established by derivation with no gravitational input; the remaining eight, connecting any of those four to the holographic/RT description, hold conditional on the standard Bekenstein-Hawking coefficient $1/4G$ taken as an external premise, not derived from GTE structure. The MDL cosmological initial state requires only $\log_2 3 \approx 1.585$ bits of initial data. At leading order the primordial spectral index satisfies $n_s = 1$ exactly; a seven-step conditional derivation chain yields

$$n_s \;=\; 1 - \frac{\ln 2}{2\pi^2} \;=\; 0.96488,$$

matching Planck 2018 at -0.005σ when the EFE-bridge step is established (5 of 7 steps closed). The tensor-to-scalar ratio satisfies

$$r \;=\; 0$$

exactly, constituting the primary falsifiable prediction for the LiteBIRD satellite ($\sigma_r \approx 0.001$).

D.10 The Complete Φ_{MDL} Framework

Papers P42 and P43 [46, 21] constitute the capstone of the 1+1D programme. P42 establishes four principal results for Φ_{MDL} as a field theory: exact Poincaré invariance ($\mathbf{A}_{\text{Lean}}$), the Born rule from field amplitudes ($\mathbf{A}_{\text{Lean}}$), the uniqueness of Φ_{MDL} among all $\mathbb{Z}_N \times \mathbb{Z}_M$ sine-Gordon

fields satisfying PSC-axiom-selected constraints (MDL-proved), and the no-CA-replica theorem (`no_finite_ca_exact_lorentz_replica` + `phimdl_is_unique_exact_lorentz_model`, $\mathbf{A}_{\text{Lean}}$).

P43 closes the framework with three capstone results:

1. **Algebraic Necessity Theorem** ($\mathbf{A}_{\text{Lean}}$, `AlgebraicNecessityTheorem.lean`): MDL minimality forces the unique $\mathbb{Z}_7 \times \mathbb{Z}_3$ structure over all competitors.
2. **Complete Lagrangian derivation:** the Φ_{MDL} Lagrangian is fully derived from F_{21} symmetry with zero free parameters. The coupling form $V_{\text{coupling}} = \varepsilon|\phi|^2(D_\mu\chi)^2$ and coupling constant $\varepsilon = N_7/N_3^2 = 7/9 \approx 0.778$ (in the BPS range $[0.444, 0.800]$) are both Lean-certified ($\mathbf{A}_{\text{Lean}}$). The field mass $m_\varphi = m_\tau = 1776.86$ MeV follows from the F_{21} leptonic-sector self-consistency condition. The kink–Higgs coupling $g_{hKK} = 4/7^4 = M_{\text{kink}}/(v_H/\sqrt{2})$ and the tau-lepton Yukawa $y_\tau = c_V/N_{\text{mod2}} = 1/98$ are both algebraically exact consequences of $c_V = 1/49$ (CatA).
3. **Newton's constant at high precision** ($\mathbf{A}/\mathbf{D}$, 0.040%, upgraded in P43).

The complete two-level algebraic structure is established by the interplay of two theorems: the **Algebraic Lifting Theorem** (Lean: `algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$) carries algebraic results from discrete to physical scale; the **Algebraic Descent Theorem** (Lean: `AlgebraicDescentTheorem.lean`, $\mathbf{A}_{\text{Lean}}$) shows that all F_{21}-dependent properties hold at every lattice resolution. Results proved for the continuous Φ_{MDL} are simultaneously results about the discrete f_{MDL} CA — and vice versa — with no additional proof required.

Papers P44–P47 [52, 68, 34, 23] constitute the 3+1D extension, quantum-gravity completion, and cosmological predictions. P44 (quantum gravity completeness) establishes UV finiteness on curved backgrounds and derives the primary cosmological falsifiable predictions. P45 (three-tape CMCA) proves the Dimensional Protocol Principle that identifies the minimal causal structure producing 3+1D spacetime from 1+1D building blocks, derives the SM particle spectrum from three-tape winding arithmetic, and establishes all structural features of 3+1D physics (SR, gravity, QM, confinement, Bell nonlocality). P46 (polynomial UFT) establishes MDL uniqueness of $p(L, C, R)$ and derives its five physical roles, unifying the algebraic and physical perspectives. P47 (cosmological predictions) derives the dark-energy fraction via two independent routes ($\Omega_\Lambda = 0.6899$ from PSC epoch count; $\Omega_\Lambda = 3\pi/14 = 0.6732$ from three-tape holographic mode count): (1) **PSC epoch-count route**: $\Omega_\Lambda = (\ln 2/N_{\text{gen}}\,\pi)\log_2(2000/3) = 0.6899$ from the PSC reflexive-closure orbit count, which admits a six-step first-principles derivation (CatAD): $|\mathcal{O}_{\text{GTE}}| = D^2 N_{\text{fam}}^{N_{\text{gen}}}/N_{\text{gen}} = (N_{\text{gen}}+1)^2 \cdot N_{\text{fam}}^{N_{\text{gen}}}/N_{\text{gen}} = 4^2 \cdot 5^3/3 = 2000/3$, with $D = 4$ from DPP ($\mathbf{A}_{\text{Lean}}$), $N_{\text{fam}} = 5$ from P01 ($\mathbf{A}_{\text{Lean}}$), and $D^2 = 16$ as the gravitational orbit discriminant; (2) **Three-tape holographic route**: $\Omega_\Lambda = 3\pi/14 = 0.6732$ from the three-tape mode count $3L$ vs. volume L^3 (holographic mode density $3L/L^3 \to 0$ as $L \to \infty$; the vacuum mode fraction gives $\Omega_\Lambda = \tau \cdot \pi/2 = (3/7)(\pi/2) = 3\pi/14$). Both routes bracket the observed Planck 2018 value 0.6889 ± 0.0056 from above and below, with zero free parameters ($\mathbf{A}/\mathbf{D}$, CatAD).

PSC-Adjudication Theorem (CatAD, P47). Given the PSC axioms (Two-Layer PSC Theorem, $\mathbf{A}_{\text{Lean}}$), MFRR [T], `op9_catal_unconditional` ($\mathbf{A}_{\text{Lean}}$), DPP ($\mathbf{A}_{\text{Lean}}$), and gauge spectrum ($\mathbf{A}_{\text{Lean}}$), the cosmological-constant fraction satisfies $\Omega_\Lambda \in [3\pi/14,\ 0.6899]$ with no free parameters. The Planck 2018 value 0.6889 ± 0.0056 lies within this bracket ($+0.18\sigma$ from the PSC route). The EW scale v_{PSC} does not enter Ω_Λ numerically; it enters only the energy density ρ_Λ as a kink mass reference. The SRRG role is *logical*: $g^* = 1/\varphi$ certifies the MDL-optimal and PSC-adjudication attractor — the same fixed point that selects the EW scale also certifies the CC bracket. This is strictly stronger than anthropic fine-tuning: the constraint applies to any PSC-consistent self-referential physical system, not only observer-containing universes. The bracket structure is itself a

theorem of the adjudication architecture: the irreducible residual of the PMDL action is the realized diagonal record ledger, a left-computably-enumerable real of Turing degree exactly $0'$, so Ω_Λ is limit-computable but not computable, and the two structural routes are precisely its computable bracket evaluations — the census-capacity ceiling above and the carrier-price floor below (CatAD, conditional on a named record-ledger premise bundle; P47, P51 [23, 47]).

$N_{\text{gen}} = 3$ is the *unique* integer for which both structural Ω_Λ routes fall within 5σ of the Planck 2018 observed value (all other integers: at least one route deviates by $> 35\sigma$; CatAD). Since PSC selects $N_{\text{gen}} = 3$ independently through the Two-Layer PSC Theorem, this constitutes a non-trivial structural consistency of the framework. A measurement-free counterpart sharpens this: under the derived orientation of the two routes — the holographic evaluation prices the MDL-minimal self-description carrier (the floor), while the census evaluation prices that carrier's information capacity (the ceiling) — the admissible dark-energy interval is non-empty only for $N_{\text{gen}} \leq 3$; for every $N \geq 4$ the floor exceeds the ceiling and no admissible dark-energy fraction exists at all, with zero observational input. Combined with the Layer I bound $N_{\text{gen}} \geq 3$ of the Two-Layer PSC Theorem, this determines $N_{\text{gen}} = 3$ without invoking Presentation Invariance (CatAD, conditional on the same structural premise bundle; P47, P48 [23, 20]).

The classical cosmological constant vanishes exactly ($\langle V \rangle = 0$ in the PSC vacuum, Lean-certified), quantum vacuum degeneracy is protected by $\mathbb{Z}_7$ anomaly-freedom in the pure-φ sector ($\mathbf{A}_{\text{Lean}}$; see §D.9), and one-loop corrections are suppressed by the holographic mode count ($\sim 10^{-42}$). The same coupling structure resolves the cosmological domain-wall problem: the $\mathbb{Z}_7$ vacuum symmetry, restored at $T_G \approx 0.70$ GeV after reheating, produces a transient Kibble wall foam that annihilates via the canonical coupling bias within $\sim 10^{-23}$ s — roughly $700\times$ before BBN — leaving zero surviving defect network, with relic signals $\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz and $\Delta N_{\text{eff}} \approx 0$ (a parameter-free falsifiable prediction). Any confirmed cosmological $\mathbb{Z}_7$ wall relic would falsify the annihilation mechanism and the associated thermal-history epoch, though not the substrate identification or the particle-spectrum derivations [23]. The paper also derives the CMB spectral index $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ certified by fourteen zero-sorry Lean theorems, the CKM CP phase $\delta_{CP} = 68.51^\circ$ (0.017% from PDG) from an S_3 subgroup chain, and the neutrino mass sum $\Sigma m_\nu = 59.4$ meV, all from zero free parameters with Lean 4 machine-verified steps. Together, P44–P47 close the programme's open questions on 3+1D emergence, quantum gravity perturbativity, the Single-Source Principle, and first-principles cosmological predictions.

Paper P48 [20] is the synthesis capstone: a 340+ page monograph presenting the complete PSC-to-cosmology derivation chain in fourteen chapters, integrating every result from P01–P47 with each step labeled by certification level. It traces substrate selection ($\mathbb{Z}_7 \times \mathbb{Z}_3$ MDL-minimal), the Φ_{MDL} field and particle spectrum, all four SM interactions, three-tape spacetime and quantum gravity, Born-rule quantum mechanics, and zero-parameter cosmology, with explicit comparison to eight alternative frameworks and a complete falsification table. A dedicated chapter, *The Mathematical Substrate*, develops the UGP/GTE arithmetic in full — including the UWCA, the substrate's native computer — and gives the normative statement of the ontological ladder introduced in §D.2; the Selection chapter presents the two MDL selections as sequential, the rule selection consuming the seed selection's orbit (Direct-Interpolation Lift Theorem, $\mathbf{A}_{\text{Lean}}$; full treatment in P49).

Paper P49 [40] presents the first systematic study of $p(L, C, R) = C + R - CR - LCR$ over $\mathbb{Z}_7$ as a standalone dynamical system, provides the Wolfram framing of GTE's MDL selection result (the MDL uniqueness theorem T96-02 is the selection principle the Wolfram Physics Project has lacked — a chain machine-certified link by link, $\mathbf{A}_{\text{Lean}}$: the cross-family coding inequality forcing $k = 7$, plus within-class uniqueness via the Direct-Interpolation Lift Theorem, under which the UGP generation orbit's total-parity shadow with vacuum transparency forces p uniquely among all 7^8 multilinear rules, Rule 110 arising as the binary-restriction corollary), and proves five new

algebraic theorems: Invariant Subset Uniqueness ($\mathbf{A}_{\text{Lean}}$), QNR Binary Floor ($\mathbf{A}_{\text{Lean}}$), Polynomial Uniqueness ($\mathbf{A}$), f_{MDL} non-polynomial ($\mathbf{A}$, Schwartz–Zippel), and 40% DPP cross-tape fraction ($\mathbf{A/D}$). It also presents the first $k = 7$ polynomial CA spacetime diagrams.

The arithmetic of the polynomial admits a striking unification. The diagonal self-consistency equation $p(x, x, x) = x$ factors as $-x(x^2 + x - 1)$ over $\mathbb{Z}$, revealing a **master quadratic** $x^2 + x - 1$ whose discriminant equals the family count $N_{\text{fam}} = 5$. This single $\mathbb{Z}$-quadratic has three arithmetic completions: over $\mathbb{R}$ it gives the SRRG fixed point that seeds the Higgs VEV ($x = 1/\varphi$, the golden-ratio conjugate; P27 [59]); over GF(7) it is irreducible (5 is a quadratic non-residue mod 7), which is exactly why the binary floor $\{0, 1\}$ is unique and Rule 110 is forced; and in polynomial dynamics it controls vacuum uniqueness ($\text{Fix}(T_n) = \{\text{vacuum}\}$ for every ring size n) — the electroweak scale and the unique computational floor are the two arithmetic fibers of one equation ($\mathbf{A}_{\text{Lean}}$; P49, P48 [40, 20]). Moreover, the two quadratic rings the framework employs — the golden ring $\mathbb{Z}[\varphi]$ and the Eisenstein ring $\mathbb{Z}[\omega]$, whose norm-7 residue arithmetic realises the substrate group as $F_{21} \cong (\mathbb{Z}[\omega]/(3 + \omega))^+ \rtimes \mu_3$ — are the two quadratic subfields of the single biquadratic compositum $K = \mathbb{Q}(\sqrt{-3}, \sqrt{5})$, of conductor $15 = N_{\text{gen}} \cdot N_{\text{fam}}$; membership of the GTE alphabet prime class reduces to a single Artin-symbol condition on K [40].

Paper P50 [60] realises the GTE polynomial as a 7-state spin lattice model with a phase transition in the 3-state Potts universality class (falsifiable critical exponents $\nu = 5/6$, $\eta = 4/15$), and Paper P51 [47] establishes the polynomial certificate of transputation: the Three-Level MDL Unification Theorem (theory selection, field dynamics, and event adjudication as nested instances of one MDL formula) and the computability classification of the adjudication function (§D.7).

P50's continuum-limit analysis sharpens the lattice–certificate correspondence into quantitative lattice field theory. The thermal ensemble of the spin-7 chain is the maximum-entropy shadow of the CMCA tape, and its spectral gap obeys an exact chiral gap law $\Delta(\beta) = e^{-3\beta/2}(1 + O(e^{-\beta/2}))$ ($\mathbf{A/D}$, with the exact correction tower): the half-integer slope 3/2 is the geometric mean of the directed wall energies of the pair digraph ($\mathbf{A}_{\text{Lean}}$) — impossible for any parity-symmetric chain, hence a quantitative parity-violation signature of the substrate chirality, the thermodynamic trace of the AGL(1,7) chiral reflection — and the amplitude is exactly 1, derived by two independent routes. The gap defines a lattice-QFT calibration trajectory whose physical point is selected by the Tape Saturation Theorem ($\mathbf{A/D}$, conditional on the named Compton-Support Criterion, proven minimal via a Boundary Equivalence Theorem): the matter-sector tape spacing saturates at $a = \hbar c/\Lambda_{\text{GTE}} \approx 0.0972$ fm, with exact fixed-point signature $\xi(\beta^*) = |\mathbb{Z}_7| = 7$ and lattice dictionary $aM_{\text{kink}} = 1/7$, $am_\varphi = 7/8$ — the bare mass the P42 lattice campaigns used, now derived rather than chosen. The unique empirical discriminator is the substrate-MC kink correlation length, $\xi_{\text{kink}} = 7$ cells (band 6.78–7.23 ± 0.54); a landing at 7 cells would make the physical point unconditional.

D.11 Physical Predictions and Cross-Checks

Table 6 collects key quantitative predictions against PDG measurements. Additional physical cross-checks include: neutrino mass structure and oscillation parameters [48]; the mirror-branch Braid Atlas [41] (P29) derives a parameter-free dark sector from the $\mathbb{Z}_7$ orbit mirror branch: three dark-lepton generations and $\text{SU}(3)_{\text{dark}}$, with the lightest dark state at 211.9 MeV, constituting a full set of falsifiable dark-sector predictions; nuclear binding energetics [25]; and P33 [24] derives additional arithmetic consequences of the framework: the EW boson mass staircase, the photon-as-vacuum identification, prime-lock structure, and the arithmetic closure of SM parameters beyond those already derived in P01.

Table 6: Selected GTE/UGP predictions vs. PDG measurements. **Derived**: analytically derived with zero free parameters. $\mathbf{A}_{\text{Lean}}$: machine-certified in Lean 4 with zero sorry.

Quantity	GTE/UGP value	PDG value	Accuracy	Status
$\sin^2\theta_W$ (tree)	$3/13 = 0.23077$	0.23129(4) (PDG 2024)	-13.0σ	$\mathbf{A}_{\text{Lean}}$
$\sin^2\theta_W$ (threshold)	384729/1664000	0.23122(3) (PDG 2022)	-0.42σ	$\mathbf{A}_{\text{Lean}}$
$\sin^2\theta_W$ (2-loop)	0.23129154	0.23129(4) (PDG 2024)	$+0.038\sigma$	**B**
M_Z	91.914 GeV	91.188(2) GeV	+0.80%	**A**/**D**
M_W	80.614 GeV	80.369(13) GeV	+0.30%	**A**/**D**
λ_{CKM}	$9/40 = 0.225$	0.22500(67)	0.0σ	$\mathbf{A}_{\text{Lean}}$
$\alpha_s(M_Z)$	0.11822	0.1180(9)	$+0.24\sigma$	$\mathbf{A}_{\text{Lean}}$
α_{EM}^{-1}	137	137.036 …	derived	$\mathbf{A}_{\text{Lean}}$
b_0 (QCD β_0)	$7 = \lvert\mathbb{Z}_7\rvert$	7 (exact)	exact	$\mathbf{A}_{\text{Lean}}$
θ_{QCD}	0 (exact)	$< 10^{-10}$	exact	$\mathbf{A}_{\text{Lean}}$
M_{Pl}/m_τ	$F_{21}^{10}\cdot 7^7/2$	PDG ratio	0.040%	**A**/**D**
θ_P (η–η' mixing)	$-13.08° \pm 3.74°$	$[-14.3°, -10.7°]$	in range	**A**/**D**
N_{gen}	3 (derived)	3	exact	$\mathbf{A}_{\text{Lean}}$
$\Lambda_{\text{classical}}$	0 (exact)	$\approx 10^{-52}$ m^{-2} (quantum)	classical exact	**A**/**D**
n_s (CMB tilt)	$1 - \ln(2)/(2\pi^2) = 0.96488$	0.9649 ± 0.0042 (Planck 2018)	-0.005σ	**A**/**D** (cond.)
r (tensor-to-scalar)	0 (exact)	unmeasured ($\sigma_r \approx 0.001$ at LiteBIRD)	primary falsifiable	**A**
y_τ (tau Yukawa)	$1/98 = 1/(2\times 7^2)$	SM value	exact structural	CatA
m_τ (from v_H, no mass input)	1776.57 MeV	1776.86 MeV	0.016%	CatA
m_μ (Koide from y_τ)	105.6 MeV	105.658 MeV	0.022%	CatA
m_e (Koide from y_τ)	0.511 MeV	0.51100 MeV	0.023%	CatA
g_{hKK} (kink–Higgs coupling)	$4/7^4 = M_{\text{kink}}/(v_H/\sqrt{2})$	—	structural	CatA
Ω_Λ (PSC route, P47)	$(\ln 2/3\pi)\log_2(2000/3) = 0.6899$	0.6889 ± 0.0056 (Planck 2018)	$+0.18\sigma$	CatAD
Ω_Λ (holographic, P47)	$3\pi/14 = 0.6732$	0.6889 (Planck 2018)	bracket	CatAD
Ω_Λ PSC bracket	$[3\pi/14,\ 0.6899]$; N_{gen}=3 unique	0.6889 (Planck 2018)	within bracket	CatAD
C_{Gorard}	$N_{\text{spatial}}/(2D^2) = 3/32$ exact	—	exact	$\mathbf{A}_{\text{Lean}}$ (`c_gorard_eq_n_spatial_over_two_dsq`)
δ_{CP} (CKM S_3 chain, P47)	$\pi/2 - 3/8 = 68.51°$	68.517° (PDG)	0.017%	CatAD
Σm_ν (P47)	59.4 meV	< 120 meV (Planck)	within bound	CatAD
New machine-certified results (2026)				
PMNS $\sin^2\theta_{12}$ (P21)	$4/13 = 0.3077$	0.308 ± 0.011 (NuFIT 6.0 IC24 NH)	$+0.01\sigma$	$\mathbf{A}_{\text{Lean}}$
PMNS $\sin^2\theta_{23}$ (P21)	$19/42 = 0.4524$	0.470 ± 0.013 (NuFIT 6.0 IC24 NH; 43.3°)	-1.35σ; within 3σ	$\mathbf{A}_{\text{Lean}}$
PMNS $\sin\theta_{13}$ (P21)	$11/73 = 0.1507$	0.1489 ± 0.0007 (NuFIT 6.0 IC24 NH)	$+1.0\sigma$	$\mathbf{A}_{\text{Lean}}$
Jarlskog J_{GTE} (P21)	-1.468×10^{-2} (from orbit ratios)	1.78×10^{-2} (NuFIT 6.0 IC24 NH)	see P21	$\mathbf{A}_{\text{Lean}}$
Higgs mass m_H (P27, P35)	125.2499 GeV (SRRG-corrected λ)	125.20 ± 0.11 GeV	-0.0003σ	$\mathbf{A}_{\text{Lean}}$
η_B (P47, P42; FKTT)	6.109×10^{-10}	$(6.10 \pm 0.06) \times 10^{-10}$	$+0.15\sigma$	$\mathbf{A}_{\text{Lean}}$
H_0 (P47)	67.95 km/s/Mpc (first-principles via GTB kink-overlap cert; epoch-dependent)	67.66 ± 0.42 km/s/Mpc	$+0.7\sigma$	CatAD
Quantum kink mass M^Q (P38, P42)	281 ± 21 MeV (interface dim-reg, two routes)	—	see P42	CatA
FGCI (P01)	$b_{L_1} = 73$ doubly determined; unique at $N_c = 3$	—	exact	$\mathbf{A}_{\text{Lean}}$
$\Omega_{\text{DM}}h^2$ (P29)	0.11994 (via $D_{\text{top}} = e^{-1/N_c}$, machine-certified $\mathbf{A}_{\text{Lean}}$)	0.1200 ± 0.0012	−0.044%	CatAD

D.12 Experimental Programme

The framework makes six sharp falsifiable predictions testable by currently running or near-term precision experiments, collected in Table 7. Proton stability provides the sharpest topological test: the Lean 4 theorem `proton_decay_dim4_forbidden` ($\mathbf{A}_{\text{Lean}}$, zero sorry) proves that all *dimension-4* baryon-number-violating vertices are forbidden by $\mathbb{Z}_7$ winding conservation — a direct test of the interaction skeleton theorem. Any confirmed dimension-4 proton decay would falsify the $\mathbb{Z}_7$ winding conservation law; the programme's other derivations (mass spectrum, cosmological predictions, three-generation proof) would be unaffected. Notably, standard GUT dimension-6 proton decay (e.g. $p \to e^+\pi^0$, the primary mode searched at Hyper-Kamiokande and DUNE) is *not* forbidden by the framework and, if observed, would be completely consistent with the programme's predictions.

Table 7: Six falsifiable predictions of the GTE/UGP programme with active or near-term experimental tests. The "Falsifying Condition" column specifies the observation that, if confirmed at the stated significance, would require structural revision of the framework.

Observable	GTE Prediction	Experiment(s)	Timeline	Falsifying Condition
Proton stability (dim-4)	No *dimension-4* baryon-number-violating decay; $\mathbb{Z}_7$ winding conservation forbids all dim-4 SU(5) vertices (`proton_decay_dim4_forbidden`, $\mathbf{A}_{\text{Lean}}$; P22, P33). *Note:* standard GUT dim-6 modes ($p \to e^+\pi^0$, $p \to \bar{\nu}K^+$) are **not** forbidden and would be consistent with this framework if observed.	Hyper-Kamiokande, DUNE	2027^+	Any confirmed *dimension-4* baryon-violating event (not dim-6 GUT modes such as $p \to e^+\pi^0$, which are not forbidden)
GTE-P7 dark particle	Neutral pseudoscalar at 211.9 MeV, $Q = 0$; mirror-branch $\mathbb{Z}_7$ prediction, machine-certified in Lean 4 (P02, P29)	Belle II (mono-photon / missing energy)	Running	Non-observation at $50\,\text{ab}^{-1}$
PMNS Dirac CP phase	$\delta_{CP}^{\text{PMNS}} = 205.71^\circ = \frac{4}{7} \times 360^\circ$; from $\mathbb{Z}_7$ winding arithmetic (P45)	DUNE, Hyper-Kamiokande	2027^+	Measurement outside $205^\circ \pm 15^\circ$
Tensor-to-scalar ratio	$r = 0$ exactly; no primordial gravitational waves in the GTE inflation-free cosmology (P44, P45; $\mathbf{A}_{\text{Lean}}$)	LiteBIRD	$\sim$2028	$r > 0.01$ at $> 3\sigma$
Neutrino mass sum	$\Sigma m_\nu = 59.4$ meV; normal mass ordering; individual spectrum $\{0.68, 8.6, 50.1\}$ meV (P21, P47; analytically derived)	EUCLID, CMB-S4	2026^+	$\Sigma m_\nu < 40$ meV or > 80 meV
Dark-energy EOS	$w = -1$ exactly; cosmological-constant behavior forced by PSC boundary condition, Weinberg no-go evaded (P47; analytically derived; note: DESI 2024 data favour $w \approx -0.95$)	DESI, EUCLID	Ongoing	$w \neq -1$ at $> 3\sigma$

D.13 Open Questions

The programme is mature but not complete. The following structural gaps remain open as of this writing. A number of questions previously listed as open have been closed by P44–P47 and subsequent analytical work: 3+1D emergence has been derived (DPP, $\mathbf{A}_{\text{Lean}}$); the quantum gravity sector is perturbatively complete (P44); the Single-Source Principle is established ($\mathbf{A}_{\text{Lean}}$); $r = 0$ ($\mathbf{A}$) and $\delta_{CP} = 68.51°$ (CatAD, P47) are derived as falsifiable predictions; the Newtonian gravity force law $F \propto r^{-2}$ has been established at CatAD (three equivalent derivations); the Born rule is independently confirmed at CatAD via the Page–Wootters τ_c mechanism; the Ω_Λ bracket $[3\pi/14,\ 0.6899]$ is CatAD with no free parameters (PSC-Adjudication Theorem, P47); the SRRG–MDL equivalence OP9 is CatAL at the fixed point and CatAD for the full biconditional; the $\mathbb{Z}_7$ anomaly-freedom is $\mathbf{A}_{\text{Lean}}$; the **unconditional MDL uniqueness of** $\mathbb{Z}_7 \times \mathbb{Z}_3$ is $\mathbf{A}_{\text{Lean}}$ (`mdl_ca_rule_coding_closed`, `z5_fmdl_no_psc_kink_orbits`, `mdl_total_z7z3_strictly_beats_z5z3`, all zero sorry; the GTE polynomial over GF(5) has zero PSC kink orbits, giving total MDL gap $3 + 0 < 5 + 6$ with no SM input at any step); and cosmological predictions for Ω_Λ, n_s, and the CKM CP phase have been derived from first principles in P47.

- **Explicit glider-to-fermion map.** The existence of a canonical map from specific Rule 110 glider patterns to specific SM fermions is confirmed via the Algebraic Lifting Theorem ($\mathbf{A}_{\text{Lean}}$). The mesoscopic orbital form is machine-certified: the f_{MDL} forward chain $\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3 \to$ vacuum assigns a unique orbital position to each SM fermion generation (`zgm_mes_orbital_discriminant`, `ZGMMesInvariant.lean`, $\mathbf{A}_{\text{Lean}}$). The hard form — explicit one-to-one identification of each spatial Rule 110 glider species with a specific fermion — remains the deepest open particle-ontology question.
- **Gluon first-principles discrimination (colored W_B=0 sector).** The colorless W_B=0 sector is fully discriminated: photon, neutrino, Z, and Higgs are pairwise distinguished by a three-level first-principles labeling (`z7_zero_sector_unified_discriminant`, `Z7ZeroSectorDiscriminant.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry). Gluons also carry $W_B = 0$ but are colored ($Q_\chi \neq 0$) and are excluded from the colorless hierarchy; a complete CA-dynamics account of all eight gluon species from the F_{21} orbit structure remains open.
- **Newton's constant in SI units.** The ratio $G_N = m_\tau^2/M_{\text{Pl}}^2$ is derived to 0.040% ($\mathbf{A}/\mathbf{D}$). A purely first-principles derivation of G_N in SI units from the τ_c dimensional mechanism — without using m_τ as an input — remains open.
- **Quantum cosmological constant.** Significant progress has been made. The classical $\Lambda = 0$ is Lean-certified ($\mathbf{A}/\mathbf{D}$). Quantum vacuum degeneracy across all 7 $\mathbb{Z}_7$ sectors is an $\mathbf{A}_{\text{Lean}}$ theorem (`z7_vacuum_sectors_equiprobable`; pure-φ sector, $\mathbb{Z}_7$-periodic observables). The PSC-adjudication bracket $\Omega_\Lambda \in [3\pi/14,\ 0.6899]$ is established at CatAD with no free parameters (P47), and one-loop corrections are suppressed by $\sim 10^{-42}$ via holographic non-renormalization (CatAD). **All computable quantum corrections to Λ vanish identically in the pure-φ sector** — machine-certified by `lambda_all_order_zero_pure_phi` (P48, $\mathbf{A}_{\text{Lean}}$, zero sorry). This closes OP3 at $\mathbf{A}_{\text{Lean}}$ for the pure-field sector. What remains open is the full-Lagrangian sector (including V_{coupling}): a complete non-renormalization proof to all loop orders in the interacting theory (CatD).

D.14 Summary and Significance

The UGP Physics programme has established, through a combination of Lean 4 machine-checked proofs and precision numerical verification, that the $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL} —

uniquely selected as the physical continuum substrate by MDL minimality and PSC self-consistency from first principles, formalised as the Möbius triple $(A, e, [D])$ in P34 [32], and algebraically certified by its discrete counterpart f_{MDL} and the CMCA (P41/P45) — generates the quantitative structure of the Standard Model particle spectrum, all three gauge coupling constants, the electroweak mixing angle, the CKM matrix, the full QCD sector including colour confinement and asymptotic freedom, Turing universality, quantum mechanics with its Born rule, exact Lorentz invariance, the Einstein field equations with Newton's constant, the holographic entropy formula $S(A) = \mathrm{Area}[\Gamma_{\min}]/(4G)$, Hawking temperature unmodified by $m_\varphi = m_\tau$, the MDL-selected flat cosmological initial state that dissolves the need for inflation, and the first-principles cosmological predictions of P47: the dark-energy fraction ($\Omega_\Lambda = 0.6899$ and $3\pi/14 = 0.6732$, bracketing Planck 2018 within the PSC-adjudication band $[3\pi/14,\ 0.6899]$ at CatAD), the CMB spectral index ($n_s = 0.96488$, -0.005σ), the CKM CP phase ($\delta_{CP} = 68.51°$, 0.017% from PDG), and the neutrino mass sum ($\Sigma m_\nu = 59.4$ meV). Recent machine-certified results additionally establish, all with zero sorry: the three PMNS mixing angles ($\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 19/42$, $\sin\theta_{13} = 11/73$; $\chi^2 = 0.564$, zero free parameters) and the Jarlskog invariant $J_{\mathrm{GTE}} = -(8184\sqrt{437}/5057221)\sin(\pi/7)$; the corrected Higgs mass $m_H = 125.2499$ GeV (-0.0003σ PDG 2022) via the SRRG identity $2c_H + 1 = 27 = N_{\mathrm{gen}}^3$ ($\mathbf{A}_{\mathrm{Lean}}$); the baryon asymmetry $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$) via the FKTT mechanism ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry, zero axioms; `kink_top_coupling_eq_eps_FN`); and the Frobenius-Generation Coincidence Identity confirming that $b_{L_1} = 73$ is doubly determined, witnessing consistency of the algebraic and dynamical structures at $N_c = 3$ ($\mathbf{A}_{\mathrm{Lean}}$; `FrobeniusChain.lean`). No free parameters are introduced at any stage; the only true external input is one energy scale (G_F, which fixes $v_H = (\sqrt{2}\,G_F)^{-1/2}$). The tau-lepton Yukawa $y_\tau = 1/98$ and the structural Koide relation ($\theta = 2/9$ from $N_c = 3$) then derive all three charged-lepton masses from v_H without any fermion mass anchor.

The ugp-lean library currently comprises over 300 Lean 4 modules, including zero-sorry coverage of `FrobeniusChain.lean` (17 theorems; FGCI), `MassRelations/NeutrinoSector.lean` (PMNS angles, Jarlskog, FN rank-1 theorem), `MassRelations/HiggsQuartic.lean` ($m_H = 125.2499$ GeV, $2c_H + 1 = N_{\mathrm{gen}}^3$), `UCLMassOrdering.lean` (fermion mass ordering), and `Gravity/YukawaOverlapExponent.lean` (η_B overlap exponent). Every Category-A physics theorem in the library has axiom closure exactly {`propext`, `Classical.choice`, `Quot.sound`} — the standard Mathlib signature. No UGP-specific axioms appear in any Category-A physics theorem.

The programme's distinguishing feature is not merely quantitative agreement but logical necessity: the constants are not fitted but *forced*. The number three is forced for the generation count; the fraction 3/13 is forced for the Weinberg angle; the ratio 7 is forced for the QCD β-function coefficient; the exponential hierarchy between M_{Pl} and m_τ is forced by F_{21} orbit combinatorics; $r = 0$ and $\delta_{CP} = 68.51°$ follow from the three-tape clock architecture and CKM orbit arithmetic (P47); and the dark-energy fraction, CMB tilt, CKM CP phase, and neutrino masses all follow from the PSC holographic structure of P47. If a future precision measurement displaces any of these predictions, the framework's foundational principle — PSC (Perfect Self-Containment), with MDL as its operational expression, and F_{21} symmetry as the unique admissible internal structure — provides unambiguous targets for revision.

The 52 papers of the UGP Physics programme (P01–P52), the ugp-lean Lean 4 library, and the computational infrastructure are all archived on Zenodo and catalogued at `https://www.novaspivack.com/research` [57].

One field. One foundational principle — self-containment — from which all of physics follows.

E Master Index: Inputs, Axioms, and Physical Predictions

Complete index of framework inputs, axioms, and physical predictions across P00–P55, with Lean 4 certification status. Claim levels: $\mathbf{A}_{\text{Lean}}$ = machine-verified (zero sorry); A/D = analytically derived; CatAD = derived, conditional on open axiom; B = bridge claim; D = speculative or open. External measured inputs are marked separately from derived inputs. A derivation chain (§E.1.1) and discrepancy register (§E.5) follow the prediction tables.

E.1 Framework Inputs

E.1.1 Structural and Arithmetic Quantities (all derived — zero free parameters)

These quantities are derived from the three UGP axioms and the GTE construction. They are foundational within the cascade but are not free parameters.

Derivation chain for $N_{\text{fam}} = 5$:

`fMDL structure (MDL axiom A3)`
$\rightarrow$ `GoE orbit depth forces` $N_{\text{gen}} = 3$ `(fmdl_gen1_is_garden_of_eden,` $\mathbf{A}_{\text{Lean}}$`)`
$\rightarrow$ $N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}} = 2^3 - 3 = 5$ `(gte_master_formula_complete, z5_transitivity_uniqueness,` $\mathbf{A}_{\text{Lean}}$`)`
$\rightarrow$ $N_{\text{gen}} + N_{\text{fam}} = 2^{N_{\text{gen}}} = 8$ `(ngen_plus_nfam_eq_pow2,` $\mathbf{A}_{\text{Lean}}$`)`

N_{fam} counts the five SM fermion family types [e^-, u, d, ν_R, ν_L] — not the same as N_{gen}.

Table 8: Structural and arithmetic derived inputs (all zero free parameters).

Symbol	Value	Physical Meaning	Derivation / Source	Papers	Lean Cert
n (ridge)	10	GTE ridge level; $R_{10} = 1008$	Four independent certificates: ridge minimality, global asymptotic sparsity, divisor-count CKM cert, seed-$b_1 = 73$ uniqueness	P01 §2.2, P05	`n10_is_minimal_admissible_ridge`, `asymptotic_sparsity_universal` ($\mathbf{A}_{\text{Lean}}$)
N_{gen}	3	Number of particle generations	GoE orbit depth = 3; $b_0 = 7N_{\text{gen}}/3 \in \mathbb{Z}$; Layer I enumeration: all 12 PSC survivors have $N_{\text{gen}} = 3$ (`psc_enumeration_forces_ngen_3`, $\mathbf{A}_{\text{Lean}}$); PSC Layer II PI selects $N_{\text{gen}} = 3$ from 12 survivors. *Additional:* $N_{\text{gen}} = 3$ is the unique integer for which both structural Ω_Λ routes fall within 5σ of Planck 2018 (CatAD)	P01, P28, P43, P47	`fmdl_gen1_is_garden_of_eden`, `psc_enumeration_forces_ngen_3` ($\mathbf{A}_{\text{Lean}}$)

Symbol	Value	Physical Meaning	Derivation / Source	Papers	Lean Cert
N_{fam}	5	SM fermion family types	$N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}} = 5$; Z_5 ring structure	P01, P28, P35	`gte_master_formula_complete`, `z5_transitivity_uniqueness` ($\mathbf{A}_{\text{Lean}}$)
N_c	3	QCD colour rank	Anomaly cancellation in Z_7 winding; $N_c = N_{\text{gen}}$ forced by GTE	P01, P28, P43	`anomaly_cancellation_forces_Nc_3` ($\mathbf{A}_{\text{Lean}}$)
c_H	13	Higgs boundary parameter	$c_H = 2^{N_{\text{gen}}+1} - N_{\text{gen}} = 13$; gives $\sin^2\theta_W = N_{\text{gen}}/c_H = 3/13$	P01, P31, P35, P43	CatAL (orbit arithmetic)
$\lvert Z_7\rvert$	7	Order of winding group; $b_0 = \lvert Z_7\rvert$	Mod-7 ring at core of f_{MDL}; $b_0 = 7$	P01, P28	`b0_eq_z7_order`, `f21_substrate_beta_coefficient` ($\mathbf{A}_{\text{Lean}}$)
$F_{21} = Z_7 \rtimes Z_3$	21 elements	Substrate group: winding × colour	$F_{21}^{\text{ab}} = Z_3$ (colour); identified from MDL minimality	P28, P34, P39	CatAL (14 theorems in `SylowIndexCouplingHierarchy`)
Rule 110	(8-bit CA rule #110)	Binary CA rule governing SM generation orbit	Forced by MDL minimality on SM orbit (CUP-4)	P28, P30	CUP-4, CatAL
f_{MDL}	(Z_7 truth table)	MDL-minimal mod-7 CA rule	= Rule 110's truth table (CUP-4); unique Class-4 outer-totalistic Z_7 CA	P01, P28, P35	CUP-4, CatAL conditional
Lepton Seed	$(1, 73, 823)$	Lex-/MDL-minimal GTE triple at $n = 10$	Uniquely selected by RSUC	P01	`rsuc_theorem`, `mdl_selects_LeptonSeed` ($\mathbf{A}_{\text{Lean}}$)
$b_1 = 73$	73	Primary ladder index; N_{eff}(electron)	$b_1 = b_2 + q_2 + 7 = 42 + 24 + 7 = 73$; mirror-dual prime-locked divisor pair	P01	CatAL
FGCI: $b_{L_1} = 73$ doubly determined	73	Frobenius-Generation Coincidence: Frobenius chain $F(N_c) = N_c^4 - N_c^2 + 1$ and GTE cascade $G(N_c) = N_c^4 - (N_c^2+1)/2 - N_c$ agree iff $N_c = 3$; witnesses algebraic/dynamical consistency at physical N_c	$F(3) = G(3) = 73$ uniquely at $N_c = 3$; $b_{L_2} = 42 = 2N_c\lvert Z_7\rvert$	P01	`FrobeniusChain.lean` (17 theorems, $\mathbf{A}_{\text{Lean}}$)

Symbol	Value	Physical Meaning	Derivation / Source	Papers	Lean Cert
$F_{13} = 233$	233	13th Fibonacci number; EW boson even-step increment	$\|q_2 - q_1\| = 13$ locks even-step evolution to F_{13}	P01	—
$b_0 = 7$	7	One-loop QCD β-function coefficient	$(11N_c - 2N_f)/3 = 7 = \|Z_7\|$, $N_c = 3$, $N_f = 6$	P28, P35, P43	`f21_substrate_beta_coefficient` ($\mathbf{A}_{\text{Lean}}$)
$b_1 = 26$	26	Two-loop QCD β-function coefficient	F_{21} substrate; $N_c = 3$, $N_f = 6$, zero free parameters	P28, P35, P39	`f21_substrate_two_loop_beta_b1` (zero sorry; axioms: {propext})
$\alpha_{\text{EM}}^{-1} = 137$	137	Fine structure constant inverse	$(N_{\text{eff}}(\mu) + F_{c_H} - 1)/2 = (275 - 1)/2 = 137$	P01, P43	`alpha_inv_master` ($\mathbf{A}_{\text{Lean}}$)
$1/\alpha_{\text{EM}}(0) = 2^{\|Z_7\|} + N_{\text{gen}}^2$	$128 + 9 = 137$	Cross-sector identity	$2^7 + 3^2 = 137$ (exact)	P31	`alpha_em_inverse_gte_bundle` ($\mathbf{A}_{\text{Lean}}$)
$b_0 = N_{\text{fam}} + N_{\text{gen}} - 1$	$5+3-1 = 7$	Casimir relation	Connects b_0, N_{fam}, N_{gen}; Lean cert	P31, P34	`b0_casimir_relation` ($\mathbf{A}_{\text{Lean}}$)
IPT	≈ 1.1309	Information Processing Threshold (EW scale factor)	$1 + \ln\varphi/(2\ln 2\pi)$; derived from SRRG axioms A1–A3	P27, P43	`IPT_theorem` ($\mathbf{A}_{\text{Lean}}$)

Notes on structural inputs. **λ (Wolfenstein) is a derived output, not an input:** $\lambda(\text{CKM}) = N_{\text{gen}}^2/(2^{N_{\text{gen}}} \cdot N_{\text{fam}}) = 9/40 = 0.225$ (exact, 0.0σ from PDG), proved $\mathbf{A}_{\text{Lean}}$ in P32/P35. **$N_c = N_{\text{gen}} = 3$:** $N_c = N_{\text{gen}}$ is forced by the CA structure (P43); conceptually distinct but derived to be equal. **"Zero free parameters"** (P43, P35): all structural inputs are derivable from the three UGP axioms and $n = 10$.

E.1.2 External Measured Inputs

Not derived within the UGP framework; taken from experiment as scale anchors. These are the only true "inputs" in the traditional sense.

Table 9: External measured inputs (scale anchors only).

Symbol	Physical Meaning	PDG Value	Papers	Notes
m_e	Electron mass	0.511 MeV	P19, P01	First of two scalar inputs for Koide chain. All 9 charged-fermion masses derived from (m_e, m_μ) at 61 ppm.
m_μ	Muon mass	105.658 MeV	P19, P01	Second scalar input; m_τ predicted at 61 ppm.
G_F	Fermi constant (EW energy scale)	1.1664×10^{-5} GeV^{-2}	P01	External EW-scale anchor for W/Z mass predictions.

Symbol	Physical Meaning	PDG Value	Papers	Notes
H_0	Hubble parameter	67.95 km/s/Mpc (GTE, Path D)	P47	Derived non-circularly from $T_{\mathrm{CMB}}^{\mathrm{GTE}} = 2.7265$ K, Ω_Λ, $\Omega_{\mathrm{DM}}h^2$, and FKTT η_B (CatAD, $+0.7\sigma$ vs Planck 2018). P01 used H_0 only as an external anchor for Λ.
Δm^2_{21}, Δm^2_{31}	Neutrino mass-squared splittings	NuFIT 6.0 IC24 NH	P01	Used only for Type-I seesaw realisation; not needed for structural ratio prediction.

E.1.3 Calibrated Parameters (Category B)

Non-perturbative; not derivable from first principles within the current framework.

Table 10: Calibrated parameters (Category B).

Description	Count	Papers	Notes
QCD binding-model parameters (hadronic / baryon sector, non-perturbative confinement)	15	P01 §baryon	Encode strong confinement physics beyond one-loop perturbative QCD. Total inputs: 0 (Cat A core) + 5 (scale anchors) + 15 (Cat B) = 20.
Higgs quartic λ_H	1 (MDL-selected + SRRG-corrected)	P01 §Higgs, P27, P35	$\mathbf{A}_{\mathrm{Lean}}$: $\lambda = \varphi/(4\pi) \times (1 + \ln\varphi/(54 \ln 2\pi))$; $2c_H + 1 = 27 = N_{\mathrm{gen}}^3$; certs `higgs_quartic_corrected`, `two_cH_plus_one_eq_ngen_cubed` (zero sorry). Gives $m_H = 125.2499$ GeV (-0.0003σ).
URC correction layer parameters	$\sim$3	P01 §URC	Category A/D correction layer for sub-percent fermion mass spectrum; derived from ridge divisor count, third-generation capacity, Fibonacci structure.

E.2 Axioms Index

Axioms are assumptions not derived from deeper principles within the current framework. Items marked "Proved" or $\mathbf{A}_{\mathrm{Lean}}$ are theorems — included because they were originally conjectured or are sometimes referred to as axioms.

E.2.1 Core UGP Axioms (3 axioms — truly assumed)

Source: P01 §2.1. These three axioms generate the entire framework.

Table 11: The three core UGP axioms (A1–A3).

ID	Name	Formal Statement	Source	Lean / Status
A1	Locality	All fundamental interactions are local in the substrate's causal graph.	P01 §2.1	No dedicated Lean axiom; standard Mathlib only. Assumed.

ID	Name	Formal Statement	Source	Lean / Status
A2	Symmetry	Fundamental laws are invariant under gauge symmetry transformations; redundancies give rise to gauge principles.	P01 §2.1	No dedicated Lean axiom. Assumed.
A3	Compression (MDL)	The universe evolves according to rules that minimise the total description length of its history (Rissanen MDL), consistent with A1–A2.	P01 §2.1, P06	MDL minimality used in CUP-4 proof chain. Assumed.

Lean axiom closure: All Category A physics theorems in ugp-lean have axiom closure exactly {propext, Classical.choice, Quot.sound} — the standard Mathlib signature. No UGP-specific axioms appear in the closure of any Category A physics theorem.

E.2.2 PSC Layer I Axioms (5 axioms — AC is proved)

Source: P14 §2.1, NEMS Paper 05. The five hard PSC consistency conditions.

Table 12: PSC Layer I axioms (RC, NM*, AC, TV, SA). AC is a proved theorem.

ID	Name	Formal Statement	Source	Status
RC	Renormalisability	T is a renormalisable gauge theory in $d = 4$.	P14 §2.1; NEMS P05	Assumed
NM*	Structural Stability (No-Miracles)	T is structurally stable against small deformations; no external tuning required.	P14 §2.1; NEMS P05	Assumed
AC	Anomaly Cancellation	All gauge and mixed gravitational anomalies cancel.	P14 §2.1; P28	**Proved** from PSC+RC ($\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms). `anomaly_cancellation_psc_forced`.
TV	Topological / Thermodynamic Viability	Gauge bundle admits consistent instanton sectors.	P14 §2.1; NEMS P05	Assumed
SA	Spectral / Semantic Admissibility	Dirac spectrum distinguishes matter from anti-matter.	P14 §2.1; NEMS P05	Assumed

Note on TV/SA naming: In physics (P14): "Topological Viability" (TV) and "Spectral Asymmetry" (SA). In the Lean formalisation (`NemS/Physics/Rigidity.lean`): "Thermodynamic Viability" and "Semantic Admissibility". Same axioms, different physical realisations.

E.2.3 PSC Layer II Conditions (2 conditions — PI is proved)

Source: P14 §2.2, NEMS Paper 05. Information-theoretic conditions selecting $N_{\text{gen}} = 3$. Layer II selects $N_{\text{gen}} = 3$ uniquely from 12 Layer I survivors (0.03% of 34,560 candidates pass Layer I; `psc_12_survivors_card`, $\mathbf{A}_{\text{Lean}}$).

Table 13: PSC Layer II conditions.

ID	Name	Statement	Source	Status
PI	Presentation Invariance	Physical predictions invariant under change of theory presentation; selects $N_{\text{gen}} = 3$ as unique minimal solution. *Note:* PI is operationally equivalent to MDL minimality (axiom A3) at the universe-selection level: 'Why MDL?' terminates at PSC/PI. MDL is therefore not an independent foundational axiom; it is the operational form of PI.	P14 §2.2; NEMS P05	**Proved** for the Level-1 winding instance (`d2_axiom`, $\mathbf{A}_{\text{Lean}}$); the universal invariance form is a named axiom (`d2_universal`, CatAD; conditionality disclosed).
MDL-II	MDL Minimality (Layer II)	Among PSC-admissible universes, the minimum description-length one is selected. Equivalent to A3 applied to universe selection.	P14 §2.2; P01, P06	Embodied in CUP-4 proof chain ($\mathbf{A}_{\text{Lean}}$).

E.2.4 GTE Construction Principles (all proved)

Source: P01 §2.2–2.3, P05.

Table 14: GTE construction principles (all are proved theorems, not assumed axioms).

ID	Name	Statement	Source	Status
PL	Prime-Lock	Initial GTE state admissible only if first-generation capacity $c_1 = b_1 q_1 + 20$ is prime.	P01 §2.2	**Proved.** `rsuc_theorem` ($\mathbf{A}_{\text{Lean}}$).
MD	Mirror Duality	If (b_2, q_2) is admissible, so is (q_2, b_2); admissible solutions come in mirror pairs.	P01 §2.2	**Proved.** `rsuc_theorem` ($\mathbf{A}_{\text{Lean}}$).
RSUC	Residual Seed Uniqueness	Lepton Seed $(1, 73, 823)$ is the unique lex-/MDL-minimal survivor at $n = 10$.	P01, P05	**Proved.** `rsuc_theorem`, `mdl_selects_LeptonSeed` ($\mathbf{A}_{\text{Lean}}$, zero sorry).
GoE	Garden-of-Eden	Gen_1 has no f_{MDL} predecessor; forces $N_{\text{gen}} = 3$ from orbit depth.	P28	**Proved.** `fmdl_gen1_is_garden_of_eden` ($\mathbf{A}_{\text{Lean}}$, zero sorry).
QLL	Quarter-Lock Law	Algebraic identity: orbit spatial curvature k_{L^2} and generational curvature $k_{\text{gen2}} = -\varphi/2$. Preserves gauge coupling rationals across instantiation.	P01, P05	**Proved.** `quarterLockLaw` ($\mathbf{A}_{\text{Lean}}$, zero sorry).
CUP-4	Computational Universality 4	SM generation orbit forces Rule 110 on a 5-cell periodic ring (f_{MDL} truth table = Rule 110 truth table).	P28	**Proved.** `cup11b_z7_sum_conservation` ($\mathbf{A}_{\text{Lean}}$, zero sorry).

E.2.5 GTE–Möbius Substrate Constraints D1–D5

Source: P34 §6. The five constraints define the $[D]$-coherence measure class on the Möbius triple $(A, e, [D])$. Component A is the $\mathbb{Z}_7$ arithmetic carrier (Level-1: f_{MDL}/CMCA; Level-2: PSC-admissible kink sector of Φ_{MDL}). The physical substrate is Φ_{MDL}; discrete CAs certify the arithmetic but are not the universe (P43 no-CA-replica theorem).

Table 15: GTE–Möbius $[D]$-constraints D1–D5.

ID	Name	Statement	Source	Status
D1	Non-negativity	$D \geq 0$ for all configurations.	P34 §6	Structural (definition).
D2	PSC Invariance	D is invariant under the five PSC axioms, incl. Lorentz boosts on Φ_{MDL}.	P34 §6; P42	**Proved** for the Level-1 winding instance (`d2_axiom`, `algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$); the universal invariance form is a named axiom (`d2_universal`, CatAD; conditionality disclosed).
D3	Non-computability	Individual kink actualisation is non-computable.	P34 §6	**Proved** (forced by NEMS P09 diagonal barrier; CatAL conditional).
D4	Selection Completeness	D selects uniquely among realisations even in non-categorical settings.	P34 §6	Assumed / open formalisation.
D5	Born-Rule Consistency	$P(\text{outcome } n) = \lvert c_n \rvert^2$; Born rule for $[D]$-selected physical states.	P34 §6; P37; P42	**Proved.** `born_rule_unconditional` (`Beable WindingPartition Instance.lean`; $\mathbf{A}_{\text{Lean}}$, zero sorry, zero custom axioms).

E.2.6 SRRG Axioms

Source: P27. Governing the Self-Referential Renormalisation Group fixed point.
Core SRRG axioms A1–A3 (all three proved):

Table 16: SRRG core axioms A1–A3 (all proved from structural principles).

ID	Name	Statement	Source	Status
A1-SRRG	PSC Contraction	PSC self-model update contraction rate $= 1/\varphi$.	P27 §SRRG	**Proved.** `abs_psi_eq_inv_phi` (zero sorry).
A2-SRRG	PSC Entropy	PSC adjudication entropy $H_{\text{adj}} = \ln(2\pi)$ (Haar measure on $U(1)$).	P27 §SRRG	**Proved.** `entropy_formula_U1` (zero sorry).
A3-SRRG	Symmetry Splitting	PSC overhead splits equally between forward and backward self-model: factor 1/2.	P27 §SRRG	**Proved.** `A3_half_factor` (zero sorry).

Consequence: $\text{IPT} = 1 + \ln\varphi/(2\ln 2\pi) \approx 1.1309$ follows as a theorem from A1–A3 (`IPT_theorem`, $\mathbf{A}_{\text{Lean}}$).

SRRG B-axioms (5 named open axioms):

Table 17: SRRG B-axioms (named Lean axioms, not yet derived).

ID	Lean Name	Statement	Status
B1	`srrg_physical_fp_sustainable`	Every physical SRRG fixed point satisfies $\eta(S^*) \geq \text{IPT}$.	Named Lean axiom; not yet derived.
B2	`srrg_physical_fp_bounded_above`	Every physical SRRG fixed point satisfies $\eta(S^*) \leq 2$.	Named Lean axiom.

ID	Lean Name	Statement	Status
B3	`srrg_beta_polynomial_leading_order`	SRRG projected β-function is a degree-2 polynomial.	Named Lean axiom.
B4	`srrg_physical_is_ir_stable`	Physical SRRG fixed points are IR attractors under RG flow.	Named Lean axiom.
B5	`psc_ew_entropy_maximization`	PSC entropy maximised at EW fixed point; used in VEV derivation ($v_{\text{PSC}} = 246.16$ GeV).	**Proved** (zero sorry, zero new axioms) — `GoldstoneEntropyCorrection.lean` §5, `srrg-lean`. VEV derivation is unconditional (B5 proved).

E.2.7 P44 Quantum Gravity Axioms (architecture restrictions)

Three named architecture restrictions used in P44 derivations (not truly independent assumptions; follow from MDL minimality in each case):

Table 18: P44 architecture restrictions (quantum gravity sector).

ID	Statement	Use	Status
AR1	Graviton Fock space $\mathcal{H}_{\text{grav}}$ is separable and spanned by graviton excitations of Φ_{MDL} on a curved background.	Full nonlinear EFE derivation (P44 §3); required for perturbative expansion.	Structural; follows from $\xi = 0$ and $\mathbb{Z}_7$ separability ($\mathbf{A}_{\text{Lean}}$: $\xi = 0$); Fock construction analytical ($\mathbf{A}/\mathbf{D}$).
AR2	Thermal coherence: pre-measurement Φ_{MDL} state is the Hamiltonian thermal ensemble restricted to PSC-admissible sectors.	Derivation of MDL IC from thermal prior (P44 §5).	Follows from `phimdl_thermal_state_master` (P42/P43, $\mathbf{A}_{\text{Lean}}$).
AR3	EFE-bridge: the MDL variational principle on Φ_{MDL} in 3+1D produces the same n_s derivation as the classical EFE route.	Step 5/7 of n_s derivation chain; conditional on this step.	**Open** (CatOpen); closure upgrades n_s to fully unconditional.

E.2.8 Cook Bridge Axioms (P30) and GTE Universality Bridge Axioms (P28)

Table 19: Cook bridge axioms (P30) and GTE universality bridge axioms (P28).

ID	Lean Name	Statement	Status
Cook bridge axioms (P30)			
C1	`cook_c2_tape_bit_ax`	After 30 Rule 110 steps, decoder reads CTS-encoded bit correctly.	Partial $\mathbf{A}_{\text{Lean}}$ (words ≤ 7, slots ≤ 20); global: **CatOpen**.
C2	`cook_cts_step_sim_ax`	After M steps, regions outside word-glider range remain ether.	**Discharged** ($\mathbf{A}_{\text{Lean}}$) ✓

ID	Lean Name	Statement	Status
C3–five	(5 named bridge axioms)	Full CookCtsEvalSim for all positive-step nontrivial inputs (Cook 2004 §4).	**CatOpen**
GTE universality bridge axioms (P28)			
INF-TAPE	`rule110_simulates_computable`	GTE embeds on a single Boolean infinite tape with Rule 110 dynamics.	Named axiom (one bridge); CatAL conditional.
DUAL-TAPE	`simultaneous_dual_tape_ax`	Single InfTape simultaneously realises both SM sector and Rule 110 binary layer.	Named axiom.
CUP3D-six	(6 named bridge axioms)	Cook–Wolfram–APS encoding chain for 3D physical incompleteness (CUP3D).	Named bridge axioms.

Note: The C1/C2/C3 Cook axiom dependency is bypassed by an independent algebraic route (P40/P35, $\mathbf{A}_{\text{Lean}}$): Rule 110 update $= p(L, C, R) = C + R - CR - LCR$ over GF(7); with $C = 1$, $p(L, 1, R) = \text{NAND}(L, R)$ (zero sorry). Universality follows from Sheffer (1913) NAND completeness (`boolean_nand_complete`).

E.2.9 Axiom Summary Counts

Table 20: Axiom count summary across all UGP axiom classes.

Category	Count
UGP core axioms (A1–A3: locality, symmetry, MDL)	**3**
PSC Layer I axioms (RC, NM*, AC, TV, SA) — AC is proved	**5 (1 proved)**
PSC Layer II conditions (PI, MDL-II) — PI is proved	**2 (1 proved)**
GTE construction principles (PL, MD, RSUC, GoE, QLL, CUP-4) — all proved	**6 (all proved)**
$[D]$-constraints (D1–D5) — D2, D3, D5 proved	**5 (3 proved)**
SRRG axioms (A1–A3 proved; B5 proved; B1–B4 named open)	**8 (4 proved, 4 open)**
Cook bridge axioms (C2 discharged, C1 partial, 5 general CatOpen)	**7 (1 discharged, 6 open)**
GTE universality bridge axioms (P28; INF-TAPE, DUAL-TAPE, CUP3D-six)	**8 (all named/open)**
P44 architecture restrictions (AR1–AR3; AR1/AR2 proved, AR3 open)	**3 (2 proved, 1 open)**
Analytic number theory axioms (Dickman, CRT; not in Cat A closure)	**2**
Total true open axioms (assumed, not derived)	**≈ 28–31**
Total proved theorems (once called axioms)	**≈ 17–20**

The three UGP axioms (A1–A3) are the only truly foundational assumptions. GTE construction principles are all proved theorems. The remaining open axioms are: (1) PSC Layer I structural conditions defining framework scope; (2) SRRG B-axioms; (3) Cook bridge axioms for arbitrary CTS generality; (4) P28 bridge axioms for GTE tape embedding.

E.3 Physical Quantity Outputs

Tables 21–34 list the primary predictions with the canonical GTE formula, comparison to PDG / experiment, and Lean certification status; Tables 28 and 29 register beyond-SM particle predictions

from P02 and P29. Sequential refinements of the same derivation (e.g., the three $\sin^2\theta_W$ values) are listed in the discrepancy register, §E.5.

E.3.1 Electroweak Sector

$\sin^2\theta_W$ progression (sequential refinements): 3456/15101 ≈ 0.22886 (P01/P31, tree-level) → 3/13 = 0.23077 (P31/P35, best tree-level, $\mathbf{A}_{\text{Lean}}$) → 384729/1664000 ≈ 0.231207 (P41/P42, threshold-corrected, $\mathbf{A}_{\text{Lean}}$) → 0.23129154 (P31/P35, SM Higgs threshold + GTE $\alpha_{\text{em}} = 1/128$, CatB; $+0.038\sigma$ vs PDG 2024; canonical for electroweak comparisons).

Table 21: Electroweak sector predictions.

Quantity	Symbol	GTE Value / Formula	PDG Value	Error	Level	Source
Weinberg angle (orbit)	$\sin^2\theta_W$	$N_{\text{gen}}/c_H = 3/13 = 0.23077$	0.23129 ± 0.00004 (PDG 2024)	-0.195% (13.0σ)	$\mathbf{A}_{\text{Lean}}$ (tree-level)	P31, P33, P35
Weinberg angle (threshold)	$\sin^2\theta_W$	$3/13 + \lambda^3/(2c_H) = 384729/1664000 \approx 0.23121$	0.23122 ± 0.00003 (PDG 2022)	-0.42σ	$\mathbf{A}_{\text{Lean}}$	P41, P42
Weinberg angle (2-loop + Higgs)	$\sin^2\theta_W$	$384729/1664000 + (1/128)/(12\pi\times10/13)\times \ln(125.2499/91.629) = 0.23129154$	0.23129 ± 0.00004 (PDG 2024)	$+0.038\sigma$	CatB	P31, P35
Weinberg angle (GUT scale)	$\sin^2\theta_W^{\text{GUT}}$	$N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8 = 0.375$	0.375 (SU(5))	0.0%	$\mathbf{A}_{\text{Lean}}$	P31, P32, P35
$(M_W/M_Z)^2$	—	$1 - \sin^2\theta_W = 10/13 \approx 0.7692$	0.7736	+0.57%	$\mathbf{A}_{\text{Lean}}$	P41, P42
W mass (2-loop)	m_W	2-loop + threshold: **80.364** GeV	80.3692 ± 0.0133 GeV (PDG 2024)	-0.39σ	$\mathbf{A}_{\text{Lean}}$	P01, P22
W mass (P35 chain)	m_W	PSC energy scale $E_0 \sin^2\theta_W^{1/2}$: 80.614 GeV	80.3692 GeV	+0.30%	CatAD	P35, P41, P42
Z mass	m_Z	$\cos\theta_W$ chain: 91.914 GeV	91.188 GeV	+0.80%	CatAD	P35, P41, P42
Higgs VEV (SRRG)	v_{PSC}	SRRG fixed point: **246.16** GeV	246.22 GeV	-0.024%	A^- (conditional)	P27, P35
Higgs VEV (self-consistent)	v_{self}	$2M_W/g_2(M_W)$: **246.27** GeV	246.22 GeV	+0.02%	A/D	P01
Higgs mass	m_H	SRRG-corrected: **125.2499** GeV ($\lambda = \varphi/(4\pi) \times (1 + \ln\varphi/(54\ln 2\pi))$; $2c_H + 1 = 27 = N_{\text{gen}}^3$)	125.20 ± 0.11 GeV	$\mathbf{-0.0003\sigma}$	$\mathbf{A}_{\text{Lean}}$ (P27, P35)	P01, P27, P35
Bare gauge couplings	g_1^2, g_2^2, g_3^2	GTE cascade: 16/125, 2329/5400, 41 075 281/27 648 000	—	$\leq 1.6\%$ all	$\mathbf{A}_{\text{Lean}}$	P01
α_{EM}^{-1}	—	$(N_{\text{eff}}(\mu) + F_{c_H} - 1)/2 =$ **137**	137.036	$\sim 0.03\%$	$\mathbf{A}_{\text{Lean}}$	P01, P28

Quantity	Symbol	GTE Value / Formula	PDG Value	Error	Level	Source
EW boson c-values	$c(W,Z,H)$	Ridge factorisation: $\{11, 12, 13\}$	—	—	$\mathbf{A}_{\text{Lean}}$	P01
Tau Yukawa coupling	y_τ	$1/(2 \times 7^2) = 1/98$; $m_\tau = v_H y_\tau/\sqrt{2}$	—	0.016% (m_τ)	CatA	P18
All 3 charged lepton masses	m_e, m_μ, m_τ	Koide $+y_\tau = 1/98$: 0.511, 105.635, 1776.57 MeV	0.511, 105.658, 1776.86 MeV	0.02–0.016%	CatA	P18
Gauge couplings (g, g')	g, g'	$\sin^2\theta_W = 3/13 + \alpha_{\text{EM}}$ at M_Z: $g = 0.6500$, $g' = 0.3560$	PDG M_Z scale	CatAD	CatAD	P35
W mass (1-loop oblique)	m_W	1-loop oblique correction: **80.372** GeV (closes 98.8% of tree-level gap)	80.3692 ± 0.0133 GeV	−0.006%	CatAD	P27
Bare g_1^2/g_2^2 ratio	—	$86400/291125 \approx$ **0.2969** (ugp-lean bare triple)	0.3008 at M_Z	−1.34%	$\mathbf{A}_{\text{Lean}}$	P14
Bare g_3^2/g_2^2 ratio	—	$\approx$ **3.444** (ugp-lean bare triple)	3.510 at M_Z	−1.90%	$\mathbf{A}_{\text{Lean}}$	P14

E.3.2 CKM Matrix

Key note: $\lambda = 9/40 = N_{\text{gen}}^2/(2^{N_{\text{gen}}} \times N_{\text{fam}})$ **is a derived output, not an input.**

Table 22: CKM matrix predictions.

Quantity	Symbol	GTE Value / Formula	PDG Value	Error	Level	Source
Wolfenstein λ	λ	$N_{\text{gen}}^2/(2^{N_{\text{gen}}} N_{\text{fam}}) =$ **$9/40 = 0.22500$**	0.22500 ± 0.00037	**0.000σ**	$\mathbf{A}_{\text{Lean}}$	P01, P32, P35
Wolfenstein A	A	$\sqrt{N_{\text{eff}}(s)/N_{\text{eff}}(c)} = \sqrt{186/275} \approx 0.8224$	$\sim 0.814 \pm 0.013$	0.65σ	CatA	P01, P32
CKM UT radius	R_b	$N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8 = 0.375$	—	—	$\mathbf{A}_{\text{Lean}}$	P32
CP angle γ	γ	$\arctan\sqrt{b_b/b_s}/N_{\text{gen}} = 65.67°$	$65.68° \pm 0.67°$	**-0.023σ**	CatA	P32
b_b (Mersenne prime)	b_b	$2^{c_H} - 1 =$ **8191**	—	—	$\mathbf{A}_{\text{Lean}}$	P32
Jarlskog invariant	J	Structural CKM (P32 preferred): **2.999×10^{-5}**	3.08×10^{-5}	-1.81σ	A/D	P32
Strong CP phase	θ_{QCD}	F_{21} discrete group theory: **0** (exact)	$< 10^{-10}$ (exp. bound)	Consistent	$\mathbf{A}_{\text{Lean}}$	P33, P35, P39
All 9 CKM elements	$\lvert V_{ij}\rvert$	Wolfenstein $\mathcal{O}(\lambda^4)$ expansion	PDG table	$< 1\sigma$ all	CatA	P32

E.3.3 Strong Sector / QCD

Table 23: Strong sector / QCD predictions.

Quantity	Symbol	GTE Value	PDG / QCD	Error	Level	Source
$\alpha_s(M_Z)$ (bare)	α_s	$g_3^2/(4\pi) = \mathbf{0.11822}$	0.1179 ± 0.0010	$+0.24\sigma$	$\mathbf{A}_{\text{Lean}}$	P01
$\alpha_s(M_Z)$ (2-loop)	α_s	SC-ZZ 2-loop: **0.11790**	0.1179 ± 0.0010	$\mathbf{0.00}\sigma$	CatA+D	P22
$\alpha_s(M_Z)$ (3-loop, F_{21})	α_s	$N_f = 5$ running: **0.1193**	0.1180 ± 0.0009	+1.10%	CatA	P35, P39
β-function b_0	b_0	$(11N_c - 2N_f)/3 = \mathbf{7}$	7	exact	$\mathbf{A}_{\text{Lean}}$	P35, P39
β-function b_1	b_1	F_{21} substrate: **26**	26	exact	$\mathbf{A}_{\text{Lean}}$	P35, P39
EFT boundary scale	Λ_{GTE}	$7\,M_{\text{kink}}$ (seven-kink full-period unwinding threshold): tree $(8/7)m_\tau$ = **2030.70** MeV; pole $7M^Q = 1.970 \pm 0.146$ GeV; envelope $\mathbf{1.96 \pm 0.15}$ GeV	—	band < 10%	CatAD / CatA	P39, P42
Colour coupling at Λ_{GTE}	$e^2(\Lambda_{\text{GTE}})$	Sylow rational: **7/2**	3.74 ± 0.08 ($\overline{\text{MS}}$, run from PDG 2024 $\alpha_s(M_Z)$)	+7.4%; +1.3–1.5σ broadening-corrected	CatA	P42, P39
The $e^2(\Lambda_{\text{GTE}}) = 7/2$ identification is a scheme-aware consistency, not an exact-grade match: the +7.4% one-loop matching offset against PDG-run α_s reduces to +1.5σ (tree reading) / +1.3σ (pole reading) once the non-perturbatively measured kink charge-form-factor broadening $b = 1.189 \pm 0.049$ is included; the residual is consistent with the uncomputed two-loop matching term plus the coset threshold constant. Λ_{GTE}: mechanism and tree identity CatAD; pole envelope CatA.						
Casimir C_F	C_F	F_{21} reps: **4/3**	4/3	0%	$\mathbf{A}_{\text{Lean}}$	P35, P39
Casimir C_A	C_A	F_{21} reps: **3**	3	0%	$\mathbf{A}_{\text{Lean}}$	P35, P39
Pion decay constant	f_π^{SCC}	$M_{\text{kink}}^{\text{SCC}}/\pi = \mathbf{92.34}$ MeV	92.07 MeV	+0.30%	$\mathbf{A}_{\text{Lean}}$	P35, P39
Pion mass	m_π	GOR + $\langle\bar{\psi}\psi\rangle_{\text{GTE}}$: **139.57** MeV	139.5703 MeV (PDG 2024)	−0.001% (0.00σ)	$\mathbf{A}_{\text{Lean}}$	P35, P39
ω meson mass	m_ω	$\sqrt{72}\,f_\pi^{\text{SCC}}$ = **783.55** MeV	782.65 MeV (PDG)	+0.11%	$\mathbf{A}_{\text{Lean}}$	P34, P35
ρ meson mass	m_ρ	$\sqrt{70}\,f_\pi^{\text{SCC}}$ = **772.57** MeV	775.26 MeV (PDG)	−0.35%	$\mathbf{A}_{\text{Lean}}$	P34, P35
*Arithmetic (`g_rho_sq_eq_casimir_orbit`, `m_rho_ksrf_formula`): $\mathbf{A}_{\text{Lean}}$; physical SU(2)$_I$ mechanism: **B**. KSRF is leading-order $\mathcal{O}(p^2)$; $\Gamma_\rho = 149$ MeV makes bare-mass vs. PDG pole-mass comparison unreliable. The 0.35% deviation is the physically meaningful metric.*						
ϕ meson mass	m_ϕ	$\sqrt{84}\,f_K = \mathbf{1011.3}$ MeV	1019.46 MeV (PDG)	−0.80%	**B**	P34, P35
Kaon decay const.	f_K	GOR NLO Casimir: **110.34** MeV (NR)	110.4 MeV (PDG NR)	−0.05%	**B**	P34, P35
$f_K^{\text{GTE}} \times \sqrt{2} = 156.0$ *MeV is within* 0.9σ *of the PDG relativistic value* 155.7 *MeV.*						
η–η' mixing	θ_P	Full GTE chain: $-13.08^\circ \pm 3.74^\circ$	$[-14.3^\circ, -10.7^\circ]$	In range	CatA	P35, P39
Baryon J^P	J^P	[D]-weight/MDL: $1/2^+$	$1/2^+$	exact	$\mathbf{A}_{\text{Lean}}$	P39
Mass gap	Δ	Kink BPS energy: $\Delta > 0$	Positive (obs.)	—	$\mathbf{A}_{\text{Lean}}$	P39

Quantity	Symbol	GTE Value	PDG / QCD	Error	Level	Source
4D string tension	$\sqrt{\sigma_{4D}}$	$(N_3/N_7)\sigma_{2D}$: 440.6 MeV	440–475 MeV	In range	CatA	P35, P39
$\alpha_s(M_Z)$ (f_{quant} route)	α_s	GTE + $f_{\text{quant}} = 2^{-2/3}$: **0.1896**	0.1179 ± 0.0010	$\sim$ 60% (open)	CatA	P39
Z_7 kink S-matrix	$S_{kk}(\theta)$	$S_{kk}(\theta) = -1$ (all θ; repulsive; $\beta^2 = 49 > 8\pi$); tree amplitude $i\mathcal{M} = -im^2 \times 49$ (Z_7 fingerprint)	Zamolodchikov–Zamolodchikov 1979	—	CatAD	P35, P39
Lüscher / f_{quant} gap	f_{quant}	$f_{\text{quant}}(C\text{-ratio}) = 0.6289$ vs $f_{\text{quant}}(\sigma\text{-ratio}) = 0.6747$ (7.28% gap); Lüscher $\pi/(12R^2) = 8.26\%$ at $R = 1/m_{\text{kink}}$; $R^* = 0.913$ fm vs 0.680 fm (34% off); open	—	OPEN	CatA	P39

E.3.4 Lepton Masses

Table 24: Lepton mass predictions.

Quantity	Symbol	GTE Value	PDG Value	Error	Level	Source
Tau mass (Koide)	m_τ	From (m_e, m_μ): **1776.969** MeV	1776.860 MeV	**0.91**σ *(61 ppm)*	$\mathbf{A}_{\text{Lean}}$	P01, P18
Koide relation	Q	$Q = 2/3$ (Koide formula)	$Q_{\text{PDG}} = 2/3$	$\sim$0	$\mathbf{A}_{\text{Lean}}$	P18
Koide phase	θ	$(N_c^2 - 1)/(4N_c^2) = \mathbf{2/9}$	—	exact	$\mathbf{A}_{\text{Lean}}$	P18
All 9 charged-fermion masses	—	Dual-path verifier benchmark (locked UCL + theoretical renormalisation + URC, nine fermions + W–ρ): 0.295% RMS vs PDG 2022 targets; 0.261% vs PDG 2024	PDG 2022 / 2024	0.295% / 0.261% RMS	A/D	P01
Tau mass (VEV route)	m_τ	$y_\tau = 1/98$: $m_\tau = v_H/(98\sqrt{2}) =$ **1776.57** MeV	1776.86 MeV	**0.016**%	CatA	P18

E.3.5 Quark Masses

Table 25: Quark mass predictions.

Quantity	Symbol	GTE Value	PDG Value (2024)	Error	Level	Source
Up quark	m_u	2.157 MeV	$2.16^{+0.49}_{-0.26}$ MeV (PDG 2024)	-0.009σ	**A/D** (`m_u_pdg_interval`)	P39
Down quark	m_d	4.647 MeV	$4.70^{+0.50}_{-0.17}$ MeV (PDG 2024)	-0.156σ	**A/D** (`m_d_pdg_interval`)	P39
Strange quark	m_s	92.74 MeV	$93.5^{+8.6}_{-3.1}$ MeV (PDG 2024)	-0.129σ	**A/D** (`m_s_pdg_interval`)	P39
Charm quark	m_c	1270.7 MeV	1273 ± 9 MeV (PDG 2024)	-0.26σ	**A/D** (`m_c_pdg_interval`)	P39
Bottom quark (VV)	m_b	4184.2 MeV	4183 ± 8 MeV (PDG 2024)	$+0.16\sigma$	**A/D**	P32
Top quark (VV)	m_t	172 610 MeV	172 570 $\pm$ 290 MeV (PDG 2024)	$+0.14\sigma$	**A/D**	P32
m_t/m_b ratio	b_t/b_b	337920/8191 = 41.255	41.257 (PDG 2024)	0.072%	$\mathbf{A}_{\text{Lean}}$	P32
Nucleon b-values	b_p, b_n	$b_p = 11459$, $b_n = 11441$	—	exact	$\mathbf{A}_{\text{Lean}}$	P01, P23

E.3.6 Neutrino Sector

Table 26: Neutrino sector predictions.

Quantity	Symbol	GTE Value	PDG / NuFIT	Error	Level	Source
Mass-squared ratio	$\Delta m^2_{21}/\Delta m^2_{31}$	b-values $\{5, 11, 19\}$ with seesaw exp. 29/9: **0.02936**	0.02951 ± 0.00098	$\mathbf{0.16}\sigma$	$\mathbf{A}_{\text{Lean}}$	P21, P01
Mass ordering	—	Normal (from $b^{29/9}$ structure)	Preferred at 2.5σ	Consistent	$\mathbf{A}_{\text{Lean}}$	P21
Mass ordering (triality)	—	Octonionic triality: $b_{R,1}<b_{R,2}<b_{R,3}$ forces Normal Ordering (`seesaw_normal_ordering_from_seed_ordering`, $\mathbf{A}_{\text{Lean}}$); corroborates P21 via independent route; JUNO-falsifiable	NH preferred $\Delta\chi^2$=6.1	Corroborates P21	$\mathbf{A}_{\text{Lean}}$	P55
Seesaw exponent	$29/9 = N_c + \theta$	$N_c + \theta$ topological	—	Three independent decompositions	$\mathbf{A}_{\text{Lean}}$	P01, P21

Quantity	Symbol	GTE Value	PDG / NuFIT	Error	Level	Source
Sum Σm_ν	Σm_ν	**59.4** meV (GTE seesaw + three-tape winding)	$<$ 120 meV (Planck)	Within bound	CatAD	P01, P47
Δm^2_{21} (absolute)	Δm^2_{21}	Orbit-ratio formula: $\mathbf{7.370 \times 10^{-5}}$ eV^2 (`gte_delta_m21_sq_bounds`)	$(7.49 \pm 0.19) \times 10^{-5}$ eV^2 (NuFIT 6.0 IC24 NH)	0.63σ	**A/D**	P21
Δm^2_{31} (absolute)	Δm^2_{31}	Orbit-ratio formula: $\mathbf{2.511 \times 10^{-3}}$ eV^2 (`gte_delta_m31_sq_bounds`)	$(2.515 \pm 0.027) \times 10^{-3}$ eV^2 (NuFIT 6.0 IC24 NH)	-0.2% ($< 1\sigma$)	**A/D**	P21
RHN mass scale (GUT)	M_R	GTE orbit + EW: $\mathbf{1.90 \times 10^{16}}$ GeV ($\approx 0.90\, M_{\rm GUT}$)	$M_{\rm GUT}$(unif.) $\approx 2.1 \times 10^{16}$ GeV	-10%	CatAD	P21
RHN leptogenesis scale	$M_{R,1}$	BPS instanton suppression: $\mathbf{1.11 \times 10^{13}}$ GeV ($M_R e^{-\pi}/(19/5)^{29/9}$; `mr1_mr_gut_ratio_bounds`)	Calibrated from oscillations: 1.11×10^{13} GeV	0.04%	CatAD	P21, P47
$m_3(\nu)$ (P33)	m_3	$\alpha_{\rm em}^{N_{\rm fam}} \times E_{\rm ether} \approx 0.059$ eV	> 0.050 eV	In range	CatAD	P33, P35
RHN b-values	$\{b_{\nu R}\}$	Braid Atlas: $\{5, 11, 19\}$	—	exact	$\mathbf{A}_{\rm Lean}$	P17, P21
PMNS $\sin^2\theta_{12}$	$\sin^2\theta_{12}$	Orbit ratio $b^2_{\rm strand}/c_H = 4/13 = \mathbf{0.3077}$	0.308 ± 0.011 (NuFIT 6.0 IC24 NH)	$+0.01\sigma$	$\mathbf{A}_{\rm Lean}$	P21
PMNS $\sin^2\theta_{23}$	$\sin^2\theta_{23}$	Orbit ratio $b_{R_3}/b_{L_2} = 19/42 = \mathbf{0.4524}$	0.470 ± 0.013 (NuFIT 6.0 IC24 NH)	-1.35σ; within 3σ	$\mathbf{A}_{\rm Lean}$	P21
$\sin^2\theta_{23}$ (NLO)	$\sin^2\theta_{23}^{\rm NLO}$	NLO candidate $209/441 = \mathbf{0.4739}$ (`pmns_atm_nlo_candidate`, `two_b_R2_eq_F21_succ`)	0.470 ± 0.013 (NuFIT 6.0 IC24 NH)	$+0.23\sigma$ (within 1σ)	CatB (phys); $\mathbf{A}_{\rm Lean}$ (arith)	P35
PMNS $\sin\theta_{13}$	$\sin\theta_{13}$	Orbit ratio $b_{R_2}/b_{L_1} = 11/73 = \mathbf{0.1507}$	0.1489 ± 0.0007 (NuFIT 6.0 IC24 NH)	$+1.0\sigma$	$\mathbf{A}_{\rm Lean}$	P21
Jarlskog $J_{\rm GTE}$	J	$-(8184\sqrt{437}/5057221)\sin(\pi/7) = -\mathbf{1.47 \times 10^{-2}}$; $5057221 = 13 \times 73^3$	1.78×10^{-2} (NuFIT 6.0 IC24 NH)	see P21	$\mathbf{A}_{\rm Lean}$	P21
PMNS Dirac CP phase	δ_{CP}	$\mathbb{Z}_7$ winding arithmetic: $\mathbf{205.71}^\circ$	212° (NuFIT 6.0 IC24 NH)	$\mathbf{-0.15}\sigma$	CatA	P45
Neutrino fund. coupling	$g_{\rm fund}$	$7^2/2^{N_c^2} = 49/512 \approx \mathbf{0.0957}$ (provisional)	—	—	CatAD	P35, P46
Dark-ring ratio	$\Gamma_{\rm dark}$	$g^2_{\rm fund} = 7^4/2^{18}$	—	—	CatAD	P35, P46
Neutrino spectrum	$\{m_{\nu_i}\}$	$g_{\rm fund}$ cascade: $\{0.68, 8.6, 50.1\}$ meV; $\Sigma = 59.4$ meV	$\Sigma < 120$ meV (Planck)	Within bound	CatAD	P35, P46, P47

Quantity	Symbol	GTE Value	PDG / NuFIT	Error	Level	Source
CKM CP phase (S_3 chain)	$\delta_{CP}^{\mathrm{CKM}}$	$\pi/2 - 3/8 = \mathbf{68.51}^\circ$ (P47 S_3 subgroup chain)	68.517° (CKMfitter/PDG)	**0.017**%	CatAD	P47

E.3.7 Gravitational and Cosmological

CatAL upgrade (P43): The Planck/τ hierarchy was CatAD in P35/P38. P43 upgrades to $\mathbf{A}_{\mathrm{Lean}}$ via `z3_invariant_entropy_closes_hierarchy` (zero sorry). The formula and value (0.040% error) are identical across P35, P38, and P43.

Table 27: Gravitational and cosmological predictions.

Quantity	Symbol	GTE Value	PDG / Exp	Error	Level	Src.
Planck/τ hierarchy	M_{Pl}/m_τ	$\lvert F_{21}\rvert^{10} \times \lvert Z_7\rvert^{7/2} = \mathbf{6.868 \times 10^{18}}$	6.871×10^{18}	**0.040**%	$\mathbf{A}_{\mathrm{Lean}}$ (P43)	P43, P38, P35
Spacetime dimension	D	$N_{\mathrm{gen}} + 1 = \mathbf{4}$	4	exact	$\mathbf{A}_{\mathrm{Lean}}{}^\dagger$	P34, P35, P36, P43
Spectral dimension	d_s	Thermodynamic limit: **4**	4	exact	$\mathbf{A}_{\mathrm{Lean}}$	P35, P36, P43
Cosmological constant	Λ	$(\ln 2/\pi)\cdot L_{\mathrm{model}}\cdot H_0^2/c^2 = 1.097 \times 10^{-52}$ m^{-2}	1.088×10^{-52} m^{-2}	**0.31**σ	A/D	P01, P36
Λ (classical, tree)	Λ	$V(\Phi_k)$ at Z_7 vacua: **0** (exact)	$< 10^{-47}$ GeV^4 (tree)	0 (tree)	$\mathbf{A}_{\mathrm{Lean}}$	P38, P42, P43
Λ quantum corrections (pure-φ)	$\delta\Lambda$	All computable loop corrections vanish identically in the pure-φ sector: $\delta\Lambda = 0$ exact (`lambda_all_order_zero_pure_phi`, zero sorry)	$< 10^{-47}$ GeV^4	0	$\mathbf{A}_{\mathrm{Lean}}$	P48
Strong CP	θ_{QCD}	F_{21} group theory: **0** (exact)	$< 10^{-10}$	Consistent	$\mathbf{A}_{\mathrm{Lean}}$	P33, P35, P39
Baryon-to-photon ratio	η_B	$\sin^2\theta_W \times \alpha_{\mathrm{em}}^4 = \mathbf{6.54 \times 10^{-10}}$	$(6.10 \pm 0.06) \times 10^{-10}$	+7.2%	CatA	P33, P35
η_B (kink-overlap, sech upper bound)	η_B	Kink overlap suppression $\alpha = N_c - 1 = 2$: $\mathbf{6.36 \times 10^{-10}}$ (asymptotic sech upper bound)	$(6.10 \pm 0.06) \times 10^{-10}$	$+4.3\sigma$ (bracket; superseded)	$\mathbf{A}_{\mathrm{Lean}}$ cond.	P47
η_B (FKTT: $g_{\text{kink-top}} = \varepsilon_{\mathrm{FN}}$)	η_B	BPS instanton + FN coupling identity: $\mathbf{6.109 \times 10^{-10}}$ (`kink_top_coupling_eq_eps_FN`, `phi_mdl_kink_bps_saturation`; zero axioms)	$(6.10 \pm 0.06) \times 10^{-10}$	$+\mathbf{0.15}\sigma$	$\mathbf{A}_{\mathrm{Lean}}$	P47, P42

Quantity	Symbol	GTE Value	PDG / Exp	Error	Level	Src.
Einstein field equations	$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$	MDL-Lovelock + variational (linearised)	(GR)	—	CatAD	P36, P38, P43
Curved-background Lagrangian	$\mathcal{L}[\Phi_{\rm MDL};\, g_{\mu\nu}]$	$\xi = 0$ forced: $\frac{1}{2}g^{\mu\nu}\partial_\mu\Phi\,\partial_\nu\Phi - V(\Phi)$	(GR minimal coupling)	—	CatAD ($\mathbf{A}_{\rm Lean}$: ξ=0)	P38, P43
Holographic entropy	$S(A)$	Area$[\Gamma_{\rm min}]/(4G)$ (Wald + RS-QEC routes; both conditional on the Bekenstein–Hawking coefficient $1/4G$ as an external premise, not derived from GTE structure)	Ryu–Takayanagi	—	CatAD cond.	P38, P43, P44
Hawking temperature	T_H	$M_{\rm Pl}^2/(8\pi M_{\rm BH})$; unchanged by $m_\phi = m_\tau$	(Hawking 1975)	—	CatAD	P38, P43
Critical BH mass	$M_{\rm crit}$	$M_{\rm Pl}^2/(8\pi m_\tau)$ = $\mathbf{3.34\times10^{39}}$ MeV	—	—	CatAD	P38, P43
MDL cosmological IC	$k, \Phi_0, \dot\Phi_0$	k=0, Φ_0=0, $\dot\Phi_0$=$M_{\rm Pl}$; uniform; K=1.585 bits	(flat universe)	consistent	CatAD	P38, P43
GTE bounce	$\rho_{\rm max}$	$\rho_{\rm Pl}$; Friedmann correction $f_C(\rho/\rho_{\rm Pl})$	—	—	CatA	P38, P43
CMB spectral index	n_s	$1 - \ln(2)/(2\pi^2)$ = **0.96488**; algebraic chain 5/7 certified; gate OQ-QG-1-Z_2-EFT (weaker than full OQ-QG-1)	0.9649 ± 0.0042 (Planck 2018)	$\mathbf{-0.005\sigma}$	CatAD	P44
Tensor-to-scalar ratio	r	**0** (exact; no primordial GWs)	unmeasured ($\sigma_r \approx 0.001$, LiteBIRD)	primary falsifiable	$\mathbf{A}_{\rm Lean}$/A	P44, P45
UV finiteness (curved background)	C_i	DeWitt–Schwinger: $C_i \sim 10^{-5}\ln(M_{\rm Pl}/m) \approx 41.76$	—	no residual UV problem	A/D	P44
Dark-energy fraction	Ω_Λ	R1 (PSC): $(\ln 2/3\pi)\log_2(2000/3) = 0.6899$ (CatAD); R2 (holo): $3\pi/14 = 0.6732$; bracket obs. 0.6889; $\rho_\Lambda = (9/112)M_{\rm Pl}^2H_0^2$; $w=-1$; loop suppression $\sim 10^{-42}$	0.689 ± 0.006 Pl18	$+0.18\sigma$/ bracket	CatAD	P44, P47
PSC-Adj. Theorem	$\Omega_\Lambda \in [3\pi/14,\ 0.690]$	PSC + SRRG + MFRR[T] + DPP + gauge spectrum ⇒ bracket, no free params; $N_{\rm gen}$=3 unique ($<5\sigma$ both routes; all other N: $>35\sigma$); spread 2.4%	0.689 ± 0.006 Pl18	bracket ($+0.18\sigma$)	CatAD	P47

Quantity	Symbol	GTE Value	PDG / Exp	Error	Level	Src.
Grav. D^2 unification	$D^2 = 16$	D^2=(N_{gen}+1)^2=16: CC epoch, CC magnitude, Gorard ($\mathbf{A}_{\text{Lean}}$), E–H normalization (D from DPP, $\mathbf{A}_{\text{Lean}}$); absent from $\Omega_\Lambda^{\text{holo}}$, F_{21}, $\sin^2\theta_W$	—	structural	CatAD	P47
3+1D spacetime from DPP	$D = 3{+}1$	Shared clock τ_c^{out} necessary and sufficient (DPP, $\mathbf{A}_{\text{Lean}}$)	observed	exact	$\mathbf{A}_{\text{Lean}}$	P45
Lorentz algebra (3+1D)	$\mathfrak{so}(1,3)$	All 12 commutation relations machine-certified	(GR/QFT)	exact	$\mathbf{A}_{\text{Lean}}$	P45
Gravity (3-tape, continuum limit)	$F = G_{\text{eff}} M/(4\pi b^2)$	Physical Minimum Description Length (PMDL) Poisson $\nabla^2\Phi = G_{\text{eff}}\, p(w_x, w_y, w_z)$ ($\mathbf{A}_{\text{Lean}}$); 3D Poisson Green's function + multipole expansion gives exactly $F \propto r^{-2}$ in the continuum limit ($b \gg \sigma_{\text{AL}}$) (**A/D**); prior discrete deviations ($F \propto b^{-2.19\ldots-2.30}$) are finite-source artifacts: correction $O(\sigma_{\text{AL}}/b)^2 \to 0$ as $b/\sigma_{\text{AL}} \to \infty$	$F \propto r^{-2}$ (Newton)	exact in continuum limit	CatAD	P45
Z_7 field potential cap	$V_{\max}$	$2M_{\text{kink}}^2/49 = \mathbf{3.44 \times 10^{-3}}$ GeV^4; 79 orders of magnitude below M_{Pl}^4	—	structural suppression	CatAD	P44
Kink emission threshold	$M_{\text{kink}}^{\text{crit}}$	$(49/8)M_{\text{crit}}$: $\mathbf{2.04 \times 10^{40}}$ MeV; $T_H = m_{\text{kink}}$ (Hawking–kink duality)	—	—	CatAD	P44
FKTT structural identity	$g_{\text{kink-top}} = \varepsilon_{\text{FN}}$	Top-kink Yukawa = FN suppression: $g_{\text{kink-top}} = \varepsilon_{\text{FN}}$ (`kink_top_coupling_eq_eps_FN`, zero sorry, zero axioms); closes η_B leptogenesis chain at $+0.15\sigma$	—	Structural	$\mathbf{A}_{\text{Lean}}$	P42, P47
Gorard bridge coefficient	C_{Gorard}	**Exact formula:** $C_{\text{Gorard}} = N_{\text{spatial}}/(2D^2) = 3/(2{\times}16) = \mathbf{3/32}$; $N_{\text{spatial}} = N_{\text{gen}} = 3$ (DPP, $\mathbf{A}_{\text{Lean}}$), $D = N_{\text{gen}} + 1 = 4$ (DPP, $\mathbf{A}_{\text{Lean}}$)	—	exact	$\mathbf{A}_{\text{Lean}}$	P47

Quantity	Symbol	GTE Value	PDG / Exp	Error	Level	Src.
Gorard discrete-smooth bridge (G normalisation)	k_{SD}	Ollivier–Ricci curvature chain: $k_{\mathrm{SD}} = 10/13$	—	—	CatAD	P47
Neutrino mass sum	Σm_ν	59.4 meV (from GTE seesaw + three-tape winding)	< 120 meV (Planck CMB)	Within bound	CatAD	P21, P47
H_0 (non-circular from T_{CMB})	H_0	$H_0(T_{\mathrm{CMB}}; \Omega_\Lambda, \eta_B, \Omega_{\mathrm{DM}}h^2, G_N) =$ **67.95** km/s/Mpc; $T_{\mathrm{CMB}} = 2.7255$ K one external input; Path D fully first-principles: T_{CMB}, Ω_Λ, η_B, $\Omega_{\mathrm{DM}}h^2$ all GTE-derived $\to H_0$ without circular anchor (CatAD)	67.66 ± 0.42 km/s/Mpc (Planck 2018)	$+0.7\sigma$	CatAD	P47
$\Omega_{\mathrm{DM}}h^2$ (topological dilution)	$\Omega_{\mathrm{DM}}h^2$	ADM raw $0.167 \to \times D_{\mathrm{top}} = e^{-1/N_c}$ ($\mathbf{A}_{\mathrm{Lean}}$; `d_top_derivation_chain_catal`) = **0.11994**; D_{top} exponent: $q_{\mathrm{dark}}/(\lvert Z_7\rvert - 1) = 2/6 = 1/N_c$ ($\mathbf{A}_{\mathrm{Lean}}$); full chain machine-certified in Lean 4, zero sorry	0.1200 ± 0.0012 (Planck 2018)	$-\mathbf{0.044}\%$	CatAD	P29, P47
CMB temperature	T_{CMB}	$T_{\mathrm{CMB}}^{\mathrm{GTE}} = 2.7265$ K (P47 radiation-equation route)	2.72548 ± 0.00057 K	0.037%	CatB	P47

E.3.8 Novel GTE Particle Predictions (Beyond SM)

Pre-registered discovery run (P02): 1,000,035 candidates in `papers/02_GTE_spectrum/candidates.csv`. Nine genuinely novel stable candidates (GTE-P1, P2, P3, P6–P11); GTE-P4 and GTE-P5 are trajectory-path variants of canonical SM triples and are excluded from this register (P02 trajectory-path multiplicity; GTE-P4, GTE-P5). All listed masses are PCHIP-calibrated; confidence from the canonical v4 run.

Table 28: Novel GTE particle predictions (P02 register; 9 candidates).

Label	n	Mass (MeV)	Conf.	Test status	Level	Src.
GTE-P1	4,935	**2.97**	0.860	○ Open	B	P02
GTE-P2	4,702	**107.4**	0.859	○ Open	B	P02
GTE-P3	∼8K	**801** (band)	0.856	○ Open	B	P02
GTE-P6	6,333	**30.9**	0.860	○ Open	B	P02
GTE-P7	5,383	**211.9**	0.859	○ Open	B/$\mathrm{A}_{\mathrm{Lean}}$	P02, P29
GTE-P8	5,867	**298.0**	0.858	○ Open	B	P02
GTE-P9	275	**561.0**	0.857	○ Open	B	P02

Label	n	Mass (MeV)	Conf.	Test status	Level	Src.
GTE-P10	∼496K	**1100** (band)	0.855	○ Open	B	P02
GTE-P11	∼1.6M	**1600** (band)	0.853	○ Open	B	P02

E.3.9 Dark Sector Predictions

Table 29: Mirror-branch dark sector predictions (P29; GTE-P7 cross-listed with P02).

State	Mass / value	$Q_{\rm EM}$	Comparison / venue	Status	Level	Src.
GTE-P7 (DM cand.)	211.9 MeV	0	Mono-photon / missing energy (Belle II)	○ Open	$\mathrm{A_{Lean}}$/B	P02, P29
Dark lepton G1 (χ_1)	0.541 MeV	0	Sub-MeV DM; CMB spectral distortion	○ Open	CatA	P29
Dark lepton G2 (χ_2)	24.47 MeV	0	MeV-scale Higgs-portal DM	○ Open	CatA	P29
Dark lepton G3 (χ_3)	3604.7 MeV	0	LHC invisible Higgs / mono-jet	○ Open	CatA	P29
Dark photon A'	**Does not exist**	—	NA64/BaBar/ Belle II null = consistent	○ Open	$\mathbf{A}_{\rm Lean}$	P29
$Q = 0$ (all dark states)	exact	0	Lean-certified (mirror branch)	○ Open	$\mathbf{A}_{\rm Lean}$	P29
m_{G2}/m_{G1}	≈ 45.4	—	Two-peak direct-detection signature	○ Open	CatA	P29
$R_{\rm dark}$	0.2080	—	Dark radiation (precision cosmology)	○ Open	CatAD	P29
$\Lambda_{\rm dark}$	210–1700 MeV	—	Dark hadron searches ($N_{f,\rm dark}$ range)	○ Open	$\mathbf{A}_{\rm Lean}$/B	P29
η_χ (asym. DM)	$(2/7)\,\eta_{B+L,\rm pre}$	—	$\approx 1.13 \times 10^{-6}$ (structural)	○ Open	CatA	P29
$\sin^2\theta_W^{\rm dark}$	$2/7 \approx 0.286$	—	vs. SM 3/13; dark EW mixing	○ Open	CatAD	P29, P35

E.3.10 Nuclear and Hadronic

Table 30: Nuclear and hadronic predictions.

Quantity	Symbol	GTE Value	Experimental	Error	Level	Src.
Nuclear pairing constant	$\Delta_{\rm pair}$	$a_{\rm seed}^{g/2} = 5^{3/2} =$ **11.18** MeV	∼11.18 MeV	**0.003**%	B (bridge)	P03
Nuclear magic numbers	—	GTE $f_\pi + m_\pi \to$ spin-orbit: all 7 reproduced	$\{2, 8, 20, 28, 50, 82, 126\}$	Matches	B (bridge)	P03
Nilsson κ / IPT match	$\kappa_{\rm emp}/\kappa_{\rm min}$ (N=50)	**1.149** ≈ IPT=1.1309	IPT	1.6%	B (bridge)	P03

Quantity	Symbol	GTE Value	Experimental	Error	Level	Src.
Predicted magic number	N	**184** (shell-gap extrapolation)	unconfirmed	∘ Open	B (bridge)	P03
F_{10} parity feature	—	$(Z \bmod 2)\,\delta_{\text{pair}}\,A^{-3/4}$; motivated by odd $b_p = 11459$ ($\mathbf{A}_{\text{Lean}}$)	—	structural input	B (empirical fit)	P03
Binding energy (ML)	BE/A	50 GTE features: 3.17 MeV MAE	AME2020	3.17 MeV MAE	A (features)	P03
Nuclear stability classifier	—	6-term GTE law vs. NUBASE2020 labels: **94.5**% 5-fold CV	NUBASE2020	+19.5% above baseline	A (empirical)	P03
Proton mass	M_p	$3M_{\text{kink}} + G_{\text{inter}} \times \lvert p(2,2,6)\rvert/6 = 870.3 + 14.57 \times 4.67 =$ **938.27** MeV; $G_{\text{inter}} = 14.57$ MeV (empirical; analytic derivation open)	938.272 MeV	**0.000**%	CatA	P03, P35, P39

E.3.11 Quantum Mechanics and Information

Table 31: Quantum mechanics and information-theoretic predictions.

Quantity	Symbol	GTE Value	Reference	Error	Level	Src.
Physical Hilbert space dim	$\dim \mathcal{H}_{\text{phys}}$	f_{MDL} orbit: **1** (vacuum only, $E_0 = 0$)	—	—	CatA	P37
Born rule	$P(k) = \lvert c_k\rvert^2$	Z_7-KG kink superselection: derived	QM postulate	—	$\mathbf{A}_{\text{Lean}}$	P37, P41–P43
Born rule (PW/τ_c)	$P(k\lvert\tau)$	τ_c (diagonal-Hamiltonian clock) satisfies all Page–Wootters prerequisites; PW derivation yields Born rule at each τ independently of clock state distribution (classical or quantum τ_c give identical conditional probabilities)	QM postulate	—	CatAD	P45
K_{CA} lower bound (CMCA)	K_{CA}	6-channel description length: **19** bits	—	—	$\mathbf{A}_{\text{Lean}}$	P41
SR time dilation (CMCA)	$\tau_{\text{glider}}/\tau_{\text{ether}}$	CMCA inner clock: 1.563 at $M = 7$	$1/\sqrt{1 - v^2/c^2}$	5.79% (Nyquist floor at $M = 7$; $\to 0$ as $M \to \infty$)	CatA	P41

Quantity	Symbol	GTE Value	Reference	Error	Level	Src.
Lorentz invariance (KG)	$\omega^2 = k^2 + m^2$	KG equation: Lorentz-invariant	(GR/QFT)	—	$\mathbf{A}_{\text{Lean}}$	P41, P42, P43
No finite CA Lorentz replica	$\varepsilon_0(M) = \pi^2/(3M^2) > 0$	Unique at $M \to \infty$	—	—	$\mathbf{A}_{\text{Lean}}$	P43
Geodesic theorem	—	PSC orbit curvature: τ_c prefers geodesic	(GR)	—	$\mathbf{A}_{\text{Lean}}$	P43
Bell nonlocality (3-tape)	S	Cross-tape coupling: $\mathbf{S = 2.4459}$ (86.5% Tsirelson $2\sqrt{2}$)	Tsirelson bound $2\sqrt{2} \approx 2.828$	CatA	CatA	P45, P46
Entanglement negativity	$\mathcal{N}$	Three-tape: $\mathcal{N} = \mathbf{0.38}$	—	—	CatA	P45, P46
SR time dilation (3-tape)	$\tau_{\text{inner}}/\tau_{\text{outer}}$	$3/7 \approx \mathbf{0.4286}$	$1/\sqrt{1 - v^2/c^2}$	CatA (clock ratio)	CatA	P45
CGLMP d=7 inequality	I_7	Full $\mathbb{Z}_7^3$ diagonal H_{grav}: $I_7 \leq \mathbf{1.97}$ (PPT-entangled but classically bounded; $\mathcal{N} \leq 0.32$; CHSH ≤ 2)	Classical bound $I_7^{\text{cl}} = 2$	Honest negative	CatA	P45, P46
Gorard normalization gap	$\log_{10} \Delta_{\text{Gorard}}$	**77.46**; target 77.5 (GTE holographic)	77.5 (target)	0.05	CatAD	P45
Discrete Schrödinger	$U(n)$	$U(n) = e^{-iH_{\text{cyc}}\, n\tau_c}$; composition, unitarity, step recurrence certified (`Cogwheel DynamicsG21.lean`, zero sorry)	QM (continuous)	struct.	$\mathbf{A}_{\text{Lean}}$	P45, P37
NEMS–GTE Bell bridge	—	NEMS semantic nonlocality ($\mathbf{A}_{\text{Lean}}$) and GTE Bell nonlocality (CatA) both arise from PSC-forced global-to-local information loss (structural CatAD)	—	—	CatAD	P45, P46

E.3.12 Computational Universality

Table 32: Computational universality results.

Result	Statement	Level	Source	Lean Cert
CUP-4: SM orbit → Rule 110	SM orbit + vacuum-transparency forces Rule 110	$\mathbf{A}_{\text{Lean}}$	P28	Exhaustive proof
Rule 110 GF(7) polynomial	$p(L,C,R) = C + R - CR - LCR$ over GF(7)	$\mathbf{A}_{\text{Lean}}$	P40	`rule110_z7_poly_rep` (native_decide)
Rule 110 NAND identity	$p(L,1,R) = 1 - LR = \text{NAND}(L,R)$	$\mathbf{A}_{\text{Lean}}$	P40	`rule110_center1_is_nand`

Result	Statement	Level	Source	Lean Cert
Finite NAND functional completeness	Rule 110 at center cell realizes any finite two-input Boolean function (Cook-independent; distinct from Turing universality)	$\mathbf{A}_{\text{Lean}}$	P40	`z7_bool3_finite_functional_completeness` (one axiom: `boolean_nand_complete`)
f_{MDL} uniqueness	MDL-unique among 7^{325} completions	$\mathbf{A}_{\text{Lean}}$	P28	`fmdl_mdl_uniqueness` (76-bit certificate)
Algebraic Lifting Theorem	Every PSC-admissible SM config algebraically realised in f_{MDL}	$\mathbf{A}_{\text{Lean}}$	P41, P42, P43	`LiftingTheorem.lean` (zero sorry)
Unified physical exclusion	Any $[D]$-weighted state: confined, massive, anomaly-free, PSC-admissible	$\mathbf{A}_{\text{Lean}}$	ugp-lean	Zero axioms
TPC strict containment	Decidable $\subsetneq$ TPC $\subsetneq$ Hypercomputation; $N_{\text{gen}} = 3$ levels	$\mathbf{A}_{\text{Lean}}$	P34	Diagonal argument
Polynomial MDL uniqueness	$p(L, C, R) = C + R - CR - LCR$ is the unique degree-3 multilinear GF(7) object satisfying $\mathbb{Z}_7$, Rule 110, 19-bit MDL	$\mathbf{A}_{\text{Lean}}$	P46	All five roles have zero extra specification cost
$\mathbb{Z}_7 \times \mathbb{Z}_3$ MDL uniqueness (unconditional)	Non-circular MDL derivation selects $\mathbb{Z}_7 \times \mathbb{Z}_3$ with zero SM input; GTE polynomial over GF(5) has 0 PSC kink orbits; total MDL gap 3+0 < 5+6	$\mathbf{A}_{\text{Lean}}$	ugp-lean	`mdl_ca_rule_coding_closed`, `z5_fmdl_no_psc_kink_orbits`, `mdl_total_z7z3_strictly_beats_z5z3`; all zero sorry
Direct-Interpolation Lift (UGP–p triangle spine)	Total-parity shadow of the canonical generation-orbit triples + MDL-equivalent vacuum transparency is full-rank over GF(7): exactly one multilinear rule among all 7^8 survives — p itself; Rule 110 = binary restriction (corollary); sparsity floor ≥ 4 monomials, equality realized by p; chirality census over all 120 family orderings: survivor set exactly {Rule 110, Rule 124}; consilience $\delta(N_c) = q_{\min}(N_c)$ uniquely at $N_c = 3$	$\mathbf{A}_{\text{Lean}}$	P49, P48	`gte_orbit_parity_provenance`, `ugp_orbit_interpolation_lift` (+ structural route), `interpolation_lift_binary_corollary`, `gf7_rule110_sparsity_floor`, `orbit_chirality_census`, `delta_qmin_coincidence_at_three`; all zero sorry
Color confinement (MDL)	$\Delta K = \log_2(N_c^2) = \log_2 9 \approx 3.17$ bits gap forbids free quarks	$\mathbf{A}_{\text{Lean}}$	P45, P46	From MDL description-length argument
Baryon number topological	$B = (1/3)\sum_j \chi_q(w_j)$; $B = 1/3$ per quark from $N_{\text{tapes}} = 3$	$\mathbf{A}_{\text{Lean}}$	P45, P46	Three-tape architecture
Holographic 3-tape scaling	$3L/L^3 \to 0$ (area/volume); Bekenstein–Hawking realized	$\mathbf{A}_{\text{Lean}}$	P45	Three-tape holographic area scaling
Fermionic statistics	From $\mathbb{Z}_7^*$ non-primitive roots (fermion/boson split)	$\mathbf{A}_{\text{Lean}}$	P45, P46	Algebraic root structure of $\mathbb{Z}_7^*$

E.3.13 PhiMDL Field Structure

Table 33: Φ_{MDL} field structure predictions.

Quantity	Symbol	GTE Value	PDG / Exp	Error	Level	Source
Φ_{MDL} mass (SCC)	m_φ	Self-Consistency: $m_\tau = \mathbf{1776.86}$ MeV	1776.86 MeV	0.0%	$\mathbf{A}_{\mathrm{Lean}}$	P34, P35, P38, P42, P43
BPS kink mass	M_{kink}	$(8/49)m_\tau = \mathbf{290.10}$ MeV	287 MeV (calib.)	+1.1%	$\mathbf{A}_{\mathrm{Lean}}$	P35, P38, P42, P43, P45
Quantum kink mass (one-loop)	M^Q	GJQW interface dim-reg, two routes: $\mathbf{281 \pm 21}$ MeV ($\Delta M = -7$ to -10 MeV, on-shell / $\overline{\mathrm{MS}}@m_\varphi$)	— (no PDG counterpart)	scheme envelope	CatA	P38, P42
PSC admissible sectors	$\{0, 2, 3, 4, 6\}$	PSC sieve: 5 sectors; $\{1, 5\}$ forbidden	—	—	$\mathbf{A}_{\mathrm{Lean}}$	P42
Vacuum stress-energy	$T_{\mu\nu}\|_{\mathrm{vac}}$	At all 7 Z_7 vacua: $\mathbf{0}$ (exact)	—	—	$\mathbf{A}_{\mathrm{Lean}}$	P38, P42, P43
Vacuum stability (polynomial)	$p(0,0,0) = 0$	$w = 0$ unique fixed point of $p(x,x,x) \equiv x \pmod 7$ ($\mathbf{A}_{\mathrm{Lean}}$)	—	gravitational attraction = restoring force to vacuum	$\mathbf{A}_{\mathrm{Lean}}$	P45, P46
SRRG golden-ratio bridge	$1/\varphi$	Unique positive root of $p(x,x,x) = x$ over $\mathbb{R}$ ($\mathbf{A}_{\mathrm{Lean}}$)	—	connects MDL polynomial to SRRG fixed point	$\mathbf{A}_{\mathrm{Lean}}$	P46
W-boson propagator	$G_W(r)$	$e^{-m_W r}/(4\pi r)$ from universal propagator $(-\Delta)^{-1} = 1/(4\pi r)$	(massive Yukawa)	A/D	A/D	P46
Chiral gap law (spin-7 chain)	$\Delta(\beta)$	$e^{-3\beta/2}(1 + O(e^{-\beta/2}))$, exact correction tower; slope 3/2 = geometric mean of directed wall energies (`spin7_directed_wall_energies`, $\mathbf{A}_{\mathrm{Lean}}$); half-integer slope = parity-violation signature of substrate chirality; amplitude exactly 1 (doubly derived)	—	exact	**A/D**	P50

Quantity	Symbol	GTE Value	PDG / Exp	Error	Level	Source
Substrate tape spacing (matter-sector reading)	a	$\hbar c/\Lambda_{\mathrm{GTE}} =$ **0.0972** fm (tree) / 0.1007 ± 0.0077 fm (envelope); Tape Saturation Theorem (`tape_saturation_theorem`), conditional on the named Compton-Support Criterion, proven minimal (Boundary Equivalence Theorem); lattice dictionary $aM_{\mathrm{kink}} = 1/7$, $am_\varphi = 7/8$	—	derived (cond.)	**A/D**	P50, P42, P39
Kink correlation length at physical point	ξ_{kink}	$\lvert\mathbb{Z}_7\rvert =$ **7** cells exactly (saturation $\kappa = 1$); substrate-MC falsifiable target, band 6.78–7.23 ± 0.54; a measurement outside the band falsifies the MDL-saturation selection	— (in-silico target)	falsifiable	**A/D**	P50, P42

E.3.14 Structural and Combinatorial Quantities

Table 34: Key structural and combinatorial quantities.

Quantity	Value	Level	Source	Notes
Ridge n	10	$\mathbf{A}_{\mathrm{Lean}}$	P01	`n10_is_minimal_admissible_ridge`
Ridge R_{10}	$2^{10} - 16 = 1008$	$\mathbf{A}_{\mathrm{Lean}}$	P01	$\tau(1008) = 30$ used in CKM θ_{23}
Lepton Seed	$(1, 73, 823)$	$\mathbf{A}_{\mathrm{Lean}}$	P01	`rsuc_theorem`
N_{gen}	3 (PSC Layer I/II)	$\mathbf{A}_{\mathrm{Lean}}$	P01, P05	`psc_enumeration_forces_ngen_3`
N_{fam}	5 (Z_5 ring)	$\mathbf{A}_{\mathrm{Lean}}$	P01	
Higgs c-value	$c_H = 13$	$\mathbf{A}_{\mathrm{Lean}}$	P01	EW boson set $\{11, 12, 13\}$
Strange baryon c-values	$c = \pm 1023$	$\mathbf{A}_{\mathrm{Lean}}$	P01	`ugp_strange_baryon_c_values` (zero sorry)
Wolfenstein λ	$9/40 = 0.225$	$\mathbf{A}_{\mathrm{Lean}}$	P32, P35	Derived output; 0.000σ
Orbit depth = ether period	7 steps max	$\mathbf{A}_{\mathrm{Lean}}$	P41	`OrbitDepthEtherPeriod.lean`

E.3.15 Experimental Programme and Falsifiable Predictions

The following six predictions are directly testable by currently running or near-term precision experiments. Each entry specifies a prediction, the responsible experiment(s), the approximate timeline, and the falsifying condition. Any confirmed falsification would require structural revision of the GTE framework. Proton stability provides the sharpest topological test: a single confirmed *dimension-4* baryon-number-violating decay event would falsify the $\mathbb{Z}_7$ winding conservation law. Standard GUT *dimension-6* proton decay (e.g. $p \to e^+\pi^0$, the primary mode searched at Hyper-Kamiokande) is **not** forbidden by this framework and, if observed, would be fully consistent with the programme's predictions. Cross-references to the relevant prediction tables above: GTE-P7 is cross-listed in Table 29; r=0 and Σm_ν appear in Table 27; $\delta_{CP}^{\mathrm{PMNS}}$ appears in Table 26.

Table 35: Experimental programme: falsifiable predictions indexed to active or near-term experiments. Any confirmed falsifying condition requires structural revision of the framework.

Observable	GTE Prediction	Experiment(s)	Timeline	Falsifying Condition	Level	Src.
Proton stability (dim-4)	No *dimension-4* baryon-number-violating decay; $\mathbb{Z}_7$ winding conservation forbids all dim-4 SU(5) vertices (`proton_decay_dim4_forbidden`, $\mathbf{A}_{\mathrm{Lean}}$). *Note:* dim-6 modes ($p \to e^+\pi^0$, $p \to \bar{\nu}K^+$) are **not forbidden** and would be consistent with this framework	Hyper-Kamiokande, DUNE	2027^+	Any confirmed *dimension-4* baryon-violating event (not dim-6 GUT modes such as $p \to e^+\pi^0$, which are not forbidden by this framework)	$\mathbf{A}_{\mathrm{Lean}}$	P22, P33
GTE-P7 dark particle	Neutral pseudoscalar at **211.9** MeV, Q=0; mirror-branch machine-certified prediction	Belle II (monophoton / missing energy)	Running	Non-observation at $50\,\mathrm{ab}^{-1}$	$\mathrm{A}_{\mathrm{Lean}}$/B	P02, P29
PMNS Dirac CP phase	$\delta_{CP}^{\mathrm{PMNS}} = \mathbf{205.71}^\circ = (4/7) \times 360^\circ$; $\mathbb{Z}_7$ winding arithmetic (-0.15σ from NuFIT 6.0 IC24 NH)	DUNE, Hyper-Kamiokande	2027^+	Measurement outside $205^\circ \pm 15^\circ$	CatA	P45
Tensor-to-scalar ratio	$r = \mathbf{0}$ exactly; no primordial gravitational waves in GTE inflation-free cosmology	LiteBIRD	$\sim$2028	$r > 0.01$ at $>3\sigma$	$\mathbf{A}_{\mathrm{Lean}}$/A	P44, P45
Neutrino mass sum	$\Sigma m_\nu = \mathbf{59.4}$ meV; normal mass ordering; spectrum $\{0.68, 8.6, 50.1\}$ meV	EUCLID, CMB-S4	2026^+	$\Sigma m_\nu < 40$ meV or > 80 meV	CatAD	P21, P47

Observable	GTE Prediction	Experiment(s)	Timeline	Falsifying Condition	Level	Src.
Neutrino mass ordering	Normal Ordering (NH) forced by seed ordering $b_{R,1}<b_{R,2}<b_{R,3}$ via GTE seesaw $m_{\nu,k} = C \cdot b_{R,k}^{29/9}$; $\Delta m^2_{21} = 7.37 \times 10^{-5}$ eV^2 (0.63σ below NuFIT), $\Delta m^2_{31} = 2.51 \times 10^{-3}$ eV^2 ($< 1\sigma$) (`seesaw_normal_ordering_from_seed_ordering`, $\mathbf{A}_{\text{Lean}}$)	JUNO; NuFIT global fits	Running (2025+)	Confirmed IH at $>3\sigma$	$\mathbf{A}_{\text{Lean}}$	P21, P55
Dark-energy EOS	$w_0 = -\mathbf{1}$, $w_a = \mathbf{0}$ exactly; cosmological-constant behavior forced by PSC boundary condition; Weinberg no-go evaded. DESI Y1: $w_a = -0.4 \pm 0.3$ (1.33σ, not yet a challenge)	DESI, EUCLID	Ongoing	$\lvert w_0 + 1\rvert > 0.040$ at 5σ or $\lvert w_a\rvert > 0.45$ at 5σ (Euclid projected: $\sigma(w_0) \approx 0.008$, $\sigma(w_a) \approx 0.09$)	CatAD	P47

E.4 Falsifiable Predictions Registry

The pre-registered, SHA-256-committed canonical list of corpus falsifiable predictions (quantities not yet measured or decided at 2026-07-04) is maintained in:

> `papers/common/predictions/ugp_falsifiable_predictions_v1.md`
> SHA-256: `d9ae2ebf2185862c7d56a5b662af35892bf3cb5f2ef262b147b2241ab2c53f25`
> Registry date: 2026-07-04

The registry covers 18 entries across three sectors: (1) physics-sector predictions (neutrino mass ordering CatAL; neutrino mass sum and individual masses CatA; PMNS $\delta_{CP}^{\text{PMNS}} = 205.71^\circ$ CatA; tensor-to-scalar ratio $r = 0$ CatAL; proton stability at dimension 4 CatAL); (2) novel visible-sector particle candidates GTE-P1, P2, P3, P6, P7, P8, P9, P10, P11 from P02 (CatAD masses); and (3) mirror-branch dark-sector particles χ_1, χ_2, χ_3 from P29 (CatA masses; CatAL $Q_{\text{EM}} = 0$). The registry file is immutable once committed; corrections produce a new version file. Verify with:

```
python3 papers/common/predictions/verify_predictions_hash.py
```

The registry is deposited on Zenodo with SHA-256 companion and OpenTimestamps notarization; the concept DOI is `doi:10.5281/zenodo.21200551`.

E.5 Known Discrepancies and Resolutions

The following register documents known numerical discrepancies between different derivation routes or between GTE predictions and PDG values, together with their resolutions. In no case listed here is the discrepancy indicative of an error; each has a documented structural explanation.

D1: $\sin^2\theta_W$ — three derivation routes at different precision levels

Table 36: $\sin^2\theta_W$ sequential refinements.

Route	Value	Source	Error vs PDG	Status
Bare g_1/g_2 ratio (tree-level)	$3456/15101 \approx 0.22886$	P01	-1.02%	$\mathbf{A}_{\text{Lean}}$ (tree-level)
$N_{\text{gen}}/c_H = 3/13$	**0.2308**	P31, P33, P35	-0.195% (13.0σ PDG 2024)	$\mathbf{A}_{\text{Lean}}$ (tree-level)
$3/13 + \lambda^3/(2c_H)$ threshold	$384729/1664000 \approx 0.23121$	P31, P41	-0.42σ (PDG 2022)	$\mathbf{A}_{\text{Lean}}$
+ SM Higgs threshold, GTE α_{em}	0.23129154	P31, P35	$+0.038\sigma$ (PDG 2024)	**B** (canonical)

Resolution: Sequential refinements of the same derivation. The tree-level ratio 3/13 is a scheme-independent integer: it is derived from a discrete orbit count with no continuous scale parameter, and is not an $\overline{\text{MS}}$, on-shell, or Wilsonian quantity. The 13σ deviation of the tree-level value from the PDG 2024 $\overline{\text{MS}}$ measurement reflects standard SM electroweak radiative corrections (+0.22% fractional shift); the two-loop GTE computation closes the full gap to $+0.038\sigma$. The dark sector value $2/7 \approx 0.286$ (P29) uses entirely distinct dark inputs ($N_{\text{gen}}^{\text{dark}} = 4$, $c_H^{\text{dark}} = 14$); it is not the same quantity and there is no discrepancy between P29 and P35.

D2: m_W — two distinct derivation chains

Table 37: m_W derivation routes.

Chain	Value	Error	Source
Tree-level (bare g_2)	$\sim$109 GeV	$+36\sigma$	P01 (disclosed blind falsification)
1-loop	80.314 GeV	-4.88σ	P01
1-loop oblique correction	**80.372** GeV	$\mathbf{-0.006}\%$	P27 (closes 98.8% of tree-level gap; CatAD)
2-loop + threshold	**80.364** GeV	$\mathbf{-0.42}\sigma$	P01, P22
Schwinger mechanism (P33)	80.405–80.456 GeV	$\sim -0.002\%$	P33
PSC energy scale E_0 (P35/P41/P42)	80.614 GeV	+0.30%	P35, P41, P42

Resolution: P33 (Schwinger mechanism) and P35/P41/P42 (PSC scale E_0) are distinct chains at different physics hierarchy levels, not errors. The 2-loop P22 result (-0.42σ) is the best precision result.

D3: f_π — two values within P35

$m_{\text{kink}}/\pi = 91.35$ MeV (-0.81%, CatA; uncalibrated); $M_{\text{kink}}^{\text{SCC}}/\pi = 92.34$ **MeV (**$+0.30\%$**,** $\mathbf{A}_{\text{Lean}}$**; SCC-calibrated, preferred)**. Editorial inconsistency within P35, not a cross-paper discrepancy.

D4: α_s — same bare coupling at three loop orders

Table 38: $\alpha_s(M_Z)$ loop-level refinements.

Version	Value	Error	Source
Bare $g_3^2/(4\pi)$	0.11822	$+0.24\sigma$	P01
2-loop SC-ZZ running	**0.11790**	$\mathbf{0.00\sigma}$	P22
3-loop F_{21} chain	0.1193	$+1.10\%$	P35, P39

Resolution: Progressive refinements. The 2-loop P22 result (0.00σ) is the best precision result.

D5: Higgs VEV — three mutually consistent determinations

Bare VEV $v_{\text{bare}} = 244.8$ GeV (-0.6%, tree-level) | Self-consistent $v_{\text{self}} =$ **246.27** GeV ($+$**0.02**%, anchors on PDG m_W) | SRRG PSC $v_{\text{PSC}} =$ **246.16** GeV ($-$**0.024**%, more fundamental; B5-SRRG proved — conditional on B1–B4 only). The two refined values agree with PDG at $< 0.1\%$.

D6: Jarlskog J — two CKM methods

Texture-zero QLC ansatz (P01): $J = 2.75 \times 10^{-5}$ (10.8%, A/D). **Structural CKM N_{eff} approach (P32):** $J = 2.999 \times 10^{-5}$ **(**-1.81σ**, A/D; canonical**.

D7: Gravitational hierarchy CatAL upgrade

P35 and P38 label M_{Pl}/m_τ as CatAD. P43 upgrades to $\mathbf{A}_{\text{Lean}}$ via `z3_invariant_entropy_closes_hierarchy` (zero sorry). Formula and numerical value (0.040%) identical throughout. Canonical status: $\mathbf{A}_{\text{Lean}}$ (P43).

D8: Neutrino mass notation (P33 vs P35)

P33 writes $\alpha_{\text{em}}^5 E_{\text{ether}}$ in one place, $\alpha_{\text{em}}^{N_{\text{fam}}}$ in another. Since $N_{\text{fam}} = 5$, these are identical. No discrepancy.

D9: n_s value (P44)

P44 derives $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ (to 5 significant figures). Any appearance of $n_s = 0.96489$ in earlier versions is a rounding artefact; the canonical value is **0.96488**. The derivation is conditional: 5 of 7 steps are closed; the EFE-bridge step (AR3) remains open. Planck 2018 gives $n_s = 0.9649 \pm 0.0042$; the GTE value sits at -0.005σ from this central value (essentially zero, with the GTE value just below the Planck 2018 central value).

D10: η_B — sech bracket vs. FKTT canonical value

The asymptotic sech-overlap upper bound $\eta_B = 6.36 \times 10^{-10}$ lies $+4.3\sigma$ above Planck CMB+BBN ($(6.10 \pm 0.06) \times 10^{-10}$). The exact-sech lower bound gives 5.72×10^{-10} (-6.4σ). Planck sits between these bracket endpoints. The **canonical GTE headline prediction** is the FKTT mechanism: $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$; $\mathbf{A}_{\text{Lean}}$, `kink_top_coupling_eq_eps_FN`, zero axioms).

E.6 Particle Ontology Summary

E.6.1 Foundational Clarifications

Two distinct Z_7 quantities (must not be conflated):

- W_B (Level B SM winding): interaction charge from winding-conservation ($\mathbf{A}_{\text{Lean}}$, P28). Constant per SM species: $W_B(\text{lepton}) = 4$ for all three generations.
- W_A (Level A orbit-sum): $\sum$(orbit cell values) mod 7. Level A beable charge: $W_A(\text{gen}_1) = 4$, $W_A(\text{gen}_2) = 4$, $W_A(\text{gen}_3) = 3$, $W_A(\text{vac}) = 0$.

The "gen$_3$ anomaly" ($W_A = 3 \neq 4$) is resolved: $W_A(\text{gen}_3) = 3$ and $W_B(\tau) = 4$ are distinct quantities (P28 line 1137 explicitly treats gen$_3$ orbit-sum $\neq 4$ as a feature).

E.6.2 Particle Identification Table

Table 39: GTE particle identification: beable orbits, winding charges, and confidence.

Species	**Beable (orbit)**	W_B	W_A	Φ_{MDL} **kink** Q	**Confidence**
$e^-/\mu/\tau$ ($\text{gen}_1/\text{gen}_2/\text{gen}_3$)	$\text{gen}_1 = [1,5,2,2,1]$ / $\text{gen}_2 = [2,5,2,0,2]$ / $\text{gen}_3 = [5,6,5,3,5]$	4	4/4/3	$(4,1)/(4,2)/(4,1)$	$\mathbf{A}_{\text{Lean}}$ (charges); glider map open
$\nu_e/\nu_\mu/\nu_\tau$	Orbit by generation	0	—	$Q = 0$ + triple	$\mathbf{A}_{\text{Lean}}$ (discrimination); masses CatD
u, c, t	By generation	2	—	Kink cascade gen-linked	$\mathbf{A}_{\text{Lean}}$ (winding)
d, s, b	By generation	6	—	Same	$\mathbf{A}_{\text{Lean}}$
γ (photon)	$[0,0,0,0,0]$ (vacuum)	0	0	$Q = 0$; null modes	$\mathbf{A}_{\text{Lean}}$: $f_{\text{MDL}}(0,0,0) = 0$; is ether
W^+	Orbit vertex	3	—	Z_3 emission (gauge)	$\mathbf{A}_{\text{Lean}}$ (vertex); propagation CatA
W^-	Hilbert shadow	4	—	Open	$\mathbf{A}_{\text{Lean}}$ + CatAD
Z boson	Excitation above vac	0	—	$Q = 0$ + massive excitation	$\mathbf{A}_{\text{Lean}}$ (GTE triple $\neq \gamma$); mass CatAD
Gluons	Colour Z_3	0	—	Colourful kinks / flux tubes	CatAD; full QCD open
Higgs	Scalar boundary $c_H = 13$	0	—	Kink/Higgs as boundary	Triple $\mathbf{A}_{\text{Lean}}$; mass/VEV CatAD

E.6.3 Level A/B Species Map

Species formula: $W_B = 4k \bmod 7$ for a k-occupation wavepacket of gen_1 beables.

Table 40: Level A/B species map (provisional).

k **(occupation)**	$W_B = 4k \bmod 7$	**SM species**	**Colour**
1	4	$e/\mu/\tau$ (charged lepton)	Singlet (0)
4	2	$u/c/t$ (up-quark)	R/G/B
5	6	$d/s/b$ (down-quark)	R/G/B
$7 \equiv 0$	0	ν (neutrino)	Singlet
6	3	W^+ (gauge boson)	—
2, 3	1, 5	— (no SM fermion)	—

E.6.4 Open Questions in Particle Ontology

Table 41: Open problems and resolved items in particle ontology.

Status	**Item**
Open	Which specific Rule 110 / Φ_{MDL} pattern corresponds to which SM fermion?
Open	$\nu/\gamma/Z$ discrimination: $Z_7 = 0$ alone insufficient; GTE triple required.
Open	Φ_{MDL} gauge structure: upgrade to $U(1) \times Z_3$ gauge theory.
✓	$\text{SU}(2)_L$: MDL gauging machine-certified ($\mathbf{A}_{\text{Lean}}$, zero named axioms: `phimdl_potential_su2l_invariant`, `su2l_covariant_derivative_minimal`, `su2l_wpm_generator_algebra`; master bundle `su2l_l2_from_phimdl_potential_catad`, P46). Lepton-W universality from shared $\mathbb{Z}_7$ winding sectors (P45/P46). Weak-force propagator $G_W(r) = e^{-m_W r}/(4\pi r)$ derived (P46, A/D).
✓	Unified physical exclusion ($\mathbf{A}_{\text{Lean}}$, zero axioms).
✓	Strong CP: $\theta_{\text{QCD}} = 0$ ($\mathbf{A}_{\text{Lean}}$).
✓	Vector meson nonet (CatA; *note: CatA only, not* $\boldsymbol{A}_{\text{Lean}}$).
✓	Causal path axiom ($\mathbf{A}_{\text{Lean}}$).
✓	gen_3 winding anomaly resolved (ROBUST).

E.7 Index of Lean Certifications ($\mathbf{A}_{\text{Lean}}$)

All results below are Lean 4, zero sorry. Grouped by sector.
Caveat: Claimed $\boldsymbol{A}_{\text{Lean}}$ *results not in repo.* `vector_meson_nonet_jp1` (CatAL for vector meson nonet) is **not in repo — CatA only**. The $\mathbf{A}_{\text{Lean}}$ certification for the vector meson nonet is not present in the Lean repo; CatA only.

A.1 Arithmetic / Structural Core

Theorem / Module	**Statement**	**File**	**Notes**
`n10_is_minimal_admissible_ridge`	$n = 10$ minimal admissible ridge	ugp-lean	Zero sorry

Theorem / Module	Statement	File	Notes
`rsuc_theorem`	Lepton Seed $(1, 73, 823)$ unique at $n = 10$	ugp-lean	Zero sorry
`mdl_selects_LeptonSeed`	MDL uniquely selects Lepton Seed	ugp-lean	Zero sorry
`fmdl_gen1_is_garden_of_eden`	Gen_1 has no f_{MDL} predecessor	ugp-lean	Zero sorry
`gte_master_formula_complete`	$N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}} = 5$	ugp-lean	
`quarterLockLaw`	Quarter-Lock Law (algebraic identity)	ugp-lean	Zero sorry
`asymptotic_sparsity_universal`	Global asymptotic sparsity at $n = 10$	ugp-lean	

A.2 QCD / Strong Sector

Theorem / Module	Statement	File	Notes
`b0_eq_z7_order`	$b_0 = 7 = \lvert Z_7 \rvert$	ugp-lean	
`f21_substrate_beta_coefficient`	F_{21} determines b_0, C_F, C_A, T_F	ugp-lean	14 theorems
`f21_substrate_two_loop_beta_b1`	$b_1 = 26$ from F_{21}	ugp-lean	Zero sorry
`anomaly_cancellation_psc_forced`	AC proved from PSC+RC	ugp-lean	Zero axioms
`psc_layer_i_certificate`	Two-Layer PSC Layer I filter certificate	`ScanCertificate`	native_decide
`psc_12_survivors_card`	Exactly 12 universes pass Layer I PSC filters	`ScanCertificate`	native_decide
`psc_enumeration_forces_ngen_3`	Every Layer I PSC survivor has $N_{\text{gen}} = 3$	`ScanCertificate`	native_decide
`psc_12_survivors_have_ngen_3`	All 12 Layer I survivors have $N_{\text{gen}} = 3$ (explicit cert)	`ScanCertificate`	native_decide
`psc_admissible_forces_sm_gauge`	All Layer I survivors carry SU(3)×SU(2)×U(1) in d=4	`ScanCertificate`	native_decide
`universe_params_card`	Full 34,560-universe parameter space is a `Fintype` instance	`ScanCertificate`	native_decide
`no_psc_admissible_single_quark`	No single-quark PSC state	ugp-lean	native_decide
`certified_mass_gap_positive`	Mass gap $\Delta > 0$	ugp-lean	Zero sorry
`jpspin_kinks_are_spin_half`, `gte_physical_baryon_octet_is_ms`	Baryon $J^P = 1/2^+$ derived	ugp-lean	

A.3 Electroweak Sector

Theorem / Module	Statement	File	
weinberg_angle_closure	$\sin^2\theta_W = 3/13$	ugp-lean	
weinberg_two_term_prediction	$\sin^2\theta_W = 384729/1664000$	P41/P42	
gut_weinberg_angle_pow2	$\sin^2\theta_W^{\mathrm{GUT}} = 3/8$	ugp-lean	
alpha_inv_master	$\alpha_{\mathrm{EM}}^{-1} = 137$	ugp-lean	
BraidAtlas.EWBosons	EW boson c-values $\{11, 12, 13\}$	ugp-lean	
IPT_theorem	IPT $= 1 + \ln\varphi/(2\ln 2\pi)$	ugp-lean	
sm_gauge_group_certificate	SM gauge bundle SU(3) $\times$ SU(2)$_L$ $\times$ U(1)$_Y$ via three Z_7 channel identifications	SMGaugeGroup	Zero sorry (CatAD: Burnside axiom)
phimdl_angular_momentum_conserved	SO(3) Noether angular momentum conserved	NoetherAngularMomentum	CatAD
delta_rho_positive	$\delta\rho > 0$ from top-bottom doublet oblique correction	GaugeMDL	Zero sorry
m_W_one_loop_larger	$m_W^{\text{1-loop}} > m_W^{\text{tree}}$	GaugeMDL	Zero sorry
m_W_gap_fraction_certified	Oblique correction closes 98.8% of m_W gap; 4.7 MeV residual	GaugeMDL	Zero sorry

A.4 CKM / Flavor

Theorem / Module	Statement	File	
wolfenstein_lambda_formula	$\lambda = 9/40$ derived	ugp-lean	
ckm_theta23_ratio_uniqueness	CKM θ_{23} scaling ratio $= 15/8$	ugp-lean	Zero sorry
ckm_unitarity_triangle_radius_eq_gut_weinberg	$R_b = \sin^2\theta_W^{\mathrm{GUT}} = 3/8$	ugp-lean	
bb_bs_product_not_square	$b_b \times b_s$ non-square $\Rightarrow \tan\gamma$ irrational	ugp-lean	
neff_b_eq_mersenne	$b_b = 8191$ (Mersenne prime) certified	ugp-lean	
neff_u_eq_ngen_sq	$b_u = N_{\mathrm{gen}}^2 = 9$	ugp-lean	

A.5 Lepton / Quark Masses

Theorem / Module	Statement	File	
koide_angle_from_N_c_pure	Koide phase $\theta = 2/9$ from $N_c = 3$	ugp-lean	Zero sorry
two_plus_sqrt3_eq	$2 + \sqrt{3} = 4\cos^2(\pi/12)$	ugp-lean	Zero sorry
proton_b_correction_agreement	$b_p = 11459$, $b_n = 11441$	ugp-lean	Zero sorry
ugp_strange_baryon_c_values	Strange baryon c-values $= \pm 1023$	ugp-lean	Zero sorry
UgpLean.GTE.NuclearPairing	Nuclear pairing $\Delta = 5^{3/2}$	ugp-lean	Zero sorry
koide_Q_iff_amplitude	$Q = 2/3 \Leftrightarrow b = \sqrt{2}$ (Yukawa amplitude)	KoideYukawaAmplitude	Zero sorry
koide_cv_one_iff_amplitude	$\mathrm{CV}(\sqrt{m}) = 1 \Leftrightarrow Q = 2/3$	KoideEqualNorm Reformulation	Zero sorry

A.6 Neutrino Sector

Theorem / Module	Statement	File	
Seesaw exponent	$29/9 = N_c + \theta$ (three independent decompositions)	ugp-lean	Zero sorry
Normal hierarchy	Falls out of $b^{29/9}$ structure	ugp-lean	
z7_no_bion_criterion	No Z_7 bion state satisfies PSC dark-ring sector constraints (rules out bion mass mechanism)	NeutrinoVacuum SectorL2.lean	Zero sorry
dark_ring_fourth_qn_count	$Z_7^4 = (w_x, w_y, w_z, w_\tau)$ dark-ring sector has exactly 4 quantum numbers	NeutrinoVacuum SectorL2.lean	Zero sorry

A.7 Quantum Mechanics / Born Rule

Theorem / Module	Statement	File	
born_rule_unconditional	Born rule $P(k) = \lvert c_k \rvert^2$ unconditionally	BeableWinding Partition Instance.lean	Zero custom axioms
phimdl_thermal_state_master	Thermal state PSC-admissible	P42/P43	
dark_sector_orbit_structure	Dark sector dim= 5 (three period-2 cycles)	ugp-lean	
CMCAMDLMinimality.lean	K_{CA} lower bound = 19 bits	P41	

A.8 Computational Universality

Theorem / Module	Statement	File	
rule110_z7_poly_rep	$p(L,C,R) = C + R - CR - LCR$ over GF(7)	P40	native_decide
rule110_center1_is_nand	$p(L,1,R) = \text{NAND}(L,R)$	P40	
z7_bool3_finite_functional_completeness	Rule 110 finite NAND functional completeness (Cook-independent; not Turing universality)	P40	1 axiom
fmdl_mdl_uniqueness	f_{MDL} unique among 7^{325} completions	ugp-lean	76-bit cert
LiftingTheorem.lean	Every PSC-admissible SM config in f_{MDL}	P28/P43	Zero sorry
Algebraic DescentTheorem.lean	F_{21} conservation holds at all resolutions	P41/P42	
unified_physical_exclusion	$[D]$-weighted state: confined, massive, anomaly-free	ugp-lean	Zero axioms
gte_orbit_vectors_are_rs_codewords	$\text{Gen}_{1,2,3}$ orbit vectors are $[5,3,3]_7$ RS codewords over GF(7)	RSCodeOrbit.lean	Zero sorry
orbit_vectors_linearly_independent	$\text{Gen}_{1,2,3}$ orbit vectors linearly independent over GF(7)	RSCodeOrbit.lean	Zero sorry

A.9 Gravity / Spacetime

Theorem / Module	Statement	File	
z3_invariant_entropy_closes_hierarchy	Planck/τ hierarchy $= 21^{10} \times 7^{7/2}$	P43	Zero sorry
gte_spacetime_dimension	$D = N_{\text{gen}} + 1 = 4$	ugp-lean	Spatial $D_s = 3$ cert
causal_graph_spectral_dim_thermodynamic_limit	$d_s = 4$ in thermodynamic limit	ugp-lean	
phimdl_potential_at_vacuum_zero	$\Lambda = 0$ at tree level	ugp-lean	
phimdl_tmunu_vacuum_zero	$T_{\mu\nu}\|_{\text{vac}} = 0$ at all 7 Z_7 vacua	ugp-lean	
kg_dispersion_lorentz_invariant	KG equation Lorentz-invariant	ugp-lean	
poincare_invariance_of_kg	KG equation Poincaré-invariant	ugp-lean	
no_finite_ca_exact_lorentz_replica	$\varepsilon_0(M) > 0$ for all finite M	P43	

Theorem / Module	Statement	File	
psc_orbit_is_curvature_geodesic	Geodesic theorem from PSC orbit curvature	P43	
planck_density_bound_via_lifting	$7^8 > e^{4\pi}$; CMCA Planck density bound via Lifting Thm	PlanckDensityBound.lean	Zero sorry
minimal_coupling_is_mdl_minimal	$\xi = 0$ uniquely MDL-minimal; no curvature coupling	MinimalCoupling.lean	Zero sorry
z7_superselection_preserved_by_flat_metric	Z_7 superselection preserved on curved backgrounds	MinimalCoupling.lean	Zero sorry
flrw_phimdl_equation_well_defined	Z_7-KG on FLRW: existence + uniqueness (Picard–Lindelöf)	FLRWFieldEquation.lean	Zero sorry
causal_path_exists	Causal path theorem (directed/undirected)	SpatiallyExtended Lifting.lean	Zero custom axioms
kink_bps_mass_formula	$M_{\text{kink}} = 8m/49$ (analytic BPS mass; exact)	PMDLGravityTheorems	Zero sorry
gorard_bridge_coefficient	Gorard $\kappa_{\text{SD}}/(8\pi) \in (0,1)$; $C_{\text{Gorard}} \in (0,1)$ (bound)	GorardRicciFlatVacuum	Zero sorry
c_gorard_eq_n_spatial_over_two_dsq	$C_{\text{Gorard}} = N_{\text{spatial}}/(2D^2) = 3/32$ **exact** (not approximation)	TemporalVoxelCC (P47)	Zero sorry ($\mathbf{A}_{\text{Lean}}$)

A.10 SRRG / VEV

Theorem / Module	Statement	File	
abs_psi_eq_inv_phi	SRRG contraction rate $= 1/\varphi$	ugp-lean	Zero sorry
entropy_formula_U1	PSC entropy $H_{\text{adj}} = \ln(2\pi)$	ugp-lean	Zero sorry
A3_half_factor	Symmetry splitting factor $= 1/2$	ugp-lean	Zero sorry
SrrgLean.VEVProof.*	$v_{\text{PSC}} = 246.16$ GeV from SRRG	srrg-lean	Zero sorry; zero open axioms (B5 proved); conditional on B1–B4
srrg_beta_zero_iff_kCMCA_minimum	SRRG β-function zero $\Leftrightarrow$ MDL K_{CMCA} minimum; $g^* = 1/\varphi$	SRRGCABridge	Zero sorry
fca_attractor_diagonal_fp_equals_srrg_fp	FCA attractor diagonal fixed point = SRRG fixed point at $g^* = 1/\varphi$	SRRGCABridge	Zero sorry

Theorem / Module	Statement	File	
op9_catal_unconditional	$K_{\text{alg}}(g^*) = 0$; unique K_{alg} zero on $(0, \infty)$; strict positivity on $(0, g^*)$; OP9 $\mathbf{A}_{\text{Lean}}$ for canonical SRRG scalar profile	srrg-lean/ Bridges/ ToMDL.lean	Zero sorry, unconditional
srrg_op9_k_alg_biconditional	IsSrrgFixedPoint $\Leftrightarrow$ K_{alg} minimiser; full OP9 biconditional for algebraic MDL profile	srrg-lean/ Bridges/ ToMDL.lean	Zero sorry, no h_{ugp} hypothesis
srrg_mdl_functional_identity	descLen $= B -$ Viability; $K[S] = B - F[S]$	ToMDL.lean	Zero sorry
argmax_viability_iff_argmin_descLen	argmax $F[S]$ = argmin $K[S]$ (OP9 algebraic core)	ToMDL.lean	Zero sorry
srrg_fp_is_mdl_minimizer	SRRG fixed point minimises description length	ToMDL.lean	Zero sorry

A.11 Cyclotomic $\mathbb{Z}_7$ / Galois Structure

Theorem / Module	Statement	File	
z7_galois_orbit_partition	σ_2-orbits: $\{1, 2, 4\}$ (u-type) and $\{3, 5, 6\}$ (d-type)	CyclotomicZ7Galois.lean	Zero sorry
seven_not_divides_120	$7 \nmid 120$; $\mathbb{Q}(\zeta_7) \not\subset \mathbb{Q}(\zeta_{120})$	CyclotomicZ7Galois.lean	Zero sorry
gte_arithmetic_sectors_disjoint	GTE mass and vacuum arithmetic sectors disjoint	CyclotomicZ7Galois.lean	Zero sorry
cyclotomic_z7_master_bundle	$\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) = \mathbb{Z}_2 \times \mathbb{Z}_3$; CPT as Galois automorphism; $\mathbb{Z}_3$ orbits = GTE flavour families	CyclotomicZ7Galois.lean	Zero sorry

A.12 P44–P47: Quantum Gravity, Three-Tape CMCA, Polynomial UFT, Cosmological Predictions

Theorem / Module	Statement	File
minimal_coupling_is_mdl_minimal	$\xi = 0$ uniquely MDL-minimal; certified in P43 and confirmed for P44	MinimalCoupling.lean
z7_superselection_preserved_by_flat_metric	$\mathbb{Z}_7$ superselection preserved on curved backgrounds	MinimalCoupling.lean
gte_orbit_vectors_are_rs_codewords	$\text{Gen}_{1,2,3}$ orbit vectors are $[5, 3, 3]_7$ RS codewords; GTE Holographic Encoding route	RSCodeOrbit.lean
cyclotomic_z7_master_bundle	Galois $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) = \mathbb{Z}_2 \times \mathbb{Z}_3$; CPT as Galois automorphism	CyclotomicZ7Galois.lean

Theorem / Module	Statement	File	
DPP (Dimensional Protocol Principle)	Shared clock τ_c^{out} necessary and sufficient for 3+1D dynamics	P45	
Lorentz algebra ($\mathfrak{so}(1,3)$)	All 12 $\mathfrak{so}(1,3)$ commutation relations machine-certified	P45	
Color confinement MDL	$\Delta K = \log_2 9$ description-length gap forbids free quarks	P45, P46	
Baryon number topological	$B = (1/3)\sum \chi_q(w_j)$; derived from $N_{\text{tapes}} = 3$	P45, P46	
Holographic 3-tape	$3L/L^3 \to 0$; Bekenstein–Hawking area scaling realized	P45	
Fermionic statistics	From $\mathbb{Z}_7^*$ non-primitive roots	P45, P46	
Polynomial MDL uniqueness	$p(L,C,R)$ unique degree-3 multilinear over GF(7) satisfying $\mathbb{Z}_7$, Rule 110, 19-bit MDL	P46	
$\mathbb{Z}_7\times\mathbb{Z}_3$ MDL uniqueness (unconditional)	Non-circular derivation MDL $\to \mathbb{Z}_7\times\mathbb{Z}_3$ closed $\mathbf{A}_{\text{Lean}}$: `mdl_ca_rule_coding_closed`, `z5_fmdl_no_psc_kink_orbits` (GF(5) poly, winding-based), `mdl_total_z7z3_strictly_beats_z5z3`; zero SM input	ugp-lean	All zero sorry
SRRG golden ratio	$1/\varphi$ is unique positive root of $p(x,x,x) = x$ over $\mathbb{R}$	P46	
Vacuum stability (poly)	$p(0,0,0) = 0$; $w = 0$ unique fixed point of $p(x,x,x) \equiv x \pmod 7$	P45, P46	
`baryon_number_l1_l2_correspondence_certified`	$B_{L1} = B_{L2}$ (exact baryon number preservation across levels)	`BaryonNumber`	Zero sorry
`phimdl_axial_current_topological`	$\partial_\mu J_A^\mu = 0$ topological conservation at Level 2	`ChiralCurrentL2`	Zero sorry
`gibbs_sector_unique_minimizer`	Gibbs state uniquely minimises free energy at each Z_7 sector	`C2CoherenceG40`	Zero sorry
`gte_polynomial_five_labelled_roles`	GTE polynomial p has exactly 5 labelled semantic roles	`FiveRolesPolynomial`	Zero sorry
`phimdl_free_propagator_formula`	Φ_{MDL} free scalar propagator $G(p) = 1/(p^2 + m_{\text{kink}}^2)$	`PhiMDLPropagator`	Zero sorry
`phimdl_quartic_coupling`	Quartic vertex $\lambda_4 = m^2 \times 7^2$ (from Z_7 symmetry)	`PhiMDLPropagator`	Zero sorry
`phimdl_sextic_coupling`	Sextic vertex $\lambda_6 = m^2 \times 7^4$ (from Z_7 symmetry)	`PhiMDLPropagator`	Zero sorry

Theorem / Module	Statement	File	
`phimdl_satisfies_wightman_axioms`	Φ_{MDL} satisfies Wightman axioms (structural; OS reconstruction + distributional smearing named axioms)	`WightmanAxioms`	Structural CatAD
`gte_bell_violation_at_half_coupling`	GTE qutrit sector achieves $S_{\mathrm{CHSH}} > 2$ at half-coupling	`BellViolationGTE`	Named axiom
`gte_hgrav_diagonal_nontrivial`	H_{grav} diagonal part nontrivial (condition for CHSH violation)	`BellViolationGTE`	Zero sorry
`discrete_schrodinger_composition`, `discrete_evolution_unitary`, `discrete_schrodinger_step_recurrence`	Discrete Schrödinger $U(n) = e^{-iH_{\mathrm{cyc}} n \tau_c}$: composition, unitarity, step recurrence	`CogwheelDynamicsG21`	Composition, recurrence zero sorry; unitarity CatAD (two pending lemmas)
n_s binary entropy (P47)	$n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ from holographic running; 14 certified theorems	P47	Zero sorry
Ω_Λ PSC epoch count (P47)	$\Omega_\Lambda = (\ln 2/3\pi)\log_2(2000/3) = 0.6899$; orbit count $2000/3 = D^2 N_{\mathrm{fam}}^{N_{\mathrm{gen}}}/N_{\mathrm{gen}}$ from DPP first principles	P47	CatAD
Ω_Λ holographic (P47)	$\Omega_\Lambda = 3\pi/14 = 0.6732$ from three-tape mode count; $\tau = 3/7$ Rule-110 ether rate	P47	CatAD
$w = -1$ (P47)	Equation of state fixed by PSC boundary condition; Weinberg no-go evaded	P47	CatAD
CKM δ_{CP} via S_3 (P47)	$\delta_{CP} = \pi/2 - 3/8 = 68.51°$; 0.017% from PDG; S_3 subgroup chain	P47	CatAD
`c_gorard_eq_n_spatial_over_two_dsq`	$C_{\mathrm{Gorard}} = N_{\mathrm{spatial}}/(2D^2) = 3/32$ exact; $D = 4$ from DPP	`TemporalVoxelCC` (P47)	Zero sorry ($\mathbf{A}_{\mathrm{Lean}}$)
`z7_shift_measure_preserving`	$\mathbb{Z}_7$ shift $\phi \to \phi + 2\pi/7$ is measure-preserving; scalar anomaly-free	`Z7AnomalyFree.lean` (ugp-lean)	Zero sorry
`z7_jacobian_eq_one`	Radon–Nikodym derivative of shifted measure = 1 a.e. (no Jacobian anomaly)	`Z7AnomalyFree.lean`	Zero sorry
`lebesgue_shift_invariant`	Lebesgue measure shift-invariant: $\forall c$, map$(\cdot + c) =$ volume	`Z7AnomalyFree.lean`	Zero sorry
`z7_vacuum_sectors_equiprobable`	All 7 $\mathbb{Z}_7$ vacuum sector integrals equal (quantum vacuum degeneracy; pure-φ sector, $\mathbb{Z}_7$-periodic observables)	`Z7AnomalyFree.lean`	Zero sorry

Theorem / Module	Statement	File	
phimdl_potential_su2l_invariant	L_2 norm preserved under $SU(2)$ rotation on within-tape doublet	GaugeMDL.lean	$\mathbf{A}_{\text{Lean}}$
su2l_covariant_derivative_minimal	Covariant derivative is MDL-minimal upgrade of $\partial_\mu\lvert\Psi\rvert^2$	GaugeMDL.lean	$\mathbf{A}_{\text{Lean}}$
su2l_wpm_generator_algebra	$W^\pm$ generators satisfy $SU(2)$ Lie algebra	GaugeMDL.lean	$\mathbf{A}_{\text{Lean}}$
su2l_l2_from_phimdl_potential_catad	Master bundle: full $SU(2)_L$ MDL gauging chain	GaugeMDL.lean	$\mathbf{A}_{\text{Lean}}$

A.13 New Machine Certifications (2026)

Theorem / Module	Statement	File	Notes
fgci_unique_at_nc, fgci_bundle + 18 theorems	$F(N_c) = G(N_c) = 73$ uniquely at $N_c = 3$ (FGCI); $b_{L_2} = 42 = 2N_c\lvert Z_7\rvert$	FrobeniusChain.lean (20 thms)	Zero sorry
elegant_kernel_full_certification	All 9 UCL Elegant Kernel coefficients packaged in one master theorem	ElegantKernel/Unconditional/MasterCertification.lean	Zero sorry
ucl_fermion_mass_ordering	$m(\text{gen}_1) < m(\text{gen}_2) < m(\text{gen}_3)$ for lepton, up, down sectors	UCLMassOrdering.lean	Zero sorry
ucl_tier3_lepton_koide	Full Koide identity $Q = 2/3$ from phase + amplitude separately certified	UCLKoide.lean	Zero sorry
lepton_koide_amplitude_pinned	$b = \sqrt{2}$ from S_3 equal-irrep-norm cone condition forces $Q = 2/3$ for all θ	KoideYukawaAmplitude.lean	Zero sorry
higgs_quartic_corrected	SRRG-corrected λ; $m_H = 125.2499$ GeV (-0.0003σ)	MassRelations/HiggsQuartic.lean	Zero sorry
two_cH_plus_one_eq_ngen_cubed	$2c_H + 1 = 27 = N_{\text{gen}}^3$; unique at $N_{\text{gen}} = 3$	MassRelations/HiggsQuartic.lean	Zero sorry
pmns_solar_sin_sq	$\sin^2\theta_{12} = 4/13$ from orbit ratios	MassRelations/NeutrinoSector.lean	Zero sorry
pmns_atm_sin_sq	$\sin^2\theta_{23} = 19/42$ from orbit ratios	MassRelations/NeutrinoSector.lean	Zero sorry
pmns_reactor_sin_val	$\sin\theta_{13} = 11/73$ from orbit ratios	MassRelations/NeutrinoSector.lean	Zero sorry
gte_jarlskog	$J_{\text{GTE}} = -(8184\sqrt{437}/5057221)\sin(\pi/7)$	MassRelations/NeutrinoSector.lean	Zero sorry

Theorem / Module	Statement	File	Notes
pmns_cp_phase_from_z7_winding	$\delta_{CP}^{\mathrm{PMNS}} = 205.71^\circ$ from $\mathbb{Z}_7$ winding	MassRelations/ NeutrinoSector.lean	Zero sorry
fn_dirac_yukawa_rank_theorem	Rank-1 structure of Dirac Yukawa from FN charge assignment	MassRelations/ NeutrinoSector.lean	Zero sorry
yukawa_overlap_exponent_catad	Overlap exponent $\alpha = N_c - 1 = 2$ from DPP tape counting	Gravity/ YukawaOverlap Exponent.lean	Zero sorry
leptogenesis_kink_overlap_ catad	Leptogenesis η_B suppression bundle; links kink overlap to η_B	Gravity/ YukawaOverlap Exponent.lean	Zero sorry
z7_dark_baryon_correction_ identity	$q_{\mathrm{dark}}/(\lvert Z_7\rvert - 1) = 1/N_c$; $D_{\mathrm{top}} = e^{-1/N_c}$ ($\mathbf{A}_{\mathrm{Lean}}$)	ugp-lean	Zero sorry
z7_star_transitivity_under_ addition	$\mathbb{Z}_7^*$ is transitive under addition; forces equal sector distribution	GUTStructure.lean	Zero sorry
z7_symmetry_forces_equal_ sector_action	$S_{\mathrm{per}} = 2/6 = 1/N_c$; equal-weight distribution from Z_7 symmetry	GUTStructure.lean	Zero sorry
d_top_derivation_chain_catal	Full derivation chain: $D_{\mathrm{top}} = e^{-1/N_c}$ from Z_7 group theory alone; no dilute instanton gas approximation	GUTStructure.lean	Zero sorry
integral_sech_cubed	$\int_{-\infty}^{\infty} \mathrm{sech}^3 = \pi/2$ (Mathlib FTC)	Substrate/PhiMDL Fluctuation Spectrum.lean	Zero sorry
integral_sech	$\int_{-\infty}^{\infty} \mathrm{sech} = \pi$ (Mathlib FTC)	Substrate/PhiMDL Fluctuation Spectrum.lean	Zero sorry
phimdl_fluctuation_is_poschl_ teller	$V_{\mathfrak{fl}}(x) = m_\varphi^2[1 - 2\,\mathrm{sech}^2(m_\varphi x)]$ is exactly $s = 1$ Pöschl–Teller	Substrate/PhiMDL Fluctuation Spectrum.lean	Zero sorry
lambda_all_order_zero_pure_phi	All computable quantum corrections to Λ vanish identically in the pure-φ sector (closes OP3)	P48	Zero sorry ($\mathbf{A}_{\mathrm{Lean}}$)
va_op6_bundle	Quantitative V$-$A chiral coupling: $g_R = 0$, $g_A/g_V = -1$ from GTE first principles (substantially closes OP6)	P48	CatAD
f21_is_borel_psl27	F_{21} is the Borel stabilizer of ∞ in $\mathrm{PSL}(2,7)$	P52	Zero sorry ($\mathbf{A}_{\mathrm{Lean}}$)
pgl27_generated_by_singer_and_ borel	Fibonacci–Möbius generator and F_{21} generate $\mathrm{PGL}(2,7)$ of order 336	P52	Zero sorry ($\mathbf{A}_{\mathrm{Lean}}$)
psl27_is_aut_fano	$\mathrm{PSL}(2,7)$ is the full automorphism group of the Fano plane $\mathcal{F}$	P52	Zero sorry ($\mathbf{A}_{\mathrm{Lean}}$)

Theorem / Module	Statement	File	Notes
f21_regular_on_fano_flags	F_{21} acts simply transitively on all 21 incidence flags of the Fano plane	P52	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
eisenstein_a4_from_inert_2	Inert prime 2 in $\mathbb{Z}[\omega]$ gives $\mathbb{Z}[\omega]/(2) \cong \text{GF}(4)$ with additive group V_4 and multiplicative group $\mathbb{Z}_3$, assembling to $A_4 = V_4 \rtimes \mathbb{Z}_3$	P52	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
klein_quartic_genus_eq_n_gen	The Klein quartic has genus = $N_{\text{gen}} = 3$ (from the $(2,3,7)$ triangle group)	P52	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
gte_manifest_flavor_is_s3_in_a4	GTE manifest neutrino flavor symmetries μ_3 and μ–τ exchange generate $S_3 \leq A_4$	P52	Zero sorry ($\mathbf{A}_{\text{Lean}}$)

A.14 P53–P54: Framework Assessment and Synthesis

Theorem / Module	Statement	File	Notes
algebraic_lifting_theorem	Every property certified at discrete Level 1 must exist in the continuum Level 3 field Φ_{MDL}: CA theorems are physics theorems	Spacetime/LiftingTheorem.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
algebraic_descent_theorem	Every conservation law, charge-quantisation condition, and generation structure in any continuum model realising the F_{21} winding algebra must respect discrete Level 1 constraints	Universality/AlgebraicDescentTheorem.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
no_finite_ca_exact_lorentz_replica	No finite cellular automaton can exactly replicate Lorentz-invariant dynamics; any CA with finite lattice spacing carries Lorentz-violating artefacts	Framework/CMCAContinuumLimit.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
ngen_universality_seven_constraints	Seven independent structural constraints (PSC-PI, DPP, CMCA, TPC, Gorard, GTE cascade, MDL orbit) are each individually certified at $\mathbf{A}_{\text{Lean}}$ and all force $N_{\text{gen}} = 3$ unconditionally	Classification/NgenUniqueness.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
ngen_universality_master	All seven structural constraints plus the bracket-admissibility bound together force $N_{\text{gen}} \leq 3$; combined with the GoE lower bound, $N_{\text{gen}} = 3$ exactly	Classification/NgenUniqueness.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
f21_theta_term_vanishes	$\theta_{\text{QCD}} = 0$ is a structural identity of the F_{21} colour group; the exponent-sum condition and $\mathbb{Z}_3$ order-3 action are machine-certified	Universality/SylowIndexCouplingHierarchy.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
gte_baryon_number_topological_charge	$3B = \chi_{\text{baryon}}(w_x) + \chi_{\text{baryon}}(w_y) + \chi_{\text{baryon}}(w_z)$; baryon number is a topological charge over the three-tape winding labels	Algebra/BaryonNumber.lean	Zero sorry ($\mathbf{A}_{\text{Lean}}$)

Theorem / Module	Statement	File	Notes
`incompleteness_implies_nonzero_omega_lambda`	PSC physical incompleteness forces a positive residual $D_{\text{res}} > 0$ and thereby a nonzero $\Omega_\Lambda > 0$ from the PSC epoch count	`Gravity/PSCEpochSelection.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`past_hypothesis_from_mdl_seed`	MDL-minimal seed selects flat spatial curvature $k = 0$: Σ_{FLRW}-bits $= \log_2 3$ and $k = 0$ minimises the description length	`Cosmology/MDLInitialState.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)

Note. P53 (*The GTE Framework: A Comparative Assessment*) is a comparative survey paper; it introduces no new axioms or physical-quantity rows but presents the nine theorems above as load-bearing architectural results. P54 (*The Fire in the Equations*) is a philosophical synthesis of the GTE and NEMS frameworks; it introduces no new physical-quantity rows or Lean certifications in the UGP Physics series.

A.15 P55: Octonionic Shadow of GF(7)

Theorem / Module	Statement	File	Notes
`D_difference_set`	$D = \{1, 2, 4\}$ is the unique $(7, 3, 1)$ planar difference set in $\mathbb{Z}/7$	`OctonionShadowInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`fanoLines_are_D_translates`	The seven translates of D are the seven lines of the Fano plane	`OctonionShadowInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`pencil_card`	Exactly $\lvert D\rvert = 3$ lines pass through every point (pencil $= N_c = 3$)	`OctonionShadowInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`hurwitz_presentation_order_168`	$\lvert\text{PSL}(2, 7)\rvert = 168$ certified by 168-coset Todd–Coxeter enumeration	`HurwitzCosetCertificate.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`Nc_factorization`	$N_c = 3$ from the pencil count of the $(7, 3, 1)$ design via $F_{21} \hookrightarrow G_2$ and $\text{Stab}_{G_2}(\text{apex}) = \text{SU}(3)$	`OctonionShadowInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`weights_card_from_pencil`	The weight set cardinality equals the pencil count, identifying the color charge sector with the Fano geometry	`OctonionShadowInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`derivation_algebra_is_g2_certificate`	$\text{der}(\mathbb{O}) = \mathfrak{g}_2$ (dimension 14 exact) from first-principles bracket closure and Killing-form certificate	`G2StabilizerCertificate.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`apex_stabilizer_is_su3_certificate`	$\text{Stab}_{G_2}(\text{apex}) = \mathfrak{su}(3)$ (dimension 8 exact)	`G2StabilizerCertificate.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)
`G6_rigidity_conditional`	UGP's flavor package (μ_3, U, V_4, A_4) is equivariantly isomorphic to the Spin(8) triality package; $\text{gen}_1 \leftrightarrow V$ from Eisenstein norm uniqueness $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$	`TrialityInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)

Theorem / Module	Statement	File	Notes
`seesaw_triality_pinning_bundle`	Right-handed seed ordering $b_{R,1} < b_{R,2} < b_{R,3}$ pins mass eigenstates ν_k to generations k and predicts normal mass ordering	`TrialityInterface.lean`	Zero sorry ($\mathbf{A}_{\text{Lean}}$)

Note. P55 (*The Octonionic Shadow of GF(7)*) formalizes the division-algebra bridge in eleven zero-sorry Lean 4 modules (174 theorems total, all in `ugp-lean`). Key physical-quantity predictions corroborated by P55 via the octonionic route: $N_c = 3$ ($\mathbf{A}_{\text{Lean}}$, independent second derivation); the SM charge spectrum; $N_{\text{gen}} = 3$ from triality ($\mathbf{A}_{\text{Lean}}$); normal neutrino ordering ($\mathbf{A}_{\text{Lean}}$); Koide phase $2/9 + 7.4 \times 10^{-6}$ ($\mathbf{A}$). These quantities were first derived in P48 from GTE arithmetic; P55 supplies a structurally independent confirmation.

E.8 Summary: Quantity and Axiom Counts

Table 57: Grand total count summary for the UGP Physics corpus.

Category	Count
Structural/arithmetic inputs (derived from axioms)	**16**
External measured inputs (from experiment)	**5**
Calibrated parameters (Cat B groups)	**3 groups ($\sim$ 18 scalars)**
Total "inputs"	**25–35**
UGP core axioms (A1–A3)	**3**
PSC Layer I axioms (incl. 1 proved: AC)	**5**
PSC Layer II conditions (incl. 1 proved: PI)	**2**
GTE construction principles (all proved)	**6**
$[D]$-constraints (3 of 5 proved)	**5**
SRRG axioms (3 proved, 5 open)	**8**
Cook bridge axioms (1 discharged, 6 open)	**7**
GTE universality bridge axioms (all open)	**8**
P44 architecture restrictions (2 proved, 1 open: AR3)	**3**
Analytic number theory axioms (not in Cat A closure)	**2**
Total axioms / principles cataloged	**49**
Open (truly assumed) axioms	**$\approx$ 28–31**
Proved theorems (previously listed as axioms)	**$\approx$ 17–20**
Physical quantities (P00–P55)	**$\approx$ 280**
$\mathbf{A}_{\text{Lean}}$ (Lean-certified) quantities	**$\approx$ 163**
Discrepancies documented and resolved	**10**
New $\mathbf{A}_{\text{Lean}}$ certifications (Appendix A.13)	**29**
New $\mathbf{A}_{\text{Lean}}$ certifications (Appendix A.15, P55)	**174 theorems, 11 modules**

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